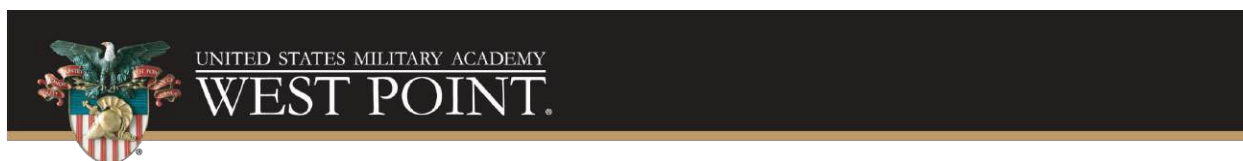




MA104

Single Variable

Calculus



MA104 SINGLE VARIABLE CALCULUS

ABET eCOURSEBOOK

AY2021-2026

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UNITED STATES MILITARY ACADEMY
WEST POINT.

SECTION I

COURSE ASSESSMENTS



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB A

AY2021 COURSE ASSESSMENT



REPLY TO
ATTENTION OF:

DEPARTMENT OF THE ARMY
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

MADN-MATH

7 June 2021

MEMORANDUM THRU COL Joseph Lindquist, Core Program Director, Department of Mathematical Sciences, United States Military Academy, West Point, NY 10996

COL Hartley, Professor and Head, Department of Mathematical Sciences, United States Military Academy, West Point, NY 10996

FOR Dean of the Academic Board, United States Military Academy, West Point, NY 10996

SUBJECT: Annual Executive Summary for MA104, AY 21-2

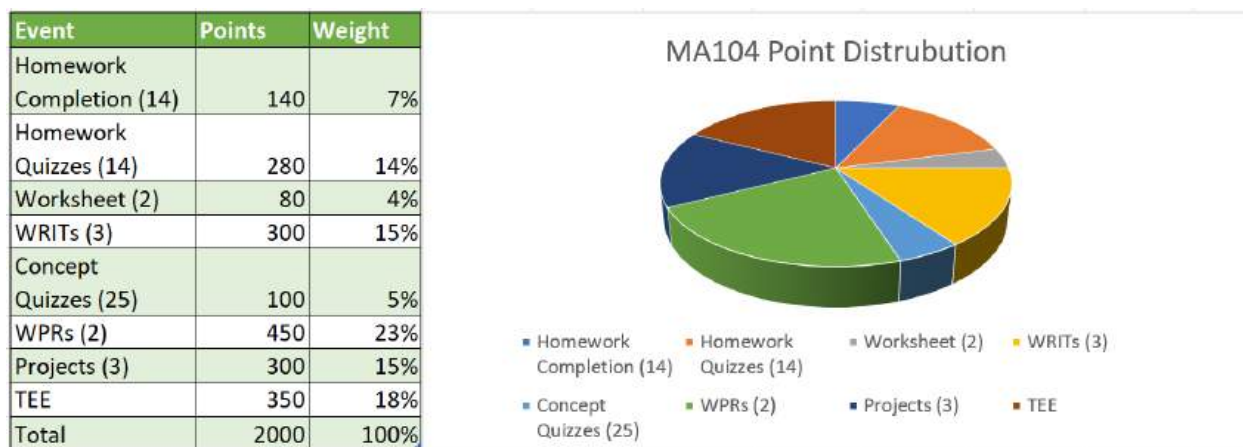
1. **Course Description.** MA104 is the second course of the mathematics core curriculum capitalizing on MA103's mathematical modeling and introduction to vectors and calculus. The goal of MA104 is for students to internalize the concept of the derivative as a rate of change, the integral as the accumulation of change, and that these two concepts are inversely related by the fundamental theorem of calculus. Students will realize that both differential and integral calculus arise from the application of the limit to an estimate. An understanding of course material is enhanced with the use of Mathematica and Microsoft Excel. MA104 services MA205 (Multivariable Calculus), MA206 (Probability and Statistics), PH205 (Physics), SS201 (Economics), the Core Engineering Sequences, and selected majors. This course is followed by MA205 or MA206.

2. **Course Composition.** The structure of MA104, AY21-2, included a 64-lesson sequence in which points are accumulated in a series of assessments of increasing value by topic. Assessments included concept quizzes, homework assignments, homework quizzes, WRITs, WPRs, and projects; and these assessments were divided into two distinct blocks during the semester. The first 9 weeks of the course explored differentiation concepts, computations, and applications while the last 8 weeks explored integration concepts, computations, and applications. Each lesson was associated with a larger grouping of lessons involving a common topic, and each common topic had its own series of assessments designed to reinforce student comprehension. The course was taught primarily in a hybrid modality, where half of the students were physically present in the classroom and the other half attended the class virtually using Microsoft Teams. Physical attendees rotated lesson to lesson in a larger attempt to achieve face to face instruction within a COVID-19 restricted environment.

a. Classroom Design. The course designed the classroom experience for students based on the mathematical ability they displayed in MA103 – Mathematical Modeling and Introduction to Calculus. This experience included four factors based on student ability: sectioning like-students together, specifying the number of students in each section (Low - 12, Intermediate - 15, High - 18), teaching modality, and deliberately assigning instructors to sections based on their experience teaching similar student populations.

Recommendations for Classroom Design. We recommend sustaining the sectioning of students by ability and attempting to engineer a classroom experience that better enables their learning of calculus. Some changes we plan to explore for future MA104 courses is to consider a more comprehensive rating of student ability when building sections. We recommend weighting FCE statistics, prior calculus experience, and performance in MA103 to get a more complete understanding of the students and gain the ability to intelligently section high performing students, as well as the marginal performing ones. We think that with a more deliberate sectioning methodology and a common teaching modality across the course, we can gain valuable insight about how to best teach our students in the future. See Appendix 1. Course Analysis based on Sectioning for details on the effects we saw in AY21-2. We support sectioning throughout AY22 to further analyze and gain a better understanding the impacts of our classroom design on students without the confounding effects of teaching modality.

b. Grading Scheme. The overall grading scheme for the course is below.



Recommendations for Grading Scheme. We recommend retaining the overall grading scheme and the course composition of low stakes assessments building into more significant tests of student learning. See Appendix 2. Quantitative Trace Analysis for details on the overall effectiveness of course assessments on student learning.

c. Homework. The course incorporated 14 homework sets throughout the term. Each homework set was due the lesson after the final topic associated with the lesson group and was submitted digitally using Blackboard. The homework sets were built to have two components for each individual lesson. The first was a set of concept questions, meant to guide students through reading of the textbook before class, and the second were two practice problems intended to be completed after class, that would reinforce what was covered that day by the instructor. When it was reasonable, the numerical solution or answer to practice problems was provided within the homework set to allow students to know when they had done a problem correctly. The homework sets were for completion credit only and accounted for 7% of course points.

Recommendations for Homework. Some recommended changes we plan to explore for future MA104 courses are the following:

- 1) In addition to the two practice problems for each lesson, include an additional optional challenge problem to encourage deeper learning and higher engagement with the course material.
- 2) Sustain providing the numerical answers or solutions to those practice problems.
- 3) Students physically submit homework and instructors score 7 of 10 points based on completion and the final 3 points on a plus/check/minus scale. This will motivate student effort and encourage better preparation.

d. Concept Quizzes. The course incorporated 30 daily WebAssign concept quizzes throughout the term – only 25 were graded to account for missing class and/or the occasional poor performance. Each 5-minute concept quiz asked 4 questions regarding topics covered in the reading from the previous night's homework concept questions. Students were allowed and encouraged to use their completed homework concept questions as a reference during these quizzes. This served to motivate nightly preparation for class and confirm the value in reading the material ahead of time. Questions were typically multiple choice, multiple select, or true/false. The prep quizzes accounted for 5% of the course points and served to motivate students to engage with the material prior to attending class.

Recommendations for Concept Quizzes. Our overall recommendation is to maintain the daily concept quizzes as they help students to keep pace with the material and establish good nightly preparation habits. See Appendix 1. Course Analysis based on Sectioning for qualitative feedback from students on the merits of the concept quizzes. Some recommended changes we plan to explore for future MA104 courses are the following:

- 1) Incorporate a single question from an earlier topic to interleave or create a distributed learning opportunity for students. Consider employing a 28-day reach-back window for topics to help students form lasting connections and encourage deep learning.
- 2) Consider making all concept quizzes 5 questions, but only worth 4 points. Allow cadets to choose which 4 problems to answer, to include the reach-back question, for each quiz. This should reduce the cadet feeling that quizzes are written in a manner that prevents them from performing well.

e. Homework Quizzes. The course assessed student learning across common topics using 14 homework quizzes, with one corresponding to each homework set. These quizzes were completed at the conclusion of each topic and were executed in a purely digital medium using student digital writing capability and Blackboard submissions. Homework quiz questions came almost directly from the practice problems within the homework sets and were completed without any references. Homework quizzes were incorporated within the course as a way to mitigate the effects of students using *Slader* or other online resources to complete homework assignments without ever truly engaging or learning the material.

From AY21-1 into AY21-2, MA104 elected to use stringent quiz rubrics to provide very blunt feedback to students in the form of low scores to communicate how well they knew the material. The course also permitted students to revise the quizzes and submit corrections for up to 14/20 points to account for the binary nature of the rubrics. Homework quiz solution videos were created by course leadership to assist students in completing revisions and understanding the material.

See Appendix 1. Course Analysis based on Sectioning for qualitative feedback from students on the effectiveness of the Homework quizzes and revisions process.

Recommendations for Homework Quizzes. Our recommendation is to sustain the homework quiz structure but incorporate some of the following changes into future MA104 courses:

- 1) Include two true/false or multiple-choice questions in addition to the computation problem to reinforce course concepts with more consistency.
- 2) Provide two computational questions and all students the choice of which one to complete.
- 3) Consider including a reach-back computational problem to interleave key course skills and create a distributed learning opportunity.
- 4) Sustain solution videos – but modify the problems coefficients or exponents to demonstrate correct mechanics, but not provide students an ability to simply directly copy the answer for points.

f. WRITs. The course consisted of three WRITs. The WRITs were timed exams that, in total, accounted for 15% of the course. The purpose of the WRIT was to provide the student with an assessment of their understanding of key concepts and their ability to conduct fundamental calculus computations. The WRITs were broken into four primary problems: a) True/False and Multiple-Choice questions, b) graphical short answer questions, c) general short answer questions, and d) computational short answer questions. WRITs were completed without the use of any student references. The frequency of the WRITs provided the students with closer recall assessments and served as an evaluation tool for instructors to identify students who may need extra assistance on certain topics.

Recommendations for WRITs. Overall recommendation is to sustain the general format and frequency of the WRITs. Some recommended changes we plan to explore for future MA104 courses are the following:

- 1) Allow the cadets to choose n problems to answer from p questions – especially within the computational short answer portion of the WRIT. This should reduce the cadet feeling that WRITs are written in a manner that prevents them from performing well.
- 2) Specific to the integration WRIT 3, telegraph the skill that students should demonstrate i.e. substitution rule, integration by parts, or to apply integration strategies. Additionally, focus on indefinite integrals within computation questions to ensure students perform the desired skills as opposed to just using their calculators to arrive at the numerical answer.

g. WPRs. The WPRs were application focused assessments designed to evaluate students' ability to apply calculus as a problem-solving tool to address real-world scenarios. Each major block (Differentiation and Integration) culminated in a WPR event that was designed to take students 55 minutes to complete. Students were allowed to use Mathematica and their textbook with any handwritten notes inside. Because of the difficulty of the problems, students were afforded 75 minutes to complete the exam and they were provided with opportunity to correct and learn from their mistakes on WPR 1 within a revision assignment. The WPRs accounted for 23% of the course points.

Within AY21-2, a high number of students were under restriction of movement due to COVID-19 testing or exposure. As a result, the course executed a completely online version of WPR 1 to account for roughly 20% of students being unable to test in-person. The WPR was administered using Blackboard, and it consisted of four question pools based on problem type, with each pool having five versions of a question. Problems were presented to the students one at a time and in a random order, and the test did not

allow students to back-track. The structure of the test proved highly effective in the prevention of cheating but was extremely challenging to grade and marginalized some students.

Also within AY21-2, we attempted to challenge high-performing students with stretch bonus questions on WPR 2 to see how thoroughly they understood the material. While this did separate high performers to some degree, many students did not even attempt those questions and so we did not feel like we got productive feedback on those problems from across the course. The confident students with high A's in class benefited from the bonus points but the optional nature of the questions failed to give us any insight into the students deeper understanding of the calculus concepts.

Recommendations for WPRs. Overall, the course recommends that we sustain the general format of the WPRs. Continue to provide simple applications of the material covered and avoid complicated scenarios. Sustain using the 75-minute Dean's Hour periods for testing and giving students alternatives in which questions they complete. Some recommended changes we plan to explore for future MA104 courses are the following:

- 1) Eliminate the use of Mathematica on WPRs. While it is a powerful tool and allows the course to approach real-world scenarios more easily in some circumstances, it seems like Mathematica presents significant challenges to some students and limits real engagement with the material in others. We suggest crafting problems that are easily accessible using the issued cadet calculators and knowledge of course concepts.
- 2) Eliminate the use of the Stewart Textbook as a reference. Again, students misjudge how helpful their textbook is or simply write strings of example problems to reference during the exam. Recommend creating an MA104 reference card with relevant formulas, problem solving steps, and calculator assistance notes for students to reference during WPRs.
- 3) Structure questions so that it is possible to assess the high performing students and allow some separation between A and A+ performances. Consider assigning 80-85% of a problems value to required skills and 15-20% to an aspect of the problem that challenges students beyond just the basic question types.

h. Projects. Three projects were designed to give students the opportunity to apply calculus as a problem-solving tool to a real-world military scenario. The projects revolved around the ongoing SpaceX Falcon Heavy and Starship experimental launches and missions. Designated lab periods with guided questions for students were allocated for instructors to be able to assist students in starting the projects.

Project 1 resulted in an individual technical report, Project 2 was a team technical report with a reflection on Project 1 mistakes, and Project 3 was an oral group presentation.

Recommendations for Projects. We recommend reducing the scope and number of projects. The SpaceX projects proved to be difficult for students to really approach and made writing a compelling technical report challenging. With a tighter scoped project, students can better demonstrate the concise and clear writing we are looking to develop. We also recommend only two projects – one technical report with revisions and another oral presentation with slide corrections. These projects could be focused on optimization and probability, which are clear and relevant applications of calculus. This structure would provide students clear opportunities to demonstrate and refine skills we want to develop within MA104 without overwhelming them with additional content. Finally, for the oral presentation of projects, we need to make very clear the expectations for works cited and acknowledgement of outside sources.

i. Term End Exam. The TEE accounted for 18% (350 Points) of the cadet grade. The TEE was a two-part assessment, the first part focused on concepts and computations and the second part focused on applications. Similar to the WRITs, no references were authorized for Part 1, and just like the WPRs, Mathematica and the Stewart Textbook was authorized for Part 2. The TEE was executed in-person and on paper. Due to the condensed TEE schedule, the MA104 TEE was a 2.5 hour exam instead of the historical 3.5 hour exam.

Recommendations for the TEE. We recommend sustaining the format and point value of the TEE. The structure of the test is familiar to students in that Part 1 is very similar to the WRITs they saw throughout the semester and Part 2 mirrors their WPRs. We recommend carrying forward the WRIT and WPR recommendations in terms of Mathematica and reference usage and providing students a range of different problems to select to demonstrate their understanding. We should plan for future TEEs being 3.5 hours; if this happens, we should consider expanding the scope (length) of the exam slightly as well as giving students some of the extra time to allow them to work through more challenging problems.

j. Course Validation. During AY 21-1 we provided first year students the opportunity to validate MA104 and proceed directly to MA205. 332 Students who were not selected for AMP but performed well in summer testing were sent an e-mail asking them to sign up to take the MA104 validation exam. 78 of these targeted students took the validation exam. Of the 78 students who took the exam, 42 passed and moved on to MA205. Of the 42 students who validated, 19 excelled (B+ or better) in MA205, 16 performed marginally well (between a C+ and B), and 7 performed poorly (C or below). Of the 36 students who did not pass the validation exam, 22 excelled (B+ or better) in

MA104, 8 performed marginally well (between a C+ and B), and 6 performed poorly (C or below).

Recommendations for the Course Validation Process. We recommend that we sustain scheduling the validation exam within the first 3 weeks of classes to maximize participation. We also intend to continue offering a live (and recorded) differentiation and integration review session for potential testers to improve their confidence and willingness to test. We will also intend to speak within an MA103 PD time to help instructors give notice to their students with validation potential.

3. Course Statistics. The course average for AY 21-2 was an 84.62%. The distribution of grades is below. NCs were given to 9 cadets who are suspected of an honor violation on Project 3, one cadet who experienced a personal issue just prior to the TEE, and one cadet who was on a cognitive profile when the TEE was administered.

MA104 AY 21-2 Grade Distribution	
Grade	Count
A+	21
A	101
A-	117
B+	109
B	141
B-	85
C+	67
C	56
C-	39
D	34
F	9
NC	11
Course Average	84.62

4. Course Goals Assessment. The course had two specific goals that will be assessed, with additional comments on two of the six primary goals of the core mathematics sequence.

Academic Program Goal	Core Math Goal	Assessment Indicators (Trace Analysis)
Graduates think critically and creatively	Acquire a body of knowledge for continued mathematical development (Para 4.c)	- Concept Quizzes - Homework Quizzes - WRITs - WPRs - Projects - TEE - Qualitative Course Survey

Graduates think critically and creatively	Students develop as competent and confident problem solvers (Para 4.d)	- Projects - Homework Quizzes - WPR Questions - TEE Questions
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- a. **Course Goal: Internalize the concept of the derivative as a rate of change, the integral as the accumulation of change, and that these two concepts are inversely related.** This course aims to develop student knowledge of differential and integral calculus. This is done through the presentation of core concepts and computations across multiple lessons, opportunities for repetitions with in-class board problems & homework sets, and multiple different types of frequent lower-stakes assessments (Concept Quizzes, Homework Quizzes, and WRITs). Overall, across the course, cadet conceptual understanding of calculus concepts is weaker than problem solving skills – Course average on the concepts portion of the TEE was a 69% and course average on the problem-solving portion of the TEE was an 86%. Improvement throughout the semester on traced problems for calculus concepts (See *Figure 11*, Appendix A) was also not strong. Some potential contributing factors are the fact that students were able to use their concept questions during concept quizzes, but no resources were allowed on subsequent assessments (WRITs and TEE) and students are better at rote learning than truly understanding the course material.
- b. **Course Goal: Understand when and how to apply differential and integral calculus to solve single-variable problems.** Applications of differential and integral calculus is the focus of this course. STEM majors need this course for follow-on courses, but all cadets need this course to help them think mathematically and be able to use mathematics to solve problems they may face as future leaders in our Army. Over one-third of the course is dedicated to the applications of calculus. Applications are reinforced with homework sets and projects and then assessed with Homework Quizzes, WPRs, and on the TEE. Most cadets walked away with a good understanding of how to apply calculus concepts to solve problems, especially if they had multiple resources at their disposal to help model the problems. The cadets who walked away with a poor understanding did so because of their weak math fundamentals, poor study habits, and the lack of grit required to overcome challenges.
- c. **Core Mathematics Goal: Acquire a Body of Knowledge for Continued Mathematical Development (Assessment: 3, Good).** Most students (~75%) walked away with a solid understanding of single variable calculus concepts, computations, and applications. The upper tier of students (~30%) showed mastery of subject material and the middle tier (~45%) showed a good understanding of course material. The upper tier of students will have no problem excelling in further mathematics courses or STEM courses. The middle tier of students can excel in further mathematics courses and STEM courses with extra effort and better study habits. The bottom tier of students (~25%) walked away

with a weak understanding of calculus concepts mainly because of their weak math fundamentals and inability to retain knowledge for extended periods of time. This bottom tier of students will struggle in further math courses and any STEM classes. If STEM majors, these bottom tier students should consider changing their major to improve their chances of successfully completing the USMA curriculum.

Progress of increased body of knowledge was tracked this semester through quantitative and qualitative data. The quantitative data consisted of student performance on fourteen problems that were tracked from assessment to assessment. Qualitative data was gathered from students through self-reported surveys on their perceived knowledge. Results for both can be found in the Appendix, with *Figure 11* summarizing the quantitative data that shows student progression (growth) over the semester. Students showed growth in body of knowledge AND a solid overall performance in most areas. Performance is seen in both the course average (85%), and the fact that 63% of students earned a B (“Good Understanding”) or better in the course. Growth can be seen from *Figure 11* where the number of students attaining a score of 85% or better increased between the first and second time they were assessed in 10 of the 14 traced problems – each increasing at a varying level. Similar increases were seen in 3 of 5 problems that were assessed on the TEE. This shows that overall, student performance improved but there are still topics that many of the students do not understand and/or only achieved short term understanding. This is why we believe we rate a “Good” in meeting this core math goal. In order to improve upon this next year, we recommend giving students more “touches” of course concepts with reach-back questions on concept quizzes and homework quizzes.

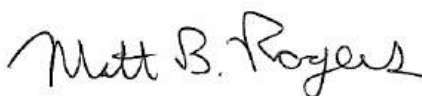
- d. **Core Mathematics Goal: Students Develop as Competent and Confident Problem Solvers (Assessment: 2, Very Good).** The main goal of this course is to create competent and confident problem solvers who can use single variable calculus concepts to solve real world problems. 29 of 64 lessons are committed to problem solving. These lessons cover applications of differential & integral calculus and three projects. Students practiced their problem-solving skills on homework sets, during board work, and during project development days. Students were assessed on their problem-solving skills during homework quizzes, WPRs, the TEE, and on the three projects. Overall, MA104 students did become more competent and confident problem solvers. Quantitative trace problems 8, 9, 13, and 14 were problems that test student’s problem-solving skills. You can see from *Figure 11* that, except for trace 9 (Max/Min Theory), the number of cadets achieving 85% or better on the problems greatly increased from assessment to assessment; student growth over the semester. A more detailed quantitative analysis on a specific topic can be seen in *Figures 48-50* of the Appendix. Qualitative analysis on the same topic, seen in *Figures 41-46*, also supports the strong problem-solving improvement by students. A performance indicator of success in achieving this goal is the 86% course

average on part 2 of the TEE, which tested students on problem solving using calculus concepts. Although we rate a “Very Good” in meeting this core math goal, we question whether or not allowing the students to use the textbook on WPRs and the TEE overinflates their scores compared to their actual knowledge. We recommend the course change resources for WPRs and the TEE to a standard reference card and not allow cadets to copy down multiple examples in hopes of being able to pattern match one to a problem on the exam.

5. **COVID-19 Impacts on MA104 AY 21-2.** COVID continued to impact MA104 during the entire academic year. Cadets were remote for over 50% of classes due to class being taught in a hybrid modality, the first two weeks being fully remote, and isolation and quarantine policies emplaced to reduce the spread of COVID. Additionally, numerous assessments were completed digitally and with a digital writing capability. Assessments administered in this manner take more time due to the digital writing and requirement to upload work onto blackboard. It is unquestionable that the above-mentioned changes to the course had a negative impact on the cadets and their ability to learn single variable calculus.

6. **Course Analysis.** The Appendix of this report provides a detailed course analysis that supports the statistics above and resulted in many of the recommended changes to the course. The analysis is based upon both quantitative and qualitative data that was gathered throughout the semester.

7. Point of contact for this memorandum is the undersigned at matthew.rogers@westpoint.edu, or by phone (845) 938-0061.



MATTHEW B. ROGERS
COL, EN

1 Enclosure

Appendix – End of Course Analysis

Associate Program Director, MA104

APPENDIX – End of Course Analysis

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1. Course Analysis based on Classroom Sectioning. Due to our decision to section our classrooms by ability, we will be analyzing the performance of the cadets in our course by looking at the performance differences between the advanced, intermediate, low, and mixed classrooms. In the following charts and graphs, the advanced, intermediate, low, and mixed students will be indicated by green, orange, red, and purple dots, respectively. To help us identify a change in performance between MA103 and MA104, we have placed orange-colored lines that give a $\pm 10\%$ band that indicates no significant change. If a dot lies outside this band, either above or below, this indicates either an improvement or degradation of that student's performance. We have also placed green and red lines in these charts indicating 80% and 65% performance respectively (See *Figure 1* for an example).

Overall Grades. We decided to use a cadet's MA103 final grade to predict the student's ability to succeed in MA104. In the end, we can see in *Figure 1* that for the most part, a cadet's MA104 final grade will be within 10% of their MA103 grade. Of note, we have quite a few students from the low sections who seem to out-perform expectations. We believe this to be a direct result from the modality of the classroom (being all in person), the smaller section sizes (12 to a classroom), and the delivery of the material specifically designed for students who struggle with mathematics.

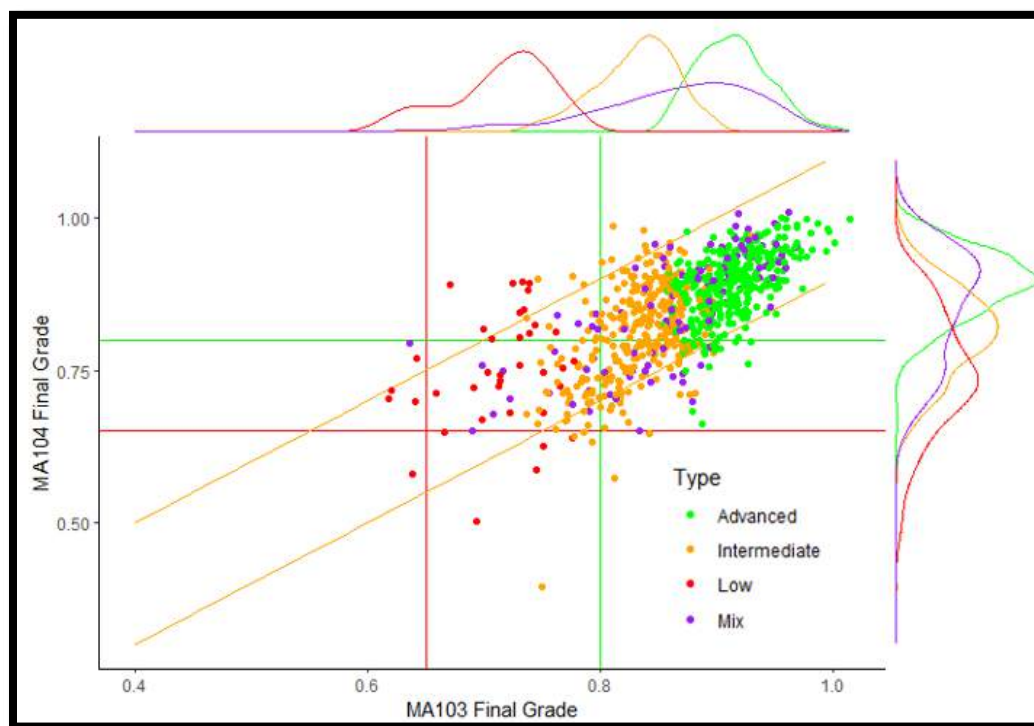


Figure 1: MA104 Performance vs. MA103 Grade

Homework Quizzes. Due to testing cadet's knowledge of homework problems in class using homework quizzes, we were able to give feedback on their actual ability to solve problems instead of work developed using additional resources. From *Figure 2* below, we can see that the students from the mixed sections (purple) performed with similar results on homework quizzes for both the advanced and the intermediate sections. However, the low students on average were able to out-perform their peers from mixed sections of similar ability.

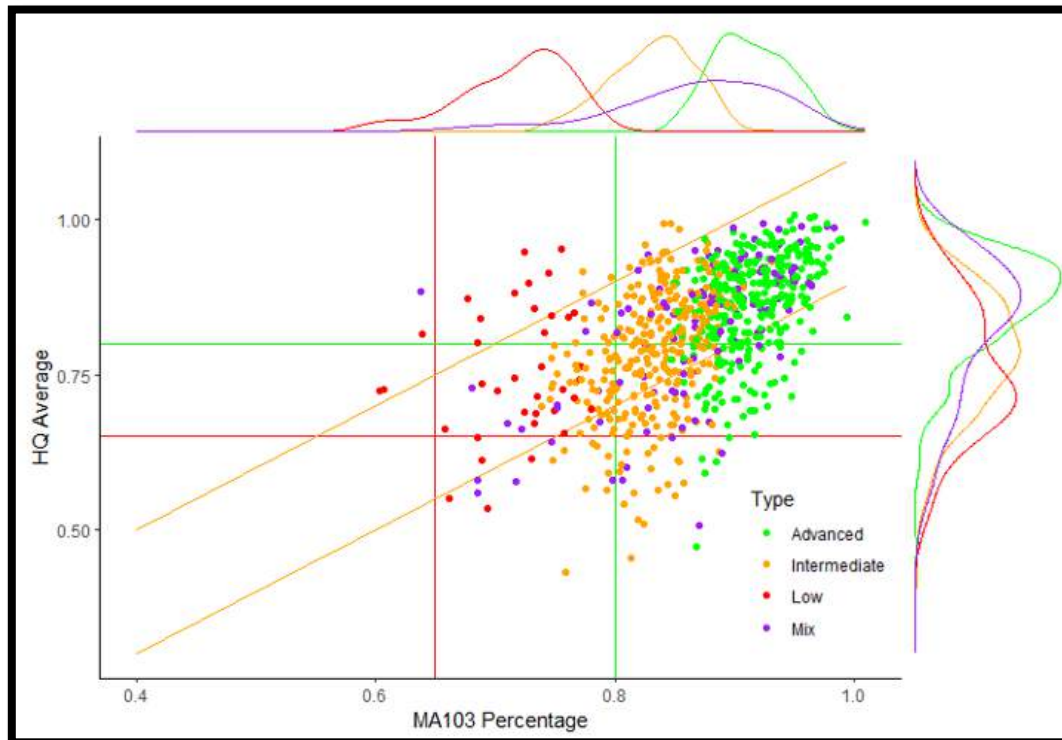


Figure 2: Homework Quiz Performance vs. MA103 Grade

Additionally, *Figures 3 & 4* show the success seen on homework quizzes seems to directly translate to success on both WRITs and WPRs. It appears that the effort students put into the homework will pay off in the higher stakes assessments.

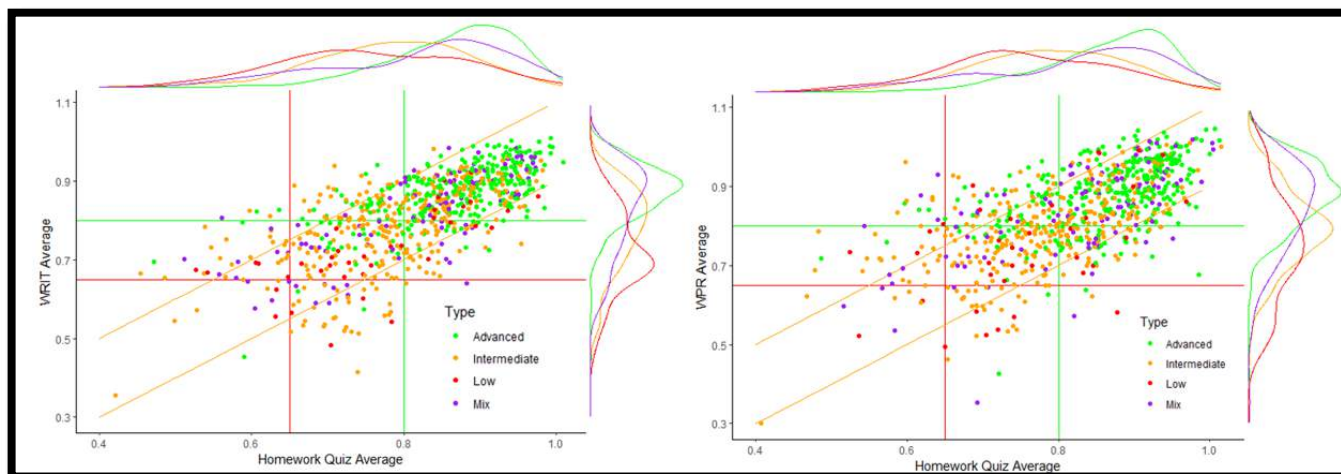


Figure 3: WRIT Performance vs. HW Quiz Avg

Figure 4: WPR Performance vs. HW Quiz Avg

When students were asked if the homework quizzes and corrections process helped them understand the material prior to major assessments, we can see that they overwhelmingly agree (See *Figure 5*).

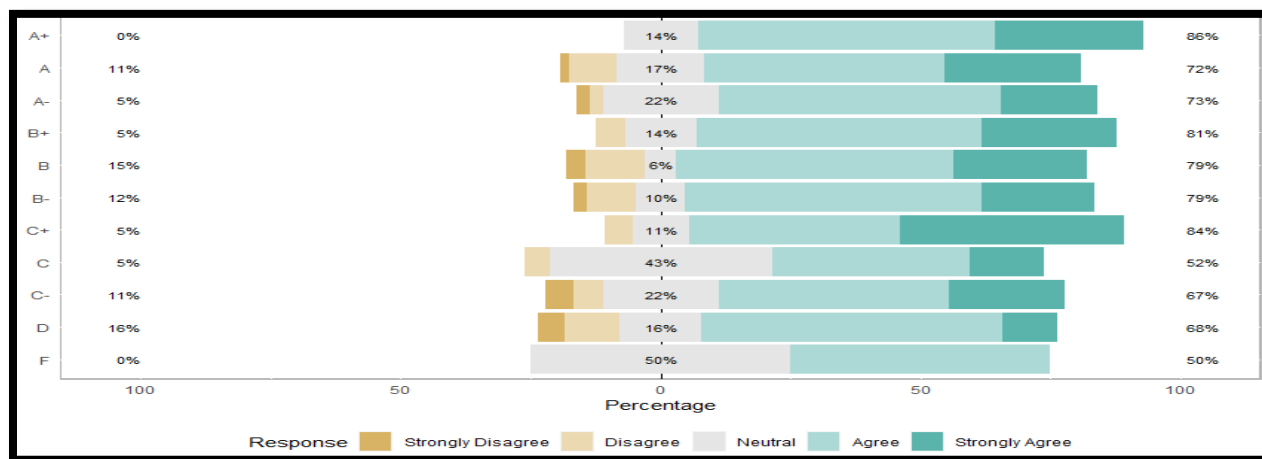


Figure 5: Student Response to Survey Question – “I believe the homework corrections process helps me to understand the material prior to WRITs and WPRs”

One of our main concerns with the homework quizzes is that the students wouldn't be receiving enough feedback on their homework. However, when asked, the majority of students agree that they receive an adequate amount of feedback on their work (See *Figure 6*). Based on this, it appears students would rather receive limited feedback on

their actual work (homework quizzes) than more feedback on work that is done using outside resources (Slader/Chegg).

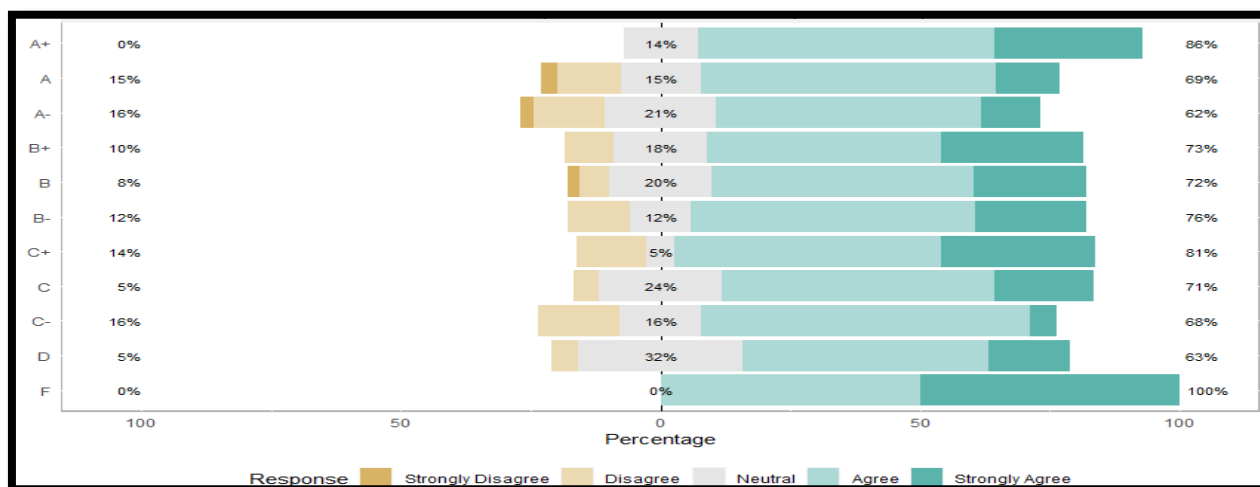


Figure 6: Student Response to Survey Question – "I believe that I received an adequate amount of feedback on my work, allowing me to develop and learn"

Concept Quizzes. The main goal of the concept quizzes is to motivate students to read the chapter and answer the concept questions from the homework prior to coming to class. Overall, we can see in *Figure 7* that the students' performance on the concept quizzes generally aligns with their ability based on their MA103 grade.

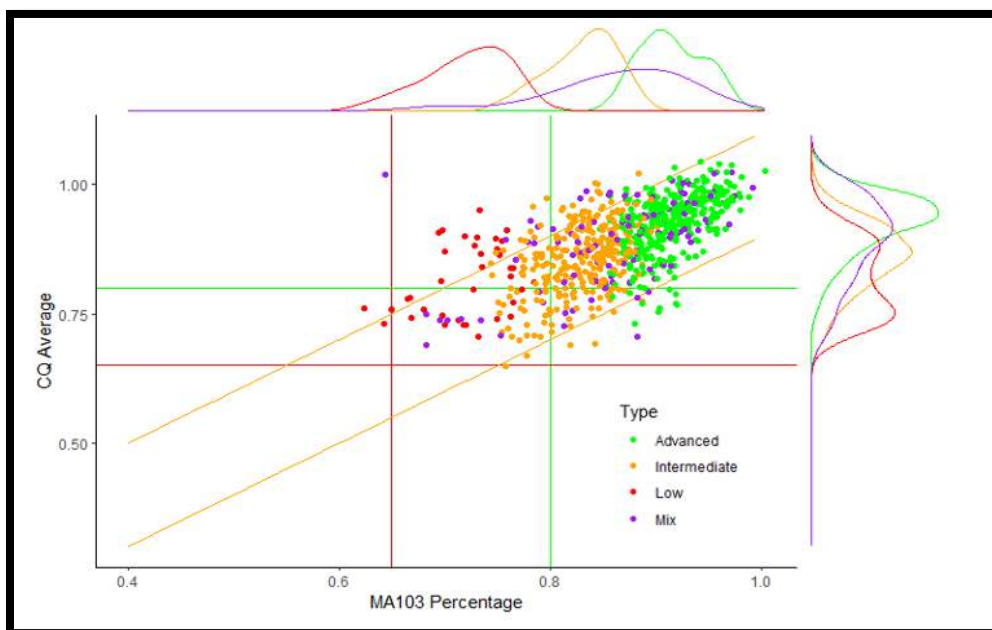


Figure 7: Concept Quiz Performance vs. MA103 Grade

Once again, we can see in *Figure 7* that the cadets from mixed sections generally perform with their peers of similar ability except with the lows. The students from low sections generally outperform their peers from the mixed sections.

Since the WRITs focus on assessing student's conceptual understanding of calculus, the students' performance on the WRITs should give us a good indication on the effectiveness of the concept questions and quizzes. In *Figure 8* we can see that for the most part, successful performance on the concept quizzes results in a successful performance on the WRITs.

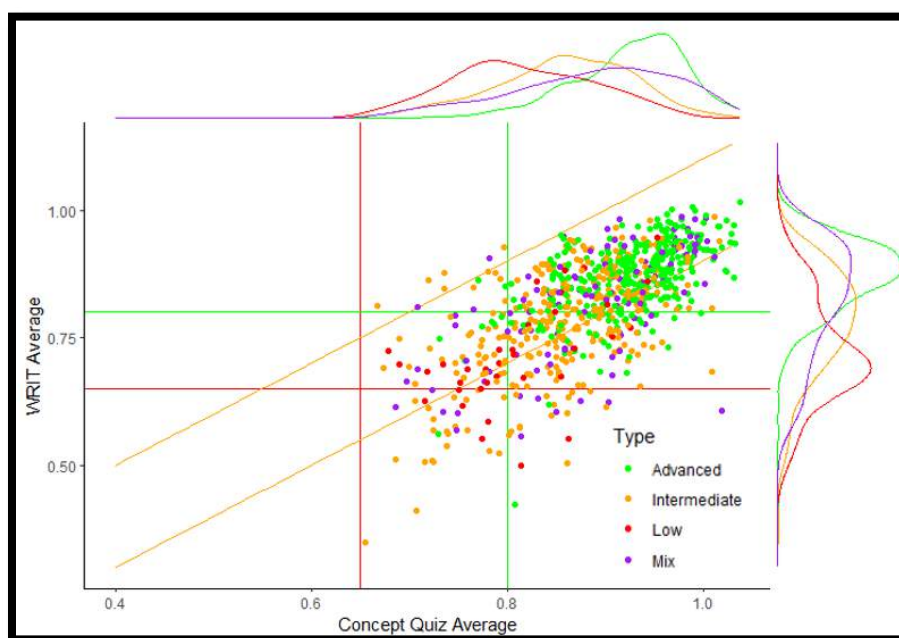


Figure 8: WRIT Performance vs. Concept Quiz Avg

When asked if the Concept and Homework Quizzes allowed the students to keep pace with the course and learn the material, we see in *Figure 9* that the students overwhelmingly agree.

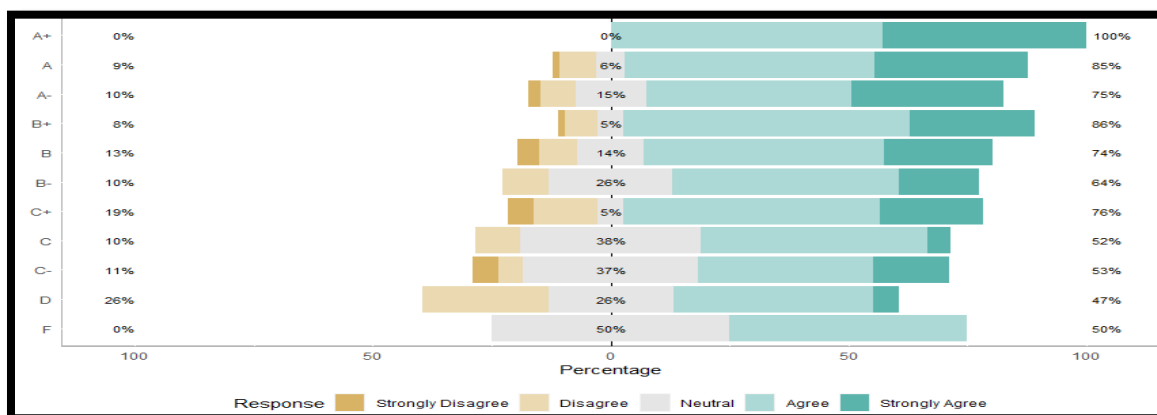


Figure 9: Student Response to Survey Question – “The daily concept and homework quizzes allow me to keep pace with the course and help me learn the material”

2. Quantitative Trace Analysis. We chose a total of 14 quantitative traces that track the progression of each student’s performance on specified concepts and tasks between the concept/homework quizzes to the WRITs/WPRs. We chose these traces specifically to help us understand our course’s success with helping our students have a good body of knowledge and be confident and competent problem solvers. We will discuss a select few of these later in the analysis. Below in *Figure 10* are the quantitative traces for the entire course:

Quantitative Trace	Type	Topic	Concept Quiz	Homework Quiz	WRIT	WPR	TEE
Trace 1	Concept	Types of discontinuities	CQ6.2B		WRT1 1B2		
Trace 2	Practice	Solving a limit		HQ2	WRT1 4A		
Trace 3	Concept	Definition of differentiability	CQ7.4A		WRT1 1B3		
Trace 4	Practice	Solving a limit definition of the derivative		HQ3	WRT1 4B		TEE P1 1B
Trace 5	Concept	Chain Rule Definition	CQ10.4A		WRT2 1B1		
Trace 6	Practice	Product/Chain Derivative		HQ4	WRT2 4B		
Trace 7	Practice	Implicit Derivative		HQ5	WRT2 3A		
Trace 8	Practice	Natural Sciences		HQ7		WPR 1: Nat Sci	TEE P2 1
Trace 9	Practice	Max/Min		HQ8		WPR 1: Max Min	
Trace 10	Concept	FTOC 1	CQ20.3A		WRT3 1B1		TEE P1 1A5
Trace 11	Practice	Reimann Sum Approximation		HQ11	WRT3 2AB		
Trace 12	Practice	Integration Strategies		HQ12	WRT3 4		TEE P1 4EFG
Trace 13	Practice	Volumes of Rotational Solids		HQ13		WPR 2: Volumes	TEE P2 5
Trace 14	Practice	Work: Pumping		HQ14		WPR 2: Work	

Figure 10: Table of Quantitative “Traced” Problems during AY 21-2

Student performance on the traced problems can be seen in *Figure 11*. The percentages in the table represent the percentage of all 779 students who achieved an 85% or better on the problem. Green and red arrows indicate improvement on subsequent assessments.

Trace	Topic	Assessment #1		Assessment #2	Assessment #3
Trace 1	Types of discontinuities	88.90%	↑	98.40%	N/A
Trace 2	Solving a limit	75.20%	↑	76.00%	N/A
Trace 3	Definition of differentiability	57.30%	↓	26.90%	N/A
Trace 4	Solving a limit definition of the derivative	58.30%	↑	62.00%	↓ 45.10%
Trace 5	Chain Rule Definition	90.90%	↑	97.10%	N/A
Trace 6	Product/Chain Derivative	60.90%	↑	74.10%	N/A
Trace 7	Implicit Derivative	77.30%	↑	81.60%	N/A
Trace 8	Natural Sciences	50.40%	↑	69.60%	↑ 90.70%
Trace 9	Max/Min	51.30%	↓	48.20%	N/A
Trace 10	FTC 1	79.10%	↓	36.00%	↑ 38.10%
Trace 11	Reimann Sum Approximation	75.30%	↓	68.60%	N/A
Trace 12	Integration Strategies	47.00%	↑	60.80%	↓ 40.10%
Trace 13	Volumes of Rotational Solids	37.50%	↑	41.10%	↑ 76.10%
Trace 14	Work: Pumping	44.80%	↑	74.20%	N/A

Figure 11: Percent of all Students who Achieved an 85% or better on Traced Problems Across Multiple Assessments

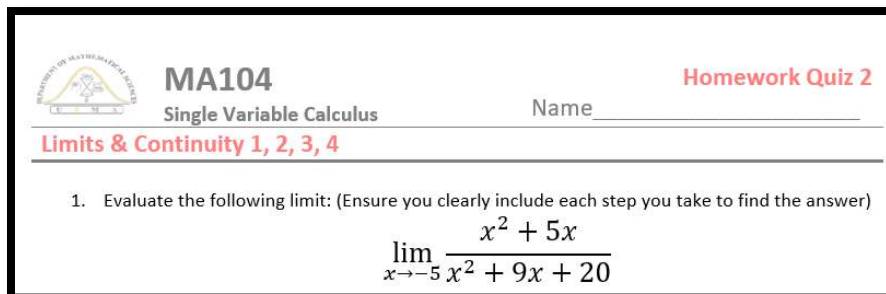
3. Qualitative Trace Analysis. We chose a total of 6 traces that track the self-reported perceived understanding of each student on specified concepts before and after completion of the specific block (Differential or Integral). We chose these traces specifically to help us understand our course's success with helping our students have a good body of knowledge and be confident and competent problem solvers. We will discuss a select few of these later in the analysis. *Figure 12* shows the qualitative traces for the entire course:

Qualitative Trace	Type	Topic	Survey 1	Survey 2
Trace 1	Derivatives	Limit Definition of the Derivative	Initial	Mid-Course
Trace 2	Derivatives	Chain Rule	Initial	Mid-Course
Trace 3	Derivatives	Related Rate Relationship Equations	Initial	Mid-Course
Trace 4	Integrals	Integration Substitution Rule	Mid-Course	EOC
Trace 5	Integrals	Applying Integration to Calculate Work	Mid-Course	EOC
Trace 6	Integrals	Graphical Interpretation of an Integral	Mid-Course	EOC

Figure 12: Table of Qualitative "Traced" Problems during AY 21-2

4. Body of Knowledge.

a. Quantitative Trace 2: Solving a Limit



MA104
Single Variable Calculus

Homework Quiz 2

Name _____

Limits & Continuity 1, 2, 3, 4

1. Evaluate the following limit: (Ensure you clearly include each step you take to find the answer)

$$\lim_{x \rightarrow -5} \frac{x^2 + 5x}{x^2 + 9x + 20}$$

Figure 13: Limit Problem on Assessments

This Trace measures the progression of our students to solve a limit between HQ2 and WRIT 1. The results from our analysis, focusing on advanced, intermediate, and low students separately are below. In this analysis by ability group, we identify those who performed proficiently on the quiz (above 85%) with green arrows or boxes and call them “good”, those who performed marginally on the quiz (between 65% and 85%) using orange arrows or boxes and call them “fair”, and those who perform poorly on the quiz (below 65%) using red arrows or boxes and call them “poor” (See Figure 14 & 15 as an Example).

Total HQ2		Total WRIT 1
75.2% ∈ (.85, 1)	→	76% ∈ (.85, 1)
7.1% ∈ (.65, .85)	→	10.5% ∈ (.65, .85)
17.6% ∈ (0, .65)	→	13.9% ∈ (0, .65)

Figure 14: All Students' Performance on Solving a Limit

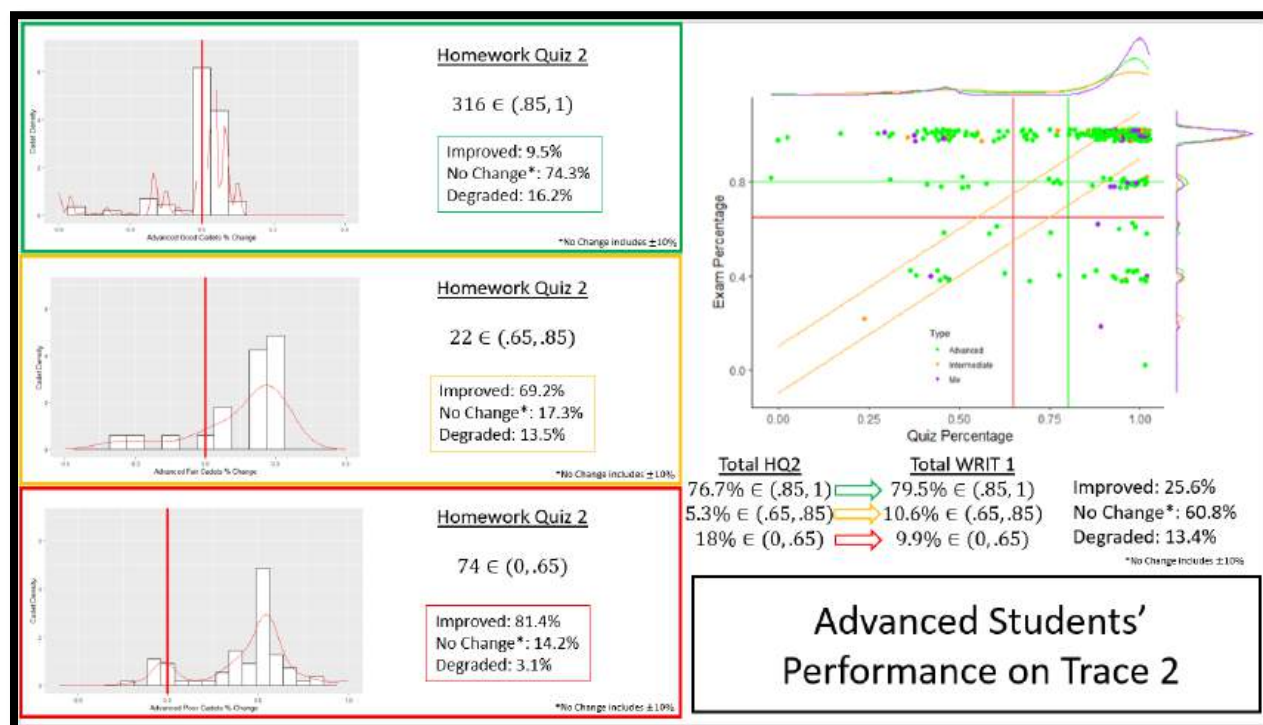


Figure 15: Advanced Students' Performance on Solving a Limit

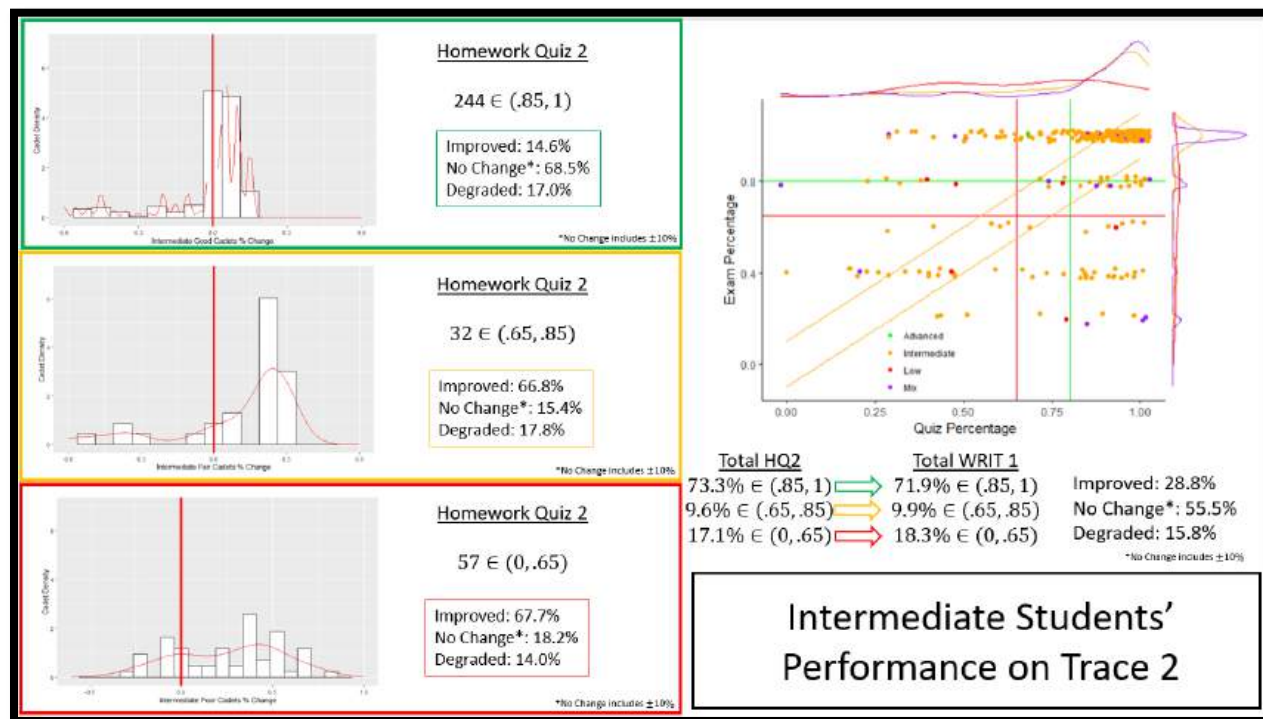


Figure 16: Intermediate Students' Performance on Solving a Limit

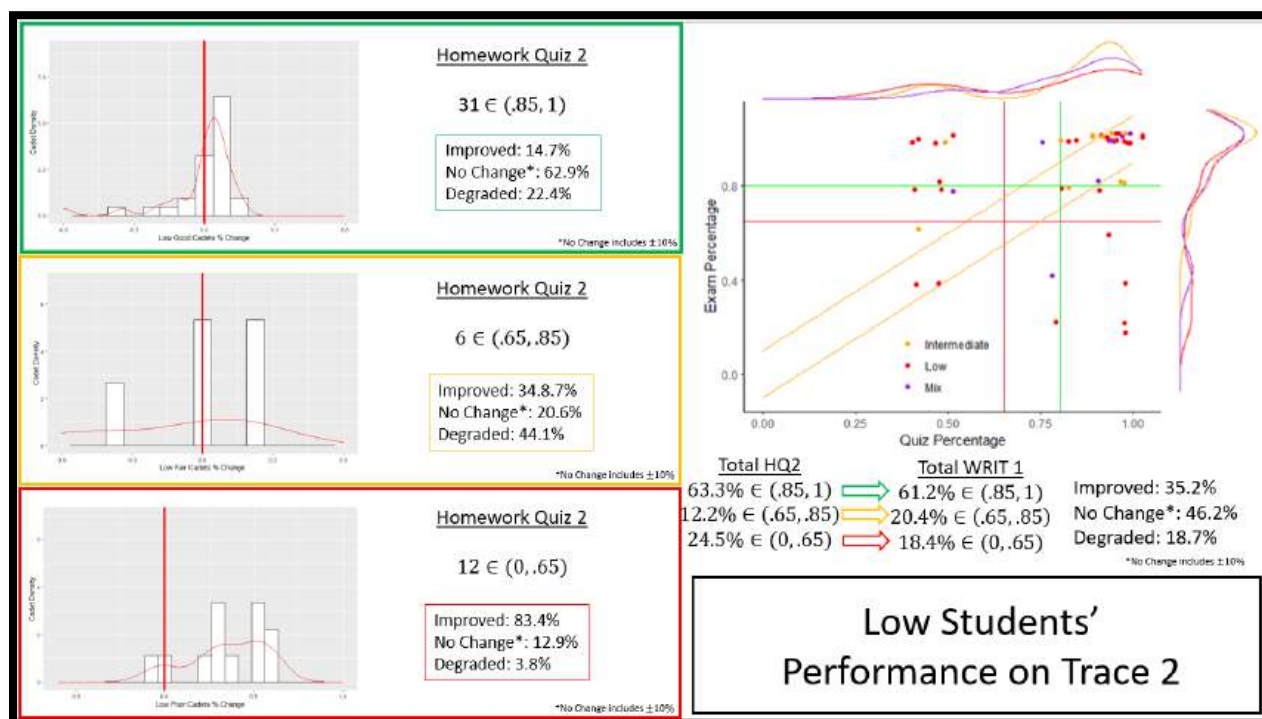


Figure 17: Low Students' Performance on Solving a Limit

Overall, students increased their ability to evaluate a limit. The only population that didn't see an overall increase are the low students who performed fairly on the quiz. Here we see that 20.6% didn't change and that 44.1% degraded. We think this is due to the low students being satisfied with a passing grade and not reviewing the material to improve.

b. Qualitative Trace 1: Limit Definition of the Derivative

Prior to being taught the content of this topic (Initial Course Survey), the cadets self-reported their understanding of the topic as shown in Figures 18-20:

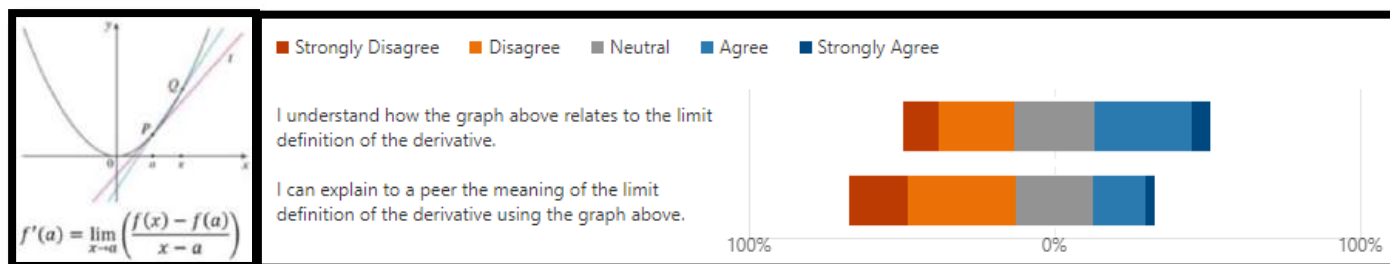


Figure 18: Student Initial Response to Survey Question - "State your opinion on how well you think you understand the limit definition of the derivative"

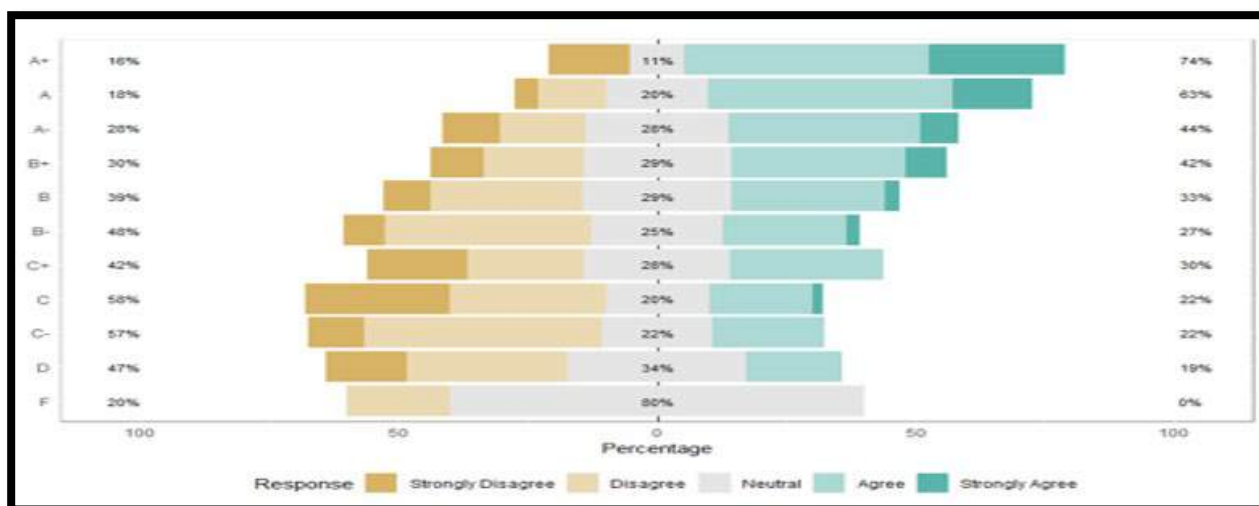


Figure 19: Student Initial Response to the Survey Question - "I understand how the graph relates to the limit definition of the derivative"

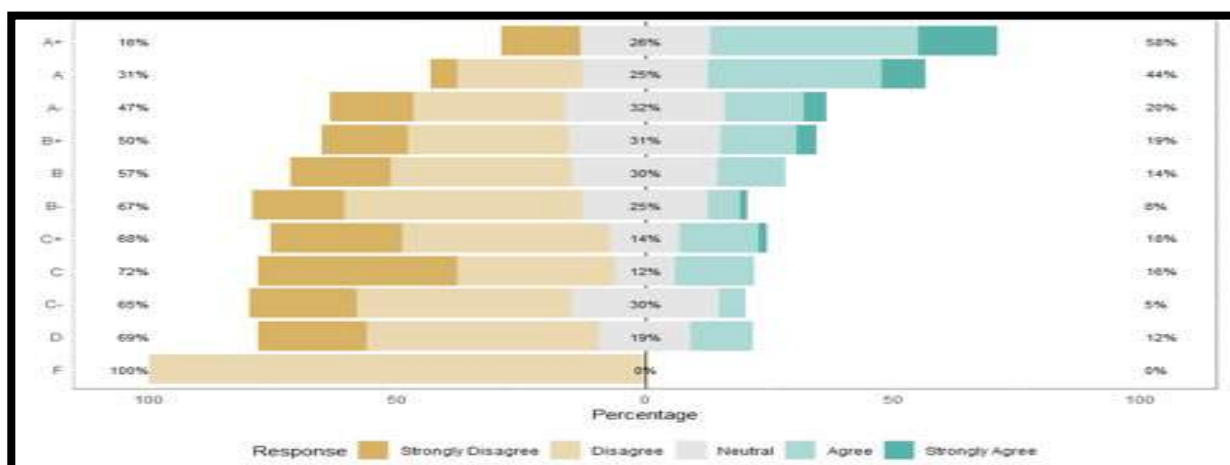


Figure 20: Student Initial Response to the Survey Question - "I can explain to a peer the meaning of the limit definition of the derivative using the graph above"

After being taught the content of this topic (Mid-Course Survey), the cadets self-reported their understanding of this topic as shown in *Figures 21-23*:

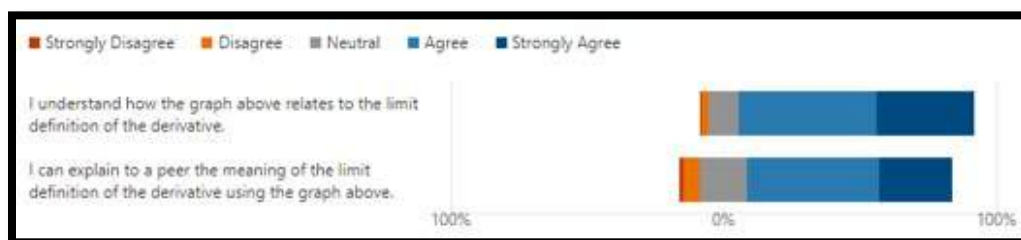


Figure 21: Student Mid-Course Response to the Survey Question - "State your opinion on how well you think you understand the limit definition of the derivative"

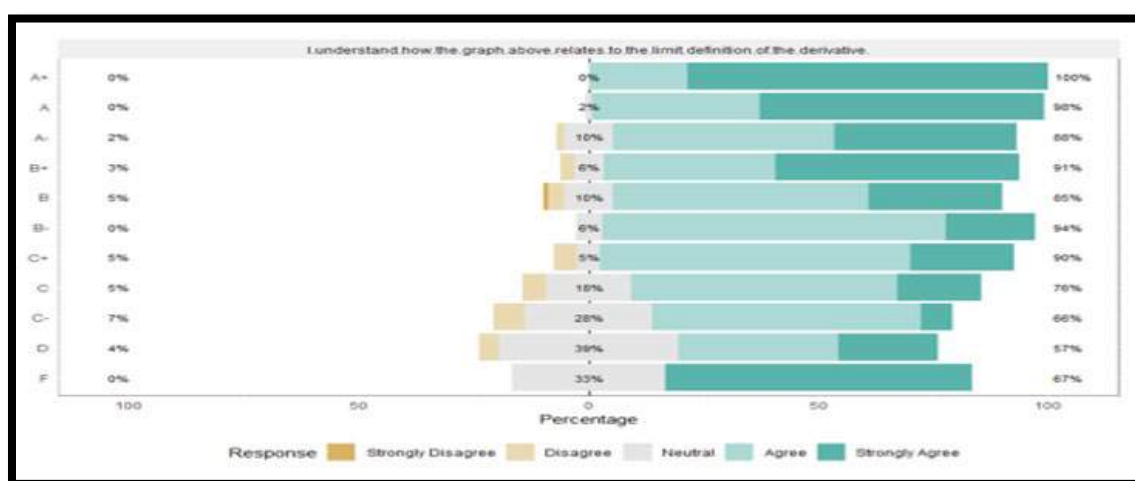


Figure 22: Student Mid-Course Response to the Survey Question - "I understand how the graph relates to the limit definition of the derivative"

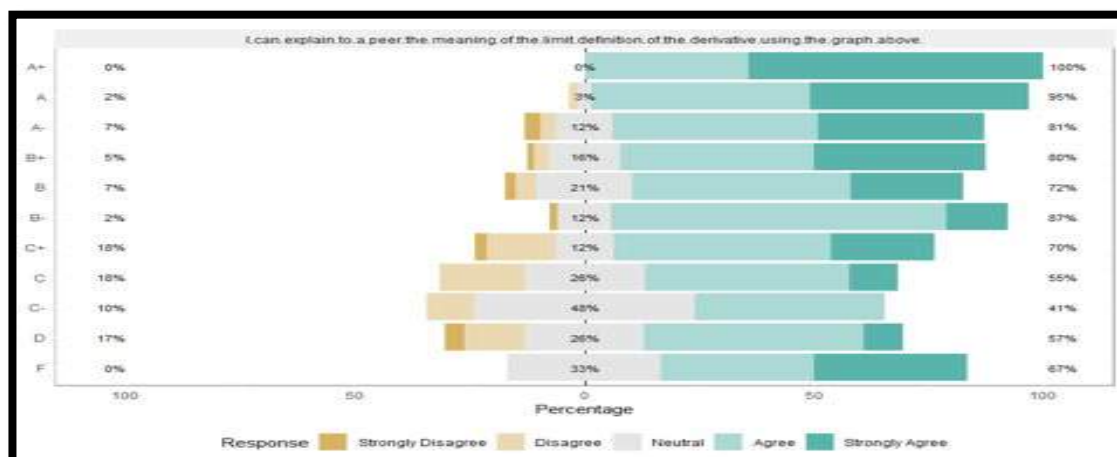
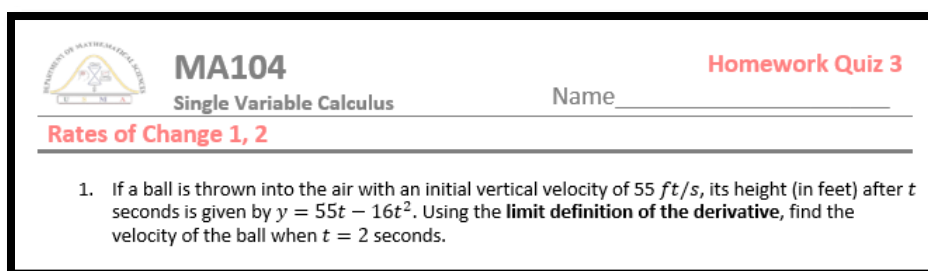


Figure 23: Student Mid-Course Response to the Survey Question - "I can explain to a peer the meaning of the limit definition of the derivative using the graph above"

Of note is the inability of failing students to accurately recognize their inability to understand this important topic. Some students from the D through C+ range all include “Disagree” when asked their understanding of this topic while those in the F range never chose this option. Overall, we see a definite improvement in the students’ self-reported understanding of this topic. These results match the Quantitative Trace below on the same topic.

c. Quantitative Trace 4: Limit Definition of the Derivative



The image shows a screenshot of a homework quiz for MA104 Single Variable Calculus. The title is "Homework Quiz 3" in red. Below it, the course name "MA104" and "Single Variable Calculus" are listed. The section is titled "Rates of Change 1, 2" in red. The problem number is 1. The text of the problem is: "If a ball is thrown into the air with an initial vertical velocity of 55 ft/s, its height (in feet) after t seconds is given by $y = 55t - 16t^2$. Using the **limit definition of the derivative**, find the velocity of the ball when $t = 2$ seconds."

Figure 24: Limit Definition of the Derivative Problem on Assessments

This Quantitative Trace measures the progression of our students to find the derivative at a point using the limit definition between HQ3 and WRIT 1 and then between WRIT 1 and the TEE. The results from our analysis, focusing on advanced, intermediate, and low students separately are below in Figures 25-28:

Total HQ3		Total WRIT 1
58.3% $\in (.85, 1)$		62% $\in (.85, 1)$
9.4% $\in (.65, .85)$		17.1% $\in (.65, .85)$
32.2% $\in (0, .65)$		21.1% $\in (0, .65)$

Figure 25: All Students' Performance on limit definition of the derivative

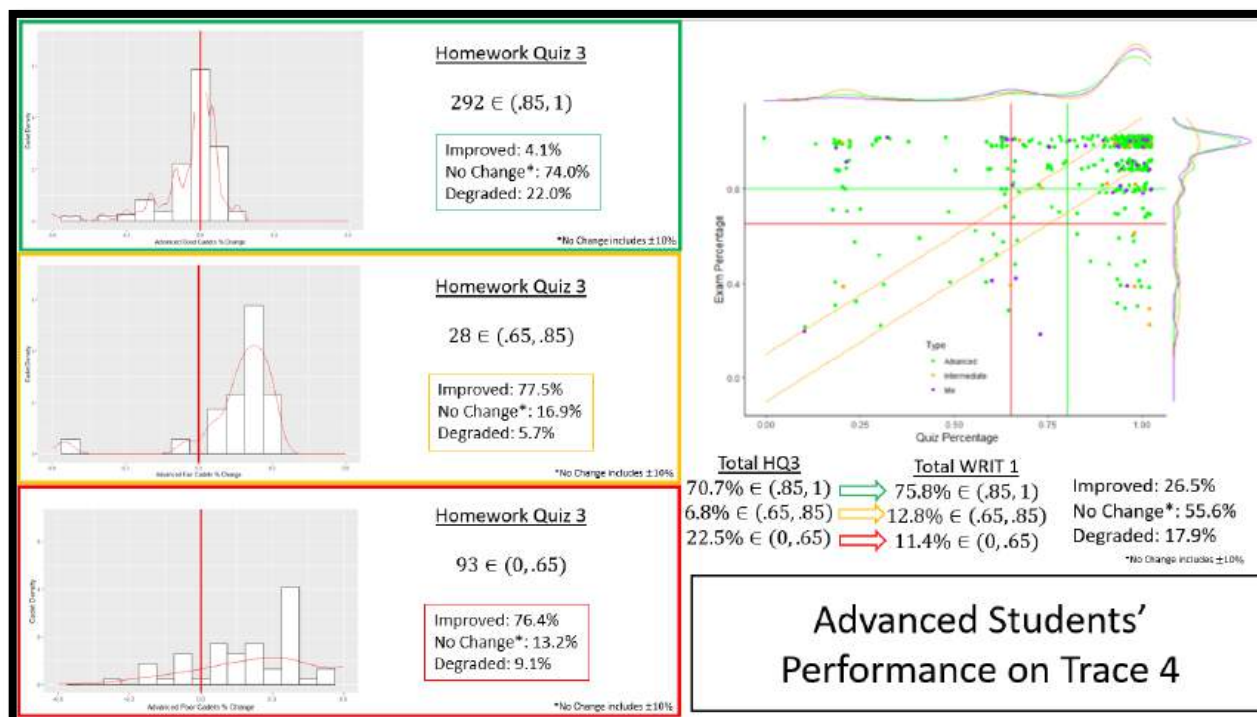


Figure 26: Advanced Students' Performance on Limit Definition of the Derivative

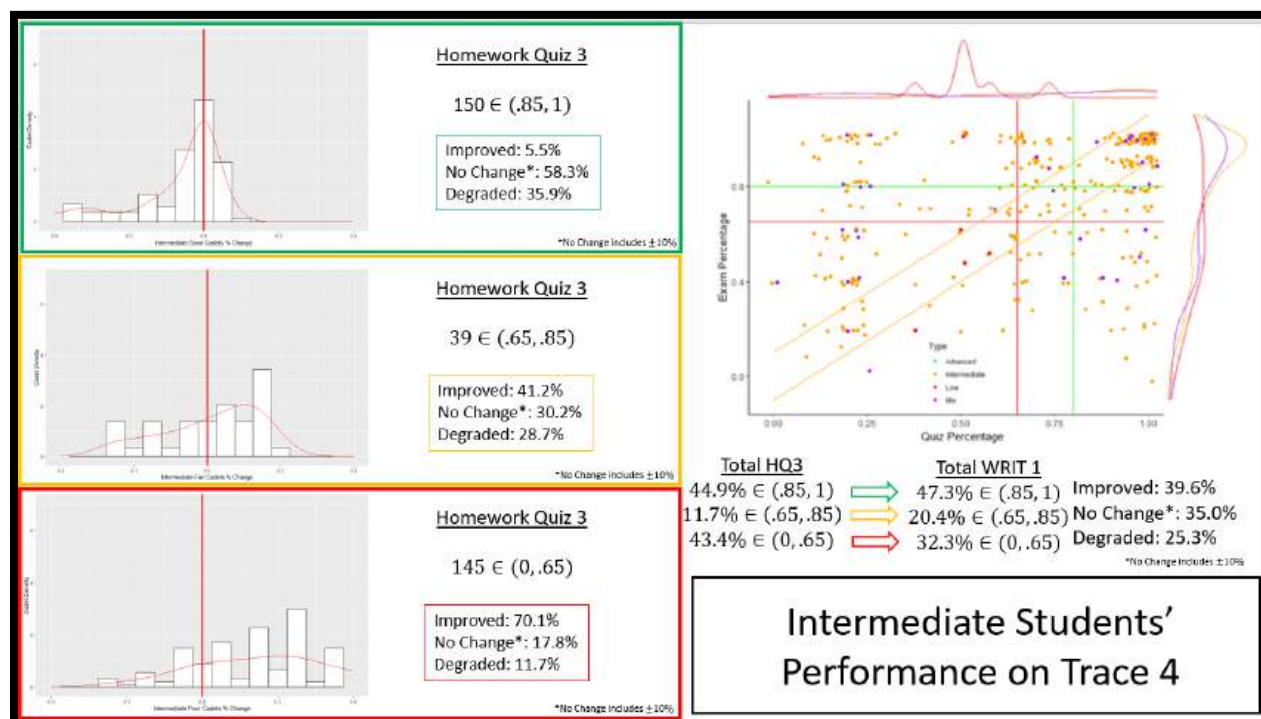


Figure 27: Intermediate Students' Performance on Solving a Limit

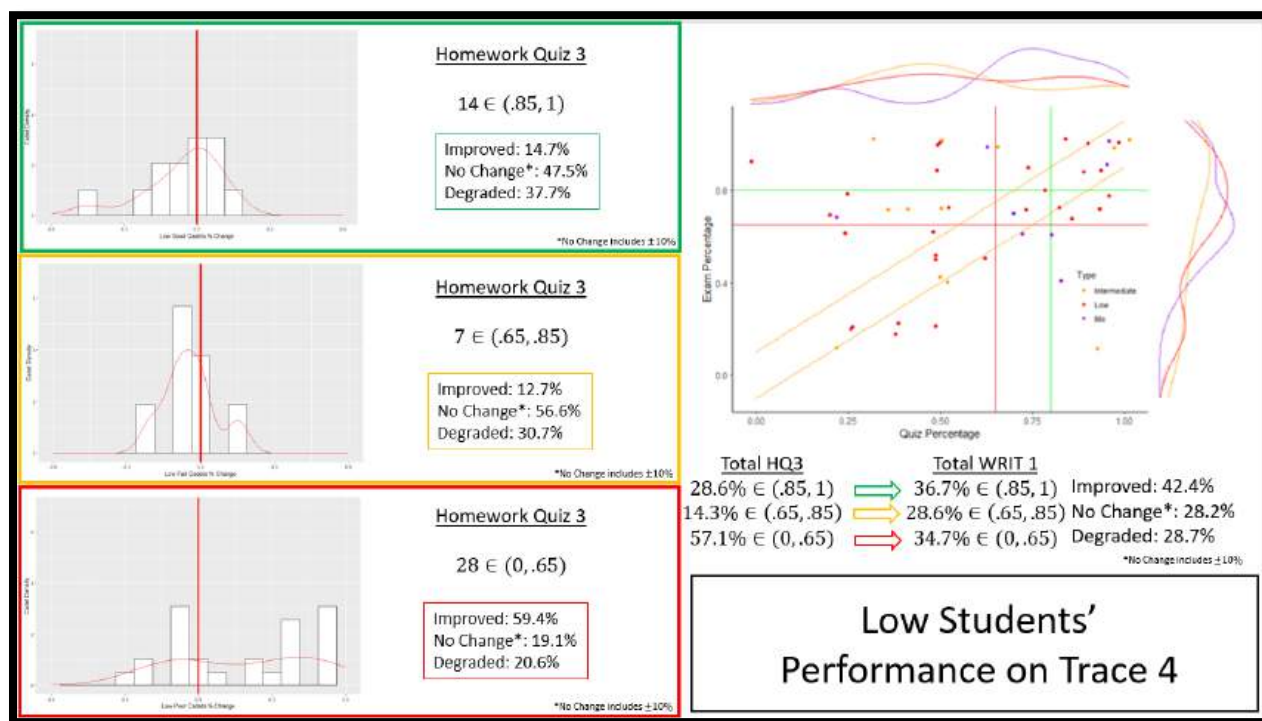


Figure 28: Low Students' Performance on Solving a Limit

Once again, the only population that doesn't improve between the Homework Quiz and the WRIT are the low students who received a fair grade on their quiz. Notice how the red-low students out-perform their purple-low peers.

Now we will see, in Figures 29-32, the analysis for the same trace problem (Limit Definition of the Derivative) between the WRIT and the TEE.

Total WRIT 1		Total TEE
61.9% $\in (.85, 1)$	$\rightarrow$	45.1% $\in (.85, 1)$
17.1% $\in (.65, .85)$	$\rightarrow$	7.1% $\in (.65, .85)$
21.1% $\in (0, .65)$	$\rightarrow$	47.7% $\in (0, .65)$

Figure 29: All Students' Performance on limit definition of the derivative (WRIT to TEE)

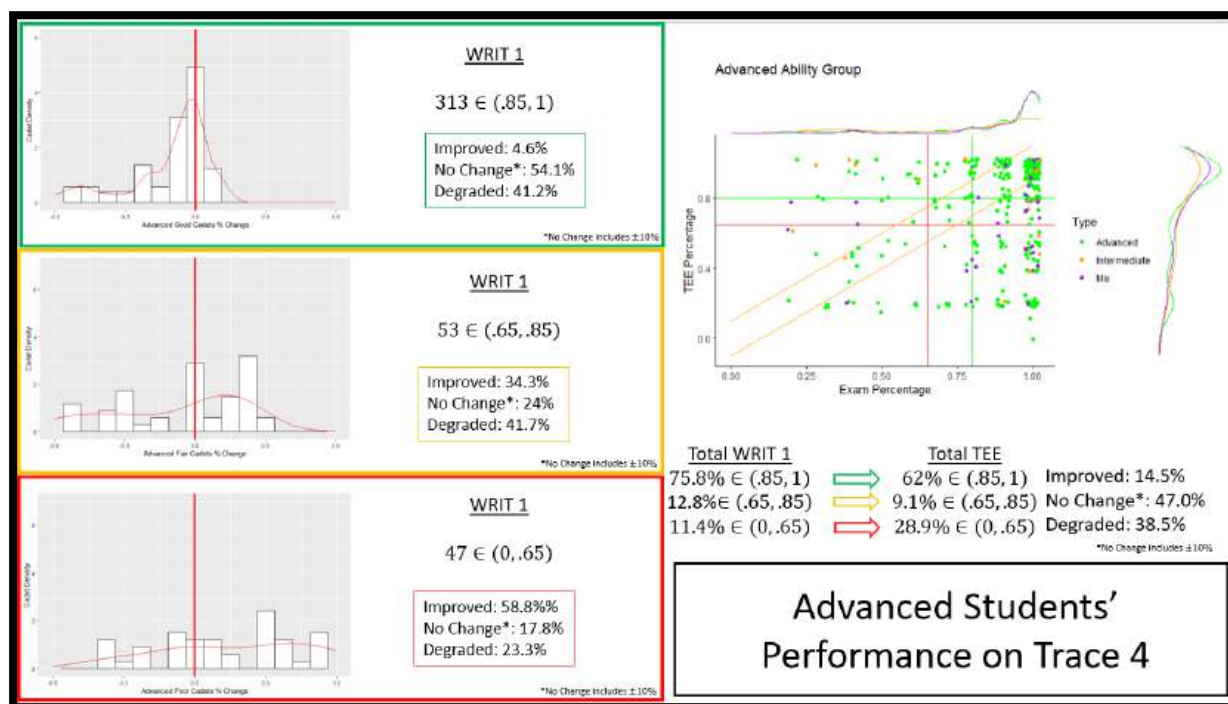


Figure 30: Advanced Students' Performance on Limit Definition of the Derivative (WRIT to TEE)

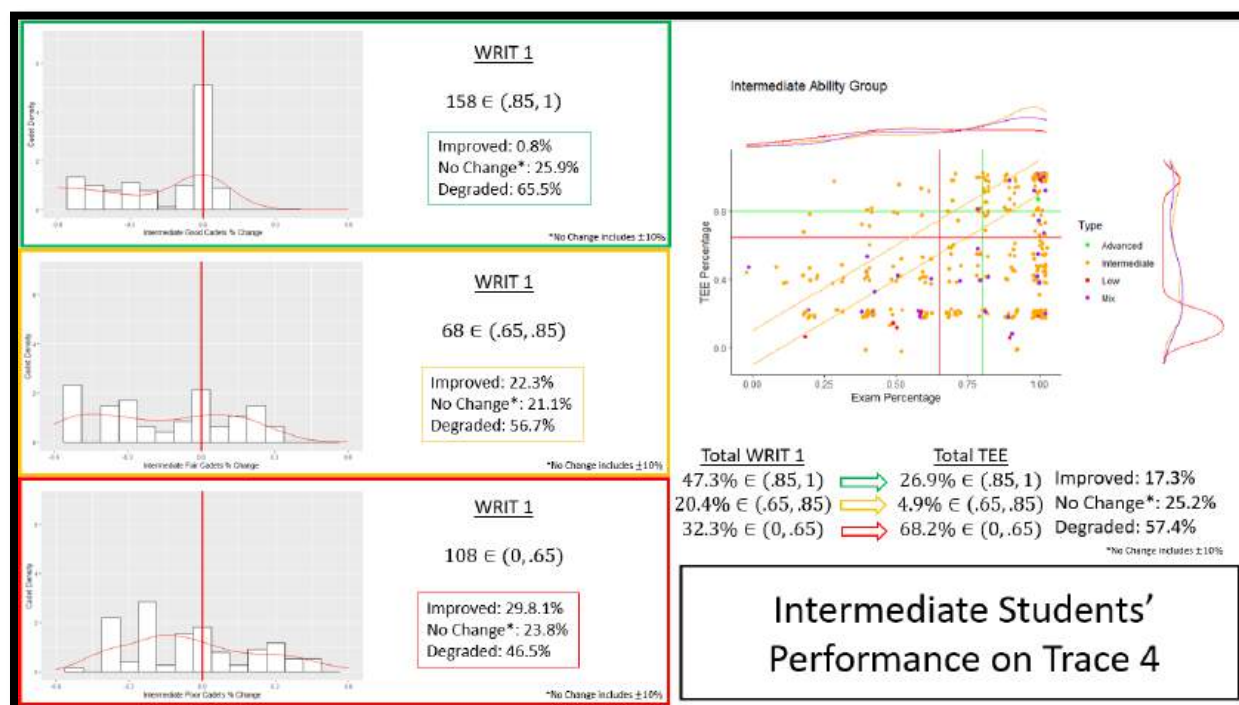


Figure 31: Intermediate Students' Performance on Limit Definition of the Derivative (WRIT to TEE)

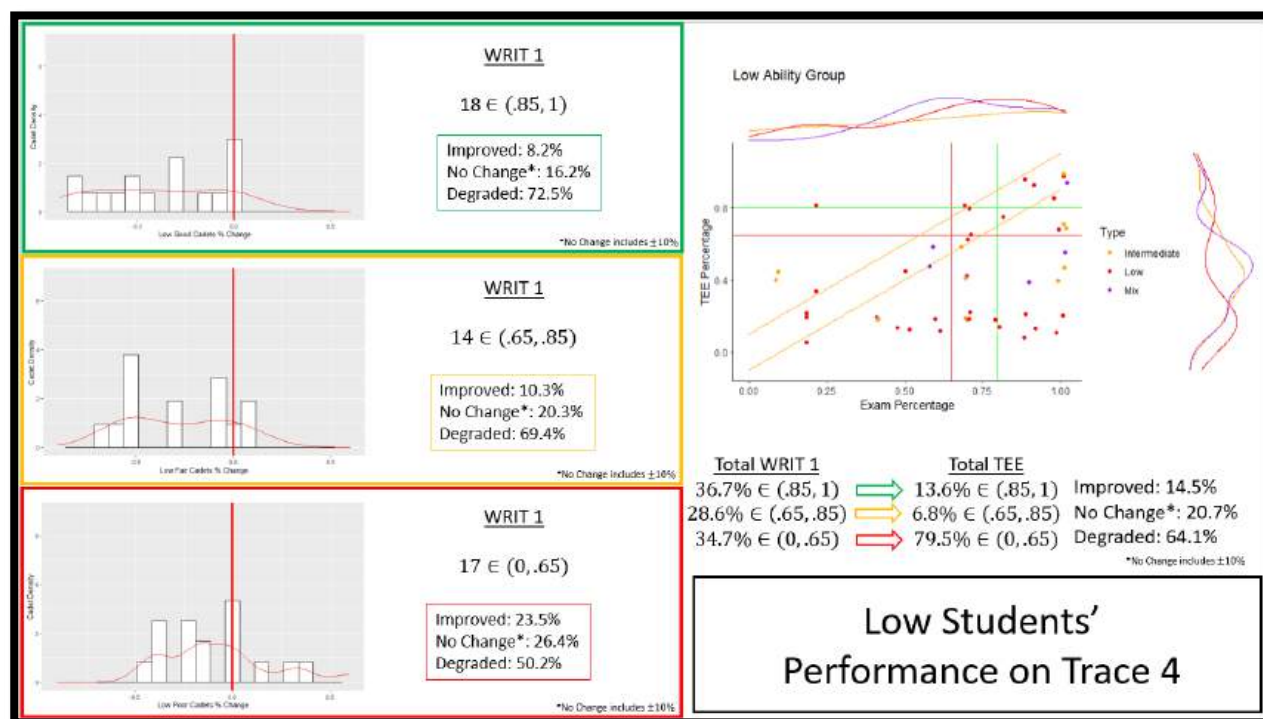


Figure 32: Low Students' Performance on Limit Definition of the Derivative (WRIT to TEE)

d. Quantitative Trace 10: Fundamental Theorem of Calculus Part 1

1. If f is continuous on $[a, b]$, then the function g defined by $g(x) = \int_a^x f(t)dt$, $a \leq x \leq b$ is continuous on $[a, b]$ and differentiable on (a, b) , and ...

a) $g(x) = f'(x)$ b) $g'(x) = f(x)$ c) $g'(x) = f(t)$ d) $g(t) = f'(x)$

Figure 33: FTC 1 Problem on Assessments

This Trace focuses on understanding the concept behind the first part of the Fundamental Theorem of Calculus. Since these concept questions are multiple choice, the students either get the question right or wrong. Based on these results seen in Figure 34, across all ability groups, the students did not understand FTC1 during concept quiz and were even worse for the WRIT.

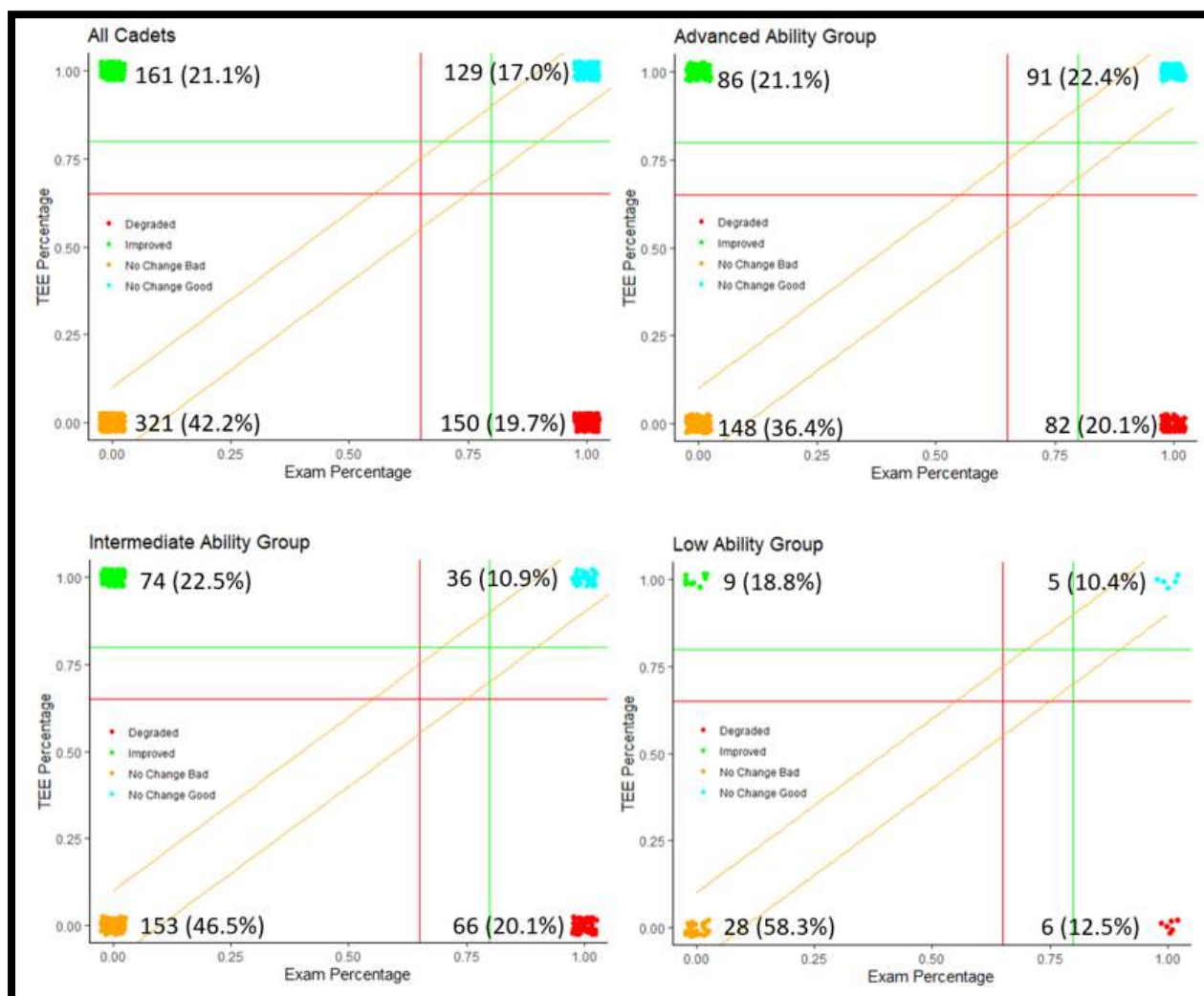


Figure 34: Student Performance of FTC 1 Problem from Quiz to WRIT

If we do the same analysis between the WRIT and the TEE, we get the following results seen in Figure 35:

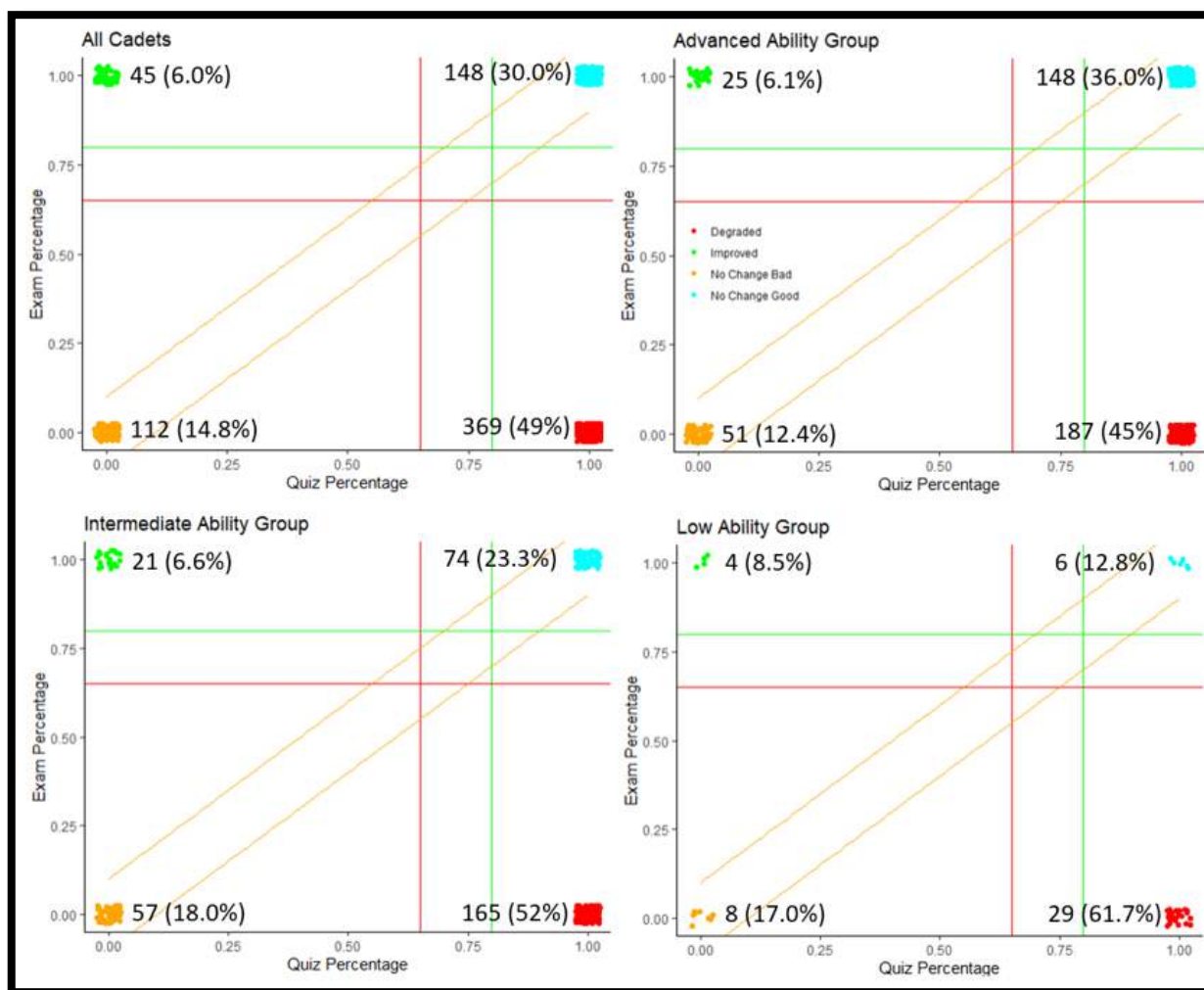
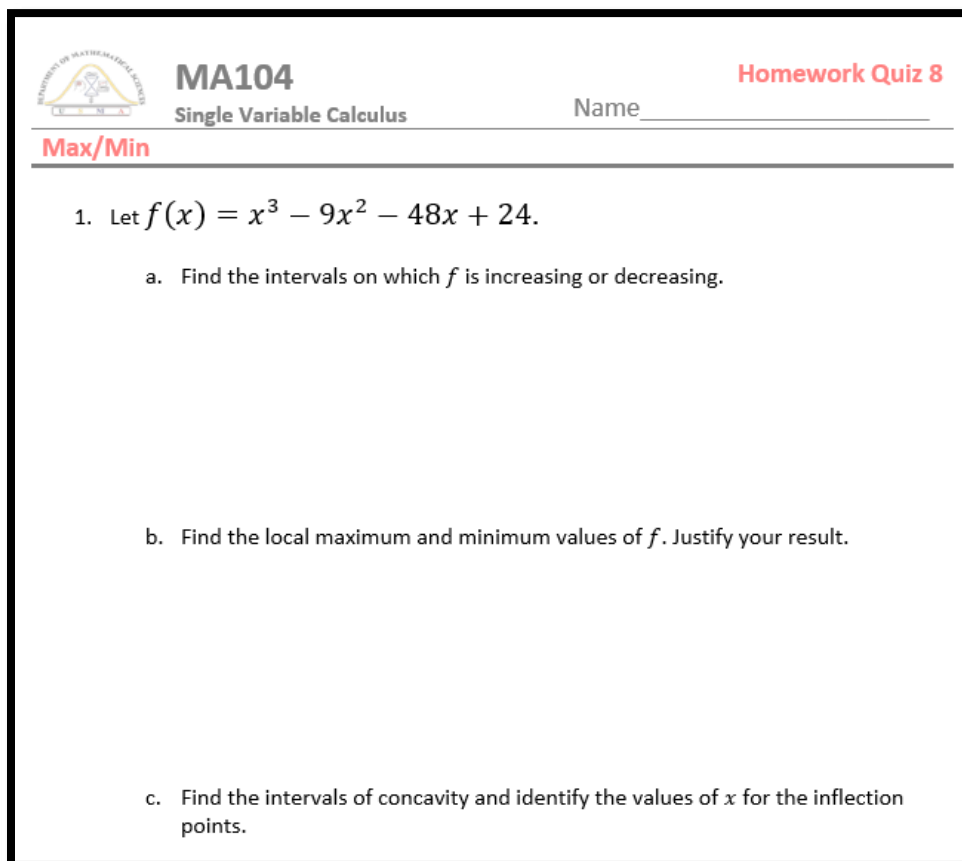


Figure 35: Student Performance of FTC 1 Problem from WRIT to TEE

As seen in Figure 11, the students' understanding of this important concept dropped from the Concept Quiz (79.1% Proficient) to the WRIT (36%) and then increased slightly to the TEE (38.1%). We found that many students who got this problem wrong chose $g'(x) = f(t)$ instead of the correct answer of $g'(x) = f(x)$ showing that they don't understand that the rate of change of an integral at a point is the value of the function at that exact point. We need to find a better way to present FTC1 to ensure the students are able to grasp the meaning of this important concept and not confuse them with the options presented in the question.

5. Confident and Competent Problem Solvers

a. Quantitative Trace 9: Max/Min Theory



 **MA104** Homework Quiz 8
Single Variable Calculus Name _____

Max/Min

1. Let $f(x) = x^3 - 9x^2 - 48x + 24$.

a. Find the intervals on which f is increasing or decreasing.

b. Find the local maximum and minimum values of f . Justify your result.

c. Find the intervals of concavity and identify the values of x for the inflection points.

Figure 36: Max/Min Theory Problem on Assessments

This Trace focuses on the students' ability to use Max/Min Theory to find critical numbers, identify intervals of increase/decrease, identify local and absolute extrema, and identify intervals of concavity with inflection points. We measured the progression of this skill between HQ8 and WPR 1. The results from our analysis, focusing on advanced, intermediate, and low students separately are below in Figures 37-40:

Total HQ8		Total WPR 1
51.3% $\in (.85, 1)$		48.2% $\in (.85, 1)$
21.2% $\in (.65, .85)$		33.9% $\in (.65, .85)$
27.5% $\in (0, .65)$		18.8% $\in (0, .65)$

Figure 37: All Students' Performance on Max/Min Theory

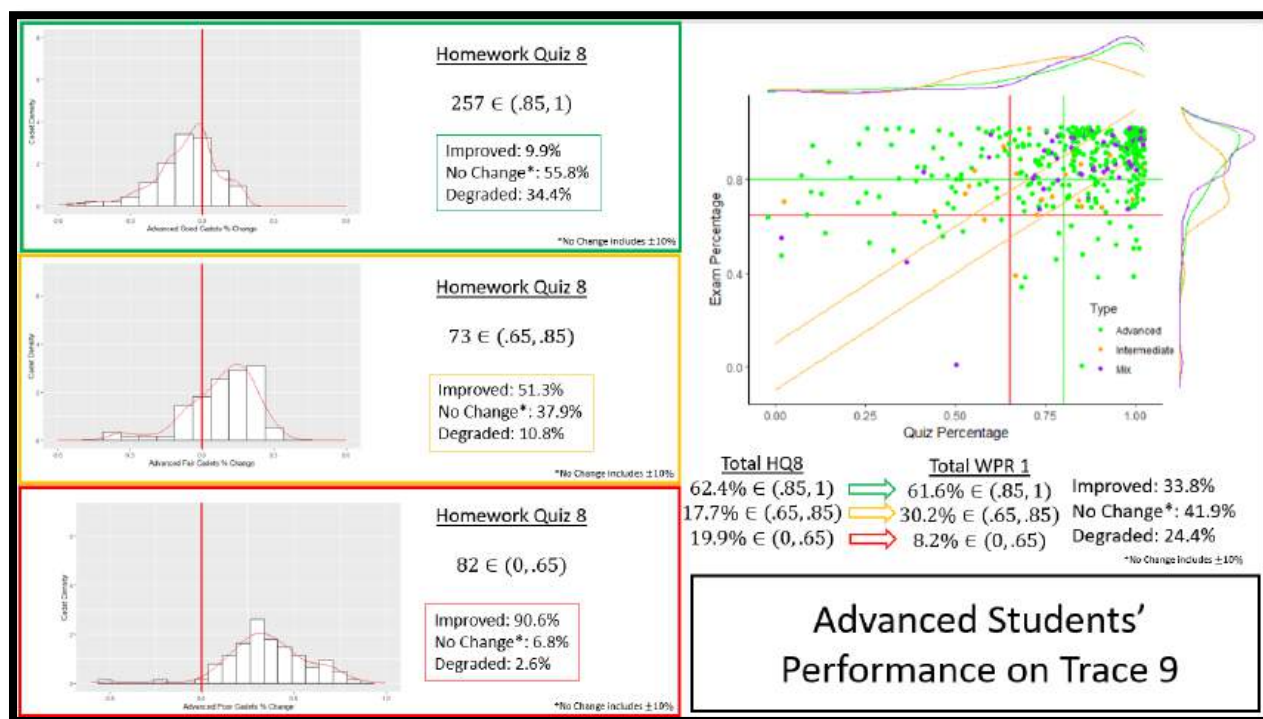


Figure 38: Advanced Students' Performance on Max/Min Theory

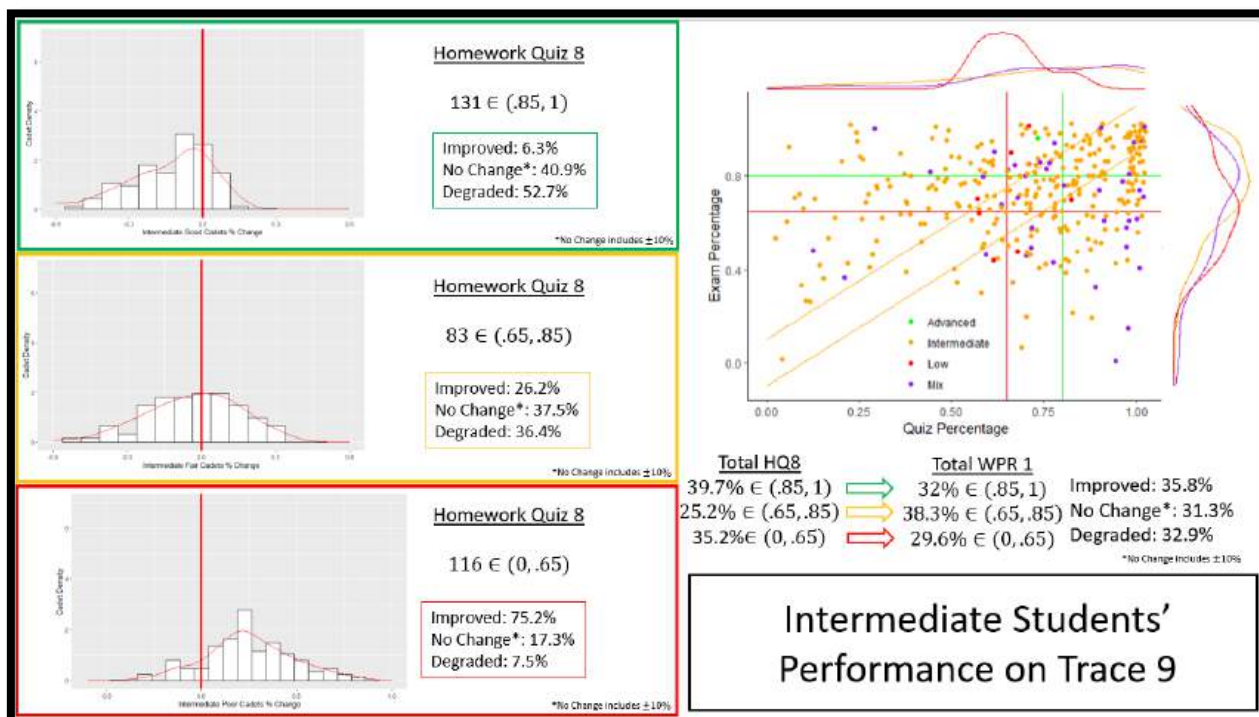


Figure 39: Intermediate Students' Performance on Max/Min Theory

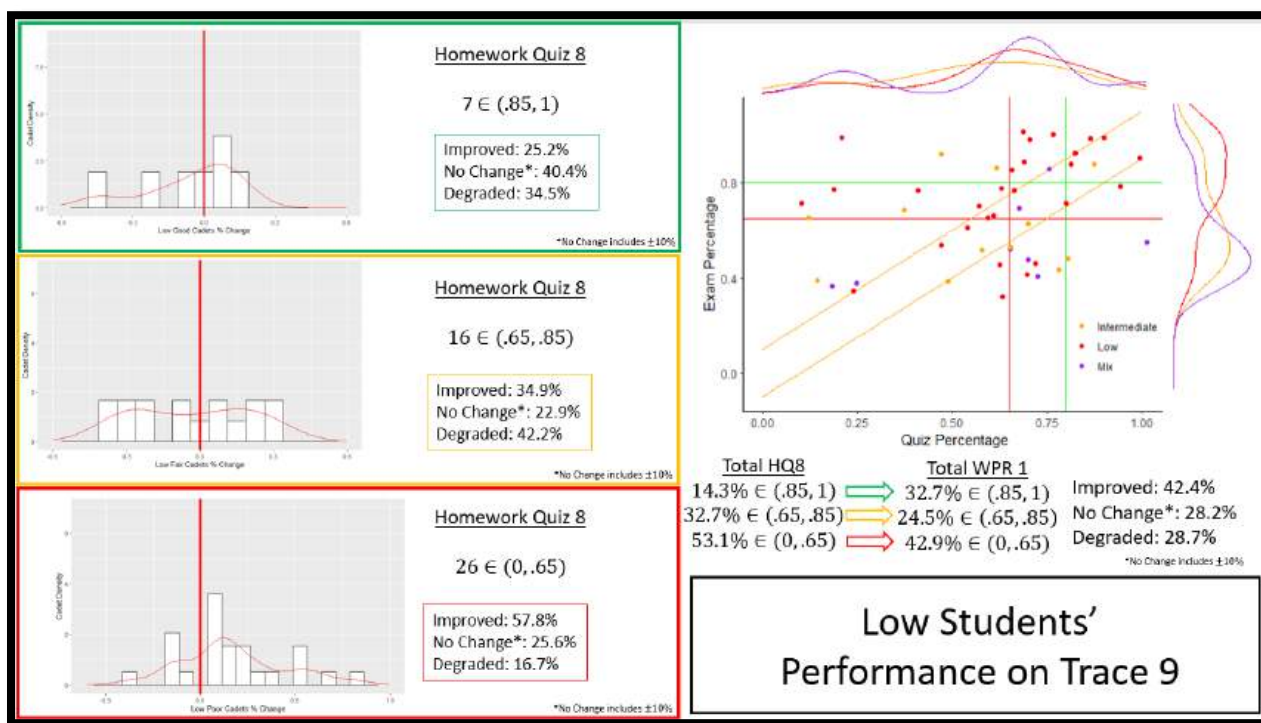


Figure 40: Low Students' Performance on Max/Min Theory

In this analysis, we see that the students in both the Advanced and the Intermediate sections tend towards a fair performance on the WPR. However, the students in the low section actually increased the number of students who did well in this trace. Also, notice how distinct the difference in performance between the red-low cadets and the purple-low cadets; the reds definitely outperformed their purple peers.

b. Qualitative Trace 4: Finding the Work to Pump Liquids out of a Tank

Prior to being taught the content of this topic (Mid-Course Survey), the cadets self-reported their understanding of this topic as follows in *Figures 41-43*:

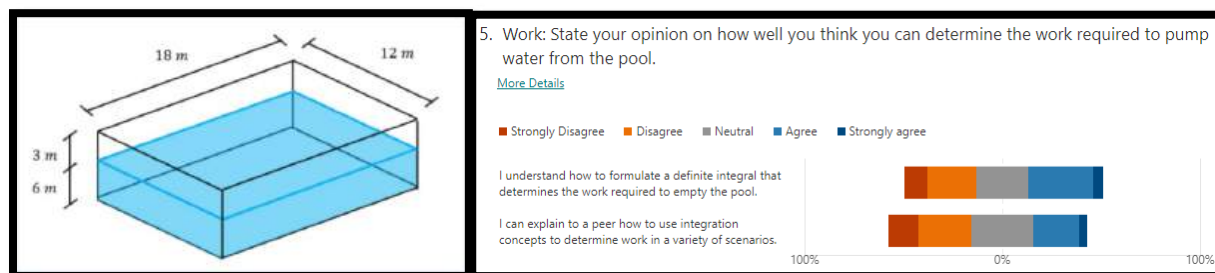


Figure 41: Student Initial Response to Survey Question - "State your opinion on how well you think you can determine the work required to pump water from the pool"

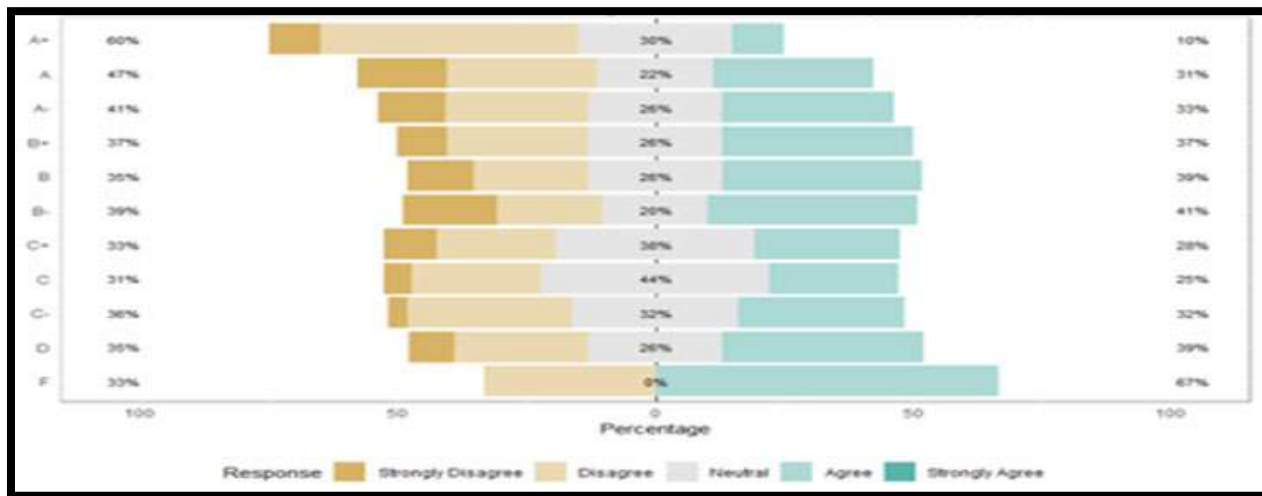


Figure 42: Student Initial Response to Survey Question – “I understand how to formulate a definite integral that determines the work required to empty the pool”

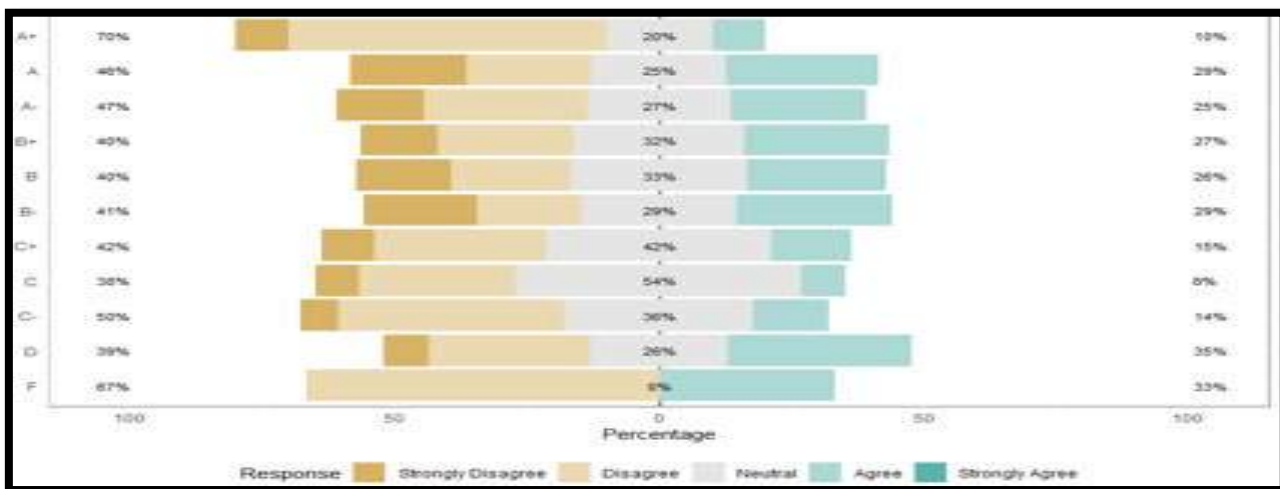


Figure 43: Student Initial Response to Survey Question – “I can explain to a peer how to use integration concepts to determine work in a variety of scenarios”

After being taught the content of this topic (Mid-Course Survey), the cadets self-reported their understanding of this topic as follows in *Figures 44-46*:

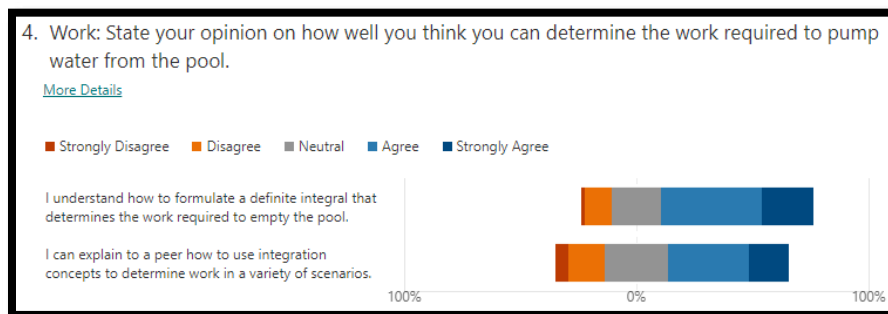


Figure 44: Student Mid-Course Response to Survey Question - “State your opinion on how well you think you can determine the work required to pump water from the pool”

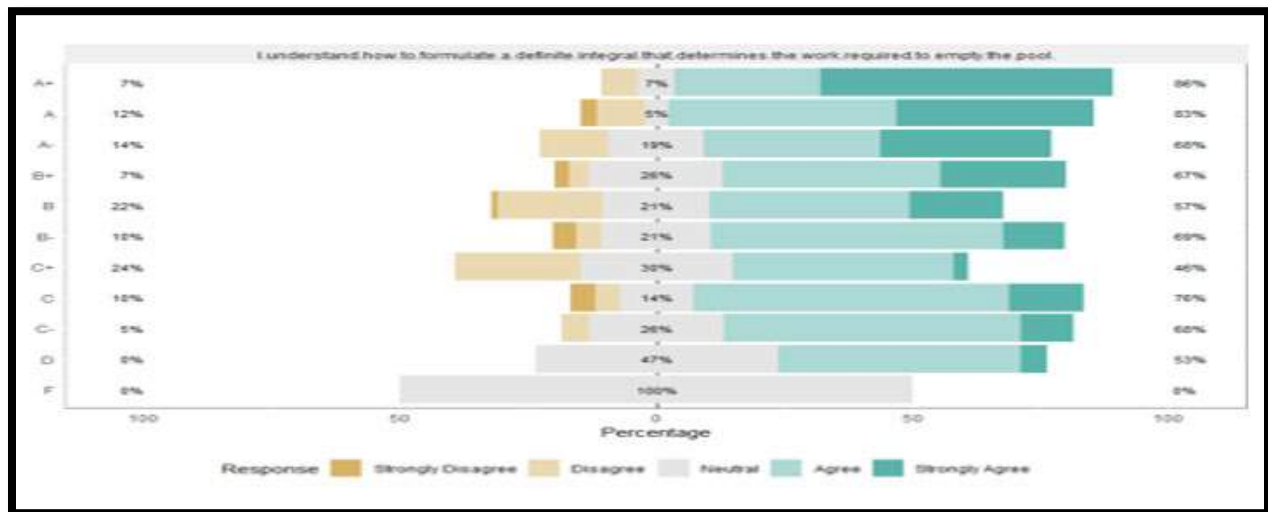


Figure 45: Student Mid-Course Response to Survey Question – "I understand how to formulate a definite integral that determines the work required to empty the pool"

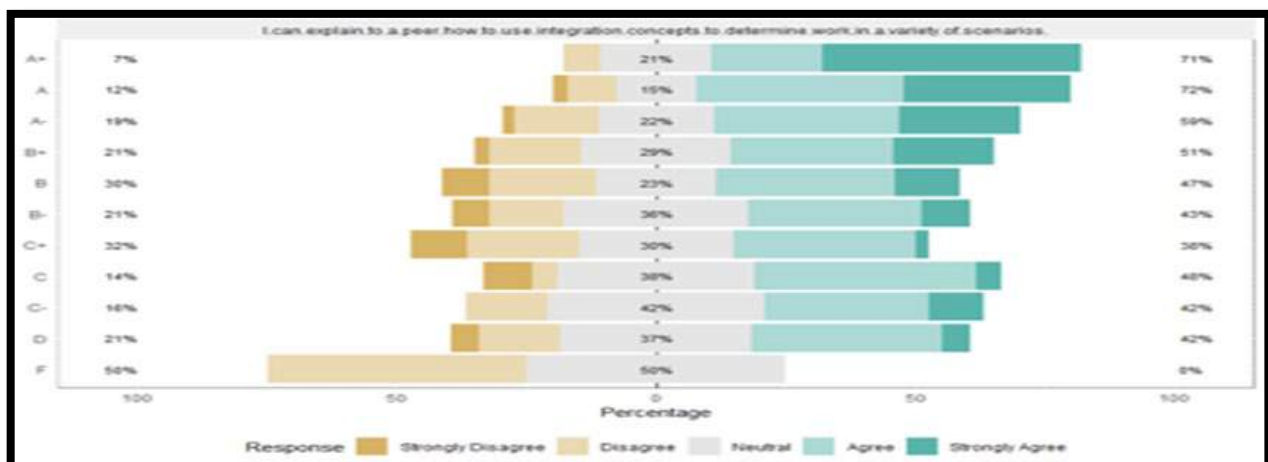


Figure 46: Student Mid-Course Response to Survey Question – "I can explain to a peer how to use integration concepts to determine work in a variety of scenarios"

Notice how the students' initial understanding is inversely related to the students' post-understanding. The higher performing students recognize that they do not understand the material initially while the poorly performing students do not understand the topic enough to even recognize their own lack of understanding.

c. Quantitative Trace 14: Finding the Work to Pump Liquids out of a Tank

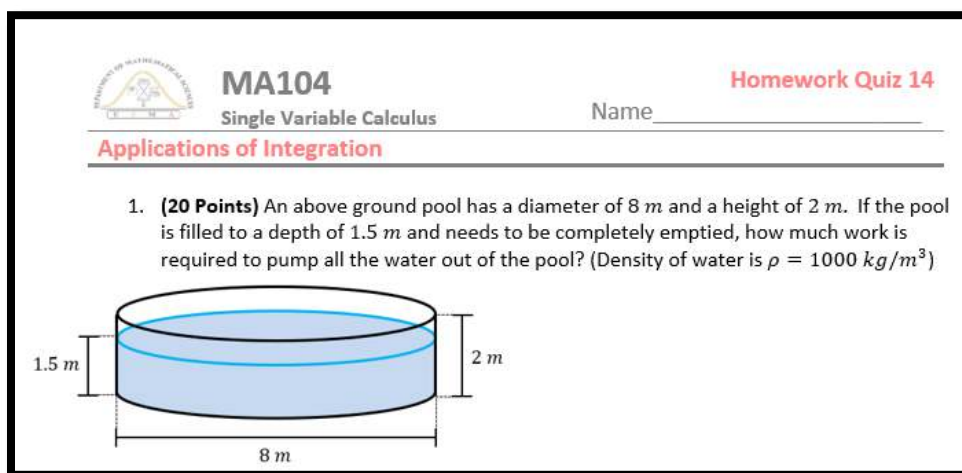


Figure 47: Work Problem on Assessments

This Trace focuses on the students' ability to determine the work (in J or $ft - lbs$) to pump a liquid out of tanks with different geometries. The cadets receive a list of steps which guide them through this problem-solving process. The results from our analysis, focusing on advanced, intermediate, and low students separately are below in Figures 48-50:

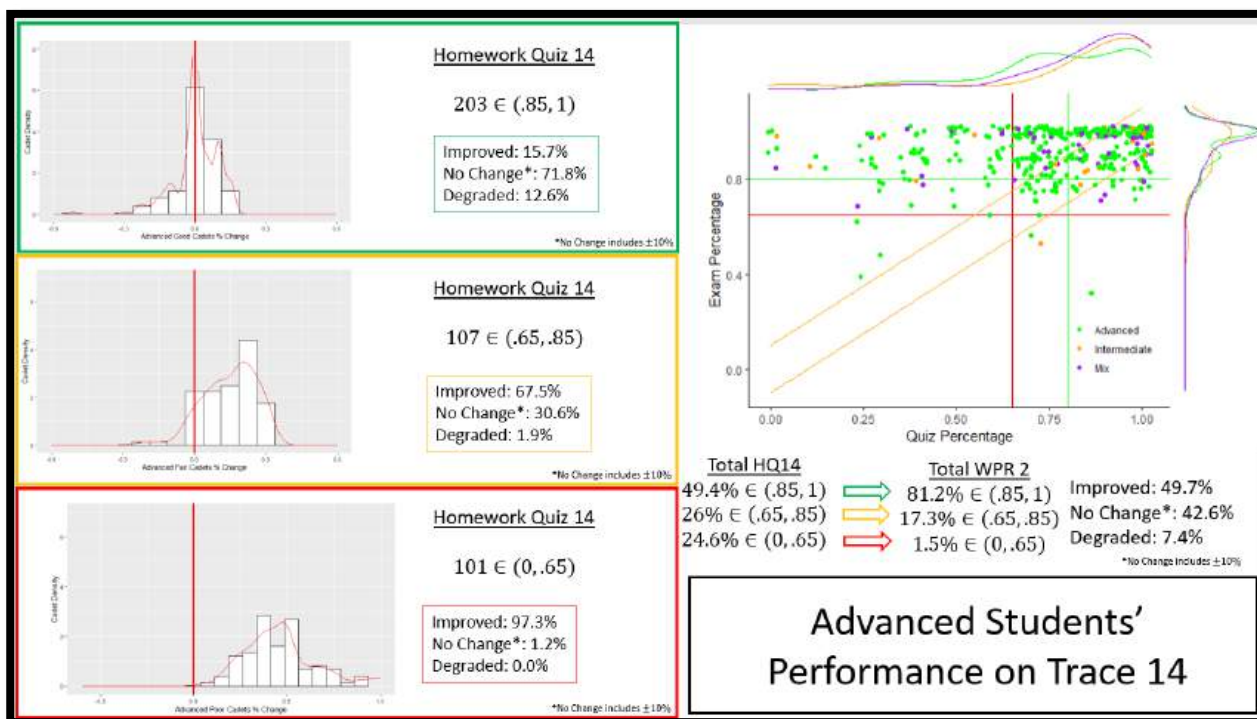


Figure 48: Advanced Students' Performance on Work

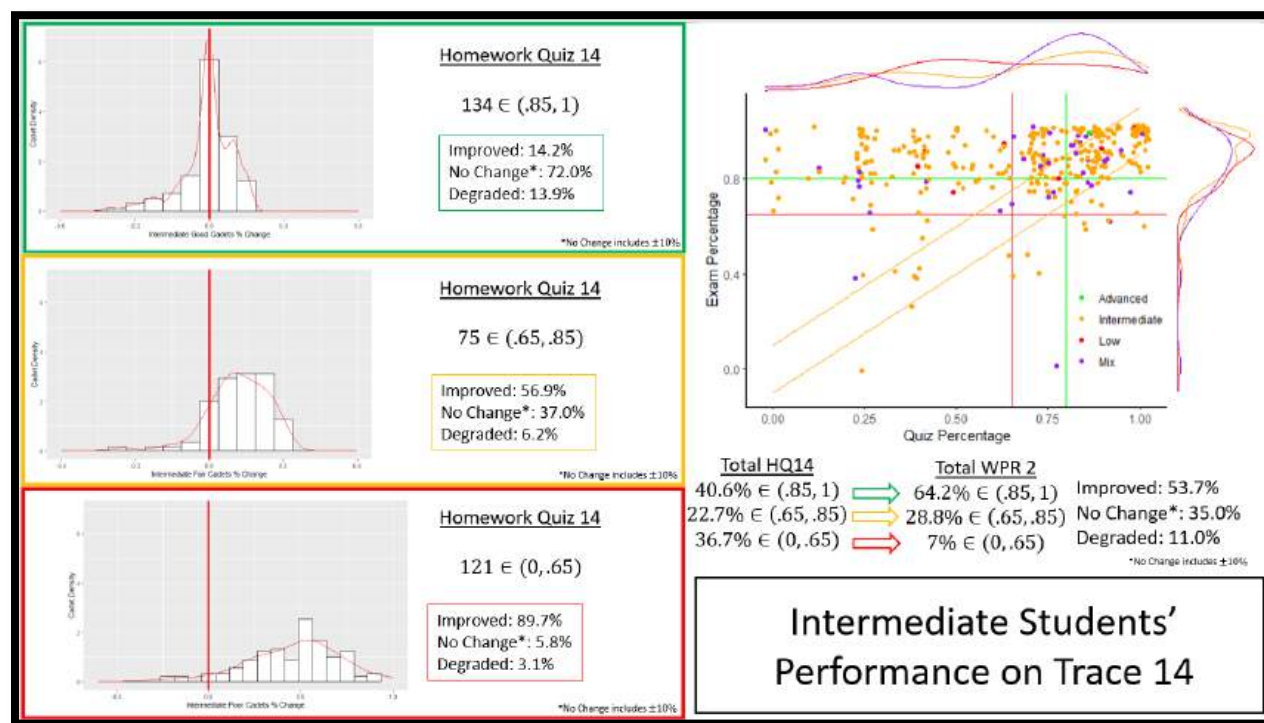


Figure 49: Intermediate Students' Performance on Work

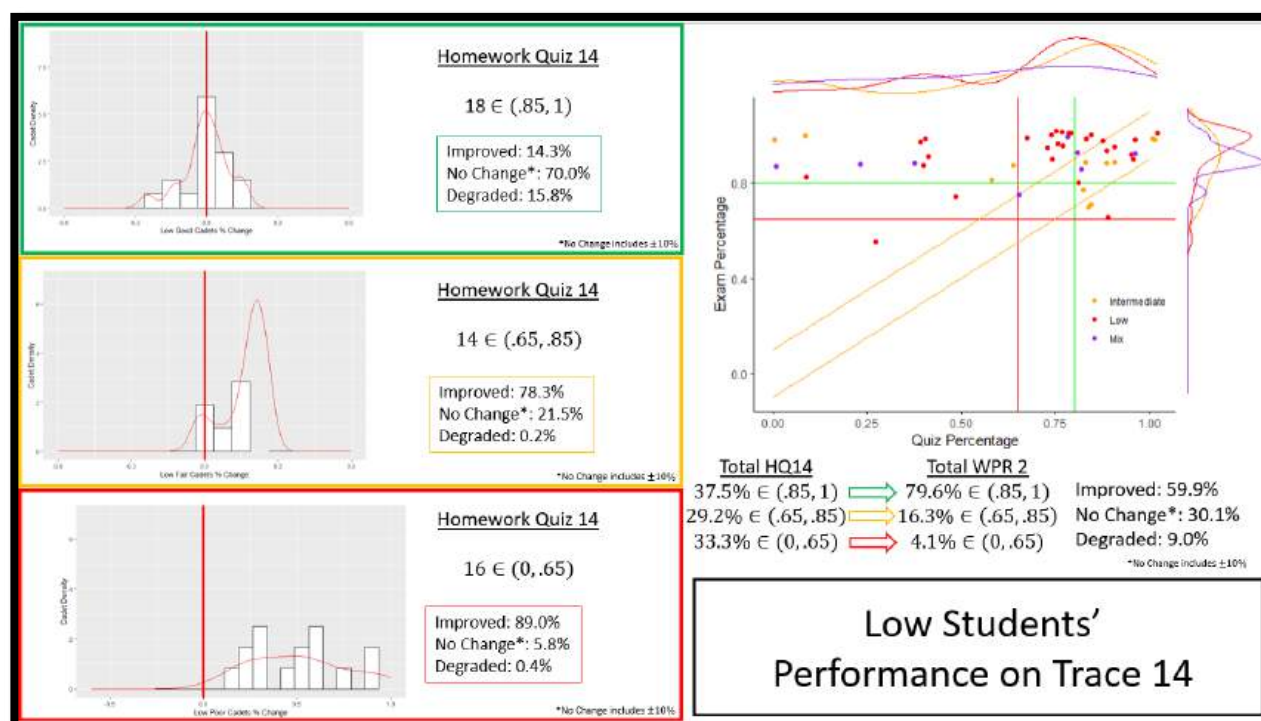


Figure 50: Low Students' Performance on Work

In this analysis, we see a remarkable improvement between HQ14 and WPR 2 across all abilities. In particular, the low students improved so much that they only had 4.1% failing this problem in WPR 2. Notice that the low-red cadets once again outperformed their low-purple peers for this Trace.

6. FCE Analysis. All cadet candidates were remotely administered the Fundamental Concepts Exam (FCE) prior to arriving on R-Day to ensure they are competent enough in their math skills to pass the USMA Core Math Curriculum and help identify cadet candidates who would benefit from taking MA100. Those who do not pass initially are given multiple opportunities to pass throughout MA103. We identified the students who never passed the exam (or passed on their 5th or 6th attempt) and tracked their performance on all major assessments in MA104 to see if their apparent lack of math fundamentals impacts their ability to be successful in MA104. In the following graphs, purple represents passing the FCE on the 5th attempt, orange represents passing the FCE on the 6th attempt, and red represents never passed the FCE.

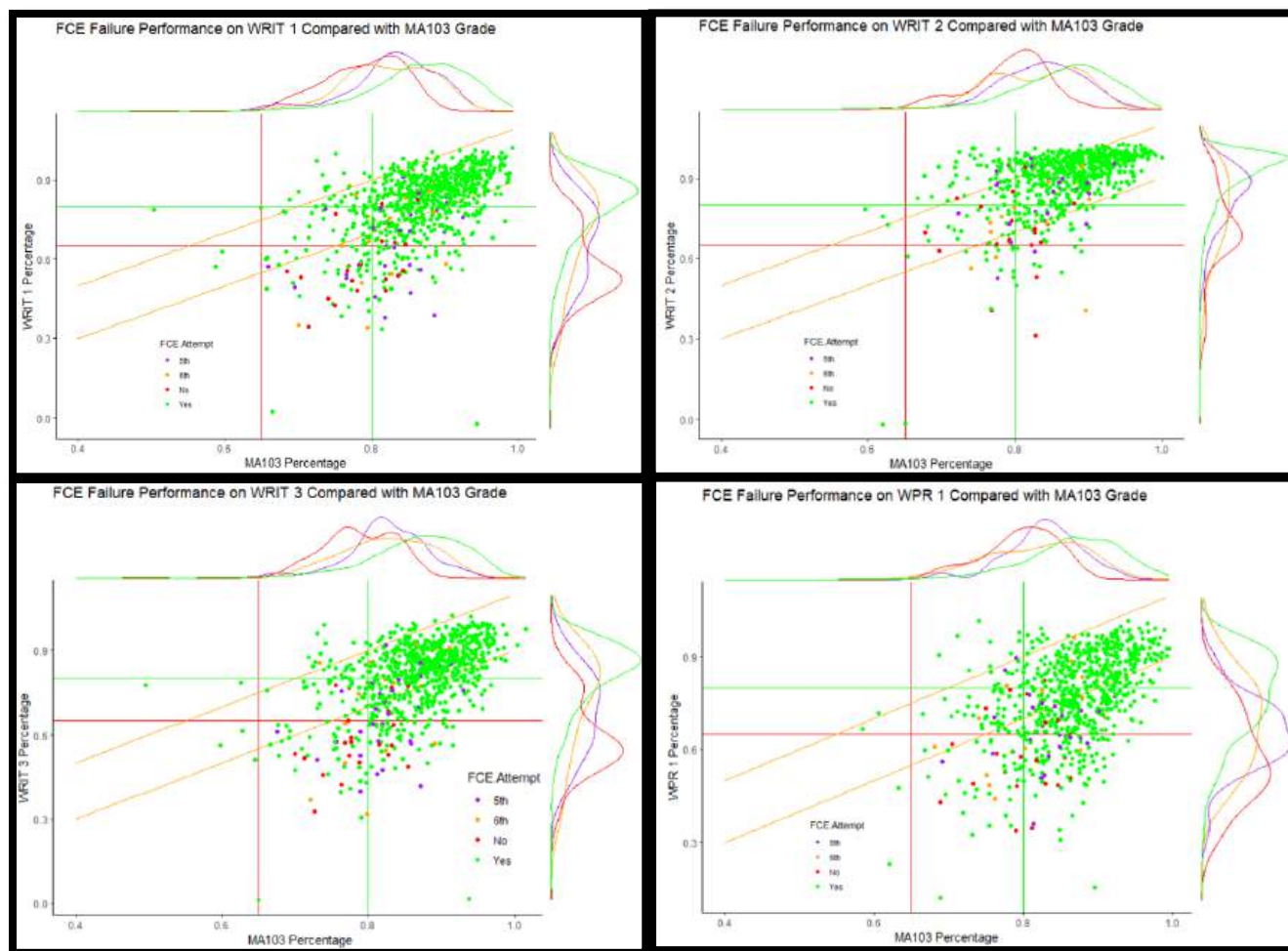


Figure 51: Cadet Performance on MA104 Assessments vs. MA103 Grade & FCE

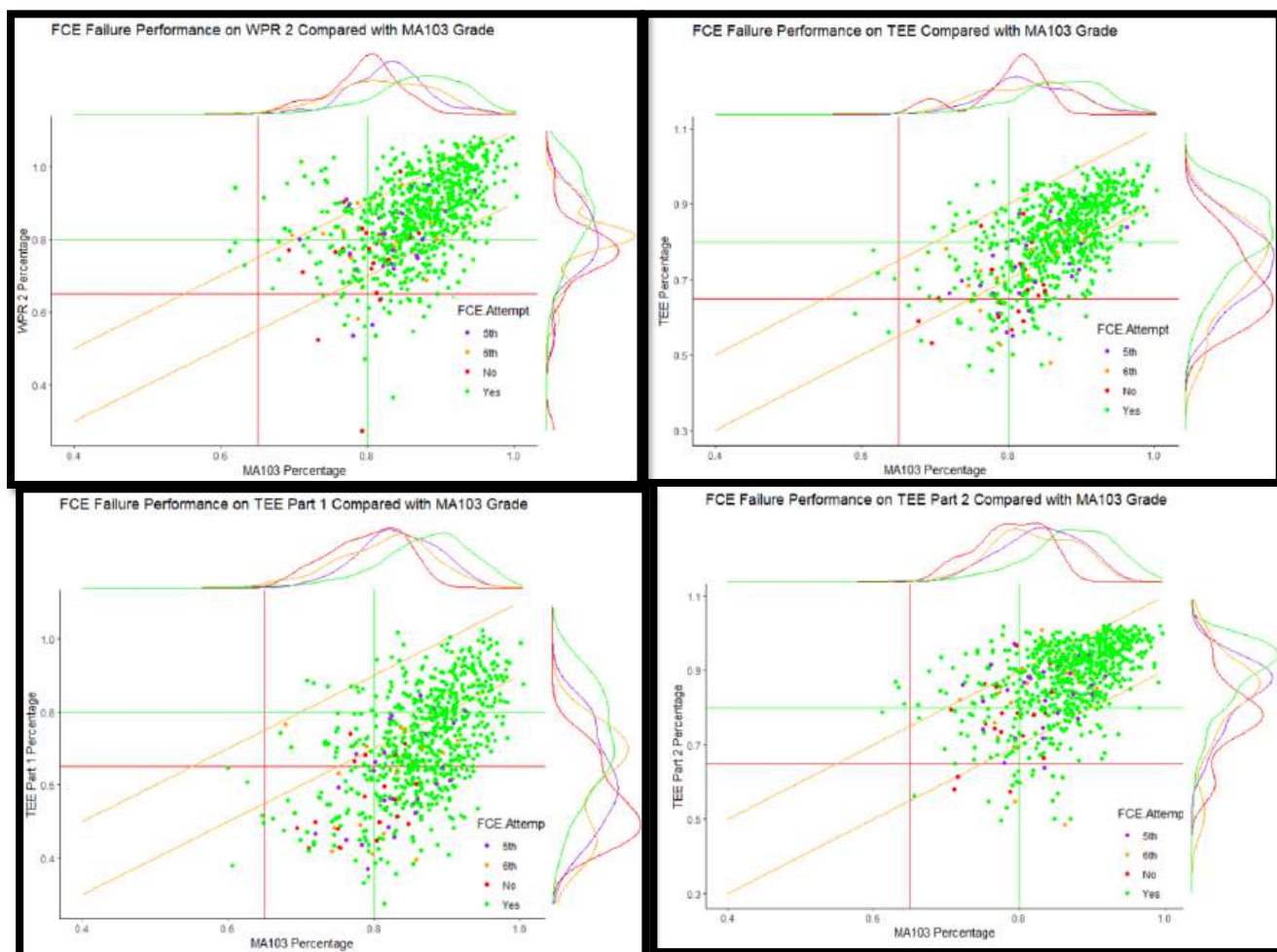


Figure 52: Cadet Performance on MA104 Assessments vs. MA103 Grade & FCE

In WRIT 1, WRIT 2, WPR 1, WRIT 3, and the TEE Part 1, any student who did not pass the FCE performed very poorly (the majority of the distribution is at or below the red failure line). In WPR 2 and the TEE Part 2, failing the FCE did not seem to impact their performance as much. We believe this is because the topics we chose for applications of integration do not depend on math fundamentals as much as the rest of the topics in MA104. However, overall, our analysis shows that if you do not pass the FCE, you will have a difficult time passing any of the MA104 major assessments. In the end, most of these FCE failures still passed the course (as indicated below). However, based on their performance on the major assessments, they did not pass with a good understanding of calculus.

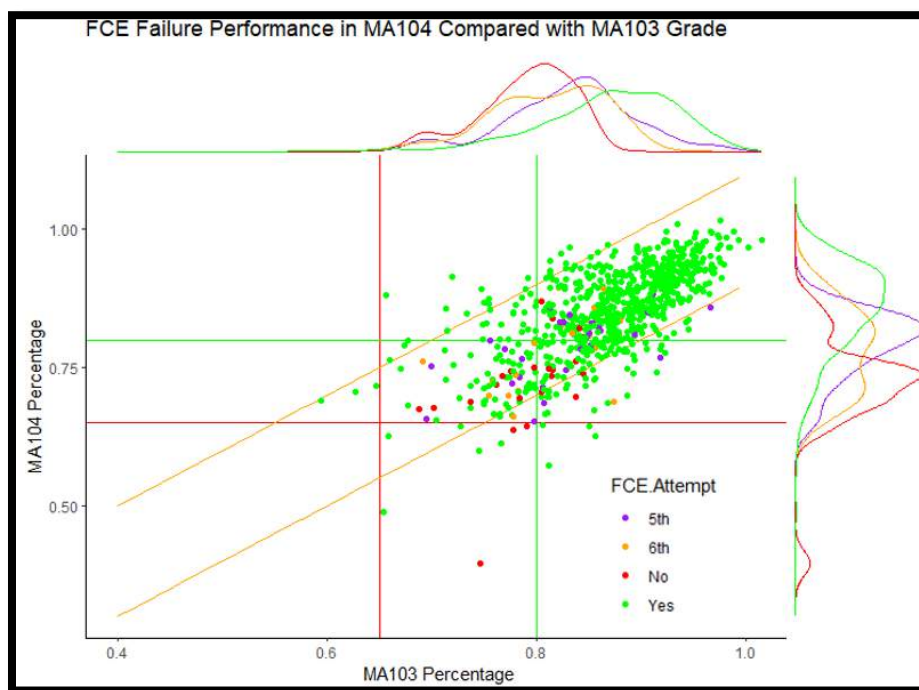


Figure 53: MA104 Performance vs. MA104 Performance And FCE

7. Reducing Student Variance in Initial Ability. Due to being a Core Math Course, all students must either validate or pass MA104 (or the advanced math track equivalent) to be eligible for graduation and ultimately commissioning. With this in mind, we receive a wide variance in the initial ability of the students entering MA104 which makes structuring the course difficult; how do we make the course approachable for those with weak fundamentals while still being rigorous enough to challenge advanced students to stretch themselves? As seen by the FCE Analysis above, our current course design allows for those with weak fundamentals to pass the course. However, for those with STEM degrees, when they arrive in MA205 they do not perform well due to their weak fundamentals. We recommend that a new course is developed (MA104S) which is a prerequisite for MA205 and necessary for any cadet with a STEM major. All cadets without STEM majors would still take MA104 with MA104S being optional if they want to pursue a more rigorous calculus experience. Then, all students would still receive a calculus course, but we would be able to ensure those moving on to MA205 receive a more rigorous calculus experience that better prepares them for multi-variable calculus. If we stick with the current course design, we recommend the following to attempt to reduce this student variance and better prepare students for MA205:

1. **Emphasize and Incentivize MA104 Validation:** In recent years, we have doubled the number of cadets who validate MA104 and move straight to MA205. This helps lower the number of students who are too advanced for the current design of MA104 and allows us to focus on the lower performing students. However, we

have a difficult time increasing our validation numbers because students without STEM degrees have no incentive to validate. If validation reduced the number of core math classes a cadet would have to take (or some other incentive), then more students would take advantage of this opportunity. At the moment, these cadets would rather take the easy A from MA104 to help their overall GPA than stretch themselves by taking MA205.

2. Continue to Section MA104 by Ability: Based on our analysis above, low performing students struggle when in a mixed classroom. The instructor prepares lessons to reach the intermediate students and the low performing students often get left behind without the instructor even realizing it. However, when sectioned by ability, the instructor can prepare lessons with the low students in mind and can more easily monitor their understanding since the more advanced students don't answer all the questions before the low students even know what is going on. In addition to deliberately sectioning the low performing students, stratify the high performers based on calculus experience and potentially STEM majoring.
3. Include more challenging problems on assessments that are weighted significantly less than typical questions. This will challenge the STEM and high-performing students to reach for the A+ scores without completely undercutting the ability of marginal students to pass the course.
4. Counsel STEM Cadets with a Poor Final Grade: After discussing MA104 student performances with the MA205 instructors, we have identified that any student who receives a C- or below generally fails to pass MA205. With this in mind, at the end of the semester any STEM cadet who is in this category should be counseled to help them recognize their potential for failing MA205 and help them consider switching to a different major.

MA104: Single Variable Calculus
AY21-2 Syllabus

Printed on: 3/24/2021

Text: Calculus - Ninth Edition, Early Transcendentals by James Stewart

Students:

All homework sets will be posted on Blackboard as an assignment and will be available for you to download. Upload each completed homework set prior to 1630 on the day it is due for completion credit.

Course Objectives:

1. Understand and internalize the role of the limit in differentiation and integration.
2. Understand, calculate, and interpret problems involving average rates of change and instantaneous rates of change.
3. Understand, calculate, and interpret problems involving the accumulation of change.
4. Evaluate derivatives and integrals.
5. Internalize the concept of the derivative as a rate of change, the integral as the accumulation of change, and that these two concepts are inversely related.
6. Understand, calculate and interpret applications of differentiation and integration.
7. Communicate calculus problem solving techniques and methodologies.

Lesson	Date	Topic (Reading)	Notes	Lesson Objectives
BLOCK 1: DERIVATIVES AND THEIR APPLICATIONS				
Functions & Mathematical Models				
Lesson 1	1/19/2021	Course Intro		Topic: Overview of the Course
Lesson 2	1/20/2021	Functions 1 (§ 1.1)	Digital Writing Assignment Due	1. Understand the definition of a function. 2. Understand and recognize the four ways to represent a function. 3. Determine the domain and range of a function. 4. Recognize the difference between even and odd functions. 5. Determine when a function is increasing and decreasing.
Lesson 3	1/21/2021	Functions 2 (§ 1.2)		1. Understand James Stewart's definition of a mathematical model. 2. Understand how empirical models are used to make a function fitting the data. 3. Starting with a verbal function, construct an algebraic function using the numerical & graphical representations of the function. 4. Recognize the shapes and basic properties of the following functions: Linear, Polynomial, Power, Rational, Algebraic, Trigonometric, Exponential, and Logarithmic.
Lesson 4	1/22/2021	Functions 3 (§ 1.3)		1. Understand how to shift functions vertically and horizontally. 2. Understand how to stretch functions vertically and horizontally. 3. Recognize and construct composite functions.
Limits & Continuity				
Lesson 5	1/25/2021	Limits/Continuity 1 (§ 2.1)	HW 1 Due	1. Understand the definition of a tangent and a secant line. 2. Determine average rates of change. 3. Estimate instantaneous rates of change. 4. Find or estimate the equation of a tangent line to a curve.
Lesson 6	1/26/2021	Limits/Continuity 2 (§ 2.2)		1. Understand the definition of a limit. 2. Find the limit of a function graphically and numerically.
Lesson 7	1/27/2021	Limits/Continuity 3 (§ 2.3)		1. Understand how to apply the properties of limits. 2. Find the limit of a function algebraically.
Lesson 8	1/28/2021	Limits/Continuity 4 (§ 2.5)		1. Understand the definition of continuity. 2. Identify where and why a function is discontinuous. 3. Recognize the types of functions that are continuous at every number in their domain.
Rates of Change				
Lesson 9	1/29/2021	Rates of Change 1 (§ 2.7)	HW 2 Due	1. Understand the definition of a tangent line. 2. Understand the limit definition of a derivative. 3. Apply the limit definition of a derivative to solve for derivatives at a point. 4. Understand the definition of velocity and how it relates to derivatives. 5. Apply the limit definition of a derivative to solve for instantaneous rates of change.
Lesson 10	2/1/2021	Rates of Change 2 (§ 2.8)		1. Understand and apply the limit definition of a derivative to a function. 2. Recognize and apply the different notation used to represent a derivative. 3. Understand how a function can fail to be differentiable. 4. Recognize and understand the relationship between differentiation and continuity.
Derivatives				
Lesson 11	2/2/2021	Derivatives 1 (§ 3.1)	HW 3 Due	1. Recognize and apply the following derivative rules: a) Derivative of a constant function, b) Power rule, c) Constant multiple rules, d) Sum rule, e) Difference rule, f) Derivative of the natural exponential function.
Lesson 12	2/4/2021	Project Development (Lab 1)		
Lesson 13	2/5/2021	Derivatives 2 (§ 3.2)		1. Recognize and apply the following derivative rules: a) Product rule, b) Quotient rule

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Lesson	Date	Topic (Reading)	Notes	Lesson Objectives
Lesson 14	2/8/2021	WRIT 1 (Deans Hour 1: 1210)		Topics: Functions, Mathematical Models, Limits, Continuity
Lesson 15	2/9/2021	Derivatives 3 (§ 3.3)		1. Recognize and apply the derivative rules of trigonometric functions. 2. Understand the relationship between the derivatives of sine and cosine functions.
Lesson 16	2/10/2021	Derivatives 4 (§ 3.4)		1. Recognize composite functions. 2. Understand when to apply the chain rule. 3. Solve derivatives that involve using the chain rule. 4. Use the chain rule to differentiate an exponential function with base b.
Implicit Differentiation				
Lesson 17	2/11/2021	Derivatives 5 (Lab 2)		
Lesson 18	2/12/2021	Project Comp Day		
Lesson 19	2/16/2021	Implicit 1 (§ 3.5: pg 209-213)	HW 4 Due	1. Recognize when a function is defined implicitly. 2. Understand how to apply implicit differentiation. 3. Solve for the equation of a tangent line at a point on any implicitly defined curve.
Lesson 20	2/17/2021	Implicit 2 (§ 3.6: pg 217-219)	Project 1 Due	1. Understand how implicit differentiation is used to determine the derivative of logarithmic functions. 2. Solve for the derivative of logarithmic functions. 3. Solve for the derivative of implicit functions with respect to time.
Lesson 21	2/18/2021	Implicit 3 (Lab 3)		
Related Rates				
Lesson 22	2/19/2021	Related Rates 1 (§ 3.9)	HW 5 Due	1. Understand how implicit differentiation and the chain rule are applied to related rates applications. 2. Understand how the USMA Mathematical Problem-Solving Process relates to Stewart's problem-solving strategy. 3. Apply the problem-solving process to related rates problems. 4. Solve for unknown rates given information about a geometric relationship and a known rate of change.
Lesson 23	2/22/2021	Related Rates 2 (§ 3.9)		1. Apply the problem-solving process to related rates problems.
Lesson 24	2/23/2021	Related Rates 3 (§ 3.9)	Derivative Worksheet Due	1. Apply the problem-solving process to related rates problems.
Lesson 25	2/25/2021	WRIT 2 (Deans Hour 1: 1210)		Topics: Average & Instantaneous Rates of Change, Derivatives, Implicit Differentiation
Lesson 26	2/26/2021	Related Rates 4 (Lab 4)		
Derivatives in the Natural Sciences				
Lesson 27	3/2/2021	Natural Sciences (§ 3.7)	HW 6 Due	1. Apply derivative concepts to physics, chemistry, biology, and economics. 2. Understand how to use derivative concepts to determine when an object in motion is slowing-down or speeding-up. 3. Understand the difference between average density and linear density. 4. Determine and understand marginal cost.
Lesson 28	3/3/2021	Project Development (Lab 5)		
Maximums & Minimums				
Lesson 29	3/4/2021	Max/Min 1 (§ 4.1)	HW 7 Due	1. Understand the difference between absolute and local extrema. 2. Apply the Extreme Value Theorem. 3. Understand how Fermat's Theorem relates to critical numbers and the cautions associated with using Fermat's Theorem. 4. Use critical numbers as part of the Closed Interval Method to determine absolute maximum/minimum values.
Lesson 30	3/5/2021	Max/Min 2 (§ 4.3)		1. Use a function's derivative to identify increasing or decreasing intervals. 2. Identify and solve for intervals where a curve is concave up or concave down. 3. Understand and identify inflection points. 4. Understand and apply the second derivative test to solve for local maximum and minimum values.
Lesson 31	3/9/2021	Max/Min 3 (§ 4.3)		1. Use information for first and second derivatives to sketch the curve of a function.
Optimization				
Lesson 32	3/11/2021	Optimization 1 (§ 4.7)	HW 8 Due	1. Understand how Stewart's optimization problem solving strategy relates to the USMA Mathematical Problem-Solving Strategy. 2. Understand and apply the first derivative test for absolute extreme values. 3. Given a scenario with objectives and constraints, solve for the maximum or minimum values.
Lesson 33	3/12/2021	Optimization 2 (§ 4.7)		1. Given a scenario with objectives and constraints, solve for the maximum or minimum values.

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Lesson	Date	Topic (Reading)	Notes	Lesson Objectives
Lesson 34	3/15/2021	Optimization 3 (§ 4.7)		1. Given a scenario with objectives and constraints, solve for the maximum or minimum values.
Lesson 35	3/16/2021	Optimization 4 (Lab 6)		
BLOCK 2: INTEGRALS AND THEIR APPLICATIONS				
Antiderivatives				
Lesson 36	3/18/2021	Antiderivatives 1 (§ 4.9)	HW 9 Due	1. Understand the definition of an antiderivative. 2. Solve for an antiderivative of a functions. 3. Understand antiderivatives as a family of functions.
Lesson 37	3/19/2021	WPR 1 (Deans Hour 1: 1210)		Topics: Applications in Natural Sciences, Related Rates, Maximum & Minimum Values, Optimization
Lesson 38	3/22/2021	Project Comp Day		
Lesson 39	3/23/2021	Antiderivatives 2 (§ 4.9)	Project 2 Due	1. Use antiderivatives to solve problems related to rectilinear motion.
Definite Integrals				
Lesson 40	3/25/2021	Definite Integral 1 (§ 5.1)	HW 10 Due	1. Understand how to estimate the area between a curve and the horizontal axis on an interval. 2. Solve area estimates using sigma notation. 3. Understand how to solve areas using the limit operator in conjunction with sigma notation. 4. Interpret distances traveled by an object as an area problem.
Lesson 41	3/26/2021	Definite Integral 2 (§ 5.2)		1. Calculate definite integrals using Definition 2 and Theorem 4 in Section 5.2. 2. Express definite integrals as a limit of a Riemann Sum and express a limit of a Riemann Sum as a definite integral. 3. Understand the mathematical notation and terminology of integrals. 4. Apply the Midpoint Rule to estimate an integral. 5. Understand an apply the properties of definite integrals.
Lesson 42	3/29/2021	Definite Integral 3 (§ 5.3)		1. Understand and apply the Fundamental Theorem of Calculus, Part 1. 2. Understand and apply the Fundamental Theorem of Calculus, Part 2. 3. Internalize differentiation and integration as inverses processes.
Integration				
Lesson 43	3/31/2021	Integration 1 (§ 5.4)	HW 11 Due	1. Distinguish between indefinite and definite integrals. 2. Determine indefinite integrals. 3. Understand and apply the Net Change Theorem.
Lesson 44	4/1/2021	Integration 2 (§ 5.5)	WPR Revisions Due	1. Understand when to apply the substitution rule. 2. Use the substitution rule to solve definite and indefinite integrals.
Lesson 45	4/2/2021	Integration 3 (§ 7.1)		1. Understand how the method of integration by parts is derived from the product rule of differentiation. 2. Understand how to select "u" and "dv" when using integration by parts. 3. Apply integration by parts to solve definite and indefinite integrals.
Lesson 46	4/6/2021	Integration 4 (§ 7.5)		1. Understand how to approach a definite and indefinite integral in order to determine the appropriate integration method(s). 2. Solve any integral by choosing the correct integration technique (simplifying w/ basic rules, u-sub, IBP)
Lesson 47	4/8/2021	Integration 5 (§ 7.5)		1. Understand how to approach a definite and indefinite integral in order to determine the appropriate integration method(s). 2. Solve any integral by choosing the correct integration technique (simplifying w/ basic rules, u-sub, IBP)
Lesson 48	4/9/2021	Integration 6 (§ 7.8)		1. Identify improper integrals as either Type 1 or Type 2 improper integrals. 2. Identify when a Type 1 improper integral is either convergent or divergent. 3. Solve Type 1 improper integrals.
Areas & Volumes				
Lesson 49	4/12/2021	Areas/Volumes 1 (§ 6.1)	HW 12 Due	1. Determine the area between two curves. 2. Interpret the meaning of the area between two curves given in context.
Lesson 50	4/13/2021	Areas/Volumes 2 (§ 6.2)		1. Understand the definition of volume. 2. Determine the volume of a solid object using integration techniques. 3. Determine the volume of a solid of revolution using the disk/washer method.
Lesson 51	4/14/2021	Exam Comp Day	No Class	
Lesson 52	4/15/2021	Areas/Volumes 3 (§ 6.2)	Integral Worksheet Due	1. Understand the definition of volume. 2. Determine the volume of a solid object using integration techniques. 3. Determine the volume of a solid of revolution using the disk/washer method.
Lesson 53	4/16/2021	Areas/Volumes 4 (§ 6.3)		1. Determine the volume of a solid of revolution using the cylindrical shell method.
Lesson 54	4/19/2021	Project Development (Lab 7)		

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Lesson	Date	Topic (Reading)	Notes	Lesson Objectives
Lesson 55	4/20/2021	Areas/Volumes 5 (§ 6.3)		1. Understand when to apply the disk/washer method and when to apply the cylindrical shell method when determining the volume of a solid of revolution.
Study Day	4/21/2021	WRIT 3 (Exam Period: 1030)		Topics: Antiderivatives, Riemann Sums, Definite & Indefinite Integrals, Integration Strategies, Improper Integrals
Applications of Integration				
Lesson 56	4/22/2021	Applications of Integration 1 (§ 6.4)	HW 13 Due	1. Understand the definition of work. 2. Solve lifting problems involving constant and variable force using both SI and US units of measure.
Lesson 57	4/23/2021	Applications of Integration 2 (§ 6.4)		1. Understand the definition of work. 2. Solve lifting problems involving constant and variable force using both SI and US units of measure.
Lesson 58	4/26/2021	Applications of Integration 3 (§ 6.4)		1. Solve pumping problems involving tanks with constant and variable cross-sectional areas using both SI and US units of measure.
Lesson 59	4/28/2021	Applications of Integration 4 (§ 6.4)		1. Solve pumping problems involving tanks with constant and variable cross-sectional areas using both SI and US units of measure.
Lesson 60	4/30/2021	Applications of Integration 5 (§ 8.5)		1. Understand the definition of a probability density function. 2. Change a density function into a PDF by solving for a scaling constant k. 3. Interpret a probability statement. 4. Solve for a probability using a probability density function.
Lesson 61	5/3/2021	Project Development (Lab 8)		
Lesson 62	5/4/2021	Project Comp Day	HW 14 Due	
Lesson 63	5/5/2021	WPR 2 (Deans Hour 1: 1210)		Topics: Applications of Integration, Work, Probability, Net Change, Areas & Volumes
Lesson 64A	5/7/2021	Project Briefs	Project 3 Due	
Lesson 64B	5/10/2021	Project Briefs		
Lesson 64C	5/11/2021	Project Briefs		



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB B

AY2022 COURSE ASSESSMENT



REPLY TO
ATTENTION OF:

DEPARTMENT OF THE ARMY
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

MADN-MATH

2 June 2022

MEMORANDUM THRU

COL Michael Yankovich, Core Program Director, Department of Mathematical Sciences, United States Military Academy, West Point, NY 10996
COL Tina Hartley, Professor and Head, Department of Mathematical Sciences, United States Military Academy, West Point, NY 10996

FOR Dean of the Academic Board, United States Military Academy, West Point, NY 10996

SUBJECT: Annual Executive Summary for MA104, AY 22-02

1. **Course Description.** MA104 is the second course of the mathematics core curriculum capitalizing on MA103's mathematical modeling and introduction to vectors and calculus. The goal of MA104 is for students to internalize the concept of the derivative as a rate of change, the integral as the accumulation of change, and that these two concepts are inversely related by the fundamental theorem of calculus. Students will realize that both differential and integral calculus arise from the application of the limit to an estimate. An understanding of course material is enhanced with the use of Mathematica and Microsoft Excel. MA104 services MA205 (Multivariable Calculus), MA206 (Probability and Statistics), PH205 (Physics), SS201 (Economics), the Core Engineering Sequences, and selected majors. This course is followed by MA205 or MA206.

2. **MA104 Goals Assessment.** The course had three Academic Program Goals (APG) – two at the “Introduce” level (Critical Thinking and Creativity, Lifelong Learning) and one at the “Reinforce” level (STEM) that directly link to three Core Math Goals. MA104 also supported the additional APG of Disciplinary Depth through the Core Math Goal of Interdisciplinary Problem Solving though that was not specifically required. The below table shows the mapping of each APG through the particular “What Graduates Can Do” (WGCD) statement into the learning objectives and assessments within MA104. See Paragraph 6 for APG and Core Math Goal assessment plan in AY23.

Table 1. Academic Program Goal Mapping into MA104 – Single Variable Calculus

Academic Program Goals (APG)	"What Graduates Can Do" (WGCD) Statement	Core Math Goals	MA104 Course Objective	MA104 Assessment
2. Critical Thinking and Creativity	2.4 (I) Reason both quantitatively and qualitatively.	Body of Knowledge	Evaluate derivatives and integrals and interpret the results in context. Communicate calculus problem solving techniques and methodologies.	Homework Quizzes, WRITs, WPRs, and Course Projects.
3. Lifelong Learning	3.1 (I) Demonstrate the willingness and ability to learn independently.	Habits of Mind	Internalize the concept of the derivative as a rate of change, the integral as the accumulation of change, and that these two concepts are inversely related.	Daily Preparation Requirements, Concept Quizzes, Revision Assignments, and Course Projects.
5. Science, Technology, Engineering, and Mathematics	5.1 (R) Apply mathematics, science, and computing to model devices, systems, processes, or behaviors.	Technology Skills	Understand, calculate, and interpret applications of differentiation and integration.	WRITs, WPRs, and Course Projects.
7. Disciplinary Depth	7.1 (NA) Apply disciplinary tools, methods of inquiry, and theoretical approaches.	Interdisciplinary Problem Solving	Understand, calculate, and interpret applications of differentiation and integration. Communicate calculus problem solving techniques and methodologies.	WPRs and Course Projects.

a. APG 2 – Critical Thinking and Creativity (Assessment: 2, Very Good). The “What Graduates Can Do” statement associated with APG 2 is that students should reason both quantitatively and qualitatively. This connects to the Core Math Goal of building a “Body of Knowledge” and the MA104 Course Objectives of evaluating derivatives and integrals, and then interpreting and communicating their results. MA104 assessed quantitative student development by tracking performance on sequential assessments of eleven key skills throughout the semester. We judged qualitative reasoning through the exploration of two course projects in which students had to apply calculus to Branch and Posting data and make recommendations to improve their outcomes. We observed that cadets demonstrate a good understanding of course material – with a course average above 80% on the TEE in 8 of 8 key skills. This suggests that they possess a good ability to think critically and creatively and that the course meets the “Introduce” Academic Program Goal for cadets. The complete analysis of this APG and associated Core Math Goal can be found in Appendix 2.

b. APG 3 – Lifelong Learning (Assessment.: 3, Good). This APG states that graduates should demonstrate the willingness and ability to learn independently which is supported by the Core Math Goal of developing strong “Habits of Mind” in our students. MA104 satisfies this requirement through the course objectives of understanding and internalizing the concepts of differentiation and integration, but more directly through the structure of the course. MA104 requires students to prepare and be assessed every day, to revisit and correct their mistakes, and to explore and investigate projects on their own. We collected student sentiment data through three surveys during the semester and we observe that most ($> 75\%$) of cadets recognize the utility of nightly preparation, consistent assessment, and revisions based on what they did not initially understand. This indicates that students have been exposed to and understand the importance of independent learning and that MA104 again meets the “Introduce” Academic Program Goal.

c. APG 5 – Science, Technology, Engineering, and Mathematics (Assessment.: 2, Very Good). This APG states that graduates should be able to apply mathematics, computing devices, and modeling systems to solve problems. This is supported by the Core Math Goal of building Technology Skills in our cadets and is executed in MA104 through the application of single variable calculus to solve problems on WRITs, WPRs, and projects using an appropriate technological tool. In AY22, MA104 sought to focus and streamline the application of technology as it has the potential to disrupt the underlying course objectives of equipping students with calculus fundamentals if done incorrectly. Throughout the semester we were deliberate about when we chose to use specific tools – whether the issued calculator, Microsoft Excel, or Mathematica based on the type of problem we were attempting to solve. After collecting quantitative and qualitative data on student performance and sentiment we observe that students are generally capable and feel comfortable executing basic commands and troubleshooting code to solve single variable calculus problems. Appendix 1, Table 6, Trace 6 – Max & Min supports this conclusion. The questions we asked on that topic forced students to select a technological tool and then use it to solve a particular problem. We see that the course average on these problems improved throughout the semester and ended above 83%, suggesting students have a good understanding of when and how to apply technology. The complete analysis of this APG and associated Core Math Goal can be found in Appendix 1 and 3. While we did not increase the amount of technology utilized in the course this academic year, we believe that we satisfy the “Reinforce” goal through our emphasis on appropriateness and that we successfully managed to balance this task without undermining our basic course objectives.

d. APG 7 – Disciplinary Depth (Assessment.: 3, Good). This goal states that graduates should apply interdisciplinary tools, methods of inquiry and theoretical approaches to solve problems. This links to our Core Math Goal of Interdisciplinary Problem Solving and is manifest in MA104 in our application and communication objectives. We used WPRs and course projects as the means to solve application problems and highlight the interdisciplinary nature of calculus. We introduced the calculus tools of optimization and probability and applied them to a variety of scenarios using a consistent problem-solving framework. We aimed the projects at relevant, but not outwardly math-heavy, problems in an attempt to demonstrate the value and utility of calculus-thinking in solving diverse and complex problems. Student performance on these projects and the sentiment data that we collected suggest that most (> 70%) of students enjoyed the context of the scenarios that motivated the projects and that they were forced to think critically to solve the problems. We believe that this indicates that the course was successful in meeting the disciplinary depth Academic Program Goal. Complete analysis of this APG can be found in Appendix 4.

3. Course Statistics. The course average for AY22-02 was an 86.3%. The distribution of grades is below. There were 6 failures in the course – 5 who scored less than 65% overall, and 1 who scored less than 50% on the TEE. There were no NCs

given in AY22-02 as the single outstanding honor case already received an “F” in the course due to performance.

Table 2. Grade Distribution

MA104 AY22-02 Grade Distribution		
Grade	Count	Course Percentage
A+	42	5.7%
A	125	16.9%
A-	118	16.0%
B+	118	16.0%
B	123	16.7%
B-	61	8.3%
C+	53	7.2%
C	59	8.0%
C-	30	4.1%
D	15	2.0%
F	6	0.8%
NC	0	0.0%
Course Average	86.30%	

4. **Course Composition and Recommendations.** The structure of MA104 AY22-2, included a 64-lesson sequence in which points were accumulated in a series of assessments of increasing value by topic. Assessments included concept quizzes, homework assignments, homework quizzes, WRITs, WPRs, and projects; and these assessments were divided into two distinct blocks during the semester. The first 8 weeks of the course explored differentiation concepts, computations, and applications while the last 8 weeks explored integration concepts, computations, and applications. Each lesson was associated with a larger grouping of lessons involving a common topic, and each common topic had its own series of assessments designed to reinforce student comprehension. The course was taught primarily in-person.

a. Classroom Design. The course designed the classroom experience for students based on the mathematical ability they displayed in MA103 – Mathematical Modeling and Introduction to Calculus, performance on the Fundamental Concepts Exam, and prior Calculus experience. Students were sectioned with like-students to form 8 - Low, 16 - Intermediate, 8 - High, and 19 - Mix sections.

Recommendations for Classroom Design. Sectioning cadets by ability was beneficial in a COVID environment, when only a limited number of students could be in the classroom. Sectioning ensured that low-mathematical ability students were able to obtain the one-on-one attention they required to perform well in the course. However, outside of the COVID environment, when all cadets were taught in-person, sectioning

did not lead to an overall improvement for low-ability students. Low-ability students in mixed sections tended to outperform their counterparts in pure low-ability sections. Similar trends could be seen in intermediate and high ability students. In the future, we recommend sectioning the highest-ability students, but allowing the other students to be sectioned with a good mixture of high, medium, and low-performing students. The hope is that by sectioning in this manner, low-performing students would recognize that they are low-performing when mixed in a class with higher-performing students. Additionally, this would allow for some peer-level learning to happen in the classroom, an aspect which seemed to be somewhat lost in pure low-ability sections during AY22.

b. Grading Scheme. The overall grading scheme for the course is below.

Table 3. MA104 Overall Grade Scale

Event	Points	Weight
Homework Sets (14)	140	7%
Homework Quizzes (14)	280	14%
Tech Tutorial	25	1.25%
Worksheets (2)	80	4%
WRITs (3)	300	15%
Concept Quizzes (25)	100	5%
WPRs (2)	450	23%
Projects (2)	225	11.25%
TEE	400	20.0%
Total	2000	100%

Recommendations for Grading Scheme. We recommend retaining the overall grading scheme and the course composition of low-stakes assessments building into more significant tests of student learning. However, we'd recommend considering weighing WRITs and WPRs equally to emphasize both the mechanics and applications of calculus in a more equivalent manner. See Appendix 1. Overview of Quantitative and Qualitative Trace for details on the overall effectiveness of course assessments on student learning.

c. Homework. The course incorporated 14 homework sets throughout the term. Each homework set was due the lesson after the final topic associated with the lesson group. The homework sets were built to have two components for each individual lesson. The first was a set of concept questions, meant to guide students through the reading of the textbook before class, and the second were two practice problems intended to be completed after class, that would reinforce what was covered that day by the instructor. When it was reasonable, the numerical solution or answer to practice problems was provided within the homework set to allow students to know when they

had done a problem correctly. The homework sets were graded approximately 7 of 10 points for completion and the final 3 points were awarded on a plus/check/minus scale. Homework assignments accounted for 7% of course points.

Recommendations for Homework. We recommend sustaining the Homework in its current format.

d. Concept Quizzes. The course incorporated 30 daily WebAssign concept quizzes throughout the term – only 25 were graded to account for missing class and/or the occasional poor performance. Each 5-minute concept quiz asked 4 or 5 questions regarding topics covered in the reading from the previous night's homework concept questions. Students were allowed and encouraged to use their completed homework concept questions as a reference during these quizzes. This served to motivate nightly preparation for class and confirm the value in reading the material ahead of time. Questions were typically multiple choice, multiple select, or true/false. The prep quizzes accounted for 5% of the course points and served to motivate students to engage with the material prior to attending class. Occasionally, single questions were devoted to previous topics to assist in interleaving topics from previous lessons.

Recommendations for Concept Quizzes. Concept quizzes were generally received positively by the students and instructors. General feedback stated that concept quizzes were effective at leading into lesson material. Other instructors mentioned that the concept questions broke up the rhythm of the class and did not offer them the flexibility to prepare their students in a personalized manner. We recommend sustaining the option of utilizing concept quizzes, but allowing instructors to choose to use concept quizzes or developing their own methods of evaluating preparation to include instructor created quizzes, board problems, and notes checks.

e. Homework Quizzes. The course assessed student learning across common topics using 14 homework quizzes, with one corresponding to each homework set. These 10–12-minute quizzes were completed at the beginning of class the lesson after the conclusion of each topic. Homework quiz questions typically contained a true/false question and a multiple-choice question that was closely related to the concept questions found on the homework. Occasionally the true/false or multiple-choice questions were a reach-back question. Additionally, homework quizzes contained a problem that was similar to a practice problem found within the homework sets and were completed with the MA104 Reference Card and issued TI-36X Pro calculator. A couple quizzes allowed the use of Mathematica. From AY22-01 into AY22-02, MA104 elected to use stringent quiz rubrics to provide very blunt feedback to students in the form of low scores to communicate their understanding of the material. The course also permitted students to revise the quizzes and submit corrections for up to half of the

missed points to account for the binary nature of the rubrics. Homework quiz solution videos were created by course leadership to assist students in completing revisions and understanding the material.

See Appendix 1. Course Analysis based on Sectioning for qualitative feedback from students on the effectiveness of the Homework quizzes and revisions process.

Recommendations for Homework Quizzes. Our recommendation is to sustain the homework quiz structure.

f. WRITs. The course consisted of three WRITs. The WRITs were timed exams that, in total, accounted for 15% of the course. The purpose of the WRIT was to provide the student with an assessment of their understanding of key concepts and their ability to conduct fundamental calculus computations. The WRITs were broken into four primary problems: a) True/False and Multiple-Choice questions, b) graphical short answer questions, c) general short answer questions, and d) computational short answer questions. Students were given alternatives in which questions they had to complete for multiple-choice and true/false questions. WRITs were completed without the use of any student references. The frequency of the WRITs provided the students with closer recall assessments and served as an evaluation tool for instructors to identify students who may need extra assistance on certain topics.

Recommendations for WRITs. Overall recommendation is to sustain the general format and frequency of the WRITs. One recommended change we plan to explore for future MA104 courses is to limit the use of calculators. Sometimes the calculators became a distraction from the Calculus and we noted that some students experienced a significant degradation of fundamental Algebra skills.

g. WPRs. The WPRs were application focused assessments designed to evaluate students' ability to apply calculus as a problem-solving tool to address real-world scenarios. Each major block (Differentiation and Integration) culminated in a WPR event that was designed to take students 55 minutes to complete. Students were allowed to use the MA104 AY22 Reference Card and their issued TI-36X Pro calculators. Because of the difficulty of the problems, students were afforded 75 minutes to complete WPR 1 and were required to complete a revision assignment to review and correct their mistakes. The additional time and revision opportunities were not granted for WPR 2. The WPRs accounted for 23% of the course points.

Recommendations for WPRs. Overall, the course recommends that we sustain the general format of the WPRs. Continue to provide simple applications of the material covered and avoid complicated scenarios. Sustain using the 55-minute Dean's Hour

periods for testing when possible. One recommended change we plan to explore for future MA104 courses is to remove the WPR 1 corrections assignment. This will remove some redundancy since students were already afforded the opportunity to do corrections for all Homework Quizzes which form the basis of the WPRs. Additionally, to structure questions so that it is possible to assess the high performing students and allow some separation between A and A+ performances. Consider assigning 80-87% of the WPR points to relate to concepts directly taught in class, 8-17% of the points to be obtained from a very good understanding of the material, and 3-5% of the points to an aspect of the concepts that challenges students beyond just the basic question types.

h. Projects. Two projects were designed to give students the opportunity to apply calculus as a problem-solving tool to a real-world military scenario. The projects revolved around student performance and their ability to obtain their first branch and post of choice. Designated lab periods with guided questions for students were allocated for instructors to be able to assist students in starting the projects. Project 1 resulted in an individual technical report focused on optimization and Project 2 was an oral group presentation focused on probability. Project 1 included a writing assignment that was meant to serve as a check on their technical writing ability before submitting the final project.

Recommendations for Projects. We recommend sustaining the number, topics, and simplicity of the projects. The projects allowed students to both apply and expand upon their knowledge of material taught during the course without overburdening them with a scenario that was difficult to understand. Regimental Tactical Officers were invited to the oral presentations of Project 2, but many did not attend. We recommend looking for opportunities to give students the ability to brief to an unfamiliar, senior leader.

i. Term End Exam. The TEE accounted for 20% (400 Points) of the cadet grade. The TEE was a two-part assessment, the first part focused on concepts and computations and the second part focused on applications. The TEE was executed in-person and on paper. The only authorized references during the TEE were the MA104 Reference Card and the issued TI36X Pro Calculator. The MA104 TEE was a 2.5 hour exam.

Recommendations for the TEE. We recommend sustaining the format and point value of the TEE. The structure of the test is familiar to students in that Part 1 is very similar to the WRITs they saw throughout the semester and Part 2 mirrors their WPRs. We recommend sustaining the authorized references of the reference card and issued calculator. We should plan for future TEEs being 3.5 hours; if this happens, we should consider expanding the scope (length) of the exam slightly as well as giving

students some of the extra time to allow them to work through more challenging problems.

j. Course Validation. During AY 22-01 we provided first year students the opportunity to validate MA104 and proceed directly to MA205. 217 Students who were not selected for AMP but performed well in summer testing were sent an e-mail asking them to sign -up to take the MA104 validation exam. We met with MA103 instructors and published a flyer inviting students to sign-up to take the validation exam to garner additional interest. We also included optional screener questions on the FCE in MA103 to identify potential validators. In total, we reached out to 245 students (217 from AMP, 19 from sign-ups, and 9 from the FCE screen) and only 36 actually took the exam. We offered the exam in consecutive weeks on study days early in September, and we held a morning and afternoon session each day in hopes of getting a large number of students to take the exam. We also published review materials, held live differentiation and integration review sessions, and made ourselves available to field questions and students worked through practice problems. Of the 36 students who tested, we recommended that 16 should be given validation credit for MA104 based on their scores.

We were disappointed with the number of students who tested and validated despite our outreach efforts, and we surveyed those who did not test to understand why. We had 140 respondents and over 60% said that they chose not to test based on a lack of confidence in high school calculus during the pandemic. While our quantity of validators was low, we did appear to select the right people as the average MA205 score for validators was 86.6% v. a full course average of 81.1%. Only three students ended with lower than a B in the class and four of the validators earned an A+.

Recommendations for the Course Validation Process. We recommend that we sustain scheduling the validation exam within the first 3 weeks of classes to maximize participation. We recommend continuing to offer a live (and recorded) differentiation and integration review session for potential testers to improve their confidence and willingness to test. We support continuing to firmly grade the validation exams and only assign credit to students with a clear potential to succeed in MA205. Observing the small number of validators and the low interest we have seen year over year does not support a full-scale effort to push more students toward MA205 without providing a gain to MA104. We support continuing to identify and validate the upper tier candidate candidates but not going to great lengths to pull in the next tier of candidates who will struggle in MA205.

MADN-MATH

SUBJECT: Annual Executive Summary for MA104, AY 22-02

5. **Course Analysis.** The Appendix of this report provides a detailed course analysis that supports the statistics above and resulted in many of the recommended changes to the course. The analysis is based upon both quantitative and qualitative data that was gathered throughout the semester.

6. **Future Goals Assessment.** The course will continue to assess the three APG – two at the “Introduce” level (Critical Thinking and Creativity, Lifelong Learning) and one at the “Reinforce” level (STEM) that directly link to three Core Math Goals. MA104 is scheduled to assess the Core Mathematics Goals of Developing Habits of Mind and Communicating Effectively. This should allow for the course to assess APG 1 and 3 in AY23.

7. Point of contact for this memorandum is the undersigned at ira.crofford@westpoint.edu, or by phone (845) 938-0061.

1 Enclosure

Appendix – End of Course Analysis

IRA L. CROFFORD, JR.

LTC, FA49

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APPENDIX – End of Course Analysis

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1. Overview of Quantitative and Qualitative Trace. MA104 has a responsibility to assess the administration of the course and the performance of students within AY22. In order to make those assessments, we deliberately collected student scores and sentiment throughout the semester for analysis and review. We selected 11 key skills to assess quantitatively and 10 topics to measure qualitatively to motivate our investigation. Those skills and how they map throughout the semester can be seen in Tables 4 and 5 below.

Table 4. Quantitative Trace Skills

Quantitative Trace					
Trace	Topic	Assessment #1	Assessment #2	Assessment #3	Recall Event
1	Graphical Limits and Discontinuities with Types	HQ2.4	WRIT1.3.B&C	TEE 1.3	HQ9.2
2	Limit Definition of the Derivative	HQ3.3	WRIT1.4.C.2	TEE 1.1.B	HQ6.1
3	Product & Chain Rule	HQ4.4	WRIT2.4.A	TEE 1.2.E	HQ5.3
4	Implicit Differentiation	HQ5.4	WRIT2.3A	TEE 1.2.F	HQ7.1
5	Natural Sciences	HQ7.3	WPR1.3B&D	N/A	N/A
6	Max & Min	HQ8.3	WPR1.1AB	TEE 7A	HQ10.1
7	Riemann Sums	HQ11.3	WRIT3.2	N/A	HQ13.1
8	Net Change / Interpreting Integrals	WRIT 3.3B	WPR2.4.B&C	N/A	N/A
9	Substitution Rule	HQ12.3	WRIT3.4.A	TEE 4E	N/A
10	Improper Integrals	HQ12.4	WRIT 3.4C	TEE 6	N/A
11	Probability	HQ14.2&3	WPR2.1B&C	TEE 8	N/A

Table 5. Qualitative Trace Topics

Qualitative Trace			
Trace	Topic	Survey #1	Survey #2
1	Limit Definition of the Derivative	Initial	Midcourse
2	Chain Rule	Initial	Midcourse
3	Related Rates	Initial	Midcourse
4	Derivatives in Mathematica	Initial	Midcourse
5	Mathematica Troubleshooting Code	Initial	Midcourse
6	Integrals in Mathematica	Midcourse	End of Course
7	Substitution Rule	Midcourse	End of Course
8	Work	Midcourse	End of Course
9	Graphical Interpretation of an Integral	Midcourse	End of Course
10	TI36X-Pro Calculator Techniques	End of Course	-

a. Quantitative Analysis. As introduced above, we deliberately collected data and student scores on key skills throughout the semester. We recorded performance on those skills with specified entries into the gradebook, and we aimed to track each skill across three assessments to assess student learning and retention of the material. We have six skills related to differential calculus and five skills that pertain to integration.

Table 6 below highlights the course average on each traced concept throughout the semester, and we can see that we saw improvement from Assessment #1 to Assessment #2 in 9 of 11 skills. We lapsed in the Riemann Sum and Net Change Theorem skills, and we assess that those declines can be largely attributed to more challenging questions on the second assessment. While the Assessment #1 to #2 presents favorably, the Assessment #2 to #3 shows a decline in 6 of 8 skills that trace completely through the semester. We attribute this shift to two factors – the first being the tendency or ability of the students to outperform their understanding on a smaller more focused assessment. Assessment #2 took place on either a WRIT or a WPR and students had the capability to really focus and demonstrate specific skills that they knew would be tested. We believe this encapsulates the natural short-term or peak understanding that our students achieve within the semester. The second factor is the nature of the third assessment. In all cases, these skills were assessed for the final time on the Term End Exam. The volume of material negated their ability to regain the peak level of understanding and the innate challenges and distractions of TEE Week prevented many students from fully committing to their preparation or testing.

Table 6. AY22-02 Quantitative Trace Results

Quantitative Trace Results				
Trace	Topic	Assessment #1	Assessment #2	Assessment #3
1	Graphical Limits and Discontinuities with Types	0.7629	0.9413	0.8349
2	Limit Definition of the Derivative	0.8848	0.8922	0.8273
3	Product & Chain Rule	0.7698	0.9280	0.8254
4	Implicit Differentiation	0.8210	0.9129	0.8126
5	Natural Sciences	0.8375	0.9629	N/A
6	Max & Min	0.7329	0.8178	0.8322
7	Riemann Sums	0.8744	0.8412	N/A
8	Net Change / Interpreting Integrals	0.8186	0.7800	N/A
9	Substitution Rule	0.7468	0.8631	0.8391
10	Improper Integrals	0.7225	0.9629	0.8182
11	Probability	0.6277	0.8111	0.9083

That said – while students did not recapture their peak short-term understanding of the material, they did retain enough information to demonstrate a good or $> 80\%$ understanding of the material on 8 of 8 skills that traced into the TEE. This is potentially an indication that some deeper learning is taking place and we think points to the success of the course design and assessment plan. Figure 1 below graphically depicts the scoring on those 8 skills that are traced into the TEE.

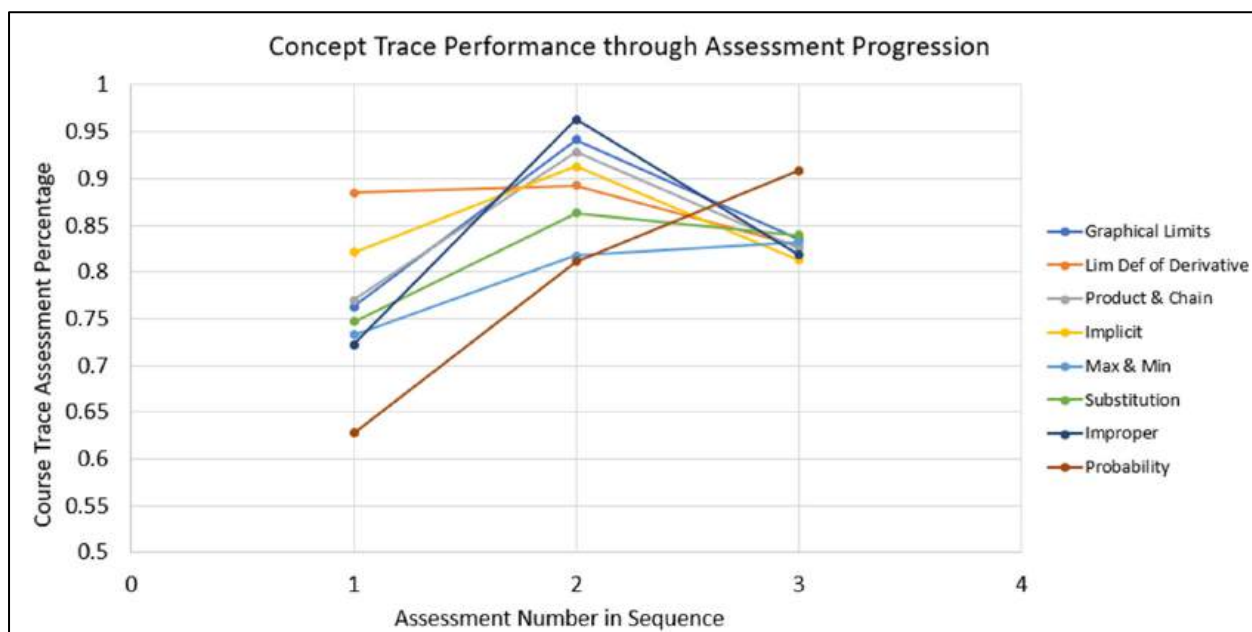


Figure 1. Concept Trace Performance on Sequential Assessments

In comparing AY22-02 and AY21-02 and the Quantitative Trace results we observed improved performance in AY22-02, which we attribute to the changes we made to the course design – references, homework quizzes, recall events, revisions, and streamlined technology applications. In the six skills that directly map from semester to semester with comparable questions and testing conditions, AY22-02 produced superior learning outcomes in the students in all cases. Student quality, teaching modality, cleaner assessment design may be confounding factors, but we believe that we witnessed a higher level of understanding versus AY21-02.

Table 7. AY21-02 Quantitative Trace Results

Trace	Topic	Assessment #1	Assessment #2	Assessment #3
Trace 1	Types of discontinuities	88.90%	98.40%	N/A
Trace 2	Solving a limit	75.20%	76.00%	N/A
Trace 3	Definition of differentiability	57.30%	26.90%	N/A
Trace 4	Solving a limit definition of the derivative	58.30%	62.00%	45.10%
Trace 5	Chain Rule Definition	90.90%	97.10%	N/A
Trace 6	Product/Chain Derivative	60.90%	74.10%	N/A
Trace 7	Implicit Derivative	77.30%	81.60%	N/A
Trace 8	Natural Sciences	50.40%	69.60%	90.70%
Trace 9	Max/Min	51.30%	48.20%	N/A
Trace 10	FTC 1	79.10%	36.00%	38.10%
Trace 11	Reimann Sum Approximation	75.30%	68.60%	N/A
Trace 12	Integration Strategies	47.00%	60.80%	40.10%
Trace 13	Volumes of Rotational Solids	37.50%	41.10%	76.10%
Trace 14	Work: Pumping	44.80%	74.20%	N/A

b. Qualitative Analysis. The collection of scores and assessed performance is helpful, but we also wanted to investigate student sentiment and better understand their perceived comfort with the material. We chose to survey students throughout the semester on both topics that relate to the quantitative traced skills as well as applications of technology within the course. In our final analysis we compared the survey responses to the final grade to students to see how the spectrum of performances relate to sentiment.

Figures 2 and 3 below show a survey question and a sample of survey results. The Likert Scale of questions ranges from Strongly Disagree to Strongly Agree and compares responses to final student grades. You can observe a significant shift from Disagree to Agree responses between the two surveys which indicates that at the conclusion of the course students believe that they are capable of solving integrals involving the substitution rule. We will further investigate the connection between how students feel about the material versus how they actually perform when those key concepts are assessed.



Figure 2. Qualitative Trace Sample Survey Question

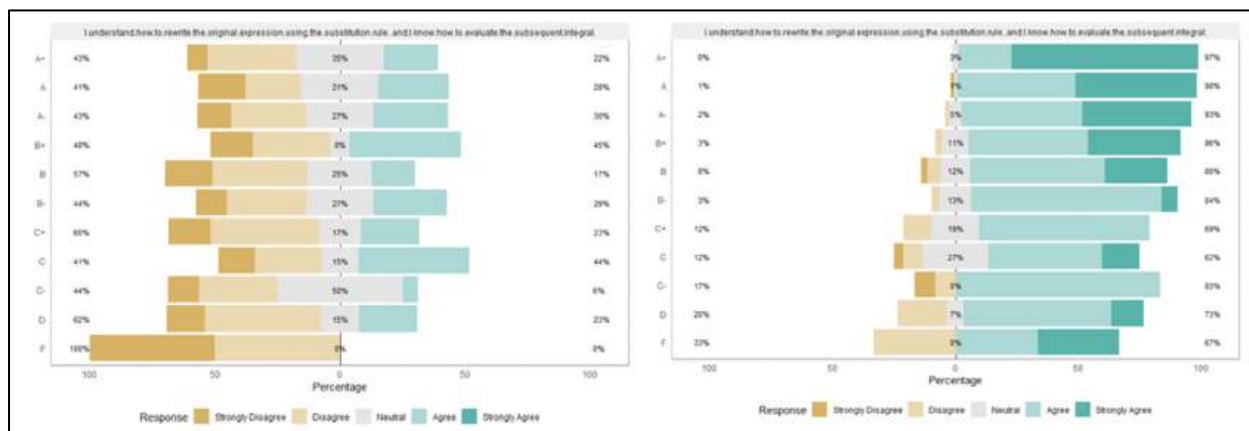


Figure 3. Qualitative Trace of Student Sentiment regarding the Substitution Rule on the Mid-Course and End of Course Surveys

2. Core Math Goal #1 – Body of Knowledge. This goal seeks to develop student breadth and depth of mathematical understanding. We sought to measure the effectiveness of our course through the analysis of the quantitative and qualitative data

we collected throughout the semester. The following is an excerpt of the total investigation that will investigate student development as it relates to using the chain and product rules to take a derivative.

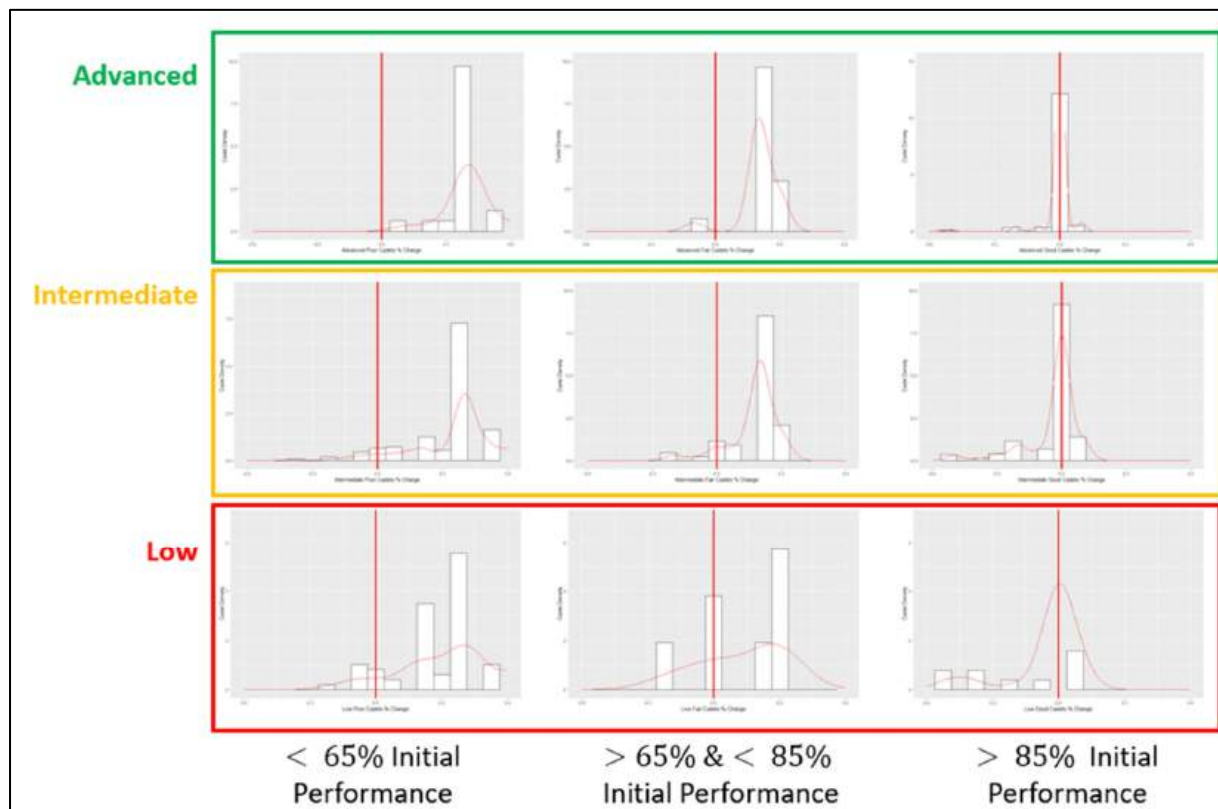


Figure 4. Trace Comparison of Performances between assessments based on Student Type.

The above Figure 4 categorizes student performance based on our original rating of their abilities. We classified students as Advanced, Intermediate, or Low based on a combination of their MA103 Score, previous calculus experience, and performance on the Fundamental Concepts Exam. Each row in Figure 4 describes the change in performance between assessments of a student's ability to employ the chain and product rules to take a derivative, and each column describes their performance on the first assessment in the sequence. Based on this analysis, we conclude that students in all populations improve assessment to assessment, but that success for weaker students is less likely to be replicated in future testing.

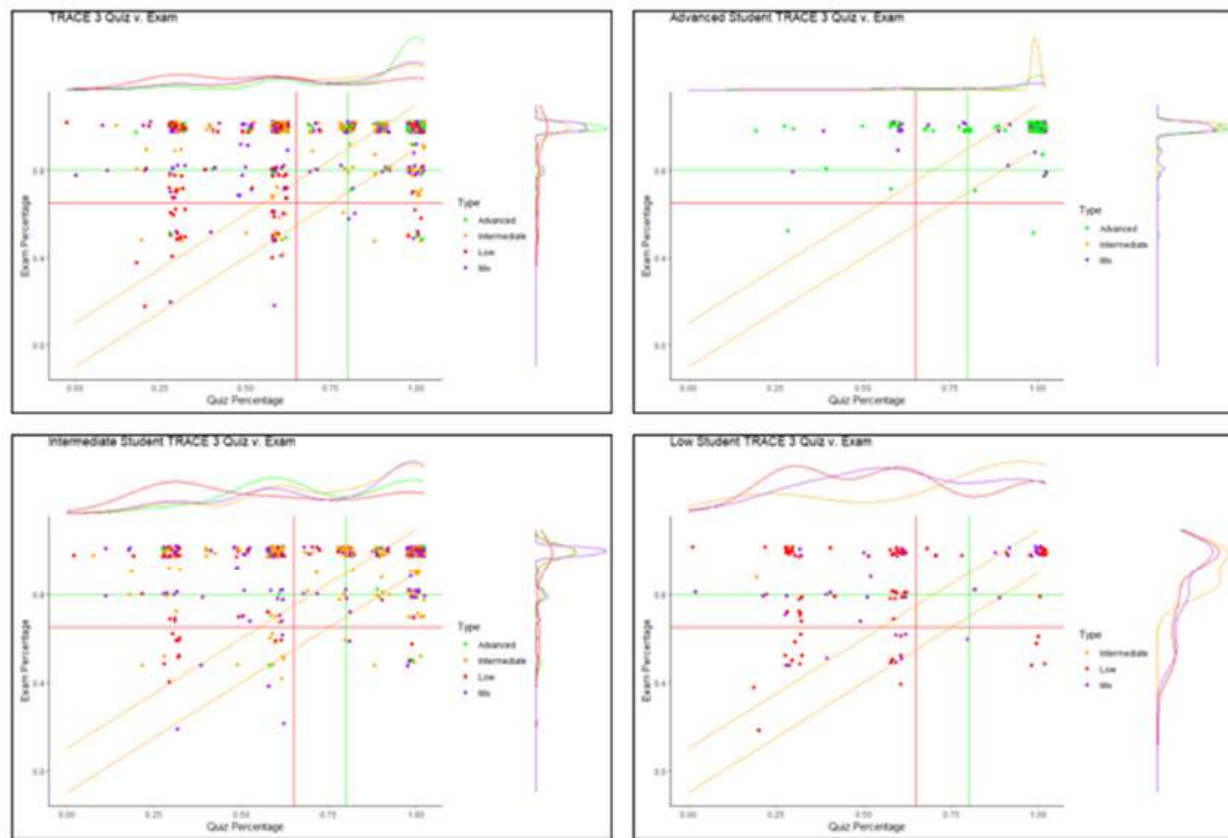


Figure 5. Trace Analysis by Student Ability

Figure 5 compares the performance of students, by type, on the first traced assessment (Homework Quiz) against the second assessment (WRIT 2). The figures above show the total course performance as well as separate graphs for each student ability level. Similar to Figure 4, we observe significant improvement between assessments in students of all abilities, with some fragility in weaker students. Figure 6 below shows the comparison in student sentiment after the first and second assessments and we again observe a significant shift in their perceived understanding and comfort with the material.

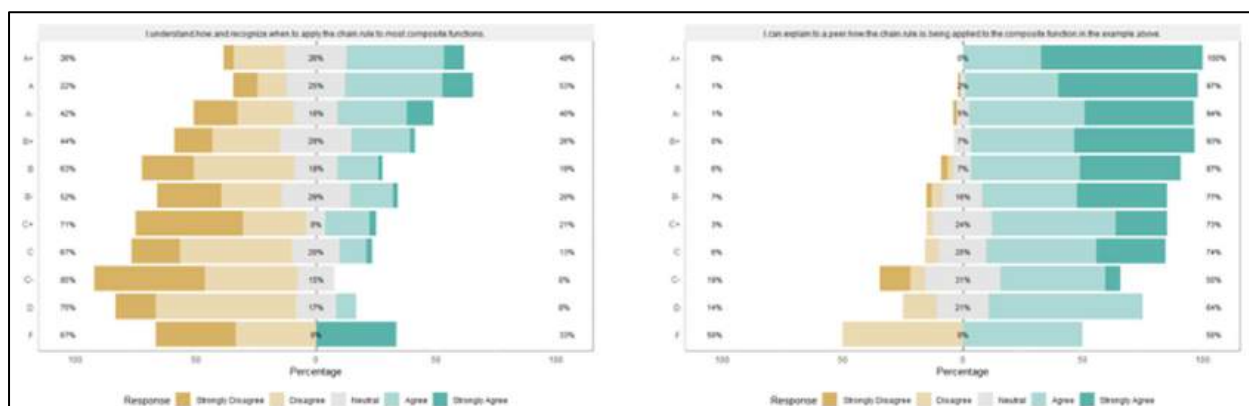


Figure 6. Qualitative Comparison of Student Sentiment regarding the Chain and Product Rules

3. Core Math Goal #2 – Application of Technology. To address this goal, we took a purposeful look at how the course was structured and reviewed the manner in which we were applying technology to solve single variable calculus problems. We came away thinking that we had come to overemphasize the use of Mathematica and it had a negative effect on our students understanding. Students had become dependent on Mathematica to solve even basic problems and they were not applying technology in a sensible manner. We found that they were simply memorizing a series of commands to solve a particular problem and that they were unable to communicate or understand the calculus concepts behind what they were doing. As such, our goal in AY22 was to refine the application of technology and employ it to solve problems of an appropriate difficulty level in an appropriate environment. We relegated Mathematica use to only the course projects and coached students to use their issued calculator to solve the single variable calculus problems that we presented in the course. This distilled down to students using calculator functions to solve for the roots of various functions and their derivatives as well as definite integrals. This required us to write the assessments deliberately so that we could be sure to assess course concepts and develop projects that required Mathematica to be used when students would be able to work with it outside of a timed environment. At the beginning of the semester we did provide a graded Technology Tutorial assignment that demonstrated all of that technology skills that would be required in the course. In total, we believe that this tighter and more deliberate use of technology produced better learning outcomes in the course as our problem-solving approach was more direct, but also that it produced students who have a better sense of when and how to apply different levels of technological tools to solve problems.

The below Figures 7 through 10 highlight the progression in student sentiment and confidence through the semester as it relates to the technology application in the course. We observed an improvement and moderation in student sentiment in all topics that we traced. We believe the moderation is particularly interesting as students seem to

be increasingly positive, but to a lesser extent, as the semester progresses; indicating that students have a better, and more confident, understanding of their own abilities.

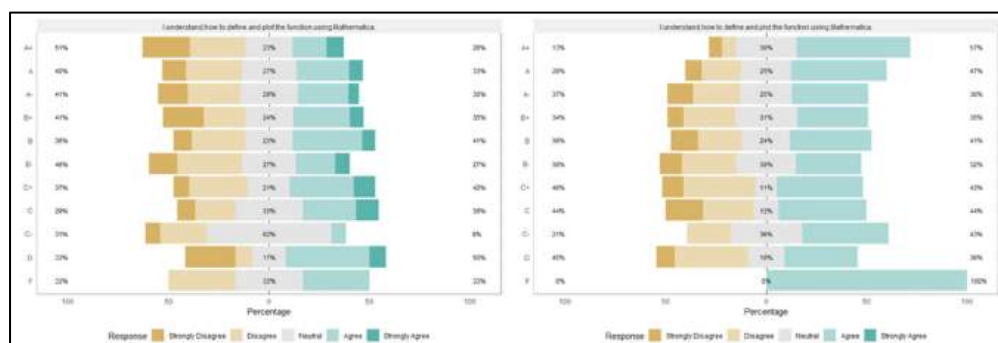


Figure 7. Qualitative Survey Results – “I understand how to define and plot the function using Mathematica.”

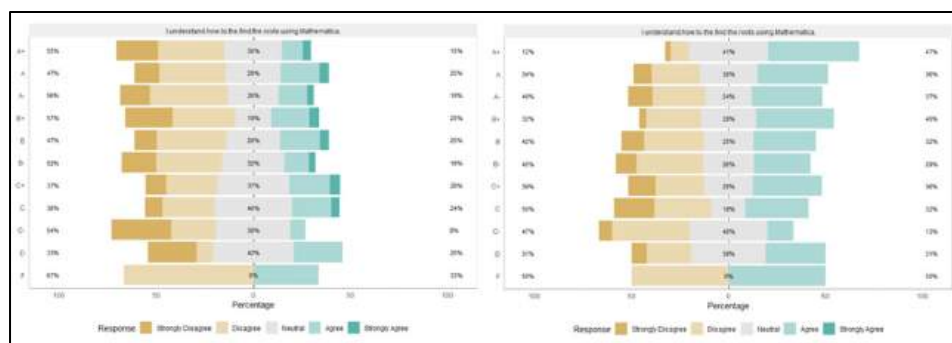


Figure 8. Qualitative Survey Results – “I understand how to find the roots of the function using Mathematica.”

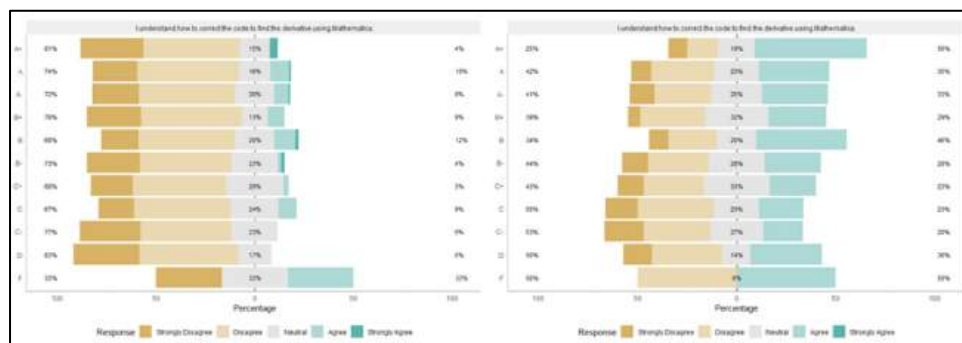


Figure 9. Qualitative Survey Results – “I understand how to correct the code to find the derivative using Mathematica.”

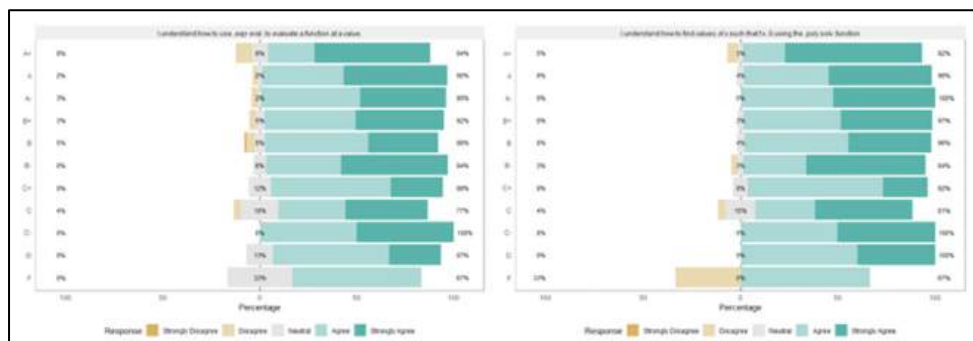


Figure 10. Qualitative Survey Results – “I understand how to use expr-eval to evaluate a function at a value [using the TI36X-Pro Calculator]” and “I understand how to find values of x such that $f(x) = 0$ using the poly-solv function [on the TI36X-Pro Calculator].”

4. Core Math Goal #3 – Interdisciplinary Problem Solvers. Similar to the technology goal, we used AY22 to focus the interdisciplinary problem solving or application of single variable calculus. Previous academic years had seen single variable calculus somewhat clumsily applied to real and complex problems on projects in MA104. While we respect and see the value in those types of projects – coming out of a pandemic year we were concerned about the foundational abilities of our students, we again aimed to simplify and be direct about how single variable calculus thinking and problem-solving can be used to approach aspects of cadet life and lay out a bridge to solving more complicated problems. We employed two projects this semester – one focused on optimization of cadet time to achieve the greatest CQPA and the second focused on using probability to make assessments about cadet branching and posting outcomes. We demonstrated the broad applicability and utility of simple ordered calculus-based thinking and student feedback, both anecdotal and surveyed, were positive as it related to those outcomes. Figure 11 below showcases those conclusions.

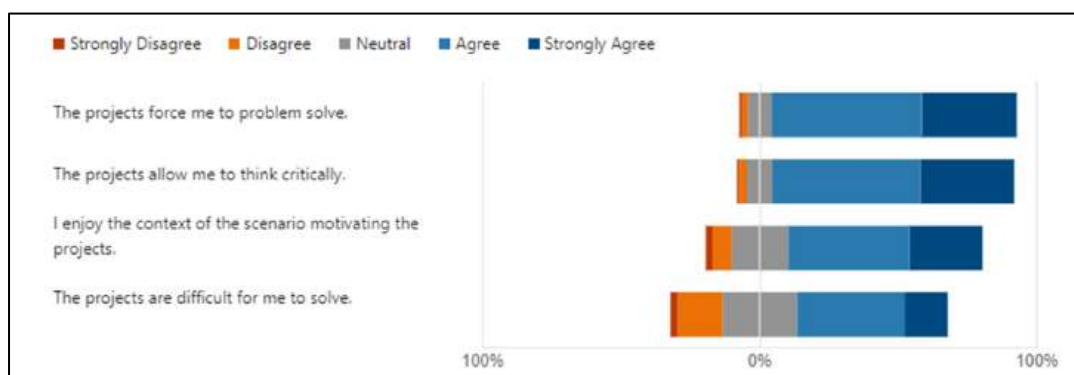


Figure 11. End of Course Survey – Project Likert Scale Questions.

5. Sectioning Analysis. In AY22-02 we chose to section students by ability and attempt to validate the conclusions presented after the AY21-02 iteration of MA104. In that semester, the course was taught in a hybrid manner, where students physically attended half of the classes and logged in remotely for the other half of the sessions. Initially we assessed that as presenting significant risk to our weaker students and so we sectioned those students into a single class and taught those sections fully in-person during that semester. We observed that weak students sectioned together and taught in-person outperformed their weaker peers in mixed sections and in AY22 we wanted to determine whether that outperformance was a product of being sectioned with like-ability students or if it could be attributed to modality of instruction each group received.

In order to make these assessments we created 8x Advanced Sections, 16x Intermediate Sections, 8x Low Sections, and 19x Mixed sections to better understand student performances. We collected qualitative trace data on their performance and then looked to interpret the results. The following section explores the conclusions of our analysis.

The first conclusion that we reached after reviewing the data is that student performance is generally as expected based on their ability. Performance in MA104 is largely dependent on the effort and amount of work students are willing to do, and with that in mind they nearly have the ability to select the grade that they are comfortable with based on their effort in the semester. If the course is adjusted to remove some of the safeguards, then we might see more clearly effects of sectioning but in AY22-02 we witnessed an almost linear relationship between our sectioning score and eventual performance in MA104.

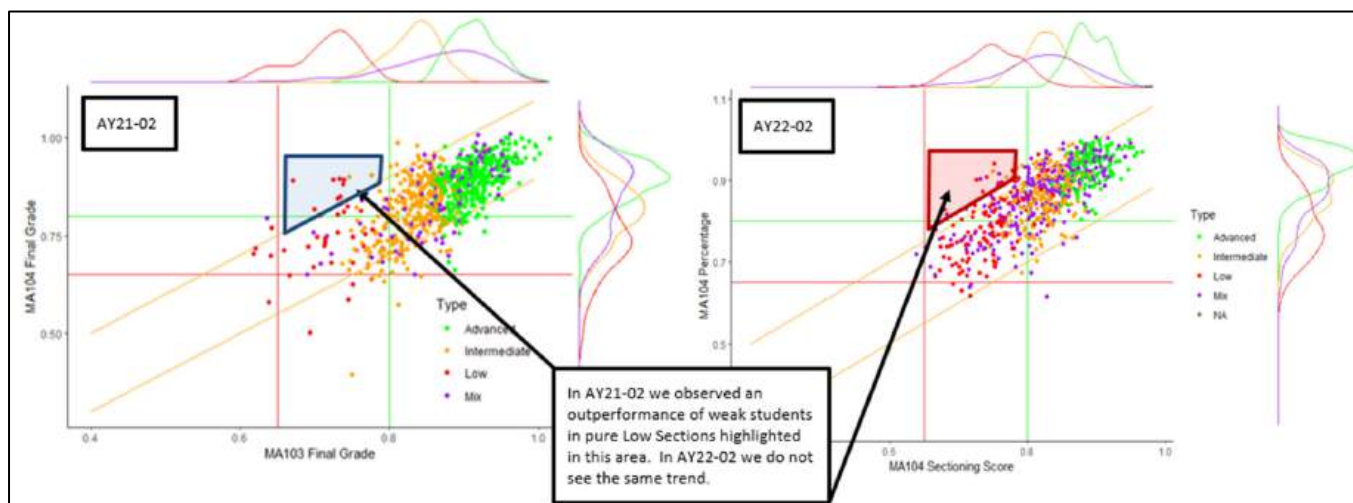


Figure 12. MA104 Course Grades v. Sectioning Score. Sectioning Score is a combination of MA104 Grade, Prior Calculus Experience, and FCE Scores.

Figure 12 above represents a comparison between AY21-02 and AY22-02 results. In AY21-02 we observed weak students sectioned together outperforming their peers in mixed sections. In AY22-02, it does not appear that the same trend holds. Our initial belief is that the in-person instruction those sections received was the principal driver of their outperformance.

As we continued to investigate the data, we began to notice that in contrast to last semester's outperformance of weak students in pure low sections, we observed that they may have in fact underperformed when compared to their peers in mixed sections. Figure 13 below shows the density of scores as being higher for weak students in mixed sections. We attribute this underperformance to students in pure sections settling into their environment and not acknowledging or perceiving how other students are clearly dominating the material. Weak students in mixed sections are confronted daily with that and are more likely to change their behavior to make up for their weakness in comparison to pure sections where students accept the norms of their particular class.

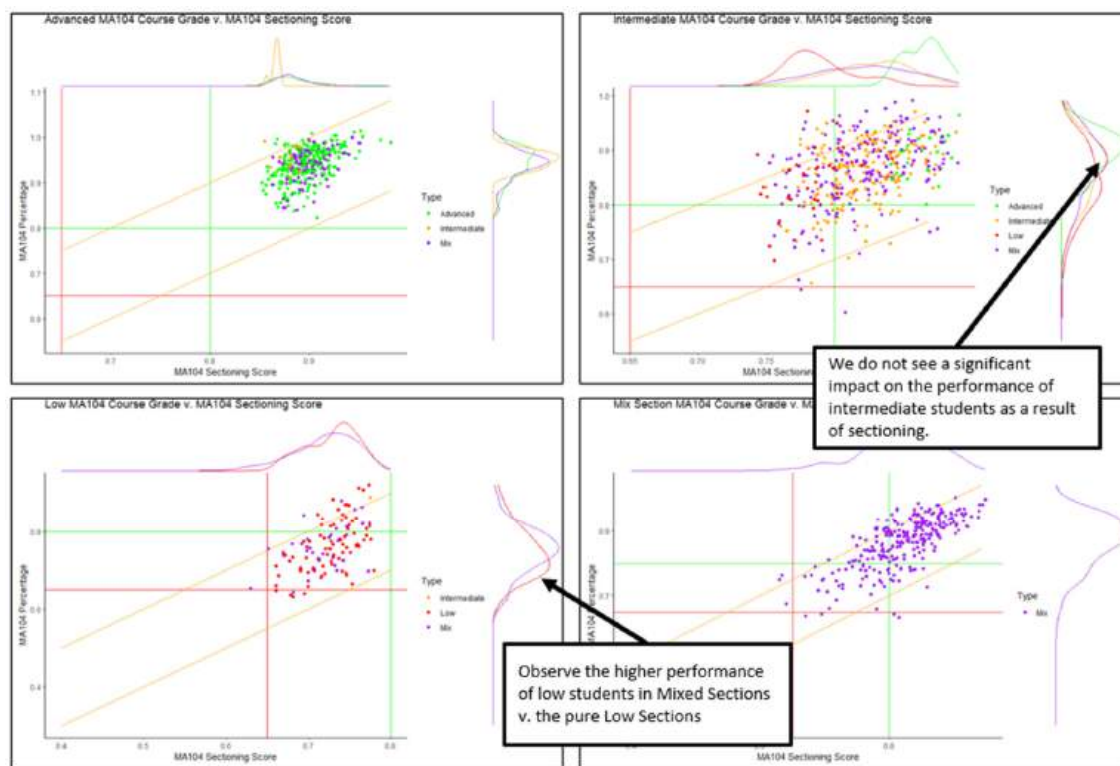


Figure 13. MA104 Course Grades v. Sectioning Score. (Specific Plots by Student Ability)

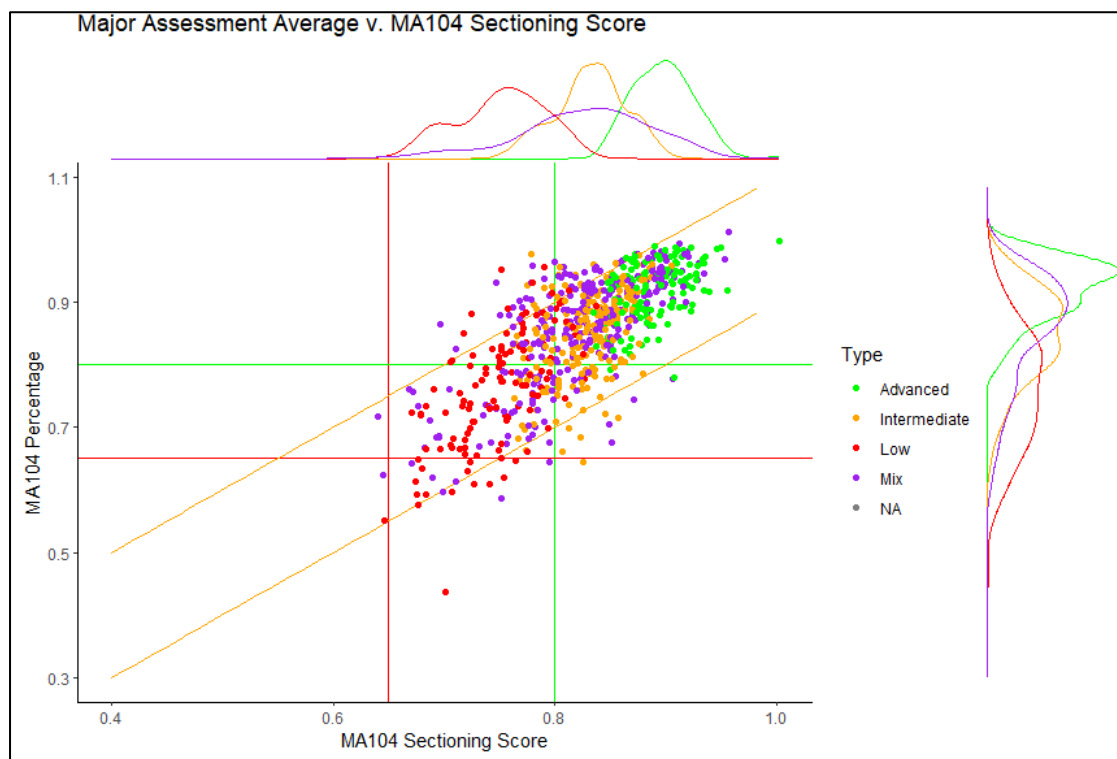


Figure 14. Major Assessment Averages v. Sectioning Score.

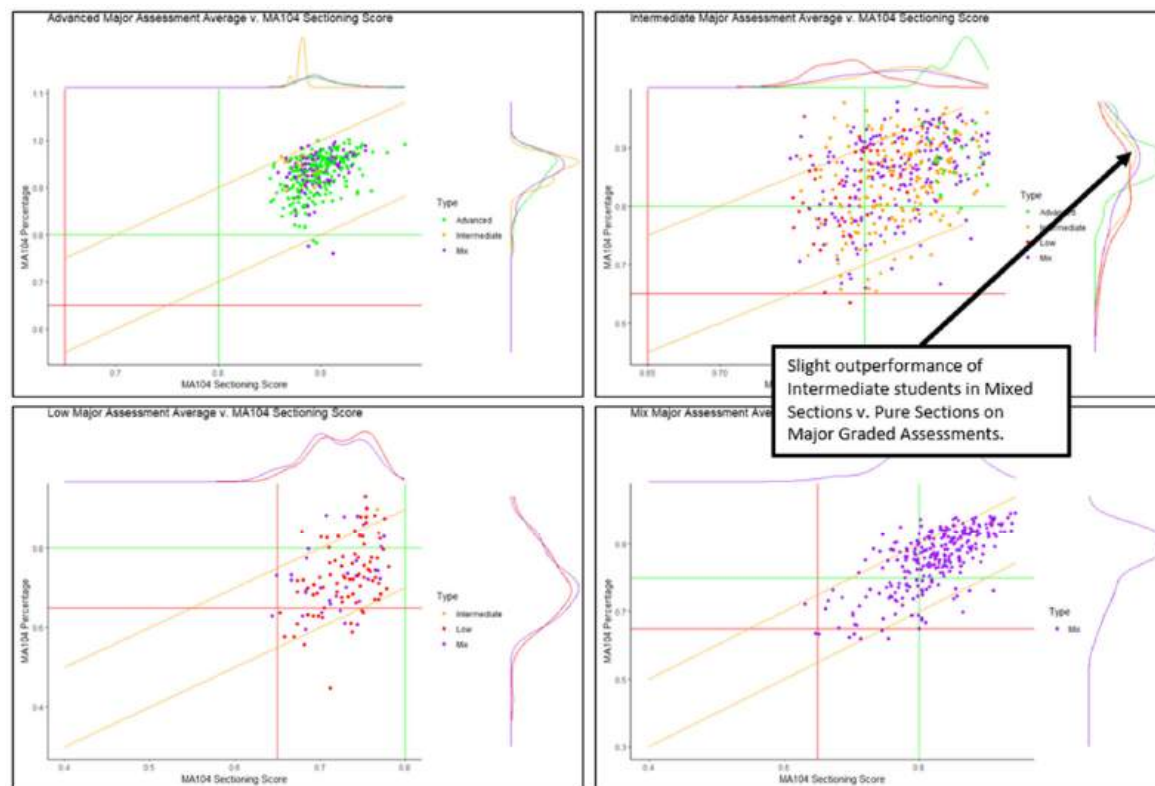


Figure 15. Major Assessment Averages v. Sectioning Score. (Specific Plots by Student Ability)

In addition to the benefits of mixed sections for weaker students, we observe some slight outperformance in intermediate students as well. We can see this trend in Figure 15 above and again in Figure 17 below. We attribute this outperformance to the likelihood that advanced students have a positive impact on their understanding and weaker students serve to improve their confidence.

Figures 16 and 17 below further highlight the discussed trends that we observed. Based on our analysis, **we believe that sectioning students by ability does not benefit any particular group of students.** If anything, it may be damaging on a number of levels. We can make an argument that it does weak students a disservice by overhelping and undermining their future growth, and it limits the development of intermediate and advanced students by subtracting opportunities for them to build confidence and lead in the classroom. There is also the instructor development perspective – teaching a homogenous section might be easier but it does not offer the challenges and dynamics required to allow instructors to reach their potential. It also forces a division of labor challenge for course leadership to manage and may result in misconceptions about degree of difficulty steering of the broader course.

After reviewing the results of AY21-02 and AY22-02, we recommend that sectioning all students into homogenous sections be suspended until it might be required based on the learning environment or Core Math Program design. We think that we should section the most advanced students together and assign to instructors that can challenge them or who has heavy external requirements. For the remainder of the population, we believe that we should engineer a specific mix of abilities into each classroom. A certain number of weak, strong, and intermediate students together to create an environment that helps them, and the instructors, to reach their potential while allowing course leadership to clearly see and steer the course.

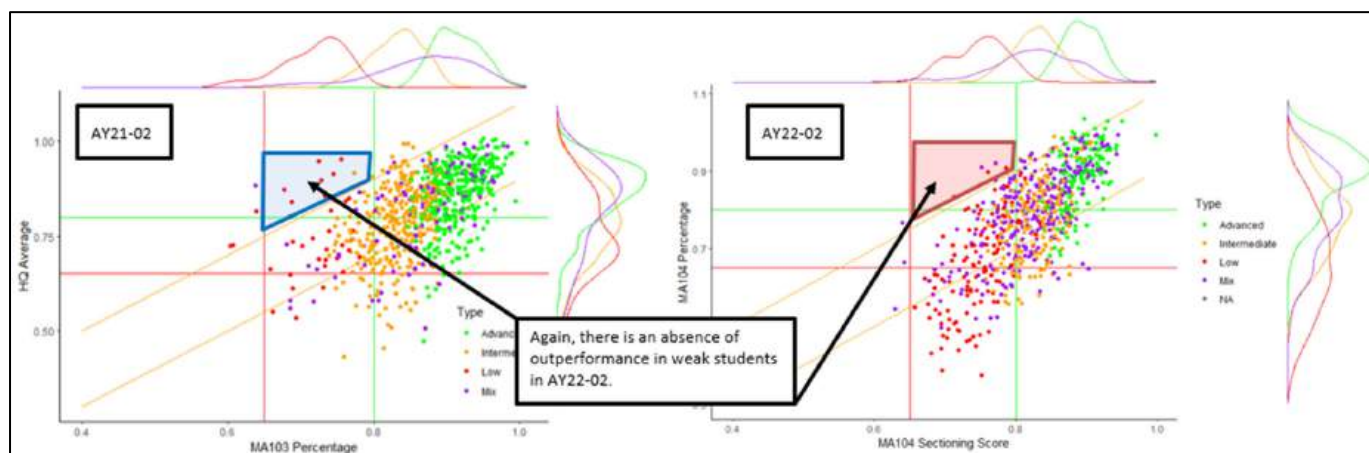


Figure 16. Homework Quiz Averages v. Sectioning Score

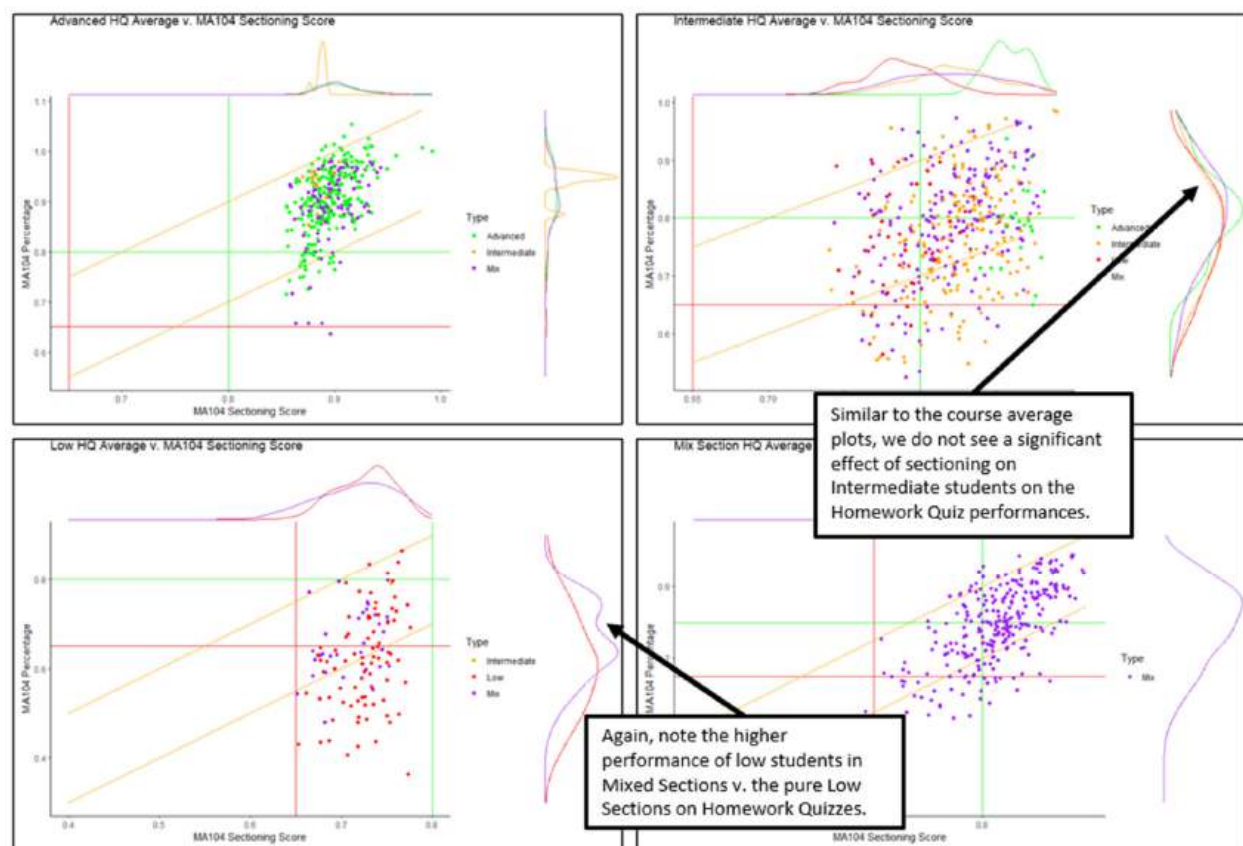


Figure 17. Homework Quiz Averages v. Sectioning Score. (Specific Plots by Student Ability)

6. Additional Instruction by Section Type Review. As we sectioned by ability for the past two academic years, we wanted to investigate the relationship between the amount of Additional Instruction provided by instructors based on their assigned section types. The goal of this review was to help the department understand the time requirement for different instructors and maybe inform a more deliberate teaching slate in upcoming years. The assumption behind this investigation is that we would continue sectioning into the future, which we do not endorse, but the results are interesting none the less.

Figure 18 below captures the trend in AI over the course of the semester by section type. We observe that initially the AI requirement for instructors is fairly uniform but as the semester progresses, weaker sections tend to require an increasing amount of time from instructors. We attribute this to the shifting confidence of students throughout the semester. Advanced students recognize that they have a good understanding of the material and trust themselves to figure it out when it is unclear, whereas weak students slowly realize that they are struggling and seek more help as the semester passes and they become more anxious about their final grades.

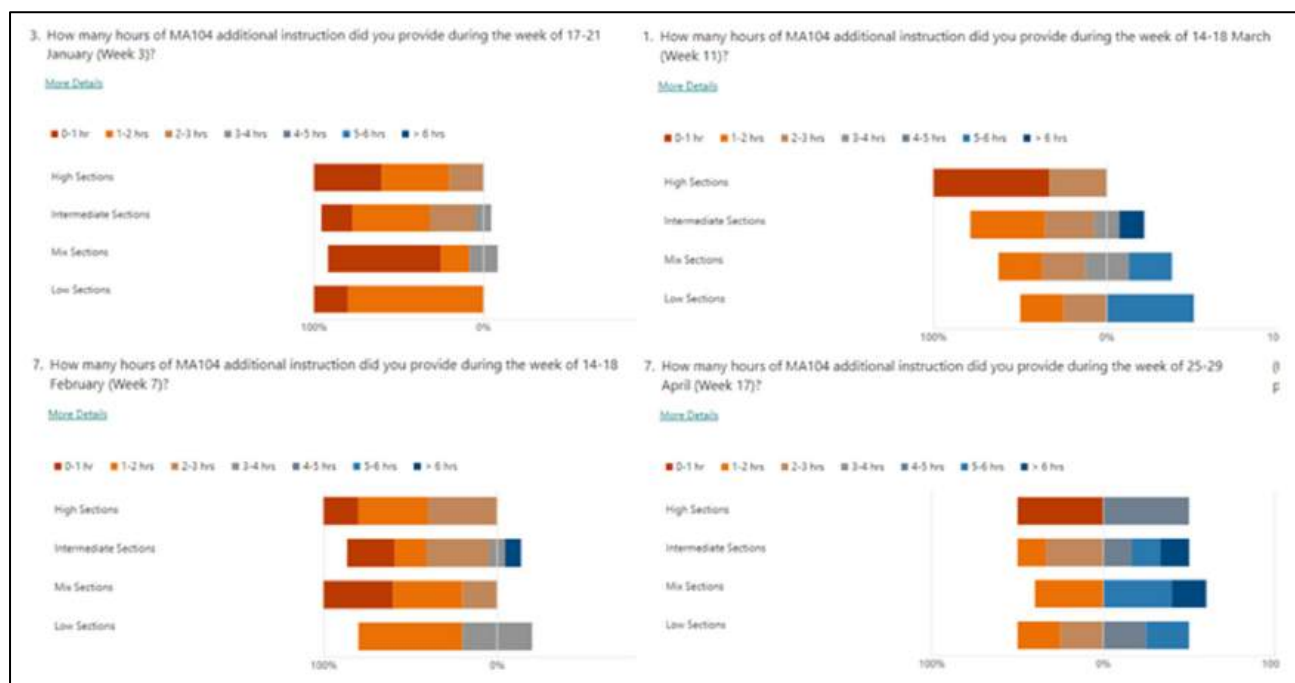


Figure 18. Weekly Instructor Additional Instruction Tracking by Section Type.

7. Homework Quiz Revision Sensitivity Analysis. In AY22-02 we allowed all students to complete revisions on their Homework Quizzes to recoup up to half of the points they lost. We had a few considerations in mind when we made this policy. First, we deliberately created harsh rubrics that directly communicated to students when they did not understand a concept or skill. We wanted to make sure students had a clear picture of their ability before assessing them on a major graded event. Second, we wanted to motivate students to revisit the material and hopefully gain a better understanding after attempting the work a second time. Assigning points to these revisions provided some extrinsic motivation at the outset that we hoped would translate into intrinsic motivation to succeed in the course and on future assignments.

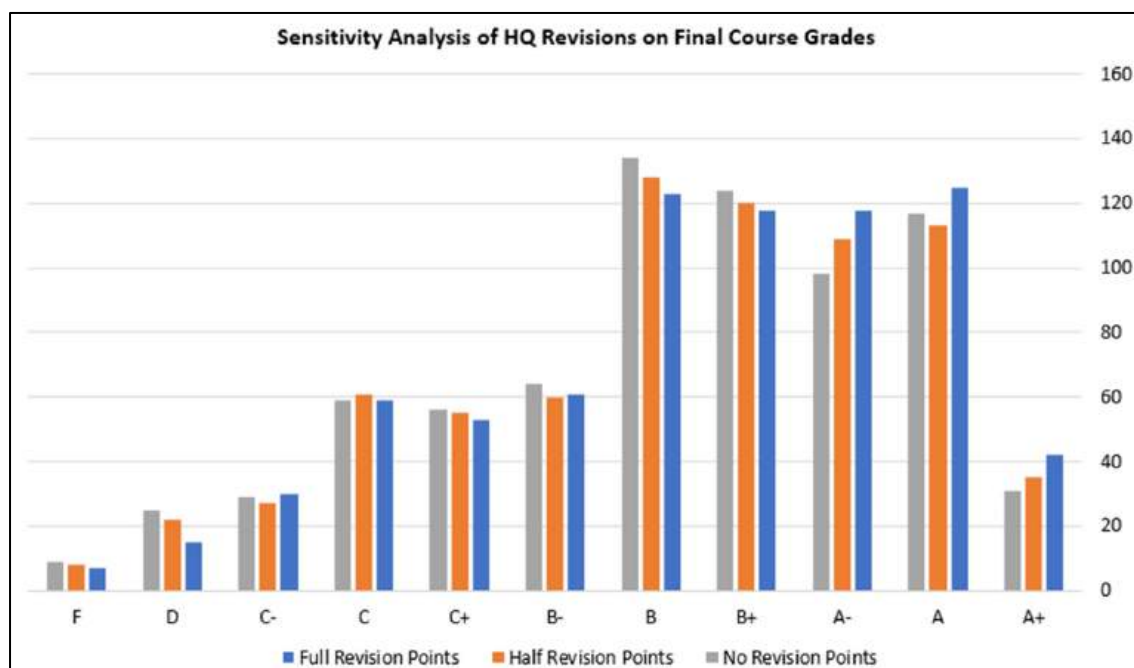
After conducting the quantitative analysis, we wanted to assess how sensitive final course scores are to changes to the number of points offered for revisions. We want to ensure that if revisions are offered in future semesters, we value them at the appropriate level to motivate learning without simply propping up poor performing students. In looking at the data, on average students earned 14 additional points (out of 2000) in the course from revisions. That translates into a 0.7% increase to the course average directly from revision points, but the larger effect of more motivated repetitions of a key skill on major assessments is more difficult to screen out. Based on the marginal impact the few extra points had in terms of our final grades, we believe that we had geared the revision points at an appropriate level.

Table 8. Sensitivity Analysis of Homework Quiz Revisions and Point Values

	Full Revision Points	Half Revision Points	No Revision Points
A+	42	35	31
A	125	113	117
A-	118	109	98
B+	118	120	124
B	123	128	134
B-	61	60	64
C+	53	55	56
C	59	61	59
C-	30	27	29
D	15	22	25
F	7	8	9

Course Average	0.8630	0.8595	0.8560
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Table 8 and Figure 19 capture the changes in the number of specific grades if we value revisions as is, if we reduced their value by half, and if we removed them from the grade entirely. The most significant shifts occur with the highest performing students, as they are committed enough to spend time and energy to recoup their points.

**Figure 19.** Sensitivity Analysis of Homework Quiz Revisions Point Values on Final Course Grade

MA104: Single Variable Calculus
AY22-02 Syllabus

Printed on: 3/28/2022

Textbook: Calculus - Ninth Edition, Early Transcendentals by James Stewart				
Students: All homework sets will be available for download on Blackboard. Submit a completed hard-copy of your work with Cover Sheet and Works Cited Page, stapled together, in class for credit. Course Objectives: 1. Understand and internalize the role of the limit in differentiation and integration. 2. Understand, calculate, and interpret problems involving average rates of change and instantaneous rates of change. 3. Understand, calculate, and interpret problems involving the accumulation of change. 4. Evaluate derivatives and integrals. 5. Internalize the concept of the derivative as a rate of change, the integral as the accumulation of change, and that these two concepts are inversely related. 6. Understand, calculate and interpret applications of differentiation and integration. 7. Communicate calculus problem solving techniques and methodologies.				
Lesson	Date	Topic (Reading)	Notes	Lesson Objectives
BLOCK 1: DERIVATIVES AND THEIR APPLICATIONS				
Functions & Mathematical Models				
Lesson 1	1/4/2022	Course Intro		Topic: Overview of the Course
Lesson 2	1/5/2022	Functions 1 (§ 1.1)		1. Understand the definition of a function. 2. Understand and recognize the four ways to represent a function. 3. Determine the domain and range of a function. 4. Recognize the difference between even and odd functions. 5. Determine when a function is increasing and decreasing.
Lesson 3	1/6/2022	Functions 2 (§ 1.2)		1. Understand James Stewart's definition of a mathematical model. 2. Understand how empirical models can be used to construct a function that "fits" the data. 3. Construct an algebraic function based on a verbal/written description of the function. 4. Recognize the shapes and basic properties of the following functions: Linear, Polynomial, Power, Rational, Algebraic, Trigonometric, Exponential, and Logarithmic.
Lesson 4	1/10/2022	Functions 3 (§ 1.3)		1. Understand how to shift functions vertically and horizontally. 2. Understand how to stretch functions vertically and horizontally. 3. Recognize and construct composite functions.
Limits & Continuity				
Lesson 5	1/11/2022	Limits/Continuity 1 (§ 2.1)	HW 1 Due	1. Understand the definition of a tangent and a secant line. 2. Determine average rates of change. 3. Estimate instantaneous rates of change. 4. Find or estimate the equation of a tangent line to a curve.
Lesson 6	1/13/2022	Limits/Continuity 2 (§ 2.2)		1. Understand the definition of a limit. 2. Find the limit of a function graphically and numerically.
Lesson 7	1/14/2022	Limits/Continuity 3 (§ 2.3 and § 2.6: pg 127-133)	Tech Tutorial Due NLT 2359	1. Understand how to apply the properties of limits. 2. Find the limit of a function algebraically. 3. Understand and evaluate limits at infinity.
Lesson 8	1/18/2022	Limits/Continuity 4 (§ 2.5)		1. Understand the definition of continuity. 2. Identify where and why a functions is discontinuous. 3. Recognize the types of functions that are continuous at every number in their domain.
Rates of Change				
Lesson 9	1/19/2022	Rates of Change 1 (§ 2.7)	HW 2 Due	1. Understand the definition of a tangent line. 2. Understand the limit definition of a derivative. 3. Apply the limit definition of a derivative to solve for derivatives at a point. 4. Understand the definition of velocity and how it relates to derivatives. 5. Apply the limit definition of a derivative to solve for instantaneous rates of change.
Lesson 10	1/20/2022	Rates of Change 2 (§ 2.8)		1. Understand and apply the limit definition of a derivative to a function. 2. Recognize and apply the different notation used to represent a derivative. 3. Understand how a function can fail to be differentiable. 4. Recognize and understand the relationship between differentiation and continuity.
Lesson 11	1/21/2022	Rates of Change (Lab 1)		
Derivatives				
Lesson 12	1/24/2022	WRIT 1	Deans Hour: 1245	Topics: Functions, Mathematical Models, Limits, Continuity
Lesson 13	1/25/2022	Derivatives 1 (§ 3.1)	HW 3 Due	1. Recognize and apply the following derivative rules: a) Derivative of a constant function, b) Power rule, c) Constant multiple rules, d) Sum rule, e) Difference rule, f) Derivative of the natural exponential function.
Lesson 14	1/27/2022	Derivatives 2 (§ 3.2)		1. Recognize and apply the following derivative rules: a) Product rule, b) Quotient rule
Lesson 15	1/28/2022	Derivatives 3 (§ 3.3)		1. Recognize and apply the derivative rules of trigonometric functions. 2. Understand the relationship between the derivatives of sine and cosine functions.
Lesson 16	1/31/2022	Derivatives 4 (§ 3.4)		1. Recognize composite functions. 2. Understand when to apply the chain rule. 3. Solve derivatives that involve using the chain rule. 4. Use the chain rule to differentiate an exponential function with a base "b".

MA104: Single Variable Calculus
AY22-02 Syllabus

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Lesson	Date	Topic (Reading)	Notes	Lesson Objectives
Implicit Differentiation				
Lesson 17	2/1/2022	Implicit 1 (§ 3.5: pg 208-213)	HW 4 Due	1. Recognize when a function is defined implicitly. 2. Understand how to apply implicit differentiation. 3. Solve for the equation of a tangent line at a point on any implicitly defined curve.
Lesson 18	2/3/2022	Implicit 2 (§ 3.6: pg 218-220)		1. Understand how implicit differentiation is used to determine the derivative of logarithmic functions. 2. Solve for the derivative of logarithmic functions. 3. Take the derivative of an implicit function with respect to time.
Lesson 19	2/4/2022	Related Rates 1 (§ 3.9)	HW 5 Due	
Related Rates				
Lesson 20	2/7/2022	Differentiation (Lab 2)	Differentiation Worksheet Due NLT 1600	
Lesson 21	2/8/2022	WRIT 2	In Class	Topics: Rates of Change, Derivatives, and Implicit Differentiation
Lesson 22	2/10/2022	Related Rates 2 (§ 3.9)		1. Apply the problem-solving process to related rates problems.
Lesson 23	2/11/2022	Related Rates 3 (§ 3.9)		1. Apply the problem-solving process to related rates problems.
Lesson 24	2/14/2022	Related Rates (Lab 3)		
Derivatives in the Natural Sciences				
Lesson 25	2/16/2022	Natural Sciences (§ 3.7)	HW 6 Due	1. Apply derivative concepts to physics applications 2. Understand how to use derivative concepts to determine when an object in motion is slowing-down or speeding-up. 3. Understand the meaning of a derivative in various scenarios.
Maximums & Minimums				
Lesson 26	2/17/2022	Max/Min 1 (§ 4.1)	HW 7 Due	1. Understand the difference between absolute and local extrema. 2. Apply the Extreme Value Theorem. 3. Understand how Fermat's Theorem relates to critical numbers and the cautions associated with using Fermat's Theorem. 4. Use critical numbers as part of the Closed Interval Method to determine absolute maximum/minimum values.
Lesson 27	2/18/2022	Max/Min 2 (§ 4.3)		1. Use a function's derivative to identify increasing or decreasing intervals. 2. Identify and solve for intervals where a curve is concave up or concave down. 3. Understand and identify inflection points. 4. Understand and apply the second derivative test to solve for local maximum and minimum values.
Lesson 28	2/22/2022	Max/Min 3 (§ 4.3)		1. Use information for first and second derivatives to sketch the curve of a function.
Optimization				
Lesson 29	2/23/2022	Optimization 1 (§ 4.7)	HW 8 Due	1. Understand how Stewart's optimization problem solving strategy relates to the USMA Mathematical Problem-Solving Strategy. 2. Understand and apply the first derivative test for absolute extreme values. 3. Given a scenario with objectives and constraints, solve for the maximum or minimum values.
Lesson 30	2/24/2022	Optimization 2 (§ 4.7)		1. Given a scenario with objectives and constraints, solve for the maximum or minimum values.
Lesson 31	2/25/2022	Project 1 Development (Lab 4)		
Lesson 32	2/28/2022	Optimization 3 (§ 4.7)	Project 1 Writing Exercise Due NLT 2359	1. Given a scenario with objectives and constraints, solve for the maximum or minimum values.
Lesson 33	3/1/2022	Optimization 4 (Lab 5)		
	3/2/2022	WPR 1	Exam Period: 1245	Topics: Natural Sciences, Related Rates, Max/Min, & Optimization
BLOCK 2: INTEGRALS AND THEIR APPLICATIONS				
Antiderivatives				
Lesson 34	3/3/2022	Exam Comp Day	No Class	1. Understand the definition of an antiderivative. 2. Solve for an antiderivative of a functions. 3. Understand antiderivatives as a family of functions.
Lesson 35	3/4/2022	Antiderivatives 1 (§ 4.9)	HW 9 Due	1. Understand the definition of an antiderivative. 2. Solve for an antiderivative of a functions. 3. Understand antiderivatives as a family of functions.
Lesson 36	3/14/2022	Antiderivatives 2 (§ 4.9)		1. Use antiderivatives to solve problems related to rectilinear motion.
Lesson 37	3/15/2022	Project 1 Comp Day	No Class	
Definite Integrals				

MA104: Single Variable Calculus
AY22-02 Syllabus

Printed on: 3/28/2022

Lesson	Date	Topic (Reading)	Notes	Lesson Objectives
Lesson 38	3/16/2022	Definite Integral 1 (§ 5.1)	Project 1 Due	1. Understand how to estimate the area between a curve and the horizontal axis on an interval. 2. Solve area estimates using sigma notation. 3. Understand how to solve areas using the limit operator in conjunction with sigma notation. 4. Interpret distances traveled by an object as an area problem.
Lesson 39	3/17/2022	Definite Integral 2 (§ 5.2)	HW 10 Due	1. Calculate definite integrals using Definition 2 and Theorem 4 in Section 5.2. 2. Express definite integrals as a limit of a Riemann Sum and express a limit of a Riemann Sum as a definite integral. 3. Understand the mathematical notation and terminology of integrals. 4. Apply the Left or Right Endpoint Method to estimate an integral. 5. Understand and apply the properties of definite integrals.
Lesson 40	3/18/2022	Definite Integral 3 (§ 5.3)		1. Understand and apply the Fundamental Theorem of Calculus, Part 1. 2. Understand and apply the Fundamental Theorem of Calculus, Part 2. 3. Internalize differentiation and integration as inverse processes.
Integration				
Lesson 41	3/21/2022	Integration 1 (§ 5.4)	HW 11 Due	1. Distinguish between indefinite and definite integrals. 2. Determine indefinite integrals. 3. Understand and apply the Net Change Theorem.
Lesson 42	3/22/2022	Integration 2 (§ 5.5)		1. Understand when to apply the substitution rule. 2. Use the substitution rule to solve definite and indefinite integrals.
Lesson 43	3/24/2022	Integration 3 (§ 7.1)	WPR Revisions Due	1. Understand how the method of integration by parts is derived from the product rule of differentiation. 2. Understand how to select "u" and "dv" when using integration by parts. 3. Apply integration by parts to solve definite and indefinite integrals.
Lesson 44	3/25/2022	Integration 4 (§ 7.5)		1. Understand how to approach a definite and indefinite integral in order to determine the appropriate integration method(s). 2. Solve any integral by choosing the correct integration technique (simplifying w/ basic rules, u-sub, IBP)
Lesson 45	3/28/2022	Integration 5 (§ 7.5)		1. Understand how to approach a definite and indefinite integral in order to determine the appropriate integration method(s). 2. Solve any integral by choosing the correct integration technique (simplifying w/ basic rules, u-sub, IBP)
Lesson 46	3/29/2022	Integration 6 (§ 7.8)	Integration Worksheet Due NLT 30 1600 MAR 22	1. Identify improper integrals as either Type 1 or Type 2 improper integrals. 2. Identify when a Type 1 improper integral is either convergent or divergent. 3. Solve Type 1 improper integrals.
Areas & Volumes				
Lesson 47	3/31/2022	Areas/Volumes 1 (§ 6.1)	HW 12 Due	1. Determine the area between two curves. 2. Interpret the meaning of the area between two curves given in context.
Lesson 48	4/1/2022	Areas/Volumes 2 (§ 6.2)		1. Understand the definition of volume. 2. Determine the volume of a solid object using integration techniques. 3. Determine the volume of a solid of revolution using the disk/washer method.
Lesson 49	4/4/2022	Integration (Lab 6)		
Lesson 50	4/5/2022	WRIT 3	In Class	Topics: Antiderivatives, Riemann Sums, Definite & Indefinite Integrals, Integration Strategies, Improper Integrals
Lesson 51	4/6/2022	Exam Comp Day	No Class	
Lesson 52	4/8/2022	Areas/Volumes 3 (§ 6.3)		1. Determine the volume of a solid of revolution using the cylindrical shell method.
Lesson 53	4/11/2022	Areas/Volumes 4 (§ 6.3)		1. Understand when to apply the disk/washer method and when to apply the cylindrical shell method when determining the volume of a solid of revolution.
Applications of Integration				
Lesson 54	4/12/2022	Applications of Integration 1 (§ 8.5)	HW 13 Due	1. Understand the definition of a probability density function. 2. Recognize when a probability density function is invalid. 3. Interpret a probability statement. 4. Solve for a probability using a probability density function. 5. Determine the average value using a probability density function.
Lesson 55	4/13/2022	Applications of Integration 2 (§ 8.5)		1. Understand the definition of a probability density function. 2. Recognize when a probability density function is invalid. 3. Interpret a probability statement. 4. Solve for a probability using a probability density function. 5. Determine the average value using a probability density function.

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Lesson	Date	Topic (Reading)	Notes	Lesson Objectives
Lesson 56	4/14/2022	Project 2 Development (Lab 7)		
Lesson 57	4/18/2022	Applications of Integration 3 (§ 6.4)		1. Understand the definition of work. 2. Solve lifting problems involving constant and variable force using both SI and US units of measure.
Lesson 58	4/19/2022	Applications of Integration 4 (§ 6.4)		1. Solve pumping problems involving tanks with constant and variable cross-sectional areas using both SI and US units of measure.
Lesson 59	4/21/2022	Applications of Integration 5 (§ 6.4)		1. Solve pumping problems involving tanks with constant and variable cross-sectional areas using both SI and US units of measure.
Lesson 60	4/22/2022	Project 2 Comp Day (No Class)		
Lesson 61	4/25/2022	Applications (Lab 8)	HW 14 Due	
Lesson 62	4/27/2022	WPR 2	Exam Period: 1245	Topics: Applications of Integration, Probability, Work, Net Change, Areas & Volumes
Lesson 63A	5/2/2022	Project Briefs		
Lesson 63B	5/3/2022	Project Briefs		
Lesson 63C	5/4/2022	Project Briefs		
Lesson 64	5/5/2022	Differentiation Review		



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB C

AY2023 COURSE ASSESSMENT



DEPARTMENT OF THE ARMY
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

MADN-MATH

23 June 2023

MEMORANDUM THRU

COL Joseph Lindquist, Calculus Program Director, Department of Mathematical Sciences, United States Military Academy, West Point, NY 10996

COL Tina Hartley, Professor and Head, Department of Mathematical Sciences, United States Military Academy, West Point, NY 10996

FOR Dean of the Academic Board, United States Military Academy, West Point, NY 10996

SUBJECT: Annual Executive Summary for MA104, AY23-02

1. **Course Description.** MA104 is the second course of the mathematics core curriculum capitalizing on MA103's mathematical modeling and introduction to vectors and calculus. The goal of MA104 is for students to internalize the concept of the derivative as a rate of change, the integral as the accumulation of change, and that these two concepts are inversely related by the fundamental theorem of calculus. Students will realize that both differential and integral calculus arise from the application of the limit to an estimate. An understanding of course material is enhanced with the use of Mathematica. MA104 services MA205 (Multivariable Calculus), MA206 (Probability and Statistics), PH201 (Physics), SS201 (Economics), the Core Engineering Sequences, and selected majors. This course is followed by MA205 or MA206.

2. **Course Composition.** The structure of MA104 AY23 included a 64-lesson sequence in which points were accumulated in a series of assessments of increasing value by topic. Assessments included concept quizzes, homework quizzes, Writs, WPRs, and projects. Assessments were divided into two distinct blocks during the semester (derivatives and integrals). The first 8 weeks of the course explored differentiation concepts, computations, and applications while the last 8 weeks explored integration concepts, computations, and applications. Each lesson was associated with a larger grouping of lessons involving a common topic, and each common topic had its own series of assessments designed to reinforce student comprehension. The course was taught entirely in-person with two lessons dedicated as "self-study," in which cadets were allocated time to study the material outside of a formal classroom setting.

3. **Classroom Design.** The course was divided into mixed-ability sections with similar section compositions between every section each hour. Ability was measured using 90% of a cadet's grade in MA103, 5% of a cadet's score on the Fundamental Concept Exam, and 5% of a cadet's prior calculus experience. Cadets were placed in sections that contained 15 cadets on average.

4. **Grading Scheme.** The overall grading scheme for the course is shown in Table 1 below. This grading scheme places a high priority on in-class assessments (87.5%). This decision was made to combat the prevalent use of computer programs such as Mathway, Photomath and other digital math equation solvers. The goal was to obtain a true assessment of a cadet's understanding of course concepts. There was also equal weighting between assessments of mechanics and applications which sought to increase the cadets' understanding of taking the derivative and finding the integral when compared to previous semesters.

Table 1: MA104 Grading Scheme

Event	Points	Weight
Concept Quizzes (25)	100	5.00%
Homework Quizzes (14)	350	17.50%
Technology Tutorial	25	1.25%
Writs (3)	400	20.00%
WPRs (2)	500	25.00%
Projects (2)	225	11.25%
TEE	400	20.00%
Total	2000	100%

5. **Course Statistics.** The course average for AY23-02 was an 82.18%. The distribution of grades is shown in Table 1. There were 25 course failures - 24 who scored less than 65% in the course and 1 that scored less than 50% on the TEE. There was 1 incomplete (NC) for an honor-related issue.

Table 2: Final Grade Distribution

Grade	Number of Cadets	Grade Scale	Percent of Course
A+	17	$\geq 96.70\%$	2.13%
A	82	$\geq 93.00\%$	10.26%
A-	96	$\geq 90.00\%$	12.02%
B+	97	$\geq 87.00\%$	12.14%
B	114	$\geq 83.00\%$	14.27%
B-	62	$\geq 79.94\%$	7.76%
C+	93	$\geq 77.00\%$	11.64%
C	88	$\geq 73.00\%$	11.01%
C-	71	$\geq 70.00\%$	8.89%
D	53	$\geq 65.00\%$	6.63%
F	25	$\geq 0.00\%$	3.13%
NC	1	Honor	0.13%

6. **Course Validation.** During AY23-01, the MA104 Validation Exam was advertised through MA103 instructors and through various flyers posted throughout Thayer Hall. This effort resulted in 36 cadets signing up to take the validation exam and 28 cadets taking the exam. We had 4 cadets validate MA104, but 1 of those cadets chose to revoke his validation credit because he had decided not to be a STEM major. Preparation for the validation exam was "self-study" focused. Cadets were given

pre-recorded videos and slideshow presentations detailing the topics of Calculus and were given the AY23 MA104 Reference Card. At the conclusion of the validation exam, cadets that did not pass the exam were given a Microsoft Forms survey to ask about their preparation for the validation exam. Most cadets mentioned that they used the preparation products provided but only studied 1 to 2 hours for the exam.

7. Faculty Development. During AY23-02, MA104 was arranged into four teams. Each team consisted of 3-4 instructors with the most experienced, junior instructor serving as the team leader for their team. At the end of AY23-01, MA104 held a team leader meeting outlining the expectations of the team leaders and their teams. Teams were assigned to design the test and rubric for a single graded event by using the tests from AY23-01 as an example. The team leader that was chosen to be the Assistant Course Director for AY24 was selected to certify the projects from AY23-01 for use during AY23-02. Teams were also assigned to create homework quizzes and to take exams from other teams for time. During Reorganization Week, MA104 held the first Program Director (PD) meeting and held an abbreviated Faculty Development Workshop during the first 20-30 minutes where instructors were made aware of the structure of the course and various course policies (these items were already briefed to the teams by their team leaders prior to the PD Meeting). Finally, new instructors were given a topic to present to the course during each PD Meeting.

8. Cadet Development. In AY23, MA104 was nested with D/Math's goals of developing "Habits of Mind" and teaching the cadets to "Communicate Effectively." The course accomplished the goal of developing habits of mind by making homework assignments optional and implementing self-study lessons. We accomplished the goal of instructing the cadets how to communicate effectively by assessing and providing feedback on written communication during Project 1 and assessing and provide feedback on verbal communication during Project 2. After WPR 1, cadets with a 68% or less in the course were counseled by course leadership. During the counseling the course leadership team discussed the various resources available to the cadets and offered cadets the opportunity to switch sections to a different instructor. Most cadets stated that their poor performance was a result of their own study habits and chose to remain with their assigned instructor, however, seven cadets chose to switch instructors.

9. Recommendations.

a. The following are changes that were implemented in MA104 that we suggest be maintained in AY24.

(1) We recommend maintaining an emphasis on in-class assessments during AY24. Decreasing the value of homework assignments was valuable in obtaining a true assessment of a cadets understanding of the material.

(2) We also recommend that all major graded events be graded as a course. We found that for some questions different instructors graded problems with high variability between scores. This was significantly reduced when exams were graded as a group. Grade calibration should be a deliberate process to ensure all instructors are grading similarly.

(3) During AY23, we significantly reduced the instructor led preparation for the MA104 validation exam and instead provided study materials. Despite the reduced effort, we obtained a similar number of cadets that took the validation exam. Good performance on the validation exam (>50%) led to good performance in MA205 as indicated by the grades of B+, B, and C for the three cadets that were placed into MA205 for AY23. We recommend sustaining this validation exam preparation model if the validation exam is offered again in the future. In AY24, it is recommended that in lieu of the validation exam, those students be considered for MA104X.

(4) During AY23, all major graded assessments (except the TEE) were limited to 55-minute exams. This allowed the course to administer the exam during class but more importantly, made it easier to schedule makeup exams during a cadets' off periods which tended to follow the 55-minute model, especially when given on a MA104 drop day. We also found that scheduling drops the day following a major exam allowed centralized time for instructors to calibrate and grade exams.

(5) We recommend sustaining the team structure for instructors. Experienced junior instructors as team leaders played a crucial role in mentoring instructors new to the course and answering questions.

b. The following are changes for MA104 we recommend be implemented during AY24.

(1) For AY24, we recommend decreasing the scope of derivative and integration applications and using laboratory days as a way of exploring the applications of calculus and having more lessons dedicated to calculus fundamentals. The course is dense with many different topics, however we found many cadets could not obtain a strong grasp of the fundamental skills of calculus to perform well in the application of these fundamentals. Additionally, cadets' fundamental skills degraded after learning how to solve problems involving the application of Calculus (see Appendix 2).

(2) We also recommend modifying the method in which the course accomplishes the goal of faculty development. Organizing instructors into diverse teams of senior and junior instructors provides an adequate amount of faculty development without having to have the instructors design exams. Oftentimes, new instructors did not know enough about the subject of the course to create well-designed assessments which in turn made it challenging to meet printing deadlines. We recommend new instructors help create makeup versions of major graded assessments and rubrics to allow flexibility in the assessment timeline while still giving them the opportunity to participate in assessment creation.

(3) During AY24, more emphasis should be placed on teaching topics to new instructors. Since most of the course material is provided to instructors, instructors should use this time to better their teaching skills. We did not provide a wide array of previous year materials for instructors to build their own classes to enable them to develop their own teaching styles, however this resulted in some instructors not covering the material that should have been covered. While performance in the course was mostly related to a cadets own effort and ability, quality instruction should always be pursued.

(4) Sectioning is challenging in the course due to the wide array of ability that arrives to the course. Some cadets arrive having completed multiple years of Calculus in high school. The discrepancy in ability was compounded by the fact that many students who would have qualified for MA100 (a class focused on fundamental mathematics skills) were put into MA104 instead due to MA100 not being offered in AY23-2. MA104 is designed for students that do not have calculus experience, which leaves the population of cadets that have already taken calculus in high school un-engaged and does not give them the opportunity to develop as students. We recommend having an intermediate-level course between MA104 and the advanced math program (MA153, MA255) that would allow cadets with both calculus experience and a desire to become a STEM major to learn more challenging topics that will ultimately help them learn the math required for their major.

(5) Since MA104 is a course filled with many topics adding a couple lessons on the method of the integrating factor was counterproductive. At the conclusion of the course very few cadets understood the method of the integrating factor because of how much material is required to be taught (that is currently outside the scope of MA104) prior to learning this technique to solve a very particular form of an ordinary differential equation. For AY24, we recommend that the method of the integrating factor, if it cannot be removed entirely, be a course concept that is taught and tested only for the small population of System Engineering majors that require this topic.

(6) For the majority of cadets, the “self-study” lessons did not inspire them to develop better study habits. Nearly all cadets gained a better understanding of the importance of self-study and discovered their bad study habits, however, the poorer performing students (whom the “self-study” lessons were meant to foster growth) tended to not use the allotted time to learn the material, while the higher performing students tended to do well because they already had good study habits. For AY24, we recommend either a couple dedicated classes meant to show cadets how to study for a math class or removal of the self-study lessons altogether.

10. The point of contact for this memorandum is the undersigned at ira.crofford@westpoint.edu or by phone (845) 938-0061.

Encl:

1. Academic Program Goals.
2. Calculus Performance.
3. Sectioning.

IRA L. CROFFORD, JR.
LTC, 49/AR
Associate Program Director, MA104

APPENDIX 1: ACADEMIC PROGRAM GOALS

1. **Core Math Goals.** During AY23, MA104 focused on the Core Math Goals of “Habits of Mind” and “Communicate Effectively.”

2. **Habits of Mind.** Learning is an inherently inefficient process. Learning how to teach oneself is a skill that requires maturity, discipline, and perseverance. In studying mathematics, students learn good scholarly habits for progressive intellectual development. Essential to developing habits of mind is the development of reasoning and critical thinking skills, creativity, work ethic, interdependent thinking, and life-long learning and curiosity. Most cadets enrolled in MA104 are first-year college students and have not fully developed the skills required to effectively teach themselves. During AY23, MA104 provided optional homework assignments, encouraged review of old material through correction assignments, incorporated “reach-back” problems on every major graded event and implemented two “self-study” lessons that allowed students to assess the effectiveness of their own study habits.

3. **Communicate Effectively.** Students learn mathematics when they construct their own mathematical understanding. Successful problem solvers must be able to clearly articulate their problem-solving process to others. During AY23, MA104 developed a standard for briefing boards, established a group project requiring a written report, and designed a group project that required a presentation. These aspects of the course required cadets to articulate mathematical ideas both through written and verbal communication.

4. **Character Development.** Cadets enrolled in MA104 during AY23, experienced character development particularly in their understanding of the importance of preparation and self-study. In the beginning of the course approximately 77% of cadets reported having the ability to learn course material outside of a formal class setting. This seemed to be over-confidence since this number went down to 68% during the mid-course brief. However, by the end of the course, 71% of the cadets stated that they were confident in their ability to self-study. 89% of the cadets also reported that Army officers should be capable of teaching themselves new concepts. Cadets were challenged daily in class with board problems, group projects (both written and verbal). Most homework was optional in the course, however doing the homework would lead to better scores on the exams.

5. **Academic Integrity and Honor.** Despite multiple discussions and acknowledgment statements, MA104 continues to see various challenges with honor. Cadets were found to be discussing testing material before academic release, using unauthorized references during in-class quizzes, and lacking citations particularly when using digital resources that solve math problems. We believe these issues may be because the cadets are inexperienced and have not fully inculcated the honor code. Additionally, many of them developed bad habits in high school which they have brought to the Academy. These factors in combination with the overall difficulty of the course challenged cadets’ honor during AY23.

6. **MA104 Goals Assessment.** The course had three Academic Program Goals (APG) – two at the “Introduce” level (Critical Thinking and Creativity, Lifelong Learning) and one at the “Reinforce” level (STEM) that directly link to three Core Math

Goals. MA104 also supported the additional APG of Communication through the Core Math Goal of Communicate Effectively though that was not specifically required. Table 3 shows the mapping of each APG through the particular “What Graduates Can Do” (WGCD) statement into the learning objectives and assessments within MA104.

Table 3: Academic Program Goal Mapping into MA104 - Single Variable Calculus

Academic Program Goal (APG)	"What Graduates Can Do" (WGCD) Statement	Core Math Goal	MA104 Course Objective	MA104 Assessment
1. Communication	1.3 (NA) Effectively convey meaningful information to diverse audiences using appropriate forms and media	Communicate Effectively	Develop verbal and written communication skills to clearly articulate calculus problem solving techniques and methodologies.	Course Projects.
2. Critical Thinking and Creativity	2.4 (I) Reason both quantitatively and qualitatively.	Body of Knowledge	Evaluate derivatives and integrals and interpret the results in context. Communicate calculus problem solving techniques and methodologies.	Homework Quizzes, WRITs, WPRs, and Course Projects.
3. Lifelong Learning	3.1 (I) Demonstrate the willingness and ability to learn independently.	Habits of Mind	Internalize the concept of the derivative as a rate of change, the integral as the accumulation of change, and that these two concepts are inversely related.	Daily Preparation Requirements, Concept Quizzes, Revision Assignments, Course Projects, and Self-Study Lessons.
5. Science, Technology, Engineering, and Mathematics	5.1 (R) Apply mathematics, science, and computing to model devices, systems, processes, or behaviors.	Technology Skills	Understand, calculate, and interpret applications of differentiation and integration.	WRITs, WPRs, and Course Projects.

7. APG 1 - Communication (Assessment: 2, Very Good) The “What Graduates Can Do” statement associated with APG 1 is that students should effectively convey meaningful information to diverse audiences using appropriate forms and media. This connects to the Core Math Goal of communicating effectively and the MA104 Course Objective to develop verbal and written communication skills to clearly articulate calculus problem solving techniques and methodologies. We assessed the cadets’ written communication skills through Project 1 and the cadets’ verbal communication skills through Project 2. We assessed the cadets to possess effective written communication skills with a course average of 87% on Project 1 and very effective verbal communication skills with a course average of 92% on Project 2.

8. APG 2 - Critical Thinking and Creativity (Assessment: 3, Good) The “What Graduates Can Do” statement associated with APG 2 is that students should reason both quantitatively and qualitatively. This connects to the Core Math Goal of building a “Body of Knowledge” and the MA104 Course Objectives of evaluating derivatives and integrals, and then interpreting and communicating their results. MA104 assessed quantitative student development by tracking performance on topics throughout the semester. We judged qualitative reasoning through the exploration of two course projects in which students had to apply calculus to running a company store and route planning. We observed that cadets possessed good quantitative reasoning skills by strong correlation between overall course performance and performance on the TEE. This suggests that they possess a good ability to think critically and creatively and that the course meets the “Introduce” Academic Program Goal for cadets. However, we did note a small decrease in fundamental calculus skills as the course progressed as shown in Appendix 2.

9. APG 3 - Lifelong Learning (Assessment: 2, Very Good) This APG states that graduates should demonstrate the willingness and ability to learn independently which

is supported by the Core Math Goal of developing strong “Habits of Mind” in our students. MA104 satisfies this requirement through the course objectives of understanding and internalizing the concepts of differentiation and integration, but more directly through the structure of the course. MA104 requires students to prepare and be assessed every day, to revisit and correct their mistakes, and to explore and investigate projects on their own. We collected student sentiment data through three surveys during the semester and we observed that most ($> 75\%$) of the cadets recognize the utility of nightly preparation, consistent assessment, and revisions based on what they did not initially understand. This indicates that students have been exposed to and understand the importance of independent learning and that MA104 again meets the “Introduce” Academic Program Goal.

10. APG 5 - Science, Technology, Engineering, and Mathematics (Assessment: 2, Very Good) This APG states that graduates should be able to apply mathematics, computing devices, and modeling systems to solve problems. This is supported by the Core Math Goal of building Technology Skills in our cadets and is executed in MA104 through the application of single variable calculus to solve problems on Writs, WPRs, and projects using an appropriate technological tool. In AY23, MA104 sought to focus and streamline the application of technology as it has the potential to disrupt the underlying course objectives of equipping students with calculus fundamentals if done incorrectly. Throughout the semester we were deliberate about when we chose to use specific tools – whether the issued calculator or Mathematica based on the type of problem we were attempting to solve. After collecting quantitative and qualitative data on student performance and sentiment we observed that students are generally capable and feel comfortable executing basic commands and troubleshooting code to solve single variable calculus problems in Mathematica although it has degraded over the years as technology has been removed from the course. However, their confidence in using their issued TI-36X Pro Calculator has increased ($> 50\%$) since the beginning of the course. A new technology topic also arose this year that the instructors embraced and led a short discussion with the students on artificial intelligence and its use in the classroom, specifically Chat Generative Pre-Trained Transformer (ChatGPT). We wanted the students, and instructors for that matter, not to be afraid of the technology, but to be aware of its limitations. We also asked that if generative AI was used, the entire “conversation” was documented.

11. Upcoming Assessment Plan In AY24, MA104 will focus on the Core Math Goals of “Acquire a Body of Knowledge” and “Build Competent and Confident Problem Solvers.” We will also continue to assess WGCD Statements 2.4, 3.1, and 5.1.

APPENDIX 2: CALCULUS PERFORMANCE

During AY23-02, equal emphasis was placed on the mechanics and the application of calculus concepts. Distributing focus in this manner tended to negatively effect the lower performing students more than the higher performing students as shown in Figure 1.

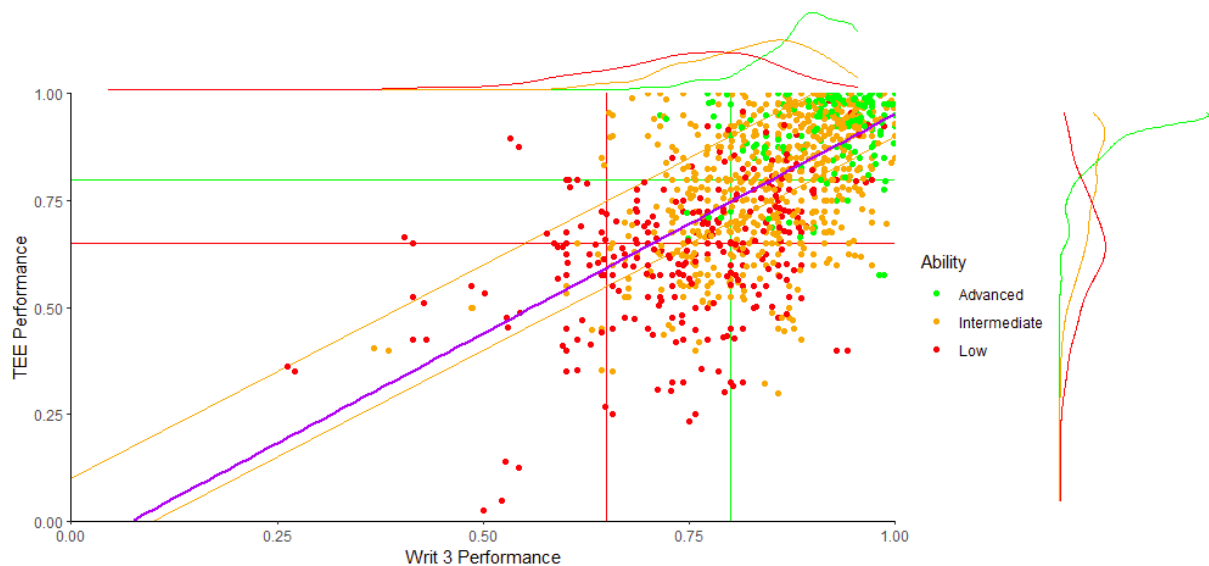


Figure 1: Writ 3 vs TEE Integral Questions

APPENDIX 3: SECTIONING

During AY23, each hour's sections were comprised of ability distributions as shown in Figure 2. As a result of this sectioning, cadets tended to perform at their predicted ability as shown in Figures 3 and 4. These results mirror what was shown during AY22 which suggests that sectioning cadets by ability has minimal effect on a cadets performance in the course.

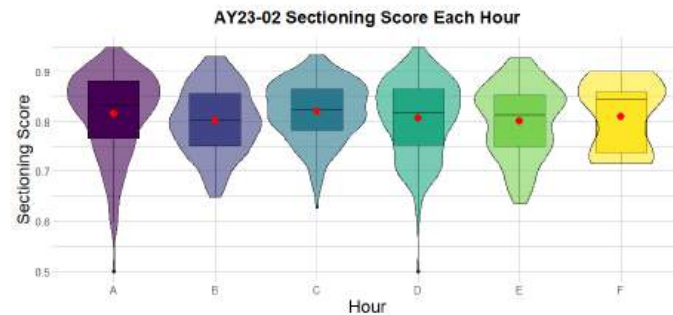


Figure 2: Approximate sectioning distribution of each section each hour.

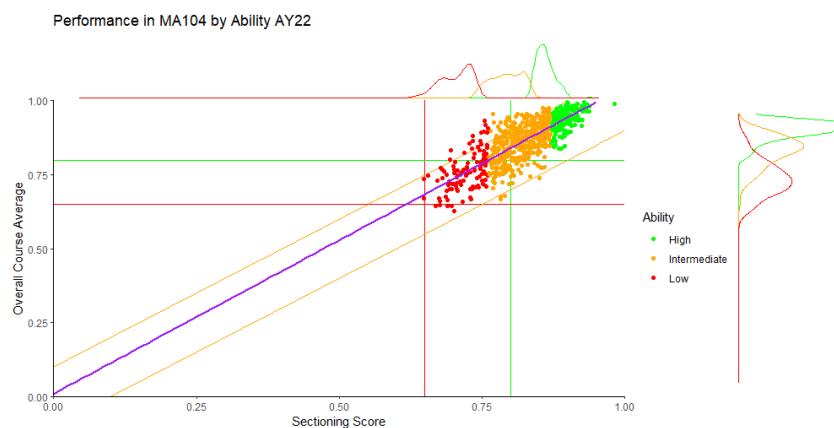


Figure 3: Course Grade vs Sectioning Score AY22-02

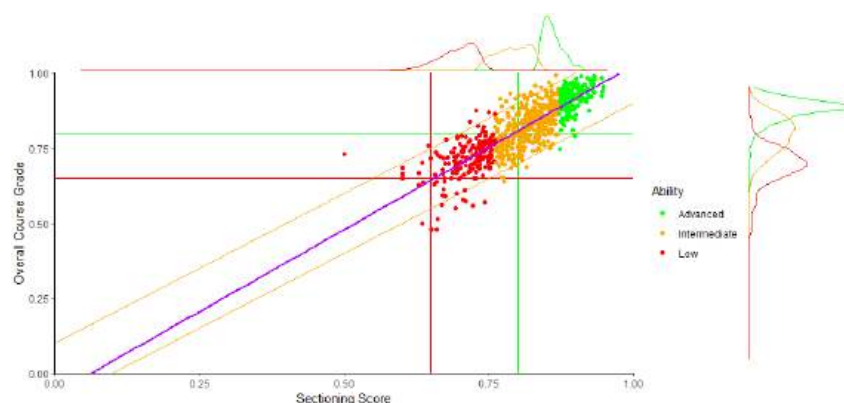


Figure 4: Course Grade vs Sectioning Score AY23-02

MA104: Single Variable Calculus
AY23-02 Syllabus

Textbook: Calculus - Ninth Edition, Early Transcendentals by James Stewart				
Students: All homework sets will be available for download on Canvas. Course Objectives: 1. Understand and internalize the role of the limit in differentiation and integration. 2. Understand the concept of the derivative as a rate of change, the integral as the accumulation of change, and that these two concepts are inversely related. 3. Develop both verbal and written communication skills to clearly articulate calculus problem solving techniques and methodologies. 4. Develop effective study practices that will enable cadets to learn new concepts without formal, classroom instruction.				
Lesson	Date	Topic (Reading)	Notes	Lesson Objectives
BLOCK 1: DERIVATIVES AND THEIR APPLICATIONS				
Functions & Mathematical Models				
Lesson 1	1/10/2023	Course Intro		Topic: Overview of the Course
Lesson 2	1/11/2023	Functions 1 (§ 1.1-1.2)		1. Understand the definition of a function. 2. Understand and recognize the four ways to represent a function. 3. Determine the domain and range of a function. 4. Construct an algebraic function based on a verbal/written description of the function. 5. Recognize the shapes and basic properties of the following functions: Linear, Polynomial, Power, Rational, Algebraic, Trigonometric, Exponential, and Logarithmic.
Lesson 3	1/12/2023	Functions 2 (§ 1.3)		1. Understand how to shift functions vertically and horizontally. 2. Understand how to stretch functions vertically and horizontally. 3. Recognize and construct composite functions.
Limits & Continuity				
Lesson 4	1/13/2023	Limits/Continuity 1 (§ 2.1)		1. Understand the definition of a tangent and a secant line. 2. Determine average rates of change. 3. Estimate instantaneous rates of change. 4. Find or estimate the equation of a tangent line to a curve.
Lesson 5	1/17/2023	Limits/Continuity 2 (§ 2.2)		1. Understand the definition of a limit. 2. Find the limit of a function graphically and numerically.
Lesson 6	1/18/2023	Limits/Continuity 3 (§ 2.3 and § 2.6: pg 127-133)		1. Understand how to apply the properties of limits. 2. Find the limit of a function algebraically. 3. Understand and evaluate limits at infinity.
Lesson 7	1/19/2023	Limits/Continuity 4 (§ 2.5)		1. Understand the definition of continuity. 2. Identify where and why a function is discontinuous. 3. Understand the Intermediate Value Theorem. 4. Recognize the types of functions that are continuous at every number in their domain.
Rates of Change				
Lesson 8	1/20/2023	Rates of Change 1 (§ 2.7)		1. Understand the definition of a tangent line. 2. Understand the limit definition of a derivative. 3. Apply the limit definition of a derivative to solve for derivatives at a point. 4. Understand the definition of velocity and how it relates to derivatives. 5. Apply the limit definition of a derivative to solve for instantaneous rates of change.
Lesson 9	1/23/2023	Rates of Change 2 (§ 2.8)		1. Understand and apply the limit definition of a derivative to a function. 2. Recognize and apply the different notation used to represent a derivative. 3. Understand how a function can fail to be differentiable. 4. Recognize and understand the relationship between differentiation and continuity.
Lesson 10	1/24/2023	Rates of Change (Lab 1)		
STUDY DAY	1/25/2023	WRIT 1	Exam Period: 1245	Topics: Functions, Mathematical Models, Limits, Continuity
Lesson 11	1/26/2023	Exam Comp Day	No Class	
Derivatives				
Lesson 12	1/27/2023	Derivatives 1 (§ 3.1)	Tech Tutorial Due	1. Recognize and apply the following derivative rules: a) Derivative of a constant function, b) Power rule, c) Constant multiple rules, d) Sum rule, e) Difference rule, f) Derivative of the natural exponential function.
Lesson 13	1/30/2023	Derivatives 2 (§ 3.2)		1. Recognize and apply the following derivative rules: a) Product rule, b) Quotient rule
Lesson 14	1/31/2023	Derivatives 3 (§ 3.3)		1. Recognize and apply the derivative rules of trigonometric functions. 2. Understand the relationship between the derivatives of sine and cosine functions.
Lesson 15	2/1/2023	Derivatives 4 (§ 3.4)		1. Recognize composite functions. 2. Understand when to apply the chain rule. 3. Solve derivatives that involve using the chain rule. 4. Use the chain rule to differentiate an exponential function with a base "b".

MA104: Single Variable Calculus
AY23-02 Syllabus

Lesson	Date	Topic (Reading)	Notes	Lesson Objectives
Implicit Differentiation				
Lesson 16	2/2/2023	Implicit 1 (§ 3.5: pg 208-213)		1. Recognize when a function is defined implicitly. 2. Understand how to apply implicit differentiation. 3. Solve for the equation of a tangent line at a point on any implicitly defined curve.
Lesson 17	2/3/2023	Implicit 2 (§ 3.6: pg 218-220)		1. Understand how implicit differentiation is used to determine the derivative of logarithmic functions. 2. Solve for the derivative of logarithmic functions. 3. Take the derivative of an implicit function with respect to time.
Related Rates				
Lesson 18	2/6/2023	Related Rates 1 (§ 3.9)		1. Apply the problem-solving process to related rates problems.
Lesson 19	2/7/2023	Differentiation (Lab 2)		
STUDY DAY	2/8/2023	WRIT 2	Exam Period: 1245	Topics: Rates of Change, Derivatives, and Implicit Differentiation
Lesson 20	2/9/2023	Exam Comp Day	No Class	
Lesson 21	2/10/2023	Related Rates 2 (§ 3.9)		1. Apply the problem-solving process to related rates problems.
Lesson 22	2/13/2023	Related Rates 3 (§ 3.9)		1. Apply the problem-solving process to related rates problems.
Lesson 23	2/14/2023	Related Rates (Lab 3)		
Derivatives in the Natural Sciences				
Lesson 24	2/16/2023	Natural Sciences (§ 3.7)	Self-Study Lesson (No Class Meeting)	1. Apply derivative concepts to physics applications 2. Understand how to use derivative concepts to determine when an object in motion is slowing-down or speeding-up. 3. Understand the meaning of a derivative in various scenarios.
Maximums & Minimums				
Lesson 25	2/17/2023	Max/Min 1 (§ 4.1)		1. Understand the difference between absolute and local extrema. 2. Apply the Extreme Value Theorem. 3. Understand how Fermat's Theorem relates to critical numbers and the cautions associated with using Fermat's Theorem. 4. Use critical numbers as part of the Closed Interval Method to determine absolute maximum/minimum values.
Lesson 26	2/21/2023	Max/Min 2 (§ 4.3)	Project 1 Writing Exercise Assigned	1. Use a function's derivative to identify increasing or decreasing intervals. 2. Identify and solve for intervals where a curve is concave up or concave down. 3. Understand and identify inflection points. 4. Understand and apply the second derivative test to solve for local maximum and minimum values.
Lesson 27	2/22/2023	Max/Min 3 (§ 4.3)		1. Use information for first and second derivatives to sketch the curve of a function.
Optimization				
Lesson 28	2/23/2023	Optimization 1 (§ 4.7)	Project 1 Writing Exercise Due	1. Understand how Stewart's optimization problem solving strategy relates to the USMA Mathematical Problem-Solving Strategy. 2. Understand and apply the first derivative test for absolute extreme values. 3. Given a scenario with objectives and constraints, solve for the maximum or minimum values.
Lesson 29	2/24/2023	Project 1 Development (Lab 4)		
Lesson 30	2/27/2023	Optimization 2 (§ 4.7)		1. Given a scenario with objectives and constraints, solve for the maximum or minimum values.
Lesson 31	2/28/2023	Optimization 3 (§ 4.7)		1. Given a scenario with objectives and constraints, solve for the maximum or minimum values.
Lesson 32	3/1/2023	Optimization (Lab 5)		
STUDY DAY	3/2/2023	WPR 1	Exam Period: Pri-0905, Alt-1055	Topics: Natural Sciences, Related Rates, Max/Min, & Optimization
BLOCK 2: INTEGRALS AND THEIR APPLICATIONS				
Antiderivatives				
Lesson 33	3/13/2023	Project 1 Comp Day	No Class	
Lesson 34	3/14/2023	Exam Comp Day	No Class	
Lesson 35	3/15/2023	Antiderivatives 1 (§ 4.9)		1. Understand the definition of an antiderivative. 2. Solve for an antiderivative of a functions. 3. Understand antiderivatives as a family of functions.
Lesson 36	3/16/2023	Antiderivatives 2 (§ 4.9)		1. Understand the definition of an antiderivative. 2. Solve for an antiderivative of a functions. 3. Understand antiderivatives as a family of functions. 4. Use antiderivatives to solve problems related to rectilinear motion.
Definite Integrals				

MA104: Single Variable Calculus
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Lesson	Date	Topic (Reading)	Notes	Lesson Objectives
Lesson 37	3/17/2023	Definite Integral 1 (§ 5.1)		1. Understand how to estimate the area between a curve and the horizontal axis on an interval. 2. Solve area estimates using sigma notation. 3. Understand how to solve areas using the limit operator in conjunction with sigma notation. 4. Interpret distances traveled by an object as an area problem.
Lesson 38	3/20/2023	Definite Integral 2 (§ 5.2)	Project 1 Due	1. Calculate definite integrals using Definition 2 and Theorem 4 in Section 5.2. 2. Express definite integrals as a limit of a Riemann Sum and express a limit of a Riemann Sum as a definite integral. 3. Understand the mathematical notation and terminology of integrals. 4. Apply the Left or Right Endpoint Method to estimate an integral. 5. Understand and apply the properties of definite integrals.
Lesson 39	3/21/2023	Definite Integral 3 (§ 5.3)		1. Understand and apply the Fundamental Theorem of Calculus, Part 1. 2. Understand and apply the Fundamental Theorem of Calculus, Part 2. 3. Internalize differentiation and integration as inverse processes.
Integration				
Lesson 40	3/23/2023	Integration 1 (§ 5.4)		1. Distinguish between indefinite and definite integrals. 2. Determine indefinite integrals. 3. Understand and apply the Net Change Theorem.
Lesson 41	3/24/2023	Integration 2 (§ 5.5)		1. Understand when to apply the substitution rule. 2. Use the substitution rule to solve definite and indefinite integrals.
Lesson 42	3/27/2023	Integration 3 (§ 7.1)		1. Understand how the method of integration by parts is derived from the product rule of differentiation. 2. Understand how to select "u" and "dv" when using integration by parts. 3. Apply integration by parts to solve definite and indefinite integrals.
Lesson 43	3/28/2023	Integration 4 (§ 7.5)		1. Understand how to approach a definite and indefinite integral in order to determine the appropriate integration method(s). 2. Solve any integral by choosing the correct integration technique (simplifying w/ basic rules, u-sub, IBP)
Lesson 44	3/30/2023	Integration 5 (§ 7.5)		1. Understand how to approach a definite and indefinite integral in order to determine the appropriate integration method(s). 2. Solve any integral by choosing the correct integration technique (simplifying w/ basic rules, u-sub, IBP)
Lesson 45	3/31/2023	Integration 6 (§ 7.8)		1. Identify improper integrals as either Type 1 or Type 2 improper integrals. 2. Identify when a Type 1 improper integral is either convergent or divergent. 3. Solve Type 1 improper integrals.
Lesson 46	4/3/2023	Integration (Lab 6)		
Lesson 47	4/4/2023	WRIT 3	In Class	Topics: Antiderivatives, Riemann Sums, Definite & Indefinite Integrals, Integration Strategies, Improper Integrals
Areas & Volumes				
Lesson 48	4/6/2023	Areas/Volumes 1 (§ 6.1)		1. Determine the area between two curves. 2. Interpret the meaning of the area between two curves given in context.
Lesson 49	4/7/2023	Areas/Volumes 2 (§ 6.2)		1. Understand the definition of volume. 2. Determine the volume of a solid object using integration techniques. 3. Determine the volume of a solid of revolution using the disk/washer method.
Lesson 50	4/10/2023	Areas/Volumes 3 (§ 6.3)		1. Determine the volume of a solid of revolution using the cylindrical shell method.

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Lesson	Date	Topic (Reading)	Notes	Lesson Objectives
Lesson 51	4/11/2023	Areas/Volumes 4 (§ 6.3)	Self-Study Lesson (No Class Meeting)	1. Understand when to apply the disk/washer method and when to apply the cylindrical shell method when determining the volume of a solid of revolution.
Applications of Integration				
Lesson 52	4/13/2023	Applications of Integration 1 (§ 8.5)		1. Understand the definition of a probability density function. 2. Recognize when a probability density function is invalid. 3. Interpret a probability statement. 4. Solve for a probability using a probability density function. 5. Determine the average value using a probability density function.
Lesson 53	4/14/2023	Applications of Integration 2 (§ 8.5)		1. Understand the definition of a probability density function. 2. Recognize when a probability density function is invalid. 3. Interpret a probability statement. 4. Solve for a probability using a probability density function. 5. Determine the average value using a probability density function.
Lesson 54	4/17/2023	Project 2 Development (Lab 7)		
Lesson 55	4/18/2023	Applications of Integration 3 (§ 6.4)		1. Understand the definition of work. 2. Solve lifting problems involving constant and variable force using both SI and US units of measure.
Lesson 56	4/20/2023	Applications of Integration 4 (§ 6.4)		1. Solve pumping problems involving tanks with constant and variable cross-sectional areas using both SI and US units of measure.
Lesson 57	4/21/2023	App. of Integration 5 (§ 9.1 pg 608-610)		1. Understand the meaning of ordinary differential equations. 2. Understand that the solution of an ordinary differential equation is an equation. 3. Find the particular solution for a first order linear ODEs given initial conditions.
Lesson 58	4/24/2023	Applications of Integration 6 (§ 9.5)		1. Solve linear differential equations in the form $y' + P(x)y = Q(x)$ using the method of the integrating factor. 2. Apply the method of the integrating factor to solve application based ODEs.
Lesson 59	4/25/2023	Applications of Integration (Lab 8)		
Lesson 60	4/26/2023	WPR 2	In Class	Topics: Apps of Integration, Probability, Work, Net Change, Areas & Volumes, ODEs
Lesson 61	4/27/2023	Project 2 Comp Day	Project 2 Due	
Lesson 62A	5/1/2023	Project Briefs	In Class	
Lesson 62B	5/2/2023	Project Briefs	In Class	
Lesson 62C	5/5/2023	Project Briefs	In Class	
Lesson 63	5/8/2023	Differentiation Review		
Lesson 64	5/10/2023	Integration Review		



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB D

AY2024 COURSE ASSESSMENT



DEPARTMENT OF THE ARMY
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

MADN-MTH

2 August 2024

MEMORANDUM THRU

COL Joseph Lindquist, Director, Calculus Program, Department of Mathematical Sciences, USMA, West Point, NY 10996

COL Michael Scioletti, Professor and Head, Department of Mathematical Sciences, USMA, West Point, NY 10996

FOR Office of the Dean, AARS, United States Military Academy, West Point, NY 10996

SUBJECT: MA104 Single Variable Calculus Executive Summary (AY24-02)

1. **Purpose:** The purpose of this memorandum is to assess the MA104 curriculum.

2. **Assessment:** The course has a three-year assessment strategy for MA104. This year's assessment examined the Course Learning Objectives (CLOs) listed below with our findings for each assessed area (In Table 1, A = exceeds the standard; B = meets the standard; C = meets the standard with some weakness; D = fails the standard (can be addressed in the short-term); F = catastrophically fails the standard (requires immediate remediation)). A list of all CLOs can be found in enclosure 1.

MA104 Course Learning Objective (CLO)	Attainment	Evidence	Overall
(a) Understand and internalize the role of the limit in differentiation and integration.	A	A	A
(b) Understand the concept of the derivative as a rate of change, the integral as the accumulation of change, and that these two concepts are inversely related.	B	A	B
(d) Engage in mindful learning instead of procedural overlearning in order to improve both comprehension, retention, and performance.	B	B	B

Table 1: **CLO Evaluations for MA104 for AY24-02**

3. **Assessment Discussion:** This section presents an overview of assessment findings for AY24. A detailed discussion of direct and indirect indicators can be found in the Course Design and Analysis Report located in Enclosure 4. The schedule for CLO evaluation can be found in Enclosure 3.

a. Status of significant changes: Changes to the course are two fold (1) the removal of select application lessons and (2) the addition of lessons called "mastery days." The reason for both changes is the same, we intended to provide students a stronger and longer lasting **body of knowledge**. We accomplished this by devoting

more class time to learning and practicing calculus skills critical to applying calculus and succeeding in follow on calculus based courses.

b. Strengths in the course: The course introduced two new ideas: mastery days and the “Big Idea” question on each major graded event (the writ, all WPRs, and the TEE). The WPRs, and specifically the performance of cadets on the “Big Idea” questions are the measures of performance and effectiveness of mastery days. Mastery days, as noted above, provided ample opportunity for students to establish a body of knowledge *in the presence of the instructor*. The “Big Idea” questions, however, directly addressed CLOs (a) and (b) by providing direct evidence for how well cadets achieved the understanding of the CLOs and by reinforcing the CLOs through repetition.

c. Areas for improvement in the course.

(1) First, the mastery days stand to be improved each semester by honing the questions presented in the problem sets to guide the cadets through asking increasingly complex questions of themselves. Buy-in on the idea of mastery days as time for students to *struggle* with critical skills in the presence of an instructor must be gained by the course leadership.

(2) The course is currently required to teach a specific method to solve a specific instance of differential equations, called “integrating factor.” This lesson must be taught at the end of the course but requires the introduction of differential equations and then skipping three chapters of solving differential equations with almost no motivation as to why we are covering just this type of differential equation. The students who will use these two lessons are a minority and won't take a relevant course until they are firsties. We offer two courses of action and recommend course of action 1.

(a) Course of Action 1. Completely remove these two lessons and spend more time on critical differentiation and integration skills (specifically the chain rule, substitution, and integration by parts). This is the easier course of action to implement but negatively impacts a select group of cadets. We believe this risk of negative impact is already negated as there is a near zero probability that the cadets required to learn integrating factor will recall learning it by the time they are firsties.

(b) Course of Action 2. Make a significant change to the course where all applications are removed and replaced with a 2 week block of differential equations at the end of the course which includes several types and methods of solving differential equations. The risk to this course of action is losing even more application lessons typically taught with this level of calculus, including natural sciences, optimization, and probability. This risk is partially mitigated with differential equations being the real framework for natural sciences anyways. The risk could further be mitigated by the calculus behind calculating probabilities using probability density functions being shown in required statistics classes.

d. Changes to be implemented in AY25: The Newton's Method lab will now be a Related Rates lab. The first week of the course will focus on algebra review more strongly, specifically using MA100 critical skills. The first week will conclude with a

MADN-MTH

SUBJECT: MA104 Single Variable Calculus Executive Summary (AY24-02)

Fundamental Concepts Exam (FCE) specifically designed with Calculus in mind. In the Spring semester, failure on WPR 1 at the six-week mark coupled with a poor effort on the FCE will result in a recommendation to drop to MA100.

e. Proposed changes to the CLOs: None.

4. The point of contact for this memorandum is MAJ Thomas Kendall, MA104 Course Director, at 845-938-2654 or thomas.p.kendall.mil@army.mil.

4 Encls

1. MA104 CLOs
2. Assessment Crosswalk
3. CLO Assessment Schedule
4. Initial Course Design

IRA L. CROFFORD, JR.
LTC, FA49/AR
Associate Program Director

Enclosure 1: Full Set of Course Learning Objectives (CLOs)

1. At the end of the course, cadets will be able to...
 - (a) Understand and internalize the role of the limit in differentiation and integration.
 - (b) Understand the concept of the derivative as a rate of change, the integral as the accumulation of change, and that these two concepts are inversely related.
 - (c) Develop both verbal and written communication skills to clearly articulate calculus problem solving techniques and methodologies.
 - (d) Engage in mindful learning instead of procedural overlearning in order to improve both comprehension, retention, and performance.
2. Anticipated changes to CLOs: None.

Enclosure 2: Mapping of Academic Program Goals to CLOs

APG	WGCD	Core Math Goals	MA104 CLO	Assessment Methods
Creative Thinking and Creativity	2.4 - Reason both quantitatively and qualitatively. (Introduce)	- Technology Skills - Problem Solving - Body of Knowledge	(b) Understand the concept of the derivative as a rate of change, the integral as the accumulation of change, and that these two concepts are inversely related.	1x Project 8x Labs 4x WPRs, 1x Writ, 11x Homework Quizzes The Big Idea Questions
Lifelong Learning	3.1 - Demonstrate the willingness and ability to learn independently. (Introduce)	- Habits of Mind - Body of Knowledge	(d) Engage in mindful learning instead of procedural overlearning in order to improve both comprehension, retention, and performance.	Daily Concept Quizzes 4x WPRs, 1x Writ, 11x Homework Quizzes 13x Mastery Lessons The Big Idea Questions
STEM	5.1 Apply mathematics, science, and computing to model devices, systems, processes, or behaviors. (Reinforce)	- Interdisciplinary Problem Solving - Body of Knowledge - Technology Skills	(a) Understand and internalize the role of the limit in differentiation and integration.	1x Project 8x Labs 4x WPRs, 1x Writ, 11x Homework Quizzes The Big Idea Questions
Communication	1.3 - Effectively convey meaningful information to a diverse audience using appropriate forms and media. (Not required)	- Communication	(c) Develop both verbal and written communication skills to clearly articulate calculus problem solving techniques and methodologies.	1x Project Presentation 8x Labs 5x Executive Summaries (Lab reports & Project)

Table 2: Assessment Crosswalk

Enclosure 3: Cyclical schedule for assessment of CLOs and Student Outcomes

a	Understand and internalize the role of the limit in differentiation and integration.	X	X		X	X
b	Understand the concept of the derivative as a rate of change, the integral as the accumulation of change, and that these two concepts are inversely related.	X		X	X	X
c	Develop both verbal and written communication skills to clearly articulate calculus problem solving techniques and methodologies.		X	X	X	
d	Engage in mindful learning instead of procedural overlearning in order to improve both comprehension, retention, and performance.	X	X	X		X

Table 3: Three-year cyclic assessment schedule for CLOs

MADN-MTH

SUBJECT: MA104 Single Variable Calculus Executive Summary (AY24-02)

Enclosure 4: Initial Course Design and Assessment (Draft)

MA104 Single Variable Calculus
AY24-2 Course Design and Assessment

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1 Introduction and Background

The importance of single variable calculus as a topic of mathematics in undergraduate education cannot be understated. Single variable calculus is perhaps one of the most important contributions of mathematics to the modern world.[4] Like many colleges and universities, the United States Military Academy (USMA) at West Point requires that students pass multivariable calculus to proceed into most science, technology, engineering, or mathematics (STEM) majors.[14, 1, 10] The USMA, because of its required engineer tracks, further requires that all students, regardless of their major, pass “MA104: Single Variable Calculus” to be eligible to graduate and commission as officers into the United States Army.[14] The instruction of calculus at USMA and many undergraduate institutions, on the other hand, has undergone numerous adjustments over the years as the result of technology in the class room and teaching reforms.[5] Along with the introduction of new technologies, the ever growing emphasis on statistics has raised the question: “Do Students really need to know how to do calculus?”[13, 19] But calculus remains at the foundation of almost all modern sciences as evidenced by the requirement for calculus by numerous STEM courses.[14] Comprehending, and we mean *really understanding*, the implications instantaneous rate of change and area under the curve are critical even today. There still exists (and we posit always will exist) problems which students who master calculus can formulate, comprehend, and solve which cannot be solved with technology, statistics, and discrete mathematics. The *ideas* of calculus are crucial to the modern world and the ideas of calculus cannot truly be known without experientially knowing them through practice. Finally, and perhaps most importantly, calculus is a tool to teach problem solving.[4, 5] Students learn that difficult problems can be modeled and, through perseverance and breaking down difficult problems into smaller, easier to manage, sub-problems, solutions can be found.

Instruction of calculus, and instruction at the collegiate level in general, has received a push towards active learning as opposed to lecture based learning.[3, 6] Many authors and teachers now advocate for new techniques to achieve retention of material. Since most single variable calculus courses exist, like MA104, to give students tools that will be used in their chosen STEM majors, retention is paramount.[14, 1, 4] Retention is achieved through active learning techniques like practice, spaced learning, overlearning, and creating significant learning events.[8, 6, 15] However, the benefits of lecturing cannot be completely dismissed and most likely should be interlaced deliberately with active learning, practice, assessment, and out of class work.[11, 8, 6] L Fink suggests that, for the goal of achieving retention, courses should be designed to maximize student engagement in the classroom by leveraging an improved Bloom’s taxonomy shown in figure 1.[6]

Single variable calculus serves as a foundation for many STEM courses, including multivariable calculus. Therefore great emphasis on the retention of key skills, or building foundational knowledge must be made.[8, 6] Focusing on retention can often lead to over-learning resulting in only short term retention unless students engage in mindful learning.[8] Mindful learning of single variable calculus can be accomplished through active learning and problem solving events that combine conceptual and procedural learning.[7, 8, 6]. Many students, whose goal is, understandably, to “pass” calculus, choose to focus on procedural techniques only.[7, 17] Unfortunately, this approach does not lead to retention, and perhaps even worse, can likely lead to the student not achieving their desired goal of “passing.”[8] Consider, for example, an aspiring weight lifter who decides they want to be great at the deadlift. This *student* of weight lifting can watch other people do the deadlift and talk about the deadlift but will never actually get better at performing the deadlift until they start practicing the deadlift through quality repetitions. However attempting this lift without guidance and instruction can lead to injury. Single variable calculus should be viewed in a similar way. Instructing students how to do calculus and showing them good calculus will only take them so far: they must get quality repetitions through



Figure 1: Fink's Taxonomy

practice and practice in class (in the presence of an instructor) to truly learn and benefit from the “lift.” [8]

Single variable calculus courses often suffer from trying to teach too many topics instead of accomplishing true understanding in a few topics, the classic breadth vs depth problem. [15] Single variable calculus at many institutions (1) relies on student’s understanding of key algebra, trigonometric, and geometry topics, (2) teaches differential and integral calculus (including, but not limited to the limit, chain rule, product rule, quotient rule, classification of extrema, the substitution rule, integration by parts, and integration strategies), and (3) attempts to justify its own existence by teaching applications (including but not limited to related rates, optimization, the natural sciences, volumes of revolution, work, and probability). [14, 18, 11] If single variable calculus as a course exists to teach students concepts and skills that are crucial in follow on STEM courses, then single variable calculus courses should focus more on depth than breadth. Consider a graduate of the typical single variable calculus course: they are likely mediocre at a broad range of both skills and (we claim *contrived*) applications. Due to limited time spent in each category, retention of either skills or applications is unlikely. In contrast, graduates of single variable calculus, through interleaved instruction, carefully designed mindful practice problems, and focused applications designed to make connections, could *master* and retain skills needed in follow on courses. [6]

Single variable calculus itself typically follows the AP Calculus format shown in table 1. [18] The main process is to divide the course over an entire academic year, teach limits, derivatives and their applications in the first semester and then teach integrals, differential equations, and their applications in the second semester. Both the United States Air Force Academy and the United States Naval Academy uses this exact model covering sequences and series and introducing vectors in the second semester as well. [21, 12, 20] USMA, however, teaches MA104: Single Variable Calculus in a single semester as shown in section 3.4. A comparable course to MA104 was taught at the Massachusetts institute of technology (MIT) in 2010 and covered the topics shown in table 2. [9]

Block / Unit	Topics
1	Limits and Continuity
2	Differentiation: Definition and Fundamental Properties
3	Differentiation: Composite, Implicit, and Inverse Functions
4	Contextual Applications of Differentiation
5	Analytical Applications of Differentiation
6	Integration and Accumulation of Change
7	Differential Equations
8	Applications of Integration
9 (BC Only)	Parametric Equations, Polar Coordinates, and Vector-Valued Functions
10 (BC Only)	Infinite Sequences and Series

Table 1: AP Calculus AB / BC Year Long Course Layout[18]

Block	Topics
1	Concepts of Function, Limits and Continuity
2	Differentiation Rules, Application to Graphing, Rates, Approximations, and Extremum Problems
3	Definite and Indefinite Integration
4	The Fundamental Theorem of Calculus
5	Applications to Geometry: Area, Volume, and Arc Length
6	Applications to Science: Average Values, Work, and Probability
7	Techniques of Integration
8	Approximation of Definite Integrals, Improper Integrals, and L'Hôpital's Rule

Table 2: MIT's Single Variable Calculus Course in a Single Semester Layout[9]

2 MA104: Single Variable Calculus at USMA

2.1 AY23-02 Redbook

The description of MA104 in the USMA redbook, which is [\[how should the redbook be described for the outside reader \(it should emphasize the attempt to connect the entire curriculum\)\]](#), is shown below.

This is the second semester of the mathematics core curriculum. It provides a foundation for the continued study of mathematics and for the subsequent study of the physical sciences, the social sciences, and engineering. MA104 covers topics in single variable differential and integral calculus, parametric equations, 3-dimensional geometry, and vectors. Throughout the course, mathematical models motivate the study of topics such as optimization, accumulation, change in one variable, motion in space, 1st order differential equations, and other topics from the natural, social, and decision sciences. An understanding of course material is enhanced through the use of computer algebra systems.[\[14\]](#)

2.1.1 Prerequisites

There is only one prerequisite for MA104 at the United States Military Academy. The prerequisite is *MA103: Mathematical Modeling and Introduction to Calculus*. However, students admitted to USMA are required to pass the fundamental concepts exam (FCE) thus requiring the mastery of basic pre-calculus skills such as algebra, basic trigonometry, and basic geometry. For cases where students' mathematical backgrounds are not strong but they are still admitted to the academy, students are enrolled in (but are not required to pass) *MA100: Pre-Calculus Mathematics*.

2.1.2 Recommended changes

[\[This section should not be published\]](#) The following items are mentioned in the redbook scope but are not and have not been covered for at least three years. All three topics are covered in multivariable calculus.[\[14\]](#)

- Parametric equations.
- Three dimensional geometry.
- Vectors.

Our suggested redbook entry is

This is the second semester of the mathematics core curriculum. It provides a foundation for the continued study of mathematics and for the subsequent study of the physical sciences, the social sciences, and engineering. MA104 covers topics in single variable differential and integral calculus. Throughout the course, mathematical models motivate the study of topics such as optimization, accumulation, change in one variable, motion in space, 1st order differential equations, and other topics from the natural, social, probability, and decision sciences. An understanding of course material is enhanced through the use of computer algebra systems.

2.2 Our Customers

Every STEM major requires MA205 or MA255. MA104 is the prerequisite for MA205. MA205 instructors need cadets to have mastered single variable calculus techniques so that they can teach and

practice multivariable techniques without constantly reviewing single variable skills. STEM majors rely on calculus concepts to develop nuanced applications in their respective fields. They need students to readily comprehend and apply the needed calculus tools.

2.3 Our Product

Every cadet must pass MA104 to graduate West Point. This requirement is likely one of the reasons why every cadet receives a bachelor of science degree. The purpose of MA104 is to lead cadets to an *understanding* of single variable calculus. The goal of our typical student is likely to simply *pass* single variable calculus, with little thought given to “understanding.” Part of our challenge is to show cadets that if their goal is our goal, then they will do better than pass. If their goal is simply to pass then they may not achieve understanding and likely won’t pass the course. If they can pass and *learn to learn* they will not only pass single variable calculus, but will likely fundamentally change and grow.[17, 6]

Our typical cadet is a second semester plebe with underdeveloped study habits and time management. They have recently graduated MA103. Due to Beast and the array of topics in MA103, cadets’ algebra skills have likely suffered attrition. Thus our course begins with a short refresher of pre-calculus skills. Plebes typically *think* they “learn” by studying all night at the last minute. Studying usually consists of looking at solutions to worksheets and previous exams instead of practicing different problems. Part of our challenge is to teach the cadets that learning occurs (1) through practice and (2) when you are well rested.[8, 3, 2]

2.4 Working Assumptions

We make the following assumptions for our course.

- Students will retain more if we cover less material. The study of too many topics serves to dilute understanding and retention of critical calculus concepts and skills.
- Follow on courses for MA104 which require MA104 provide real applications for single variable calculus.
- The average student will not benefit from being assigned homework. They will either do *whatever* it takes to complete the homework for points, making the necessary citations, or they will not do it at all. In either case, the intended benefit is not seen. The students that do the homework will do it whether it is collected or not.
- The average student in single variable calculus views the course as an obstacle to follow-on privileges and graduation instead of as an opportunity to grow and learn.
- The average student can learn to *love to learn*.[6]

3 Experiment Design

Our goal is to incorporate major concepts from *Small Teaching* by Lang. We emphasize the concepts of interleaving, connecting, retrieving, and practicing.[8] Single variable calculus itself is naturally interleaved with integration calling upon differentiation. We accomplish both retrieving and interleaving through 5 assessments each of which have questions from previous examinations. The idea is that by asking students to recall previously learned material, we improve their retention. Our laboratories are meant to focus on connecting by getting students to make connections to other disciplines such as physics and statistics. We introduce planned practice days in the classroom where the practice problems are crafted to encourage mindful practice instead of over-learning. Lastly, in order to provide quality reps at what we deem are the critical skills, we reduce the breadth of the course by removing some applications. The topics that were removed are discussed in subsection 3.2

We hypothesize that students will retain single variable calculus concepts better if they are (1) provided more days for practice and (2) taught less overall material.[8, 6]. To evaluate this hypothesis we first reduce course *breadth* by removing certain applications (see subsection 3.2). We increase *depth* by (1) adding numerous days strictly for practice which we call “mastery days” and (2) adding a few topics (see subsection 3.1).

3.1 Additions

- **Mastery Days** Most students will only lightly grasp a topic by watching the instructor do it flawlessly. Their learning will occur when they get their reps in. Mastery days exist for students to get quality reps in critical skills. Students should walk in and immediately begin working on problems. These problems should walk the student through different levels of comprehension and difficulty. The goal is for students to *master* the material through repetition (or at least be much better than they were at the beginning of the lesson). Mastery days are a direct implementation of the concept of *practice* as introduced by Lang.[8]
- **Inverse Trigonometric Derivatives.** This topic was added because multiple following chapters include practice problems involving inverse trigonometric functions.[16] Teaching this topic achieves *true* interleaving with topics taught in high school by asking students to recall their knowledge of inverse trigonometric functions.[6, 8] Furthermore, this discussion of inverse functions primes the discussion of the anti-derivative as an “inverse” derivative.
- **L’Hopital’s Rule.** This topic is introduced to remind students of the limit prior to introducing the definite integral as the limit of sums. Bringing up the limit again achieves both interleaving and connection.[8]

3.2 Removals

The following topics, typically taught in single variable calculus, were removed in order to provide time for practice and mastery days.[18, 16] Each of these topics, we acknowledge build problem solving skills, but unfortunately do so at the cost of retaining critical calculus skills. Each topic below is of equivalent difficulty as critical calculus skills to master for students, but must, by necessity, compete with other topics such as the chain rule and the substitution rule. We **do not** remove applications all together. A single application is focused upon within each block (optimization and probability) to greater effect and other applications are explored during laboratory days, such as the natural sciences.

- **Related Rates.** This topic in particular, though potentially fun to the mathematician is perhaps the worst of the contrived applications. There is no question asked in a related rates problem that (1) anyone would ever ask and (2) would ever be solved using the methods taught.

For example, When one's ladder starts sliding down a frictionless wall, they do not ask themselves "How fast is the base of the ladder moving right now?" Continuing the example, even if one was curious about the speed of the bottom of the ladder, they would likely parameterize the location of the bottom of the ladder as a function of time and apply calculus from there to answer their strange question.[16]

- **Volumes of Revolution.** This topic provides an introduction to thinking in three dimensions but is notoriously difficult for students to grapple with immediately after learning the substitution rule and integration by parts.[16] This topic provides, at best, a shortcut to methods of determining volume learned in multivariable calculus.
- **Work.** This topic, intended to provide another exploration of integration, at best provides mediocre practice at simple integration in ways contrary to how work is taught in follow on STEM courses.[16] [Need more citations on how other courses teach work.](#) [Current proof is anecdotal from personal interviews with CME instructors \(Bluman and Randow\).](#)

3.3 Laboratories

In here we will discuss how laboratories are designed to accomplish the following.

- **Connecting.**[8] Labs are our opportunity to draw connections to other courses students might be taking to help improve retention. Labs will involve technology, physics, rudimentary economics, and statistics.
- **Practice, but on army writing.**[8] Cadets are provided multiple iterations at writing assessments so that they have multiple opportunities to improve based on instructor feedback. This is done through requiring lab reports on a subset of labs. Lab reports are 1-2 page Army Memorandums, typically requiring a figure.
- **Motivating.**[8] Most labs are designed to (1) be a stretch on already learned material and (2) be team competitions with bonus points at stake. For example in the natural sciences lab, teams are (1) challenged to extend their integration understanding to 2D vector space and (2) rewarded as a team to striking one of four targets which they choose.

Each lab is discussed in further detail below.

Lab 1: Limits and Rates of Change. This lab is a day for additional practice prior to the first graded event. Instructors are encouraged to make this day competitive.

Lab 2: Optimization. Students are placed into teams and tasked to solve a harder than usual optimization problem, turning in a physical object as the result of their optimization. Teams are rewarded for being first and correct. Students will turn in a lab report for this lab.

Lab 3: Newton's Method Lab. This lab is an exploration of an application of derivatives to find roots of functions that cannot be solved by hand. Students will get into Mathematica with their instructors and develop Newton's method together. Students turn in their Mathematica notebook that night for credit for this lab.

Lab 4: Natural Sciences Part 1. This is part 1 of a 2 day lab where students are placed into teams and tasked to choose a target to hit. Further targets are worth more points (with correct math). Part 1 consists of students being introduced to the idea of performing integration on vectors and try to find the initial velocities of their projectile launchers.

- Lab 5:** Natural Sciences Part 2. This is part 2 of a 2 day lab where students are placed into teams and tasked to choose a target to hit. Further targets are worth more points (with correct math). Part 2 consists of students calculating the necessary launch angle to strike their intended target and then attempting their shots. Students will turn in a lab report for this lab.
- Lab 6:** Riemann Sums. This lab serves as a reminder that definite integrals are infinite sums. Students are introduced to probability and the Gaussian distribution. They are first asked to find the integral of ae^{-bx^2} which they will find to be impossible. They will then be tasked to find the area under the curve anyways! The best estimate receives the most points. Students will turn in a lab report for this lab.
- Lab 7:** Project Development. This lab kicks off the final project. This is the instructors opportunity to shape the quality of projects. The final project includes the final writing assessment.
- Lab 8:** Probability. This lab is a fun and competitive exploration of Probability Density Functions (PDFs). Students are given PDFs for four different vehicles and asked to place a wager (bonus points) on which car will win 2 out of 3 races (must be supported by correct mathematics). The outcomes of the races are calculated according to the PDFs and are different for each instructor and race.

3.4 Course Flow

MA104 is divided into two blocks according to table 3.4. Block I covers limits, differentiation, and a few applications of integration with a deliberate focus on optimization as an extension of finding and classifying extrema. Block II covers integration and integration applications with a focus on the natural sciences and probability.

Block	Lesson	Topic	Section	Exam
I	1	Course Intro		
	2	Algebra and Trig Review		
	3	Functions Review	§ 1.1-1.5	
	4	Limits/Continuity 1	§ 2.1-2.2	HQ1
	5	Limits/Continuity 2		
	6	Limits/Continuity 3	§ 2.5	
	7	Limits Mastery		
	8	Rates of Change	§ 2.7-2.8	HQ2
	9	Rates of Change Mastery		
	10	Limits and Rates of Change (Lab 1)		
	11	Writ (Precalculus)		Writ
	12	Derivatives of Polyn. and Exp.	§ 3.1	
	13	Product and Quotient Rules	§ 3.2	
	14	Product and Quotient Rule Mastery		
	15	Derivatives 3 (Trig)	§ 3.3	HQ3
	16	Derivatives 4 (Chain)	§ 3.4	
	17	Chain Rule Mastery		
	18	Implicit 1 (Implicit Differentiation)	§ 3.5	HQ4
	19	Implicit 2 (Log and Inverse Trig)	§ 3.6	
	20	Derivatives Mastery	§ 3.1-3.6	
	21	WPR 1 (Derivatives)		WPR 1
	22	Max/Min 1	§ 4.1	

Block	Lesson	Topic	Section	
II	23	Max/Min 2	§ 4.3	HQ5
	24	Max/Min 3	§ 4.3	
	25	Optimization 1	§ 4.7	
	26	Optimization 2	§ 4.7	
	27	Optimization (Lab 2)		
	28	Optimization 3	§ 4.7	HQ6
	29	L'Hospitals Rule	§ 4.4	
	30	Newton's Method Lab (Lab 3)		HQ6
	31	Derivative Application Mastery 1		
	32	Derivative Application Mastery 2		WPR 2
	33	WPR 2 (Derivative Applications)		
	34	Antiderivatives	§ 4.9	HQ7
	35	Natural Sciences (Lab 4)		
	36	Natural Sciences 2 (Lab 5)		
	37	Definite Integral 1	§ 5.1	
	38	Definite Integral 2	§ 5.2	
	39	Definite Integral 3	§ 5.3	HQ8
	40	Indefinite Integrals	§ 5.4	
	41	Basic Integration Mastery		HQ9
	42	Substitution Rule	§ 5.5	
	43	Substitution Rule Mastery	§ 5.5	
	44	Integration by Parts	§ 7.1	
	45	Integration by Parts Mastery	§ 7.1	
	46	Riemann Sum Lab (Lab 6)		HQ10
	47	Integration Strategies 1	§ 7.5	
	48	Integration Strategies 2	§ 7.5	
	49	Integration Strategies Mastery	§ 7.5	WPR 3
	50	WPR 3 (Integration)		
	51	Areas Between Curves	§ 6.1	HQ11
	52	Improper Integrals	§ 7.8	
	53	Improper Integrals Mastery	§ 7.8	HQ11
	54	Project Development (Lab 7)		
	55	Probability 1	§ 8.5	HQ11
	56	Probability 2	§ 8.5	
	57	Differential Equations	§ 9.1	HQ11
	58	Probability Lab (Lab 8)		
	59	Project Comp Day		WPR 4
	60	Integrating Factor	§ 9.5	
	61	Integration Application Mastery		WPR 4
	62	WPR 4 (Integration Applications)		
	63	Project Briefs		
	64	Differentiation Review		

4 Assessment or Evaluation

This section should take what the experiments were in section 3 and discuss results. This is where a few charts and figures would be handy/nice

Possible topics of interest...and build out the framework so all you have to do at the end of the semester is fill-in results.

- Learning assessment schedule
- Pedagogical techniques
- Student-centered activities
- Instructor-centered activities
- Cross-discipline collaborative efforts

5 Conclusions and Future Work

From our experiments we can conclude that the following goals are being achieved or not...

Given these results and the same planning considerations for future courses, we would recommend the following...sustains/improves. I'm sure you will get plenty of guidance form your APD and PD, but

in general, I'd like to give you my Top 10 thoughts on being a CD:

1. You represent the department in all you do, so please consider that in how you respond to 'situations', but if ever a doubt, be gracious in all responses.
2. Pursue your passions in leading this course. If we have a system or requirement that inhibits this, let's talk about it...I need you operating off of passion-fuel.
3. Your rating reflects your ability to lead, think, collaborate, and contribute to the team...if we design our courses the right way, we won't have to make a lot of changes...how you lead the course and take care of our faculty and cadets is the most important thing to me. Don't make changes to make changes, make changes because they are needed and we have the data to prove it (everything we do is for a reason).
4. If you don't need a PD meeting, don't meet...PD meetings should be like a class lesson (inclusive learning environment): motivation, discussion, active learning, questions, pre-teach of next topic...treat them as such. Faculty should be eager to attend ...if you have a cool topic invite the department (please!). Consider thoughtful invitations...
5. We don't build courses as we teach them (hence my reason for asking for this document); think critically about what learning we want to occur in the classroom and try to achieve that with the resources you have. If you need more resources, then let's talk...this is the motivation of the experimental design format.
6. Budget for faculty time: meetings, preparation, assessment development, grading, etc...don't go over budget. Think critically about how to save them time: assessments, rubrics, board sheet examples, etc...please plan for 1 or 2 faculty development team building events (low overhead) to take a day or afternoon off and enjoy.
7. Budget for cadet time: daily assignments, assessments, projects, technology requirements, etc...don't go over budget. Think critically about who you are teaching to and what resources they need: AI, tutoring, JIT, study skills, etc. My recommendation would be to integrate with CEP and BDE Academic Staff early and get them on board with your ideas...
8. Be thoughtful on how to inspire cadets to be truthful about their learning and detailing what work is their own. Embrace generative AI technologies and seek ways to help us do our jobs better.
9. When I meet with you on your course, I will ask a lot of questions...don't read into this, but if you have a question or my guidance is unclear, then just say that and ask please. Our dialogue is important to me.
10. I have your back if you make a 'human' decision for the well-being of a cadet or faculty member, do what you think is best and if you need help, ask. Normally the first report is always wrong (so investigate a little)...and bad news doesn't get better with time (elevate to APD/PD quickly). Let me know how I can help.

If there's something else to add (at the expense of something above) let me know.

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MA104: Single Variable Calculus
AY24-02 Syllabus

Textbook: Calculus - Ninth Edition, Early Transcendentals by James Stewart				
Students:				
Lesson	Date	Topic (Reading)	Notes	Lesson Objectives
BLOCK 1: DERIVATIVES AND THEIR APPLICATIONS				
Functions & Mathematical Models				
Lesson 1	1/8/2024	Course Intro		Topic: Overview of the Course
Lesson 2	1/9/2024	Algebra and Trig Review	(\$ Preface xxvi-xxx)	1. Understand basic trigonometry 2. Understand basic algebra
Lesson 3	1/10/2024	Functions Review (§ 1.1-1.5)		1. Understand the basic characteristics of a function. 2. Determine the domain and range of a function. 3. Determine when a function is increasing and decreasing. 4. Understand the shapes and properties of common functions (Linear, Polynomial, Power, Rational, Algebraic, Trigonometric, Exponential, and Logarithmic). 5. Recognize and construct composite functions.
Limits & Continuity				
Lesson 4	1/11/2024	Limits/Continuity 1 (§ 2.1-2.2)		1. Understand the definition of a tangent and a secant line. 2. Determine average rates of change. 3. Estimate instantaneous rates of change. 4. Find or estimate the equation of a tangent line to a curve. 5. Understand the definition of a limit. 6. Find the limit of a function graphically and numerically.
Lesson 5	1/16/2024	Limits/Continuity 2	(\$ § 2.3 and § 2.6: pg. 127-133)	1. Understand how to apply the properties of limits. 2. Find the limit of a function algebraically. 3. Understand and evaluate limits at infinity.
Lesson 6	1/17/2024	Limits/Continuity 3 (§ 2.5)		1. Understand the definition of continuity. 2. Identify where and why a function is discontinuous. 3. Understand the Intermediate Value Theorem. 4. Recognize the types of functions that are continuous at every number in their domain.
Lesson 7	1/18/2024	Limits Mastery		
Rates of Change				
Lesson 8	1/19/2024	Rates of Change (§ 2.7-2.8)		1. Understand the definition of a tangent line. 2. Understand and apply the limit definition of a derivative. 3. Understand the definition of velocity and how it relates to derivatives. 4. Recognize and apply the different notation used to represent a derivative. 5. Understand how a function can fail to be differentiable. 6. Recognize and understand the relationship between differentiation and continuity.
Lesson 9	1/22/2024	Limits and Rates of Change (Lab 1)		
Lesson 10	1/23/2024	Rates of Change Mastery		
Lesson 11	1/24/2024	Writ (Precalculus)	In Class	
Derivatives				
Lesson 12	1/29/2024	Derivatives of Polyn. and Exp. (§ 3.1)	Tech Tutorial Due	1. Recognize and apply the following derivative rules: a) Derivative of a constant function, b) Power rule, c) Constant multiple rules, d) Sum rule, e) Difference rule, f) Derivative of the natural exponential function.
Lesson 13	1/30/2024	Product and Quotient Rules (§ 3.2)		1. Recognize and apply the following derivative rules: a) Product rule, b) Quotient rule
Lesson 14	1/31/2024	Product and Quotient Rule Mastery		
Lesson 15	2/1/2024	Derivatives of Trig. (§ 3.3)		1. Recognize and apply the derivative rules of trigonometric functions. 2. Understand the relationship between the derivatives of sine and cosine functions.
Lesson 16	2/5/2024	Chain Rule (§ 3.4)		1. Recognize composite functions. 2. Understand when to apply the chain rule. 3. Solve derivatives that involve using the chain rule. 4. Use the chain rule to differentiate an exponential function with a base "b".
Lesson 17	2/6/2024	Chain Rule Mastery		
Implicit Differentiation				
Lesson 18	2/7/2024	Implicit 1 (Implicit Differentiation) (§ 3.5)		1. Recognize when a function is defined implicitly. 2. Understand how to apply implicit differentiation. 3. Solve for the equation of a tangent line at a point on any implicitly defined curve.
Lesson 19	2/8/2024	Implicit 2 (Log and Inverse Trig) (§ 3.6)		1. Understand how implicit differentiation is used to determine the derivative of logarithmic and inverse trig functions. 2. Solve for the derivative of logarithmic and inverse trig functions. 3. Take the derivative of an implicit function with respect to time.
Lesson 20	2/12/2024	Derivatives Mastery (§ 3.1-3.6)		
Lesson 21	2/14/2024	WPR 1 (Derivatives)	Deans Hour	
Maximums & Minimums				
Lesson 22	2/16/2024	Max/Min 1 (§ 4.1)		1. Understand the difference between absolute and local extrema. 2. Apply the Extreme Value Theorem. 3. Understand how Fermat's Theorem relates to critical numbers and the cautions associated with using Fermat's Theorem. 4. Use critical numbers as part of the Closed Interval Method to determine absolute maximum/minimum values.
Lesson 23	2/20/2024	Max/Min 2 (§ 4.3)		1. Use a function's derivative to identify increasing or decreasing intervals. 2. Identify and solve for intervals where a curve is concave up or concave down. 3. Understand and identify inflection points. 4. Understand and apply the second derivative test to solve for local maximum and minimum values.
Lesson 24	2/21/2024	Max/Min 3 (§ 4.3)		1. Use information for first and second derivatives to sketch the curve of a function. 2. Classify local extrema using first and second derivatives.

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Lesson	Date	Topic (Reading)	Notes	Lesson Objectives
Derivative Applications: Optimization and L'Hospital's Rule				
Lesson 25	2/22/2024	Optimization 1 (§ 4.7)		1. Understand how Stewart's optimization problem solving strategy relates to the USMA Mathematical Problem-Solving Strategy. 2. Understand and apply the first derivative test for absolute extreme values. 3. Given a scenario with objectives and constraints, solve for the maximum or minimum values.
Lesson 26	2/23/2024	Optimization 2 (§ 4.7)		1. Understand how Stewart's optimization problem solving strategy relates to the USMA Mathematical Problem-Solving Strategy. 2. Understand and apply the first derivative test for absolute extreme values. 3. Given a scenario with objectives and constraints, solve for the maximum or minimum values.
Lesson 27	2/26/2024	Optimization (Lab 2)		
Lesson 28	2/27/2024	Optimization 3 (§ 4.7)		1. Given a scenario with objectives and constraints, solve for the maximum or minimum values.
Lesson 29	2/28/2024	L'Hospitals Rule (§ 4.4 pg 309-313)	Lab 2 Report Due	1. Apply L'Hospitals Rule to solve indeterminate limits
Lesson 30	2/29/2024	Newton's Method Lab (Lab 3)	Lab 3 Mathematica Due	
Lesson 31	3/4/2024	Derivative Application Mastery 1		
Lesson 32	3/5/2024	Derivative Application Mastery 2		
Lesson 33	3/6/2024	WPR 2 (Derivative Applications)	In Class	Topics: L'Hospitals Rule, Max/Min, & Optimization
BLOCK 2: INTEGRALS AND THEIR APPLICATIONS				
Antiderivatives				
Lesson 34	3/8/2024	Antiderivatives (§ 4.9)		1. Understand the definition of an antiderivative. 2. Solve for an antiderivative of a functions. 3. Understand antiderivatives as a family of functions.
Lesson 35	3/11/2024	Natural Sciences 1 (Lab 4)		1. Understand the relationship between antiderivatives and projectile motion.
Lesson 36	3/12/2024	Natural Sciences 2 (Lab 5)		1. Understand the relationship between antiderivatives and projectile motion.
Definite Integrals				
Lesson 37	3/13/2024	Definite Integral 1 (§ 5.1)		1. Understand how to estimate the area between a curve and the horizontal axis on an interval. 2. Solve area estimates using sigma notation. 3. Understand how to solve areas using the limit operator in conjunction with sigma notation. 4. Interpret distances traveled by an object as an area problem.
Lesson 38	3/14/2024	Definite Integral 2 (§ 5.2)		1. Calculate definite integrals using Definition 2 and Theorem 4 in Section 5.2. 2. Express definite integrals as a limit of a Riemann Sum and express a limit of a Riemann Sum as a definite integral. 3. Understand the mathematical notation and terminology of integrals. 4. Apply the Left or Right Endpoint Method to estimate an integral. 5. Understand and apply the properties of definite integrals.
Lesson 39	3/18/2024	Definite Integral 3 (§ 5.3)	Lab 5 Report Due	1. Understand and apply the Fundamental Theorem of Calculus, Part 1. 2. Understand and apply the Fundamental Theorem of Calculus, Part 2. 3. Internalize differentiation and integration as inverses processes.
Integration				
Lesson 40	3/19/2024	Indefinite Integrals (§ 5.4)		1. Distinguish between indefinite and definite integrals. 2. Determine indefinite integrals. 3. Understand and apply the Net Change Theorem.
Lesson 41	3/20/2024	Basic Integration Mastery		
Lesson 42	3/21/2024	Substitution Rule (§ 5.5)		1. Understand when to apply the substitution rule. 2. Use the substitution rule to solve definite and indefinite integrals.
Lesson 43	4/2/2024	Substitution Rule Mastery (§ 5.5)		
Lesson 44	4/3/2024	Integration by Parts (§ 7.1)		1. Understand how the method of integration by parts is derived from the product rule of differentiation. 2. Understand how to select "f" and "g" when using integration by parts. 3. Apply integration by parts to solve definite and indefinite integrals.
Lesson 45	4/4/2024	Integration by Parts Mastery (§ 7.1)		
Lesson 46	4/5/2024	Riemann Sum Lab (Lab 6)		1. Leverage Riemann sums to solve integrals
Lesson 47	4/8/2024	Integration Strategies 1 (§ 7.5)		1. Understand how to approach a definite and indefinite integral in order to determine the appropriate integration method(s). 2. Solve any integral by choosing the correct integration technique (simplifying w/ basic rules, u-sub, IBP)
Lesson 48	4/9/2024	Integration Strategies 2 (§ 7.5)	Lab 6 Report Due	1. Understand how to approach a definite and indefinite integral in order to determine the appropriate integration method(s). 2. Solve any integral by choosing the correct integration technique (simplifying w/ basic rules, u-sub, IBP)
Lesson 49	4/10/2024	Integration Strategies Mastery (§ 7.5)		
Lesson 50	4/12/2024	WPR 3 (Integration)	Deans Hour	Topics: Basic Integration, Substitution Rule, Integration by Parts

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Lesson	Date	Topic (Reading)	Notes	Lesson Objectives
Applications of Integration				
Lesson 51	4/16/2024	Areas Between Curves (§ 6.1)		1. Determine the area between two curves. 2. Interpret the meaning of the area between two curves given in context.
Lesson 52	4/17/2024	Improper Integrals (§ 7.8)		1. Identify improper integrals as either Type 1 or Type 2 improper integrals. 2. Identify when a Type 1 improper integral is either convergent or divergent. 3. Solve Type 1 improper integrals.
Lesson 53	4/18/2024	Improper Integrals Mastery (§ 7.8)		
Lesson 54	4/19/2024	Project Development (Lab 7)		
Lesson 55	4/22/2024	Probability 1 (Hand Out and § 8.5)		1. Basic understanding of probability. 2. Understand the definition of a probability density function. 3. Recognize when a probability density function is invalid. 4. Interpret a probability statement. 5. Solve for a probability using a probability density function. 6. Determine the average value using a probability density function.
Lesson 56	4/23/2024	Probability 2 (§ 8.5)		1. Understand the definition of a probability density function. 2. Recognize when a probability density function is invalid. 3. Interpret a probability statement. 4. Solve for a probability using a probability density function. 5. Determine the average value using a probability density function.
Lesson 57	4/24/2024	Differential Equations (§ 9.1 pg. 608-610)		1. Understand the meaning of ordinary differential equations. 2. Understand that the solution of an ordinary differential equation is an equation. 3. Find the particular solution for a first order linear ODEs given initial conditions.
Lesson 58	4/25/2024	Probability Lab (Lab 8)		
ODEs				
Lesson 59	4/26/2024	Project Comp Day		
Lesson 60	4/29/2024	Integrating Factor (§ 9.5)	Project Due	1. Solve linear differential equations in the form $y' + P(x)y = Q(x)$ using the method of the integrating factor. 2. Apply the method of the integrating factor to solve application based ODEs.
Lesson 61	4/30/2024	Integration Application Mastery		
Lesson 62	5/1/2024	WPR 4 (Integration Applications)	Deans Hour	Topics: Area between curves, Probability, Integrating Factor
Lesson 63A	5/6/2024	Project Briefs		
Lesson 63B	5/7/2024	Project Briefs		
Lesson 63C	5/8/2024	Project Briefs		
Lesson 64	5/9/2024	Differentiation Review		



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB E

AY2025 COURSE ASSESSMENT



DEPARTMENT OF THE ARMY
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

MADN-MTH

30 May 2025

MEMORANDUM THRU

COL Andrew Lee, Director, Operations Research, Department of Mathematical Sciences, USMA, West Point, NY 10996

COL Michael Scioletti, Professor and Head, Department of Mathematical Sciences, USMA, West Point, NY 10996

FOR Office of the Dean, AARS, United States Military Academy, West Point, NY 10996

SUBJECT: MA104 Single Variable Calculus Executive Summary (AY25-02)

1. **Purpose:** The purpose of this memorandum is to assess the MA104 curriculum.

2. **Assessment:** The course has a three-year assessment strategy for MA104. This year's assessment examined the Student Learning Outcomes (SLOs) listed below with our findings for each assessed area (In Table 1, A = exceeds the standard; B = meets the standard; C = meets the standard with some weakness; D = fails the standard (can be addressed in the short-term); F = catastrophically fails the standard (requires immediate remediation)). A list of all SLOs can be found in enclosure 1.

MA104 Student Learning Outcome (SLO)	Attainment	Evidence	Overall
(1) Understand the derivative as a rate of change, the integral as an accumulation of change, and explain how these operations are inverse processes.	A	A	A
(3) Clearly communicate calculus problem-solving strategies and results in verbal and written form using appropriate mathematical language and notation.	B	A	B
(5) Recognize and articulate the ethical implications of applying calculus-based models in decision-making contexts.	B	A	B

Table 1: **SLO Evaluations for MA104 for AY25-02**

3. **Assessment Discussion:** This section presents an overview of assessment findings for AY25. A detailed discussion of direct and indirect indicators can be found in the Course Design and Analysis Report located in Enclosure 4. The schedule for SLO evaluation can be found in Enclosure 3.

a. Status of significant changes: The first week of the course became an algebra review week. This allowed for each cadet to refresh common algebra skills which

would be required throughout the rest of the semester. This week ended with an algebra WRIT which highlighted areas of weakness, and identified the cadets who needed to move to MA100. The course also added a related rates lab back into the course with no prior teaching. This offered the cadets the opportunity to conduct discovery learning on a topic they had not be formally instructed on.

b. Strengths in the course: The course introduced ethical considerations to each lab report and the final project. Cadets had to assess their work, as well as the implications of using their models to make decisions within the scenario. For example, in the natural sciences lab, cadets had to calculate artillery trajectories to hit enemy targets. Since they built the model, they had to assess whether or not their calculations were good enough to prevent collateral damage. Then they fired their projectiles and either gained or lost points based on their accuracy. This repeated requirement allowed the course to assess SLO 5. Also, the course continued to use the “Big Idea” question on each major graded event to assess cadet understanding of the concepts behind the numbers. This ensures that we can assess understanding as well as a cadets ability to memorize steps to solve a particular problem.

c. Areas for improvement in the course.

(1) First, the daily concept questions are supposed to incentivize class preparation, and are directly from the Stewart text book. Through discussions with cadets and instructors, it does not appear this is meeting its intended purpose. Cadets who prepare are not always rewarded with a good grade, and many cadets use reasonable deduction to get the correct answer without having prepared. Future course iterations should explore alternative methods to encourage student lesson preparation.

(2) The course is currently required to teach a specific method to solve a specific instance of differential equations, called “integrating factor.” This lesson must be taught at the end of the course but requires the introduction of differential equations and then skipping three chapters of solving differential equations with almost no motivation as to why we are covering just this type of differential equation. The students who will use these two lessons are a minority and wont take a relevant course until they are firsties. We offer two courses of action and recommend course of action 1.

(a) Course of Action 1. Completely remove these two lessons and add in an application of integration that was removed to make room. Work is a directly applicable use of integration that leads into the cadets' physics and engineering courses. This is the easier course of action to implement but negatively impacts a select group of cadets who will use integrating factor in future courses.. We could mitigate this risk by providing supplementary materials on differential equations for the few cadets who will need this topic yet do not take MA 365.

(b) Course of Action 2. Make a significant change to the course where all applications are removed and replaced with a 2 week block of differential equations at the end of the course which includes several types and methods of solving differential equations. The risk to this course of action is losing even more application lessons typically taught with this level of calculus, including natural sciences, optimization, and

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SUBJECT: MA104 Single Variable Calculus Executive Summary (AY25-02)

probability. This risk is partially mitigated with differential equations being the real framework for natural sciences anyways. The risk could further be mitigated by the calculus behind calculating probabilities using probability density functions being shown in required statistics classes.

d. Changes to be implemented in AY25: Daily quizzes will be replaced by a daily submission requirement. The quizzes are failing to provide incentive for the cadets to prepare for class. Therefore, the daily preparation will include students interacting with generative AI to enhance their independent learning skills.

e. Proposed changes to the SLOs: SLO 5 was added AY 25-01.

4. The point of contact for this memorandum is MAJ Bradford Witt, MA104 Course Director, at 845-938-2654 or bradford.l.witt.mil@army.mil.

4 Encls

1. MA104 SLOs
2. Assessment Crosswalk
3. SLO Assessment Plan
4. Course Design Assessment Report

RUSSELL NELSON
LTC, FA49/AV
Associate Program Director

MA104: Single Variable Calculus
AY25-02 Syllabus

Textbook: Calculus - Ninth Edition, Early Transcendentals by James Stewart

Students:

Lesson	Date	Topic (Reading)	Notes	Lesson Objectives
BLOCK 1: DERIVATIVES AND THEIR APPLICATIONS				
Functions & Mathematical Models				
Lesson 1	8/18/2025	Course Intro		Topic: Overview of the Course
Lesson 2	8/19/2025	Critical Skills (1-3)	Algebra Supplement	1. Execute arithmetic operations with expressions containing exponents and nth roots. 2. Simplify an n-order equation by distribution, factoring, using the quadratic formula. 3. Add, subtract, multiply, divide, and solve algebraic expressions.
Lesson 3	8/20/2025	Critical Skills(4-6)	Algebra Supplement and § 1.1-1.5	1. Determine the equation of a line algebraically and graphically using the pointslope and slope-intercept form. 2. Determine the domain and range of a function. 3. Demonstrate the relationship between exponential and logarithmic functions algebraically and graphically. 4. Recognize and construct composite functions.
Lesson 4	8/21/2025	Critical Skills (7-8)	Algebra Supplement	1. Define the six trig functions using a right triangle and on a coordinate plane. 2. Use an inverse trig function to solve for an angle. 3. Use various methods to solve systems of equations.
Lesson 5	8/22/2025	WRIT 0 (Pre-Calculus)	In Class	1. Critical Skills 1-8
Limits & Continuity				
Lesson 6	8/25/2025	Limits/Continuity 1 (§ 2.1 - 2.2)		1. Understand the definition of a tangent and a secant line. 2. Determine average rates of change. 3. Estimate instantaneous rates of change. 4. Find or estimate the equation of a tangent line to a curve. 5. Understand the definition of a limit. 6. Find the limit of a function graphically and numerically.
Lesson 7	8/26/2025	Limits/Continuity 2 (§ 2.3 and 2.6: Pg 127-133)		1. Understand how to apply the properties of limits 2. Find the limit of a function algebraically 3. Understand and evaluate limits at infinity
Lesson 8	8/27/2025	Limits/Continuity 3 (§ 2.5)		1. Understand the definition of continuity 2. Identify where and why a function is discontinuous 3. Understand the Intermediate Value Theorem 4. Recognize the types of functions that are continuous at every number in their domain
Lesson 9	8/28/2025	Limits Mastery		
Lesson 10	8/29/2025	Rates of Change (§ 2.7-2.8)		1. Understand the definition of a tangent line 2. Understand and apply the limit definition of a derivative 3. Understand the definition of velocity and how it relates to derivatives 4. Recognize and apply the different notation used to represent a derivative 5. Understand how a function can fail to be differentiable 6. Recognize and understand the relationship between differentiation and continuity
Lesson 11	9/2/2025	Rates of Change Lab (Lab 1)		
Lesson 12	9/3/2025	Rates of Change Mastery	Lab 1 Report Due	
Derivatives				
Lesson 13	9/5/2025	Derivatives of Polyn. and Exp. (§ 3.1)		1. Recognize and apply the following derivative rules: a. Derivative of a constant function b. Power rule c. Constant multiple rule d. Sum rule e. Difference rule f. Derivative of the natural exponential function
Lesson 14	9/8/2025	Product and Quotient Rules (§ 3.2)		1. Recognize and apply the following derivative rules: a. Product rule b. Quotient rule
Lesson 15	9/9/2025	Product and Quotient Rule Mastery		
Lesson 16	9/10/2025	Derivatives of Trig. (§ 3.3)		1. Recognize and apply the derivative rules of trigonometric functions 2. Understand the relationship between the derivatives of sine and cosine functions
Lesson 17	9/11/2025	Chain Rule (§ 3.4)		1. Recognize composite functions 2. Understand when to apply the chain rule 3. Solve derivatives that involve using the chain rule 4. Use the chain rule to differentiate an exponential function with a base "b"
Lesson 18	9/12/2025	Chain Rule Mastery	Tech Tutorial Due	
Implicit Differentiation				
Lesson 19	9/15/2025	Implicit 1 (Implicit Differentiation) (§ 3.5)		1. Recognize when a function is defined implicitly 2. Understand how to apply implicit differentiation 3. Solve for the equation of a tangent line at a point on any implicitly defined curve
Lesson 20	9/16/2025	Implicit 2 (Log and Inverse Trig) (§ 3.6)		1. Understand how implicit differentiation is used to determine the derivative of logarithmic and inverse trig functions 2. Solve for the derivative of logarithmic and inverse trig functions 3. Take the derivative of an implicit function with respect to time
Lesson 21	9/17/2025	Related Rates Lab (Lab 2)		
Lesson 22	9/18/2025	Derivatives Mastery (§ 3.1-3.6)		
Lesson 23	9/19/2025	WPR 1 (Derivatives)	In Class	

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Lesson	Date	Topic (Reading)	Notes	Lesson Objectives
Maximums & Minimums				
Lesson 24	9/23/2025	Max/Min 1 (§ 4.1)		1. Understand the difference between absolute and local extrema 2. Apply the Extreme Value Theorem 3. Understand how Fermat's Theorem relates to critical numbers and the cautions associated with using Fermat's Theorem 4. Use critical numbers as part of the Closed Interval Method to determine absolute maximum/minimum values
Lesson 25	9/24/2025	Max/Min 2 (§ 4.3)		1. Use a function's derivative to identify increasing or decreasing intervals 2. Identify and solve for intervals where a curve is concave up or concave down 3. Understand and identify inflection points 4. Understand and apply the second derivative test to solve for local maximum and minimum values
Lesson 26	9/25/2025	Max/Min 3 (§ 4.3)		1. Use information from first and second derivatives to sketch the curve of a function 2. Classify local extrema using first and second derivatives
Derivative Applications: Optimization and L'Hospital's Rule				
Lesson 27	9/26/2025	Optimization 1 (§ 4.7)		1. Understand how Stewart's optimization problem solving strategy relates to the USMA Mathematical Problem-Solving Strategy 2. Understand and apply the first derivative test for absolute extreme values 3. Given a scenario with objectives and constraints, solve for the maximum or minimum values
Lesson 28	9/29/2025	Optimization 2 (§ 4.7)		1. Understand how Stewart's optimization problem solving strategy relates to the USMA Mathematical Problem-Solving Strategy 2. Understand and apply the first derivative test for absolute extreme values 3. Given a scenario with objectives and constraints, solve for the maximum or minimum values
Lesson 29	9/30/2025	Optimization (Lab 3)		
Lesson 30	10/1/2025	Optimization 3 (§ 4.7)	Lab 3 Report Due	1. Given a scenario with objectives and constraints, solve for the maximum or minimum values.
Lesson 31	10/2/2025	L'Hospitals Rule (§ 4.4 pg 309-313)		1. Apply L'Hospitals Rule to solve indeterminate limits
Lesson 32	10/3/2025	Derivative Application Mastery 1		
Lesson 33	10/6/2025	Derivative Application Mastery 2		Topics: L'Hospitals Rule, Max/Min, & Optimization
Lesson 34	10/7/2025	WPR 2 (Derivative Applications)	In Class	
BLOCK 2: INTEGRALS AND THEIR APPLICATIONS				
Lesson 35	10/9/2025	Antiderivatives (§ 4.9)		1. Understand the definition of an antiderivative 2. Recognize basic antiderivative rules 3. Solve basic antiderivative problems
Integration				
Lesson 36	10/10/2025	Definite Integral 1 (§ 5.1)		1. Understand the definition of a definite integral 2. Interpret the definite integral as the net area under a curve
Lesson 37	10/14/2025	Definite Integral 2 (§ 5.2)		1. Calculate definite integrals using Definition 2 and Theorem 4 in Section 5.2. 2. Express definite integrals as a limit of a Riemann Sum and express a limit of a Riemann Sum as a definite integral. 3. Understand the mathematical notation and terminology of integrals. 4. Apply the Left or Right Endpoint Method to estimate an integral. 5. Understand and apply the properties of definite integrals.
Lesson 38	10/15/2025	Definite Integral 3 (§ 5.3)		1. Understand and apply the Fundamental Theorem of Calculus, Part 1. 2. Understand and apply the Fundamental Theorem of Calculus, Part 2. 3. Internalize differentiation and integration as inverse processes.
Lesson 39	10/16/2025	Riemann Sum Lab (Lab 4)		
Lesson 40	10/17/2025	Indefinite Integrals (§ 5.4)	Lab 4 Report Due	1. Distinguish between indefinite and definite integrals. 2. Determine indefinite integrals. 3. Understand and apply the Net Change Theorem.
Lesson 41	10/20/2025	Basic Integration Mastery		
Lesson 42	10/21/2025	Natural Sciences (Lab 5)		
Lesson 43	10/22/2025	Natural Sciences 2 (Lab 6)		
Lesson 44	10/23/2025	Substitution Rule (§ 5.5)		1. Understand when to apply the substitution rule. 2. Use the substitution rule to solve definite and indefinite integrals.
Lesson 45	10/24/2025	Substitution Rule Mastery (§ 5.5)	Lab 5-6 Report Due	
Lesson 46	10/27/2025	Integration by Parts (§ 7.1)		1. Understand how the method of integration by parts is derived from the product rule of differentiation. 2. Understand how to select "f" and "g" when using integration by parts. 3. Apply integration by parts to solve definite and indefinite integrals.
Lesson 47	10/28/2025	Integration by Parts Mastery (§ 7.1)		
Lesson 48	10/29/2025	Integration Strategies 1 (§ 7.5)		1. Understand how to approach a definite and indefinite integral in order to determine the appropriate integration method(s). 2. Solve any integral by choosing the correct integration technique (simplifying w/ basic rules, u-sub, IBP)
Lesson 49	10/30/2025	Integration Strategies 2 (§ 7.5)		1. Understand how to approach a definite and indefinite integral in order to determine the appropriate integration method(s). 2. Solve any integral by choosing the correct integration technique (simplifying w/ basic rules, u-sub, IBP)
Lesson 50	10/31/2025	Integration Strategies Mastery (§ 7.5)		
Lesson 51	11/3/2025	WPR 3 (Integration)	In Class	
Lesson 52	11/5/2025	Project Development (Lab 7)		
Applications of Integration				
Lesson 53	11/6/2025	Areas Between Curves (§ 6.1)		1. Determine the area between two curves. 2. Interpret the meaning of the area between two curves given in context.
Lesson 54	11/7/2025	Improper Integrals (§ 7.8)		1. Identify improper integrals as either Type 1 or Type 2 improper integrals. 2. Identify when a Type 1 improper integral is either convergent or divergent. 3. Solve Type 1 improper integrals.
Lesson 55	11/10/2025	Improper Integrals Mastery (§ 7.8)		

MA104: Single Variable Calculus
AY25-02 Syllabus

Lesson	Date	Topic (Reading)	Notes	Lesson Objectives
Lesson 56	11/12/2025	Probability 1 (Hand Out and § 8.5)		1. Basic understanding of probability. 2. Understand the definition of a probability density function. 3. Recognize when a probability density function is invalid. 4. Interpret a probability statement. 5. Solve for a probability using a probability density function. 6. Determine the average value using a probability density function.
Lesson 57	11/13/2025	Probability 2 (§ 8.5)		1. Understand the definition of a probability density function. 2. Recognize when a probability density function is invalid. 3. Interpret a probability statement. 4. Solve for a probability using a probability density function. 5. Determine the average value using a probability density function.
Lesson 58	11/14/2025	Probability Lab (Lab 8)		
ODEs				
Lesson 59	11/17/2025	Differential Equations (§ 9.1 pg. 608-610)		1. Understand the meaning of ordinary differential equations. 2. Understand that the solution of an ordinary differential equation is an equation. 3. Find the particular solution for a first order linear ODEs given initial conditions
Lesson 60	11/18/2025	Integrating Factor (§ 9.5)	Project Due	1. Solve linear differential equations in the form $y' + P(x)y = Q(x)$ using the method of the integrating factor. 2. Apply the method of the integrating factor to solve application based ODEs.
Lesson 61	11/19/2025	Integration Application Mastery		
Lesson 62	11/20/2025	WPR 4 (Integration Applications)	In Class	
Lesson 63A	11/24/2025	Project Briefs		
Lesson 63B	11/25/2025	Project Briefs		
Lesson 63C	11/26/2025	Project Briefs		
Lesson 64	11/28/2025	Differentiation Review		1. Review differentiation key concepts



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB F

AY2026 COURSE ASSESSMENT



DEPARTMENT OF THE ARMY
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

MADN-MTH

25 May 2026

MEMORANDUM THRU

Dr. Rachelle DeCoste, Director, Mathematical Sciences, Department of Mathematical Sciences, USMA, West Point, NY 10996
COL Michael Scioletti, Professor and Head, Department of Mathematical Sciences, USMA, West Point, NY 10996

FOR Office of the Dean, AARS, United States Military Academy, West Point, NY 10996

SUBJECT: MA104 Single Variable Calculus Executive Summary (AY26-02)

1. **Purpose:** The purpose of this memorandum is to assess the MA104 curriculum for Academic Year 2026.

2. **Assessment:** The course has a three-year assessment strategy. This year's assessment focus was Lifelong Learning (APG 3) and examined Student Learning Outcomes (SLOs) 2, 3, and 4. Findings for each assessed area are summarized in Table 1. An SLO is accomplished satisfactorily if the proportion of cadets above the threshold exceeds 80%, marginally if between 65 and 80%, and unsatisfactorily if below 65%. A full list of SLOs is provided in Enclosure 1.

MA104 Student Learning Outcome (SLO)	Assessment
(2) Apply differential and integral calculus techniques to interpret and solve real-world problems.	Marginal
(3) Develop verbal and written communication skills to effectively convey calculus problem-solving strategies using appropriate mathematical language and notation.	Satisfactory
(4) Demonstrate effective study habits and self-directed learning by using feedback to foster independent improvement and deeper understanding.	Satisfactory

Table 1: **SLO Evaluations for MA104, AY26-02**

3. **Assessment Discussion:** This section presents an overview of assessment findings for AY26. A detailed discussion of direct and indirect indicators can be found in the Course Design and Analysis Report located in Enclosure 4. The schedule for SLO evaluation can be found in Enclosure 3.

a. *Status of significant changes.* AY26 introduced several meaningful course modifications. Two additional mastery days, *Derivatives Mastery* and *Integration Applications Mastery 3*, were added to target historically difficult topics and provide structured low-stakes practice. Related Rates was transitioned from a self-guided lab to

a dedicated lesson, reinforcing its connection to implicit differentiation. Three integration applications that had been previously removed, Volume, Work, and Average Value of a Function, were restored to better prepare cadets for subsequent STEM coursework. Conversely, the algebra review was reduced from one week to a single lesson, as prior data showed it had little measurable impact on overall performance.

b. *Strengths in the course.* Working and briefing board problems continued to be the course's most consistently praised instructional element, cited by the majority of cadets in the end-of-course survey. Mastery days were similarly highlighted as a high-value sustain across all sections. Assessment data for SLO 3 showed clear improvement across the three formal communications assessments: the class mean rose from 84.2% on the Optimization Lab to 87.5% on the Natural Sciences Lab and 91.6% on the Final Project, while the standard deviation narrowed from 17.6% to 6.3%, indicating that the floor rose rather than just the mean. No cadet scored below 70% on the Final Project. A paired t -test confirmed the improvement between the two labs was statistically significant ($t = 3.282, p = 0.0011$). For SLO 4, end-of-course survey results ($n = 220$) showed that 91.4% of cadets met the threshold on Q7 ("My study habits have improved since the beginning of MA104") and 86.4% met the threshold on Q6 ("My experience in MA104 has increased my confidence in using Generative AI to generate practice problems"), both exceeding the 80% satisfactory benchmark. For SLO 2, two of six direct indicators were satisfactory, with the Optimization Lab (88.8%) and WRIT 10 (84.0%) both meeting the standard.

c. *Areas for improvement in the course.*

(1) First, SLO 2 performance on integration applications represents the largest area in need of improvement. WPR 4's integration applications section (Work, Average Value, and Volume, 45 points) produced a mean of 72.6% with only 66.1% of cadets above 70%, the weakest result of any direct indicator. The sub-question analysis revealed that Work and Average Value were the primary drivers, with means of 67.5% and 68.4% respectively. Cadets consistently demonstrated the ability to identify the correct method but lost points to algebra errors throughout the solution process that compounded into calculus errors in the application itself. This pattern persisted on the TEE, where across all six SLO 2 direct indicators combined, 71.9% of cadets met the 70% threshold, placing the overall SLO 2 result at marginal. Additional structured practice on integration applications, particularly Work and Average Value, is recommended for AY27.

(2) Second, while the Cengage homework structure was generally functional, the first question of each assignment, which required cadets to use generative AI, produced mixed results. Cadets broadly appreciated the opportunity to engage with AI as a study tool, but several noted that AI explanations for mathematical concepts were occasionally incorrect or misleading. More explicit instruction on how to critically evaluate and effectively use generative AI for conceptual learning is recommended for AY27.

(3) Third, the alignment between WRITs, mastery days, and WPRs was identified as inconsistent. Multiple cadets reported performing well on WRITs yet struggling on WPRs, and noted that mastery problems were sometimes harder than what

appeared on graded events. Improving this alignment would better calibrate cadet expectations and reduce the gap between formative and summative assessment performance.

(4) Fourth, weekly Forge meetings should be restructured to place greater emphasis on conceptual preparation for upcoming lessons. Of the topics currently covered in Forge, rubric preparation, assessment review, and grading philosophy should be sustained. In addition, the course should provide a standardized LaTeX template for each lesson's "Big Idea," giving instructors a shared starting point for explaining the conceptual foundation of the material before cadets encounter it procedurally in class. Instructors should be encouraged to adapt the template as they see fit to match their teaching style and section's needs. Instructors should walk through the Big Idea during Forge, ensuring consistent conceptual delivery across all sections and reducing variance in how the underlying mathematics is motivated and explained.

d. *TEE Performance by Exam Day.* The TEE was administered across three sittings: Wednesday (TEE 1), Thursday (TEE 2), and Saturday (TEE 3). Table 2 summarizes TEE performance by day alongside the full-course baseline.

Sitting	N	Mean	Median	Above 70%
Wednesday (TEE 1)	365	79.2%	80.8%	76.4%
Thursday (TEE 2)	97	81.5%	86.2%	76.3%
Saturday (TEE 3)	169	76.4%	77.2%	66.9%
Course Baseline (all cadets)	645	78.7%	80.8%	73.8%

Table 2: TEE performance by exam sitting. A one-way ANOVA found a statistically significant difference between sittings ($F = 5.18$, $p = 0.006$).

Wednesday and Thursday cadets performed at or above the course baseline on all measures, while Saturday cadets scored lower across the board. A one-way ANOVA confirmed the TEE difference is statistically significant ($F = 5.18$, $p = 0.006$). However, a broader analysis of semester grades reveals that Saturday cadets were not simply lower performers on the TEE alone. As shown in Figure 3, Saturday cadets also scored below Wednesday and Thursday cadets on every other grade category throughout the semester: WPRs (79.5% vs. 81.6% and 83.0%), WRITs (81.8% vs. 84.1% and 85.7%), Labs (92.6% vs. 94.8% and 92.7%), and overall course grade (82.8% vs. 85.0% and 85.2%), with statistically significant differences across all categories ($p < 0.05$ in each case). This pattern strongly suggests that the performance gap on the TEE reflects the academic profile of cadets assigned to the Saturday sitting rather than any effect of the exam day itself. Further investigation should be conducted to determine whether certain exam days yield better performance outcomes. If a meaningful difference is identified, every effort should be made to assign struggling cadets to those higher-performing days to give them the best opportunity to succeed on the TEE.

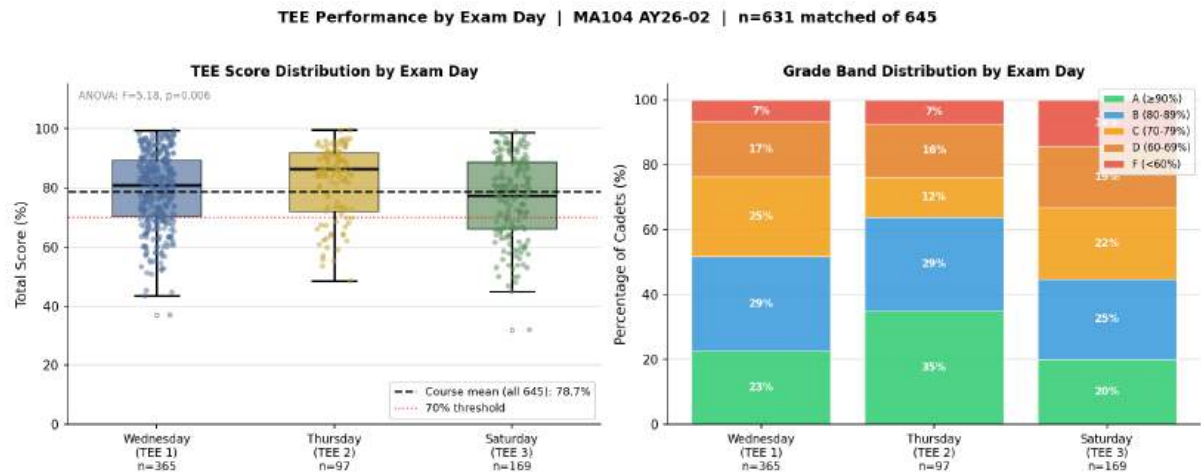


Figure 1: TEE score distribution and grade band breakdown by exam sitting. The dashed line indicates the course mean across all 645 cadets.

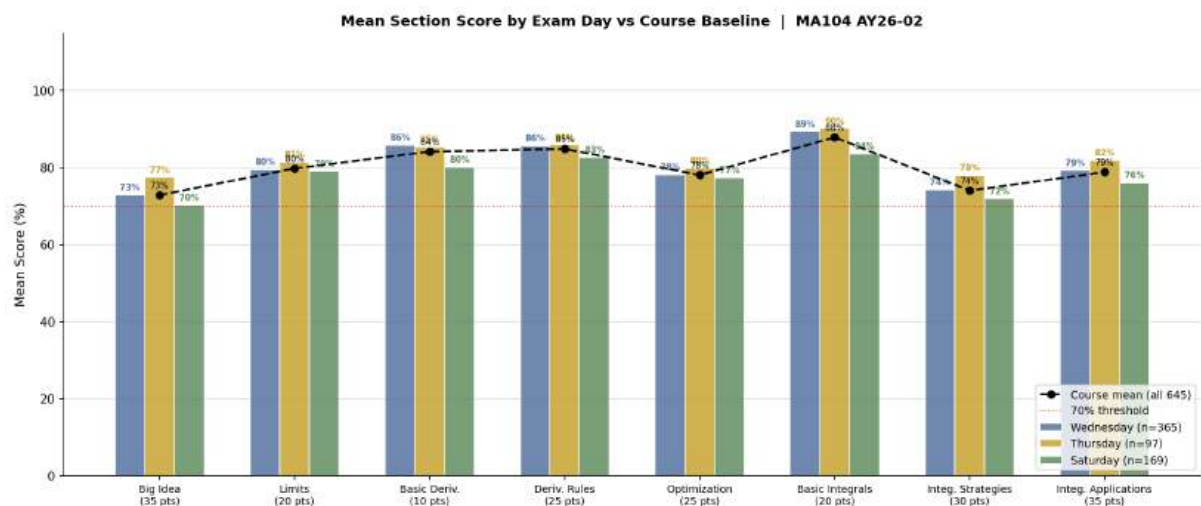


Figure 2: Mean section score by exam sitting compared to the course baseline. Each bar group represents one of the eight TEE sections; the black dashed line shows the course mean for all cadets.

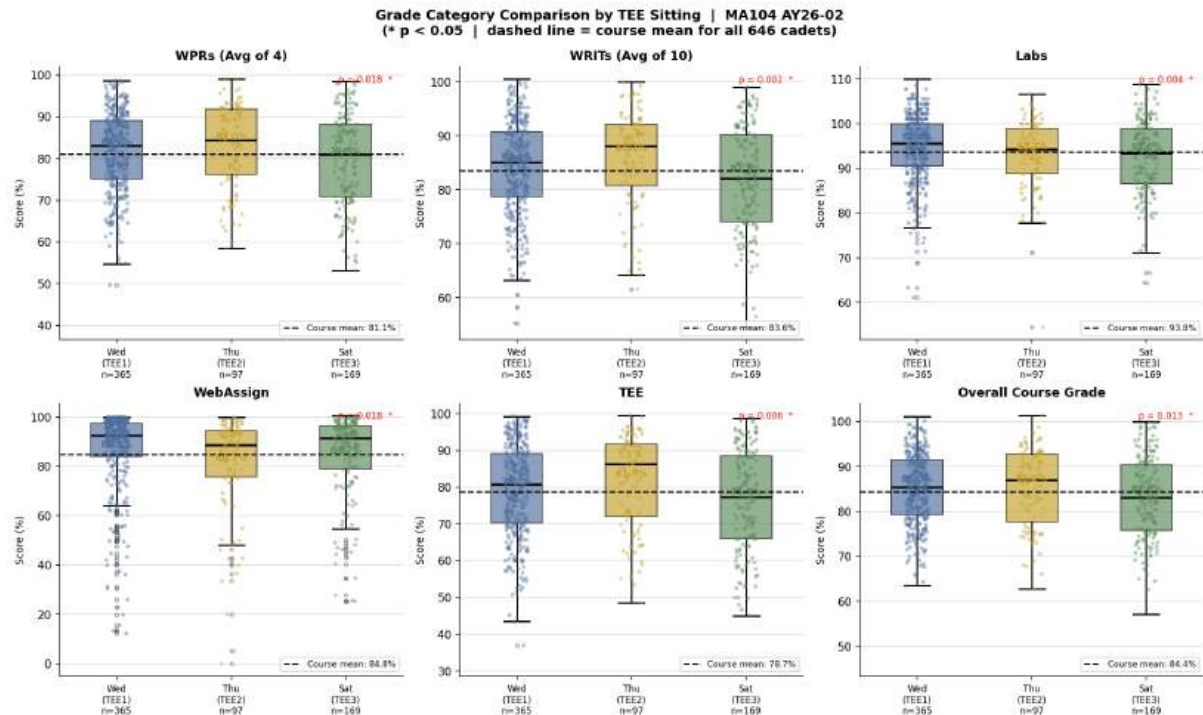


Figure 3: Grade category comparison by exam sitting across WPRs, WRITs, Labs, WebAssign, TEE, and overall course grade. The dashed line indicates the course mean for all cadets. An asterisk (*) denotes a statistically significant difference between sittings ($p < 0.05$).

e. *Proposed changes to the SLOs: None.*

4. The point of contact for this memorandum is MAJ Dylan Hyde, MA104 Course Director, at dylan.hyde@westpoint.edu.

- 4 Encls
1. MA104 SLOs
 2. Assessment Crosswalk
 3. SLO Assessment Plan
 4. Course Design Assessment Report

DYLAN HYDE
MAJ, FA49/LG
Course Director

Enclosure 1: Full Set of Student Learning Outcomes (SLOs)

At the end of the course, cadets will be able to...

- (1) Understand the derivative as a rate of change and the integral as an accumulation of change and explain how these operations are inverse processes.
- (2) Apply differential and integral calculus techniques to interpret and solve real-world problems.
- (3) Develop verbal and written communication skills to effectively convey calculus problem-solving strategies using appropriate mathematical language and notation.
- (4) Demonstrate effective study habits and self-directed learning by using feedback to foster independent improvement and deeper understanding.
- (5) Recognize and clearly articulate the ethical implications of applying calculus-based models in decision-making scenarios.

Anticipated changes to SLOs: None.

Enclosure 2: Mapping of Academic Program Goals to SLOs

Academic Program Goal	What Graduates Can Do Statement	Core Mathematics Goal	MA104 Student Learning Objectives	MA104 Assessments
1. Graduates communicate effectively with all audiences.	1.3 Effectively convey meaningful information to diverse audiences using appropriate forms and media	G4: Students demonstrate an ability to communicate mathematics effectively	3. Develop verbal and written communication skills to effectively convey calculus problem-solving strategies using appropriate mathematical language and notation. 5. Recognize and clearly articulate the ethical implications of applying calculus-based models in decision-making scenarios.	Board Problems, Lab Reports, Final Project Lab Reports
2. Graduates think critically and creatively	2.4 Reason quantitatively and qualitatively.	G5: Students demonstrate knowledge of mathematics	2. Apply differential and integral calculus techniques to interpret and solve real-world problems. 3. Develop verbal and written communication skills to effectively convey calculus problem-solving strategies using appropriate mathematical language and notation.	WPR/TEE Big Idea Questions Labs and Project
3. Graduates demonstrate the capability and desire to pursue progressive and continued intellectual development.	3.1 Demonstrate the willingness and ability to learn independently.	G1: Students display effective habits of mind in their intellectual process	4. Demonstrate effective study habits and self-directed learning by using feedback to foster independent improvement and deeper understanding. 5. Recognize and clearly articulate the ethical implications of applying calculus-based models in decision-making scenarios.	End-of-Course Survey, Daily GenAI Assignments Labs and Project
5. Graduates apply science, technology, engineering, and mathematics concepts to processes to solve complex problems.	5.1 Apply mathematics, science, and computing to model devices, systems, processes, or behaviors.	G3: When appropriate, students apply technology to solve mathematical problems	1. Understand the derivative as a rate of change, the integral as an accumulation of change, and explain how these operations are inverse processes. 2. Apply differential and integral calculus techniques to interpret and solve real-world problems.	WPR/TEE Big Idea Questions, Labs, and Project

Enclosure 3: Cyclical Schedule for Assessment of SLOs

Academic Year	Assessment Focus	
	WGCD	SLO
2025	Ethical Reasoning	1, 3, 5
2026	Lifelong Learner	2, 3, 4
2027	Communication	1, 2, 5

MA104 Single Variable Calculus
AY26-02 Course Design and Assessment

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1 Introduction and Background

College single variable calculus has traditionally followed the AP Calculus format taught to high schoolers across the country. This can be seen in Table 1 below [4]. This divides the course over an entire academic year (two semesters), teaching limits, derivatives, and their applications in the first semester, and integrals, differential equations, and their applications in the second. Both the United States Air Force Academy and the United States Naval Academy use this model for their calculus sequence [2, 1, 34]. The United States Military Academy, however, teaches MA104: Single Variable Calculus in a single semester. This semester is divided into two blocks. Block I covers limits, differentiation, and a few applications of differentiation with a deliberate focus on optimization as an extension of finding and classifying extrema. Block II covers integration and integration applications with a focus on the natural sciences and probability. A comparable course to MA104 was taught at the Massachusetts Institute of Technology (MIT) in 2010 and covered the topics shown in Table 2 [16]. The Naval Postgraduate School also offers a single course in calculus that covers similar topics to MA104 [25].

Block	Topics
1	Limits and Continuity
2	Differentiation: Definition and Fundamental Properties
3	Differentiation: Composite, Implicit, and Inverse Functions
4	Contextual Applications of Differentiation
5	Analytical Applications of Differentiation
6	Integration and Accumulation of Change
7	Differential Equations
8	Applications of Integration
9	(BC Only) Parametric Equations, Polar Coordinates, and Vector-Valued Functions
10	(BC Only) Infinite Sequences and Series

Table 1: AP Calculus AB/BC Year-Long Course Layout [4]

Until 1991, the United States Military Academy taught single-variable calculus over a traditional two-semester sequence. However, with the introduction of MA103: Discrete Dynamical Systems, the curriculum was revised, condensing single-variable calculus into a single semester [36].

Single-variable calculus courses often face the challenge of balancing breadth and depth, with many attempting to cover too many topics at the expense of developing students' deeper understanding of key concepts [29]. In order to address this issue, it is important to understand what types of students are taking calculus, as the student population should inform the goals of the course. At universities across the United States, single-variable calculus is often a requirement for engineering, mathematics, and science majors [22, 5]. At the United States Military Academy, however, all students are required to follow an engineering track, which means every cadet, regardless of major, must take and pass MA104 as a graduation requirement [35].

1.1 Student Learning Outcomes

The current Student Learning Outcomes (SLOs) for MA104: Single Variable Calculus are as follows:

1. Understand the derivative as a rate of change and the integral as an accumulation of change and explain how these operations are inverse processes.

Block	Topics
1	Concepts of Function, Limits and Continuity
2	Differentiation Rules, Application to Graphing, Rates, Approximations, and Extremum Problems
3	Definite and Indefinite Integration
4	The Fundamental Theorem of Calculus
5	Applications to Geometry: Area, Volume, and Arc Length
6	Applications to Science: Average Values, Work, and Probability
7	Techniques of Integration
8	Applications of Integration
9	Approximation of Definite Integrals, Improper Integrals, and L'Hôpital's Rule

Table 2: MIT's Single Variable Calculus Course in a Single Semester Layout [17]

2. Apply differential and integral calculus techniques to interpret and solve real-world problems.
3. Develop verbal and written communication skills to effectively convey calculus problem-solving strategies using appropriate mathematical language and notation.
4. Demonstrate effective study habits and self-directed learning by using feedback to foster independent improvement and deeper understanding.
5. Recognize and clearly articulate the ethical implications of applying calculus-based models in decision-making scenarios.

The first two SLOs are directly related to the content of the course, while the third, fourth, and fifth SLOs are related to the skills that are developed through the course but are not unique to calculus.

We can compare the MA104 course learning objectives to the Naval Academy's single-variable calculus course, which has the following learning objectives [37]:

1. Interpret, analyze, create, and communicate mathematical models involving derivatives.
2. Carry out computations involving order of growth, limits, and derivatives.
3. Describe relations between geometry, formulas, and data.
4. Recognize and apply mathematical procedures to solve applied problems, including related rates and optimization.
5. Write simple proofs of mathematical results.

The Naval Academy's learning objectives emphasize course content, whereas the MA104 learning objectives prioritize the skills students develop throughout the course. This approach is less common based on a review of the literature, as institutions such as Kent State University and Montana State University structure their objectives around course material rather than skill development [19, 24].

The Mathematical Association of America (MAA) discusses learning outcomes in terms of three key categories: content, cognitive, and affective goals. Content goals focus on the specific mathematical skills and knowledge that students are expected to acquire in the course. Cognitive goals extend beyond the course itself, emphasizing broader mathematical thinking and problem-solving abilities. Affective goals address students' attitudes and perceptions, shaping how they engage with and feel about the subject. These learning outcomes should drive course design, pedagogical practices, and assessment strategies [23]. MA104 SLOs 1 and 2 are content goals, while SLOs 3, 4, and 5 are cognitive.

1.2 Knowledge Retention

Since all students at USMA are required to take a calculus course due to the engineering track requirement—even for non-STEM majors—it is important to define the course’s objectives clearly. The primary goal should be to ensure long-term retention of key skills while building a strong conceptual foundation. However, an excessive focus on retention can lead to overlearning, which often results in only short-term recall unless students actively engage in mindful learning [20]. In order to achieve mindful learning in single-variable calculus, it is important to engage in active learning with both conceptual and procedural approaches [8, 14]. This is often at odds with students who simply want to pass the course and focus only on procedural learning, which does not lead to long-term retention [20]. To achieve lasting retention, lecturing must be interleaved with active learning, practice, assessments, and out-of-class work [20]. L. Fink also emphasizes that maximizing student engagement in the classroom is essential for effective learning [12]. The MAA identifies active learning as one of seven characteristics of a successful calculus program. Concrete examples of active learning include students working together to solve novel problems (not textbook problems), having students explain their process in written work, and reducing reliance on lecture as the primary mode of instruction [13].

Single-variable calculus is a unique college course because most students have already encountered some form of calculus in high school. In 2014, over 750,000 students took a calculus course, meaning that approximately 67% of those enrolled in college-level single-variable calculus had prior exposure to the subject [5]. Despite this widespread exposure, only about 20% of these students actually begin college calculus at an advanced level [31]. A study from Purdue University found that just under half of the students who received a 4 on their AP Calculus exam earned a B– or lower in their first college calculus course [10]. This highlights the issue that students often fail to retain their calculus knowledge long-term, even after achieving short-term success on exams.

The United States Military Academy differs from other universities in that all students are required to take some form of calculus. For the Class of 2028, approximately 54% of students reported having taken calculus before attending the Academy. However, when accounting for those placed into advanced math courses, only 28% of students in MA104 reported prior experience with calculus.

1.3 Feedback

Feedback plays a crucial role in college courses, significantly influencing students’ motivation and academic performance. Effective feedback goes beyond providing a numeric grade; it offers students clear insight into what they did well, where they went wrong, and how they can improve. When students receive constructive and detailed feedback, their motivation to learn increases, ultimately leading to better outcomes [6]. Research has shown that students who engage with feedback on assessments and are given the opportunity to revise and resubmit their work develop a deeper understanding of the material and perform better in the course, provided they take advantage of that opportunity [27].

Providing quality feedback helps shift assessments from merely measuring student performance to serving as a tool that allows instructors to “enhance the learning process and correct its trajectory” [28]. This requires instructors to ensure their students show all steps in solving a problem, not just the final answer, enabling feedback on the process and further enhancing learning outcomes [28]. One potential issue with feedback is that students may not take the time to read it. Research has shown that students often do not review feedback in class and frequently do not take the time to do so outside of class either [3]. As Levitt and Dubner argue in *Freakonomics*, “people respond to incentives” [21]. The previous assessment structure did not provide a direct incentive for cadets to engage with their feedback, and the prospect of better performance alone proved insufficient to motivate thorough review. To address this, the course introduced WRIT and WPR corrections, requiring cadets to revisit missed problems and produce corrected solutions. This structural change directly incentivizes engagement with feedback by tying a grade consequence to the act of reviewing and correcting errors..

There is some evidence suggesting that students perceive numerical feedback combined with comments similarly to receiving only a numerical grade [3], further suggesting that students often look only at their numeric score and do not engage with written comments. This can negatively affect students' perception of both the course and themselves, decreasing their motivation to invest the effort required to succeed [3].

One way to address this is a post-exam assignment, in which students review their exams and provide a corrected solution for each problem they missed, along with an explanation of the correct approach *and* a similar problem drawn from the course text, in-class assignments, or other course assessments. This ensures all students engage with the material and the feedback on their exams [18].

2 MA104: Single Variable Calculus at USMA

2.1 AY26-01 Redbook

The description of MA104 in the USMA Redbook is shown below.

MA104 introduces the core concepts of differential and integral calculus, including limits, derivatives, integrals, and their applications. Students develop the ability to analyze and model change in one variable, solve real-world optimization and accumulation problems, and interpret the results of mathematical models. The course emphasizes conceptual understanding, problem-solving, and communication of results, with applications that may be drawn from areas such as the natural sciences, social sciences, engineering, and real-world decision-making contexts.[26]

2.1.1 Prerequisites

There is one prerequisite for MA104 at the United States Military Academy: *MA103: Mathematical Modeling and Introduction to Calculus*. Students admitted to USMA are also required to take the Fundamental Concepts Exam (FCE), which requires mastery of basic pre-calculus skills such as algebra, basic trigonometry, and basic geometry. In cases where a student's mathematical background is not sufficiently strong, that student is placed with a more experienced instructor.

2.2 Student Learning Outcomes

The MA104 Student Learning Outcomes (SLOs) are:

1. Understand the derivative as a rate of change and the integral as an accumulation of change, and explain how these operations are inverse processes.
2. Apply differential and integral calculus techniques to interpret and solve real-world problems.
3. Develop verbal and written communication skills to effectively convey calculus problem-solving strategies using appropriate mathematical language and notation.
4. Demonstrate effective study habits and self-directed learning by using feedback to foster independent improvement and deeper understanding.
5. Recognize and clearly articulate the ethical implications of applying calculus-based models in decision-making scenarios.

2.3 Our Customers

Every STEM major requires MA204, MA205, or MA255. MA104 is the prerequisite for MA205. MA205 instructors need cadets to have mastered single-variable calculus techniques so that they can teach and practice multivariable techniques without constantly reviewing single-variable skills. STEM majors rely on calculus concepts to develop nuanced applications in their respective fields and need students who can readily comprehend and apply the necessary calculus tools.

2.4 Our Product

Every cadet must pass MA104 to graduate from West Point. The purpose of MA104 is to lead cadets to an *understanding* of single-variable calculus. The goal of our typical student is likely simply to *pass*, with little thought given to genuine understanding. Part of our challenge is to show cadets that if their goal is our goal, they will do better than pass. If their goal is simply to pass, they may not

achieve understanding—and may not pass at all. If they can pass *and* learn to learn, they will not only succeed in single-variable calculus but will likely experience fundamental personal and intellectual growth [32, 11].

Our typical cadet is a second-semester plebe still developing study habits and time management skills. Having recently completed MA103, in which algebra and pre-calculus receive limited reinforcement, many arrive at MA104 with atrophied foundational skills. To address this, we provide supplemental algebra materials and open the course with a diagnostic algebra assessment, ensuring cadets can identify their gaps and access resources to close them.

A deeper challenge lies in how plebes approach learning itself. Most believe they learn by cramming the night before an assessment, reviewing solutions to worksheets and old exams rather than wrestling with unfamiliar problems. Part of our mission in MA104 is to reshape that mindset: genuine learning happens through active practice and with a well-rested mind [15, 9, 7].

2.5 Working Assumptions

We make the following assumptions for our course.

- Students will rise to the expectations of the course.
- The average student in single-variable calculus views the course as an obstacle to follow-on privileges and graduation, rather than as an opportunity to grow and learn.
- The average student can learn to *love to learn* [11].

3 Assessment Design

The focus for assessments in Academic Year 2026 is Lifelong Learning (APG 3). The Redbook states that Lifelong Learning is achieved when “Graduates demonstrate the capability and desire to pursue progressive and continued intellectual development.” The down-trace What Graduates Can Do (WGCDs) include demonstrating willingness and ability to learn independently and engaging successfully in deliberate self-directed and collaborative learning experiences. SLOs 2 and 3 are also due for assessment this year from the cyclic assessment plan.

1. **SLO 2:** Apply differential and integral calculus techniques to interpret and solve real-world problems. This learning outcome is first introduced in Lesson 27 and formally assessed on WRIT 5, the Optimization Lab, and WPR 2. The application of integral calculus to real-world problems is introduced in Lesson 56 through the topic of Work, and assessed on WRIT 10 and WPR 4. Finally, the TEE comprehensively evaluates students’ ability to apply both differential and integral techniques to interpret and solve real-world problems.
2. **SLO 3:** Develop verbal and written communication skills to effectively convey calculus problem-solving strategies using appropriate mathematical language and notation. The primary method is through briefing board problems each day, ensuring good standards are maintained and that cadets develop their briefing skills daily. They are formally assessed during each lab and the final project, where they must write an executive summary describing the techniques used to solve a problem.
3. **SLO 4:** Demonstrate effective study habits and self-directed learning by using feedback to foster independent improvement and deeper understanding. Evaluation of this learning outcome will come from daily generative AI assignments and end-of-course surveys.

We will use the specific assessments on the following pages to evaluate how well the course is achieving the SLOs for this year.

SLO Evaluation Criteria: The proportion of cadets who achieve better than an Unsatisfactory score on the question is used to evaluate each SLO. If the proportion exceeds 80%, the SLO is accomplished satisfactorily. If the proportion falls between 65–80%, the SLO is accomplished marginally. If the proportion is below 65%, the SLO is accomplished unsatisfactorily.

Category	Score
Satisfactory	> 80%
Marginal	65–80%
Unsatisfactory	< 65%

3.1 Additions

- **Additional Mastery Days.** Mastery days have been consistently cited in the past two end-of-course assessments as an element worth sustaining. To build on this momentum, two additional mastery days were introduced this year—*Derivatives Mastery* and *Integration Applications Mastery 3*—targeting areas where cadets have historically struggled. Mastery days are a direct implementation of the principle of *practice* as introduced by Lang, giving cadets structured opportunities to engage with challenging material through repeated, low-stakes problem solving rather than passive review [15].
- **Related Rates Lesson.** In Academic Year 25, Related Rates was delivered as a self-guided lab. Upon reviewing student work, it became clear that many cadets did not grasp the connection

between related rates and implicit differentiation. As a result, the topic was transitioned from a lab to a dedicated lesson in order to make this relationship explicit. Teaching Related Rates in this format also achieves true interleaving with topics introduced earlier in the course while reinforcing cadets' understanding of modeling and trigonometric functions [11, 15].

- **Integration Applications.** In previous iterations of this course, the only integration applications covered were *Area Between Curves*, *Improper Integrals*, and *Probability*. *Volume*, *Work*, and *Average Value of a Function* were added to round out cadets' exposure to the topics most commonly encountered in subsequent STEM coursework.

3.2 Removals

Based on last year's success in trading commonly taught applications in single-variable calculus for additional time devoted to practice and mastery days [33, 30], the course sought to further refine which applications provide the best reinforcement of core topics. This has allowed for stronger mechanics and deeper understanding of the topics that remain.

- **Algebra Review.** While previous iterations of the course devoted several weeks to reviewing algebra skills, this approach showed little impact on cadets' overall performance. The review was therefore reduced to a single day covering the first chapter of Stewart [30].
- **Rates of Change, Related Rates, and Riemann Sums Lab.** With the broad availability of artificial intelligence tools and the grading burden on instructors, it was determined that the learning value of these labs did not justify their continued inclusion. Cadet performance and feedback suggested the labs were not producing meaningful gains relative to the time invested by both cadets and instructors.

3.3 Laboratories

Laboratories are designed to accomplish the following goals.

- **Connecting** [15]. Labs are our opportunity to draw connections to other courses students may be taking, helping to improve retention. Labs incorporate technology, physics, rudimentary economics, and statistics.
- **Technical writing practice** in an academic style and format [15]. Cadets complete multiple lab reports throughout the course, receiving iterative instructor feedback to support continuous improvement. Reports follow a standardized Department of Mathematical Sciences format (1–2 pages) and typically include a figure.
- **Motivating** [15]. Most labs are designed to (1) extend already-learned material and (2) be team competitions with bonus points at stake. For example, in the Natural Sciences Lab, teams are challenged to extend their integration understanding to 2D vector space and rewarded as a team for striking one of four targets of their choosing.

Each graded lab is discussed in further detail below.

Lab 1: Optimization. Optimization is one of the most useful applications students learn in this course, and understanding how to apply it across different frameworks will pay dividends in their careers. Cadets were asked to optimize a construction project subject to a series of binding and non-binding constraints.

Lab 2: Natural Sciences Part 1. This is the first of a two-day lab in which students are placed into teams and tasked to choose a target to hit. Targets that are farther away are worth more points (contingent on correct mathematics). Part 1 consists of students being introduced to the idea of integration over vectors and attempting to find the initial velocities of their projectile launchers.

Lab 3: Natural Sciences Part 2. This part of the lab consists of students calculating the necessary launch angle to strike their intended target and then attempting their shots.

Lab 4: Project Development. This lab initiates the final project and represents the instructor's primary opportunity to shape project quality. The final project includes the final writing assessment.

Lab 5: Probability. This lab is a competitive exploration of Probability Density Functions (PDFs). Students are given PDFs for four different vehicles and asked to place a wager (bonus points) on which car will win two out of three races, supported by correct mathematics. The race outcomes are computed according to the PDFs and differ for each instructor and race.

3.4 Course Flow

MA104 is divided into two blocks, outlined in Table 3.4. Block I covers limits, differentiation, and applications of derivatives, with deliberate emphasis on optimization as an extension of finding and classifying extrema. Block II covers integration and its applications, with focus areas including probability, work, average value, and volume.

Block	Lesson	Topic	Section	Exam
I	1	Course Intro		
	2	Functions Review	§ 1.1–1.5	
	3	Limits/Continuity 1	§ 2.1–2.2	
	4	Limits/Continuity 2	§ 2.3 and § 2.6	
	5	Limits/Continuity 3	§ 2.5	
	6	Limits Mastery		WRIT 1
	7	Rates of Change	§ 2.7–2.8	
	8	Rates of Change Mastery		WRIT 2
	9	Derivatives of Polynomial and Exp. Functions	§ 3.1	
	10	Derivatives of Trigonometric Functions	§ 3.3	WRIT 3
	11	Product and Quotient Rules	§ 3.2	
	12	Product and Quotient Rule Mastery		
	13	Chain Rule	§ 3.4	
	14	Chain Rule Mastery		WRIT 4
	15	Implicit Differentiation 1	§ 3.5	
	16	Implicit Differentiation 2	§ 3.6	
	17	Implicit Mastery		
	18	Derivative Mastery		
	19	WPR 1 (Derivatives)		WPR 1
	20	Related Rates	§ 3.9	
	21	Maximum/Minimum 1	§ 4.1	
	22	Maximum/Minimum 2	§ 4.3	
	23	Maximum/Minimum 3	§ 4.3	WRIT 5
	24	Maximum/Minimum Mastery		

Block	Lesson	Topic	Section	Exam
II	25	Optimization 1	§ 4.7	WRIT 6
	26	Optimization 2	§ 4.7	
	27	Optimization Lab		
	28	Optimization 3	§ 4.7	WRIT 7
	29	L'Hôpital's Rule	§ 4.4	
	30	Derivative Applications Mastery 1		
	31	Derivative Applications Mastery 2		WPR 2
	32	WPR 2 (Derivative Applications)		
	33	Antiderivatives	§ 4.9	WRIT 8
	34	Definite Integral 1	§ 5.1	
	35	Definite Integral 2	§ 5.2	
	36	Definite Integral 3	§ 5.3	WRIT 9
	37	Indefinite Integrals	§ 5.4	
	38	Basic Integration Mastery		
	39	Natural Sciences Lab 1		WRIT 10
	40	Natural Sciences Lab 2		
	41	Substitution Rule	§ 5.5	
	42	Substitution Rule Mastery	§ 5.5	WRIT 10
	43	Integration by Parts	§ 7.1	
	44	Integration by Parts Mastery	§ 7.1	
	45	Integration Strategies 1	§ 7.5	WRIT 9
	46	Integration Strategies 2	§ 7.5	
	47	Integration Strategies Mastery	§ 7.5	
	48	WPR 3 (Integration)		WPR 3
	49	Spring Break Lab		WRIT 10
	50	Areas Between Curves	§ 6.1	
	51	Volumes	§ 6.2	
	52	Work	§ 6.4	WRIT 10
	53	Average Value of a Function	§ 6.5	
	54	Integration Applications Mastery 1		
	55	Project Development		WRIT 10
	56	Improper Integrals	§ 7.8	
	57	Improper Integrals Mastery	§ 7.8	
	58	Probability 1	Hand Out and § 8.5	WRIT 10
	59	Probability 2	§ 8.5	
	60	Probability Lab		
	61	Integration Applications Mastery 2		WPR 4
	62	Integration Applications Mastery 3		
	63	WPR 4 (Integration Applications)		
	64A	Project Briefs		WPR 4
	64B	Project Briefs		
	64C	Project Briefs		
	65	Differentiation Review		

4 Assessment or Evaluation

During Academic Year 26, the following Student Learning Outcomes (SLOs) were evaluated.

4.1 SLO 2

Apply differential and integral calculus techniques to interpret and solve real-world problems.

This learning outcome is first introduced in Lesson 27 and formally assessed on WRIT 5, the Optimization Lab, and WPR 2. The application of integral calculus to real-world problems is introduced in Lesson 56 and assessed on WRIT 10 and WPR 4. The TEE comprehensively evaluates students' ability to apply both differential and integral techniques to interpret and solve real-world problems.

Overview of Direct Indicators

Six direct assessments were used to evaluate SLO 2 across the semester. Table 4 summarizes performance on the SLO 2-relevant questions within each assessment. The SLO evaluation criteria from Section 3 are applied: satisfactory if the proportion above 70% exceeds 80%, marginal if between 65 and 80%, and unsatisfactory if below 65%.

Assessment	N	Mean	>70%	Result
WRIT 5: Min/Max (20 pts)	632	81.0%	78.5%	Marginal
Optimization Lab (10 pts)	466	84.2%	88.8%	Satisfactory
WPR 2: Optimization Questions (30 pts)	653	82.5%	78.3%	Marginal
WRIT 10: Probability and Improper (20 pts)	630	83.7%	84.0%	Satisfactory
WPR 4: Integration Applications (45 pts)	649	72.6%	66.1%	Marginal
TEE: SLO 2 Questions Combined (60 pts)	645	78.5%	71.9%	Marginal

Table 4: SLO 2 direct indicator summary. The column “>70%” shows the proportion of cadets scoring above the unsatisfactory threshold.

Block I: Differential Calculus Applications

The first formal assessment of SLO 2 was WRIT 5, which tested the ability to find absolute and local extrema, core prerequisites for optimization. The class mean was 81.0% with 78.5% of cadets above 70%, placing performance at the **marginal** threshold. The Optimization Lab, administered the same week, produced notably stronger results: mean 84.2%, with 88.8% above 70%, earning a **satisfactory** rating. This gap is consistent with the lab's collaborative format providing additional scaffolding relative to the individual WRIT. On WPR 2, the optimization questions (Questions 4a through 4c, 30 pts) produced a mean of 82.5% but only 78.3% above 70%, again **marginal**. The distribution of WPR 2 optimization sub-scores shows a cluster at full credit alongside a long lower tail, suggesting the majority of cadets who understood the setup executed well, while a meaningful minority struggled with problem formulation.

Block II: Integral Calculus Applications

WRIT 10 assessed probability density functions and improper integrals, producing the strongest result of the Block II indicators: mean 83.7%, 84.0% above 70%, **satisfactory**. Performance degraded on WPR 4's integration applications section (Work, Average Value, and Volume, 45 pts), which produced a mean of only 72.6% with 66.1% above 70%, **marginal** and the weakest result of any SLO 2 indicator. The sub-question breakdown in Figure 2 reveals Work (b) (mean 67.5%) and Average Value

(mean 68.4%) as the primary drivers of this performance gap, while the Volume multiple choice item performed well (mean 94.2%). This suggests cadets could recognize the correct method but struggled with execution on the more algebraically demanding problems.

Term End Examination

On the TEE, the 25-point Optimization section produced a mean of 78.0% with 69.6% above 70% (**marginal**), and the 35-point Integration Applications section produced a mean of 78.8% with 74.3% above 70% (**marginal**). Combined across the 60 SLO 2 points on the TEE, 71.9% of cadets met the threshold, indicating a marginal result on the comprehensive evaluation.

SLO Evaluation

Of the six direct indicators, two (Optimization Lab and WRIT 10) were satisfactory and four (WRIT 5, WPR 2 Optimization, WPR 4 Integration Applications, and TEE SLO 2 Combined) were marginal. No indicator fell below the 65% unsatisfactory threshold. **SLO 2 was accomplished marginally.** The consistent pattern across indicators is that cadets perform well on recognition and setup tasks but lose points on multi-step execution, particularly for Work and Average Value problems where algebraic errors compound. This is an area for targeted improvement through additional integration applications practice in AY27.

Figures

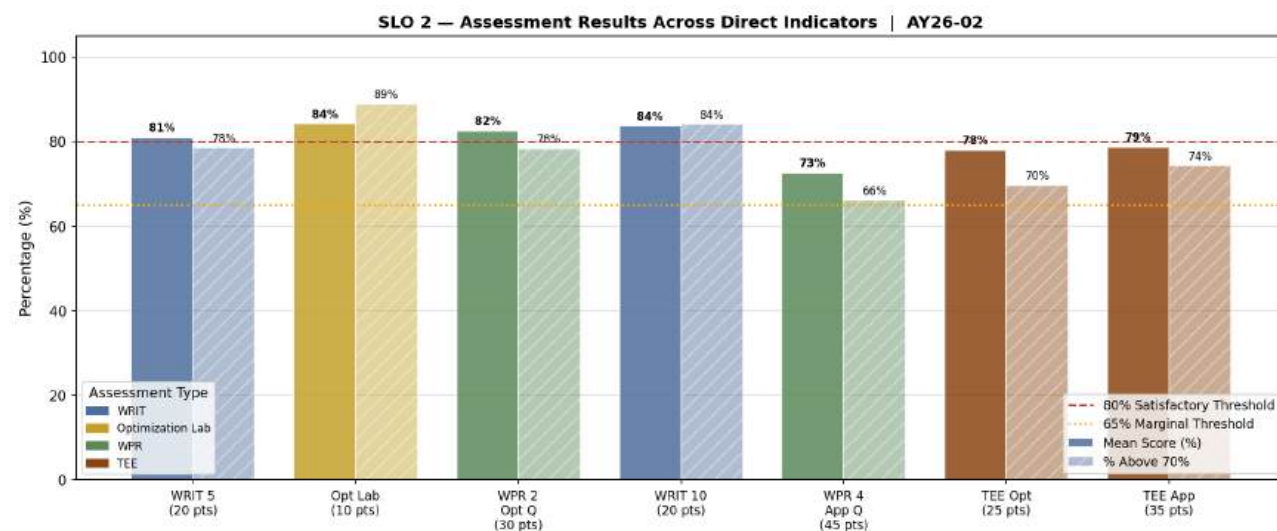


Figure 1: Mean score and proportion above 70% for each SLO 2 direct indicator. Dashed lines indicate the satisfactory (80%) and marginal (65%) thresholds.

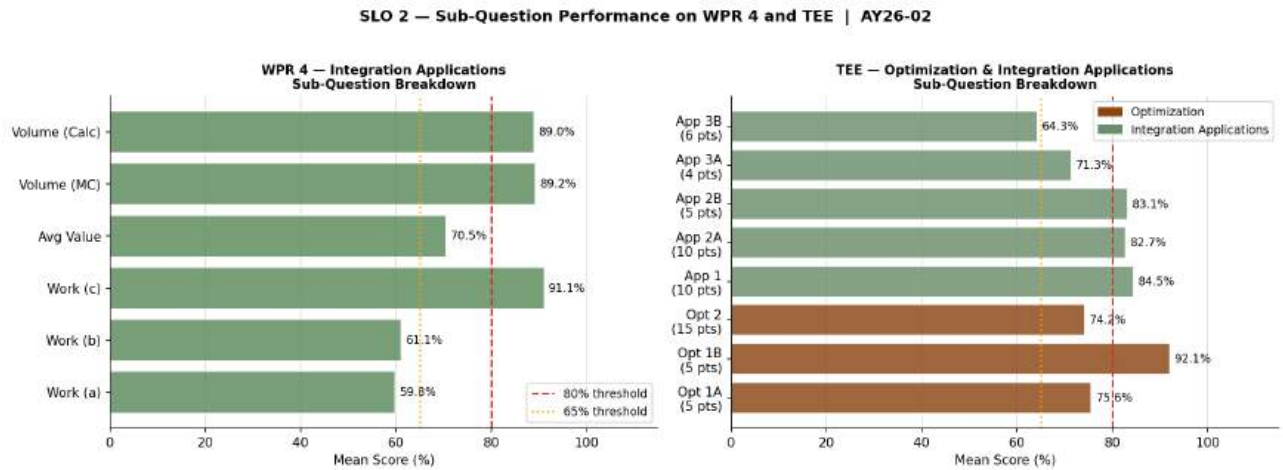


Figure 2: Mean score by sub-question on WPR 4 Integration Applications and TEE SLO 2 questions.

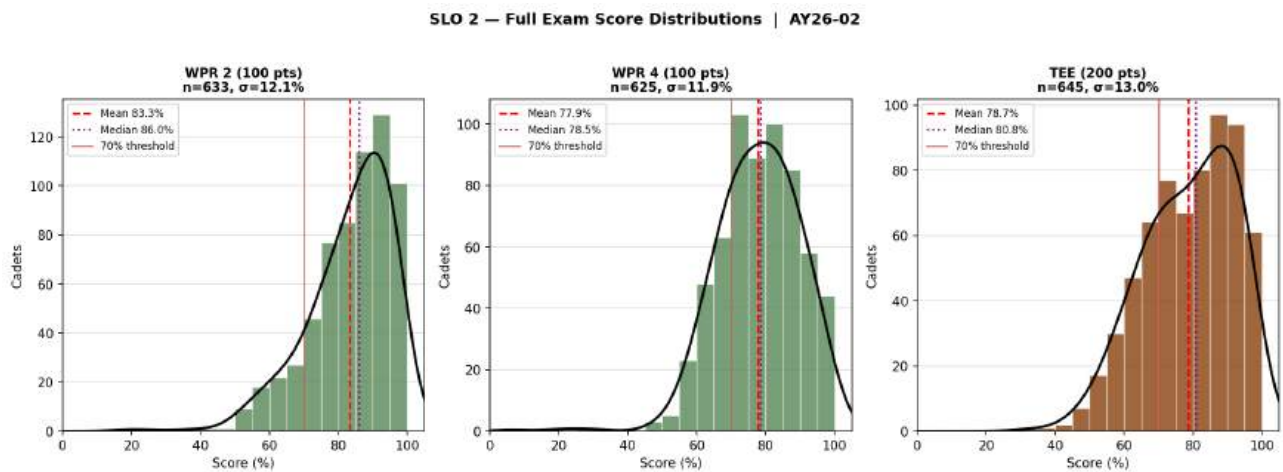


Figure 3: Full exam score distributions for WPR 2, WPR 4, and the TEE. The red vertical line indicates the 70% threshold.

4.2 SLO 3

Develop verbal and written communication skills to effectively convey calculus problem-solving strategies using appropriate mathematical language and notation.

The primary method of developing this SLO is through daily briefing board problems. Maintaining high standards each day ensures cadets build their briefing skills consistently throughout the semester. Cadets are formally assessed on this SLO during each lab and the final project, where they must write or brief an executive summary describing the techniques used to solve the problem.

Overview

Cadets in MA104 were formally assessed on two written products and only oral assessment across the semester: the **Optimization Lab** (10 pts), the **Natural Sciences Lab** (10 pts), and the **Final Project** (50 pts). Each assessment required cadets to produce an executive summary explaining the mathematical techniques used to solve an applied problem. Submissions were evaluated on methodology, results and discussion, solution correctness, writing quality, and citations.

Assessment Results

Optimization Lab (max = 10 pts, $n = 466$). The class mean was 84.2% with a median of 90.0% and a standard deviation of 17.6%. Scores ranged from 10% to 100%. Fifty-two cadets (11.2%) scored below the 70% threshold.

Natural Sciences Lab (max = 10 pts, $n = 529$). The class mean was 87.5% with a median of 90.0% and a standard deviation of 15.6%. Scores ranged from 0% to 100%. Thirty-five cadets (6.6%) scored below the 70% threshold.

Final Project (max = 50 pts, $n = 468$). The class mean was 91.6% with a median of 92.0% and a standard deviation of 6.3%. Scores ranged from 70% to 100%. No cadets scored below the 70% threshold.

Grade Distribution by Band

Assessment	A ($\geq 90\%$)	B (80–89%)	C (70–79%)	D (60–69%)	F ($< 60\%$)
Optimization Lab	278 (59.7%)	88 (18.9%)	48 (10.3%)	23 (4.9%)	29 (6.2%)
Natural Sciences Lab	362 (68.4%)	95 (18.0%)	37 (7.0%)	13 (2.5%)	22 (4.2%)
Final Project	340 (72.6%)	104 (22.2%)	24 (5.1%)	0 (0.0%)	0 (0.0%)

Table 5: Grade band counts and percentages by assessment. Values shown as n (%).

Statistical Analysis

A paired t -test compared each cadet's base score on the Natural Sciences Lab against their Optimization Lab score ($n = 345$ cadets with scores on both assessments). The improvement was statistically significant at the $\alpha = 0.05$ level ($t = 3.282$, $p = 0.0011$).

The class mean improved from 84.2% on the Optimization Lab to 87.5% on the Natural Sciences Lab and reached 91.6% on the Final Project, while the standard deviation decreased markedly across all three assessments (17.6%, 15.6%, 6.3%). It should be noted that the two labs were written assignments whereas the Final Project was an oral presentation, so the difference in delivery format likely

contributed to the lower variance on the Final Project. Nevertheless, the statistically significant improvement between the two written labs ($t = 3.282$, $p = 0.0011$) suggests that consistent daily briefing board practice produced meaningful gains in cadet communication over the course of the semester.

Applying the SLO evaluation criteria defined above, the proportion of cadets scoring above Unsatisfactory on the Final Project was 100%, indicating that **SLO 3 was accomplished satisfactorily**.

Figures

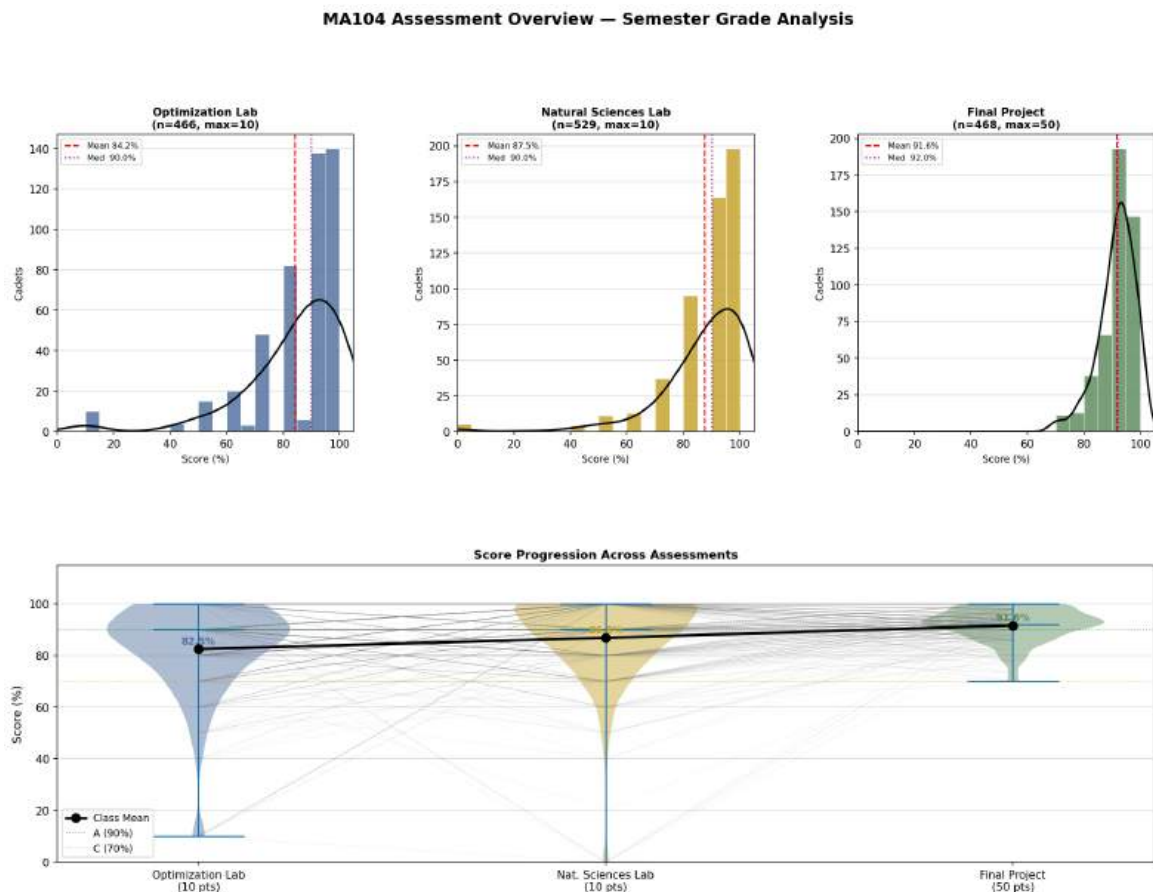


Figure 4: Score distributions and progression across all three assessments.

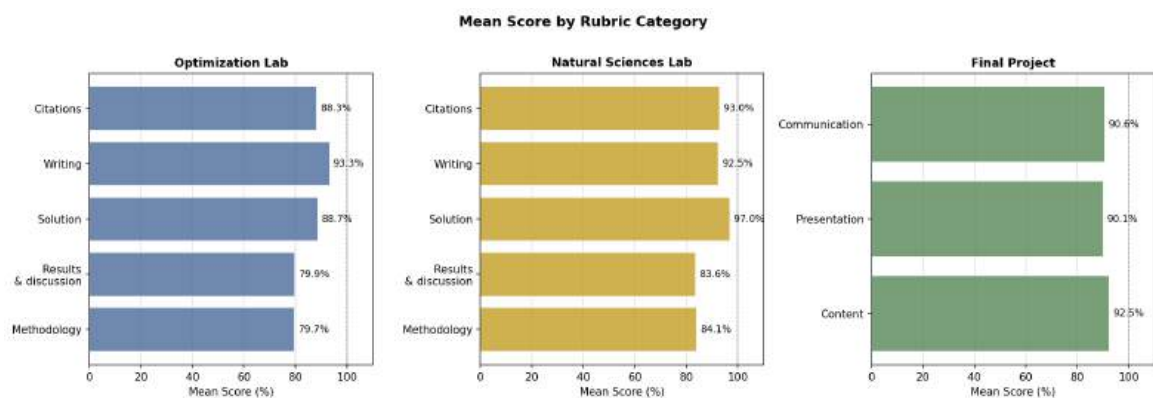


Figure 5: Mean score (%) by rubric category for each assessment.

4.3 SLO 4:

Demonstrate effective study habits and self-directed learning by using feedback to foster independent improvement and deeper understanding.

Evaluation of this learning outcome was conducted through an end-of-course survey administered during the final week of instruction ($n = 220$ respondents). The survey included three Likert-scale items rated from 1 (Strongly Disagree) to 5 (Strongly Agree):

- **Q5:** My experience in MA104 has increased my confidence in using Generative AI to *understand* problems.
- **Q6:** My experience in MA104 has increased my confidence in using Generative AI to *generate* practice problems.
- **Q7:** My study habits have improved since the beginning of MA104.

Survey Results

Q5 — GenAI confidence (understanding problems). The mean response was 3.32 ($\sigma = 1.02$, median = 3). Of the 220 respondents, 93 cadets (42.3%) agreed or strongly agreed, 87 (39.5%) responded neutrally, and 40 (18.2%) disagreed or strongly disagreed.

Q6 — GenAI confidence (generating practice problems). The mean response was 3.62 ($\sigma = 1.07$, median = 4). A majority of cadets, 129 (58.6%), agreed or strongly agreed, 61 (27.7%) responded neutrally, and 30 (13.6%) disagreed or strongly disagreed.

Q7 — Study habits improved. This item produced the strongest response, with a mean of 3.75 ($\sigma = 1.03$, median = 4). A total of 134 cadets (60.9%) agreed or strongly agreed, 67 (30.5%) responded neutrally, and only 19 (8.6%) disagreed or strongly disagreed.

Question	Mean	Median	Std Dev	Agree (4–5)	Disagree (1–2)
Q5: GenAI — understand problems	3.32	3	1.02	93 (42.3%)	40 (18.2%)
Q6: GenAI — generate problems	3.62	4	1.07	129 (58.6%)	30 (13.6%)
Q7: Study habits improved	3.75	4	1.03	134 (60.9%)	19 (8.6%)

Table 6: End-of-course survey results for SLO 4. $n = 220$.

SLO Evaluation

Applying the course evaluation criteria, cadets scoring 3 (Neutral) or above are considered to have met the minimum threshold for this SLO. Under this interpretation, 81.8%, 86.4%, and 91.4% of respondents met the threshold on Q5, Q6, and Q7, respectively — all three exceeding the 80% satisfactory benchmark. **SLO 4 was accomplished satisfactorily.**

The weakest response was on Q5, which asked about using AI to better *understand* problems. This is consistent with written feedback from several cadets who noted that while they were comfortable using generative AI as a homework aid and practice-problem generator, they found it less reliable for conceptual explanations. Q6 and Q7 both produced majority-agree responses, suggesting that the structured AI integration in WebAssign and the overall course design had a meaningful positive impact on cadets' self-directed learning behaviors.

Figures

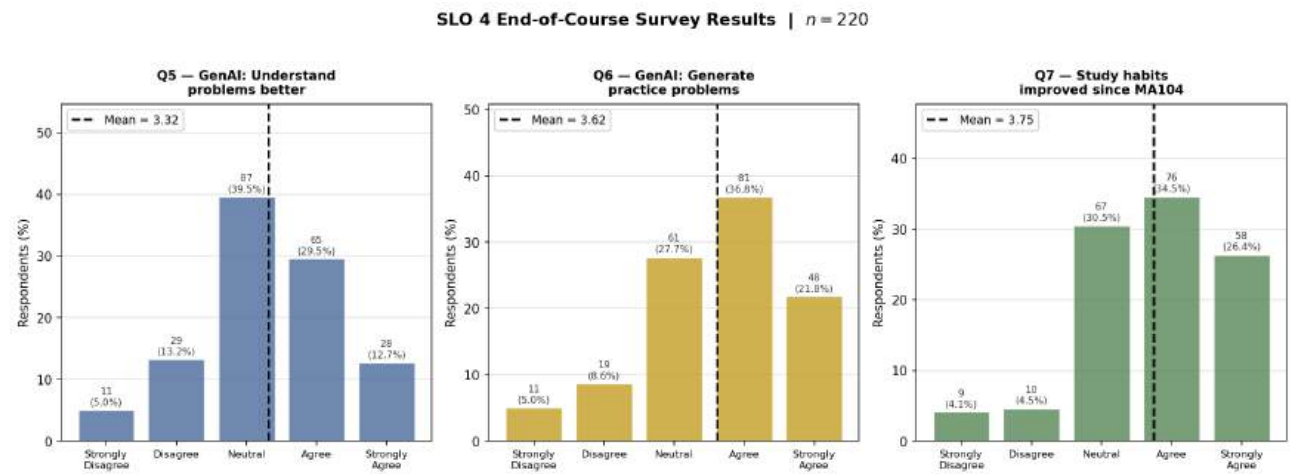


Figure 6: Response distributions for each SLO 4 survey item. Dashed line indicates the mean.

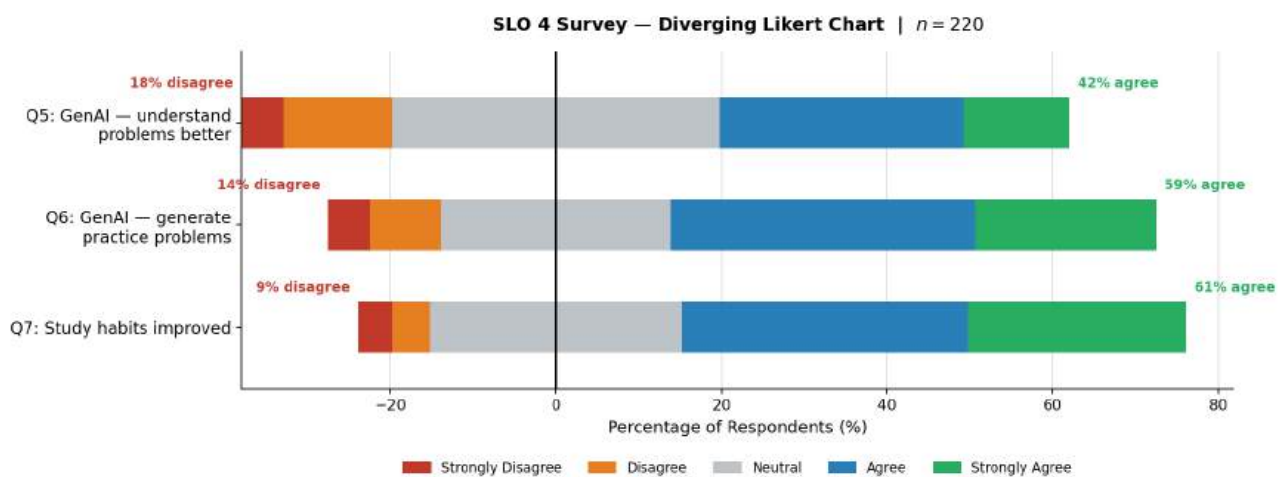


Figure 7: Diverging Likert chart showing the balance of agreement and disagreement across all three SLO 4 items.

5 Conclusions and Future Work

The course is in a good position. The SLOs are measuring the correct things, the assessments are achieving their intended purpose, and cadets are improving over time. This is not to say that there is no room for improvement, only that the current offering achieved its stated goals. The course should focus on the following updates for the next semester:

1. Increase structured practice on integration applications, particularly Work and Average Value of a Function. These two sub-topics produced the lowest scores of any direct SLO 2 indicator, with means of 67.5% and 68.4% respectively on WPR 4. Cadets demonstrated the ability to identify the correct method but consistently lost points on multi-step algebraic execution. Additional mastery day problems and WRIT questions targeting these topics should more closely reflect the algebraic complexity seen on WPRs and the TEE.
2. Improve alignment between WRITs, mastery days, and WPRs. Cadets frequently reported performing well on formative assessments yet struggling on WPRs, and the SLO 2 data corroborates this pattern. Mastery and WRIT problems should be reviewed each year to ensure they are meaningfully predictive of WPR performance.
3. Restructure weekly Forge meetings to include a dedicated review of each upcoming lesson's Big Idea. The course should provide a standardized LaTeX template for each lesson as a shared starting point, which instructors should adapt as they see fit. This should reduce variance in how the conceptual foundation of each lesson is motivated and explained across sections.
4. Investigate the Saturday TEE sitting. A statistically significant performance gap was identified between Saturday cadets and those who sat on Wednesday and Thursday ($F = 5.18$, $p = 0.006$), despite all cadets taking the same exam. The gap is distributed broadly across all eight exam sections, ruling out a content-specific cause. The composition and scheduling of the Saturday sitting should be examined in AY27.

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MA104: Single Variable Calculus
AY26-02 Syllabus

Textbook: Calculus - Ninth Edition, Early Transcendentals by James Stewart

Students:

Lesson	Date	Topic (Reading)	Notes	Lesson Objectives
BLOCK 1: DERIVATIVES AND THEIR APPLICATIONS				
Functions & Mathematical Models				
Lesson 1	1/6/2026	Course Intro		Topic: Overview of the Course
Limits & Continuity				
Lesson 2	1/7/2026	Functions Review (§ 1.1 - 1.5)		1. Understand the basic characteristics of a function. 2. Determine the domain and range of a function. 3. Determine when a function is increasing and decreasing. 4. Understand the shapes and properties of common functions (Linear, Polynomial, Power, Rational, Algebraic, Trigonometric, Exponential, and Logarithmic). 5. Recognize and construct composite functions.
Lesson 3	1/9/2026	Limits/Continuity 1 (§ 2.1 - 2.2)		1. Understand the definition of a tangent and a secant line. 2. Determine average rates of change. 3. Estimate instantaneous rates of change. 4. Find or estimate the equation of a tangent line to a curve. 5. Understand the definition of a limit. 6. Find the limit of a function graphically and numerically.
Lesson 4	1/12/2026	Limits/Continuity 2 (§ 2.3 and 2.6: Pg 127-133)		1. Understand how to apply the properties of limits. 2. Find the limit of a function algebraically. 3. Understand and evaluate limits at infinity. 4. Apply the Squeeze Theorem to solve limits.
Lesson 5	1/13/2026	Limits/Continuity 3 (§ 2.5)		1. Understand the definition of continuity. 2. Identify where and why a function is discontinuous. 3. Understand the Intermediate Value Theorem. 4. Recognize the types of functions that are continuous at every number in their domain.
Lesson 6	1/14/2026	Limits Mastery		
Derivatives				
Lesson 7	1/16/2026	Rates of Change (§ 2.7-2.8)		1. Understand the definition of a tangent line. 2. Understand and apply the limit definition of a derivative. 3. Understand the definition of velocity and how it relates to derivatives. 4. Recognize and apply the different notation used to represent a derivative. 5. Understand how a function can fail to be differentiable. 6. Recognize and understand the relationship between differentiation and continuity.
Lesson 8	1/20/2026	Rates of Change Mastery		
Lesson 9	1/21/2026	Derivatives of Polynomial and Exponential Functions (§ 3.1)		1. Recognize and apply the following derivative rules: a. Derivative of a constant function b. Power rule c. Constant multiple rule d. Sum rule e. Difference rule f. Derivative of the natural exponential function
Lesson 10	1/22/2026	Derivatives of Trigonometric Functions (§ 3.3)		1. Recognize and apply the derivative rules of trigonometric functions. 2. Understand the relationship between the derivatives of sine and cosine functions.
Lesson 11	1/23/2026	Product and Quotient Rules (§ 3.2)		1. Recognize and apply the following derivative rules: a. Product rule b. Quotient rule
Lesson 12	1/26/2026	Product and Quotient Rule Mastery		
Lesson 13	1/27/2026	Chain Rule (§ 3.4)		1. Recognize composite functions. 2. Understand when to apply the chain rule. 3. Solve derivatives that involve using the chain rule. 4. Use the chain rule to differentiate an exponential function with a base "b".
Lesson 14	1/28/2026	Chain Rule Mastery		
Implicit Differentiation				
Lesson 15	1/30/2026	Implicit Differentiation 1 (§ 3.5)		1. Recognize when a function is defined implicitly. 2. Understand how to apply implicit differentiation. 3. Solve for the equation of a tangent line at a point on any implicitly defined curve.
Lesson 16	2/2/2026	Implicit Differentiation 2 (§ 3.6)		1. Understand how implicit differentiation is used to determine the derivative of logarithmic and inverse trigonometric functions. 2. Solve for the derivative of logarithmic and inverse trigonometric functions. 3. Take the derivative of an implicit function with respect to time.
Lesson 17	2/3/2026	Implicit Mastery		
Lesson 18	2/5/2026	Derivative Mastery		
Lesson 19	2/6/2026	WPR 1 (Derivatives)	In Class	
Lesson 20	2/10/2026	Related Rates (§ 3.9)		1. Recognize that rates of change are derivatives. 2. Apply related rates reasoning to geometric models.
Maximums & Minimums				
Lesson 21	2/11/2026	Maximum/Minimum 1 (§ 4.1)		1. Understand the difference between absolute and local extrema. 2. Apply the Extreme Value Theorem. 3. Understand how Fermat's Theorem relates to critical numbers and the cautions associated with using Fermat's Theorem. 4. Use critical numbers as part of the Closed Interval Method to determine absolute maximum and minimum values.
Lesson 22	2/12/2026	Maximum/Minimum 2 (§ 4.3)		1. Use a function's derivative to identify increasing or decreasing intervals. 2. Identify and solve for intervals where a curve is concave up or concave down. 3. Understand and identify inflection points. 4. Understand and apply the second derivative test to solve for local maximum and minimum values.
Lesson 23	2/13/2026	Maximum/Minimum 3 (§ 4.3)		1. Use information from first and second derivatives to sketch the curve of a function. 2. Classify local extrema using first and second derivatives.
Lesson 24	2/17/2026	Maximum/Minimum Mastery		

MA104: Single Variable Calculus
AY26-02 Syllabus

Lesson	Date	Topic (Reading)	Notes	Lesson Objectives
Derivative Applications: Optimization and L'Hôpital's Rule				
Lesson 25	2/18/2026	Optimization 1 (§ 4.7)		1. Understand how Stewart's optimization problem-solving strategy relates to the USMA Mathematical Problem-Solving Strategy. 2. Understand and apply the first derivative test for absolute extreme values. 3. Given a scenario with objectives and constraints, solve for the maximum or minimum values.
Lesson 26	2/19/2026	Optimization 2 (§ 4.7)		1. Understand how Stewart's optimization problem-solving strategy relates to the USMA Mathematical Problem-Solving Strategy. 2. Understand and apply the first derivative test for absolute extreme values. 3. Given a scenario with objectives and constraints, solve for the maximum or minimum values.
Lesson 27	2/20/2026	Optimization Lab		
Lesson 28	2/23/2026	Optimization 3 (§ 4.7)		1. Given a scenario with objectives and constraints, solve for the maximum or minimum values.
Lesson 29	2/24/2026	L'Hôpital's Rule (§ 4.4 pg 309-313)		1. Apply L'Hôpital's Rule to evaluate indeterminate limits.
Lesson 30	2/25/2026	Derivative Applications Mastery 1	Optimization Report Due	
Lesson 31	2/26/2026	Derivative Applications Mastery 2		
Lesson 32	2/27/2026	WPR 2 (Derivative Applications)	In Class	
BLOCK 2: INTEGRALS AND THEIR APPLICATIONS				
Lesson 33	3/3/2026	Antiderivatives (§ 4.9)		1. Understand the definition of an antiderivative. 2. Recognize basic antiderivative rules. 3. Solve basic antiderivative problems.
Integration				
Lesson 34	3/4/2026	Definite Integral 1 (§ 5.1)		1. Understand how to estimate the area between a curve and the horizontal axis on an interval. 2. Solve area estimates using sigma notation. 3. Understand how to solve areas using the limit operator in conjunction with sigma notation. 4. Interpret distances traveled by an object as an area problem.
Lesson 35	3/5/2026	Definite Integral 2 (§ 5.2)		1. Calculate definite integrals using Definition 2 and Theorem 4 in Section 5.2. 2. Express definite integrals as a limit of a Riemann Sum and express a limit of a Riemann Sum as a definite integral. 3. Understand the mathematical notation and terminology of integrals. 4. Apply the Left or Right Endpoint Method to estimate an integral. 5. Understand and apply the properties of definite integrals.
Lesson 36	3/6/2026	Definite Integral 3 (§ 5.3)		1. Understand and apply the Fundamental Theorem of Calculus, Part 1. 2. Understand and apply the Fundamental Theorem of Calculus, Part 2. 3. Internalize differentiation and integration as inverse processes.
Lesson 37	3/9/2026	Indefinite Integrals (§ 5.4)		1. Distinguish between indefinite and definite integrals. 2. Determine indefinite integrals. 3. Understand and apply the Net Change Theorem.
Lesson 38	3/10/2026	Basic Integration Mastery		
Lesson 39	3/11/2026	Natural Sciences Lab 1		
Lesson 40	3/12/2026	Natural Sciences Lab 2		
Lesson 41	3/13/2026	Substitution Rule (§ 5.5)		1. Understand when to apply the substitution rule. 2. Use the substitution rule to solve definite and indefinite integrals.
Lesson 42	3/16/2026	Substitution Rule Mastery (§ 5.5)	Natural Sciences Report Due	
Lesson 43	3/17/2026	Integration by Parts (§ 7.1)		1. Understand how the method of integration by parts is derived from the product rule of differentiation. 2. Understand how to select "f" and "g" when using integration by parts. 3. Apply integration by parts to solve definite and indefinite integrals.
Lesson 44	3/18/2026	Integration by Parts Mastery (§ 7.1)		
Lesson 45	3/19/2026	Integration Strategies 1 (§ 7.5)		1. Understand how to approach a definite and indefinite integral in order to determine the appropriate integration method(s). 2. Solve any integral by choosing the correct integration technique (simplifying with basic rules, substitution, integration by parts).
Lesson 46	3/23/2026	Integration Strategies 2 (§ 7.5)		1. Understand how to approach a definite and indefinite integral in order to determine the appropriate integration method(s). 2. Solve any integral by choosing the correct integration technique (simplifying with basic rules, substitution, integration by parts).
Lesson 47	3/24/2026	Integration Strategies Mastery (§ 7.5)		
Lesson 48	3/25/2026	WPR 3 (Integration)	Dean's Hour	
Lesson 49	3/27/2026	Spring Break Lab		
Applications of Integration				
Lesson 50	4/7/2026	Areas Between Curves (§ 6.1)		1. Determine the area between two curves. 2. Interpret the meaning of the area between two curves given in context.
Lesson 51	4/8/2026	Volumes (§ 6.2)		1. Use the disk and washer methods to construct and evaluate definite integrals representing volumes of solids of revolution. 2. Identify radii, limits of integration, and interpret the final volume with correct units and context.
Lesson 52	4/9/2026	Work (§ 6.4)		1. Understand the concept of work and how calculus is used to compute work when force varies over a distance. 2. Set up and solve definite integrals to determine the work done by constant and variable forces.
Lesson 53	4/10/2026	Average Value of a Function (§ 6.5)		1. Solve for the average value of a function using a definite integral over a closed interval. 2. Interpret and apply the Mean Value Theorem for Integrals to find values where the function equals its average value.
Lesson 54	4/13/2026	Integration Applications Mastery 1		
Lesson 55	4/14/2026	Project Development		
Lesson 56	4/15/2026	Improper Integrals (§ 7.8)		1. Identify improper integrals as either Type 1 or Type 2 improper integrals. 2. Identify when a Type 1 improper integral is either convergent or divergent. 3. Solve Type 1 improper integrals.
Lesson 57	4/16/2026	Improper Integrals Mastery (§ 7.8)		

MA104: Single Variable Calculus
AY26-02 Syllabus

Lesson	Date	Topic (Reading)	Notes	Lesson Objectives
Lesson 58	4/20/2026	Probability 1 (Hand Out and § 8.5)		1. Develop a basic understanding of probability. 2. Understand the definition of a probability density function. 3. Recognize when a probability density function is invalid. 4. Interpret a probability statement. 5. Solve for a probability using a probability density function. 6. Determine the average value using a probability density function.
Lesson 59	4/21/2026	Probability 2 (§ 8.5)		1. Understand the definition of a probability density function. 2. Recognize when a probability density function is invalid. 3. Interpret a probability statement. 4. Solve for a probability using a probability density function. 5. Determine the average value using a probability density function.
Lesson 60	4/22/2026	Probability Lab		
Lesson 61	4/27/2026	Integration Applications Mastery 2	Final Project Due	
Lesson 62	4/28/2026	Integration Applications Mastery 3		
Lesson 63	4/29/2026	WPR 4 (Integration Applications)	Dean's Hour	
Lesson 64A	5/4/2026	Project Briefs		
Lesson 64B	5/5/2026	Project Briefs		
Lesson 64C	5/6/2026	Project Briefs		
Lesson 65	5/8/2026	Differentiation Review		1. Review differentiation key concepts.



UNITED STATES MILITARY ACADEMY
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SECTION II

EXAMINATIONS



UNITED STATES MILITARY ACADEMY
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TAB G

WPR #1

Outstanding Student Work

Name (First Last): Sarah Andrews
MA104 AY26-2 WPR 1 (100 Points)
Time: 55 min

C Number: C20316045
Instructor: CPT Dinkel
Section: D19

Instructions—Read Carefully

Number of Extra Sheets: 0

1. Print your name on every page and ensure you have all pages. It is highly recommended that you review all questions so that you may formulate a plan of attack.
2. **References.** You are authorized your issued calculator and *your* MA104 Reference Card. The reference card must not be used to copy down example problems or gain an unfair advantage and must be in your own handwriting. All reference cards will be turned in with each exam and returned with the graded exam. Any reference card found with example problems or information directly from the exam (ie exam questions) could result in a 0 for the exam.
3. Cell phones, smart watches, and/or music playing devices must be placed in backpacks during the exam.
4. All work written on the WPR is subject to grading unless marked through or clearly marked with words to the effect of “do not grade.”
5. Use a continuation sheet if you need more space for a problem. **Clearly indicate** at the problem and on the continuation sheet which problem is continued and be sure to put your name on the continuation sheet.
6. Always use proper notation. Assume all numbers are exact. Define any variables that you use which are not explicitly defined in the problem. Unless otherwise stated, assume all angles are given in radians. In problems that involve units of measurement, ensure that your answers include the appropriate units.
7. **SHOW ALL WORK.** Sufficient work that indicates your method of reasoning for the operations performed is required. This work must be done using *calculus concepts* learned in class. **Clearly indicate your final answer.**
8. Early departure is authorized. “Begin work” and “cease work” times will be given by your instructor.
9. **Academic Security** Cadets are not authorized to discuss the content, structure, or other information about this exam until it has been released from academic security. Discussion includes all forms of written, electronic, and verbal communication.

Question	Points
The Big Idea	20
Limits and Rates of Change	25
Derivative Techniques	40
Implicit Differentiation and Tangent Lines	15
Total:	100

This WPR will be released from academic security on **11FEB26 at 1610**. Prior to that time, I may not discuss any aspects of this exam with anyone except an MA104 Instructor.

Signature and date to acknowledge: Sarah Andrews 02/06/2026

1. **The Big Idea.** Use figure 1 to answer parts (a) through (c).

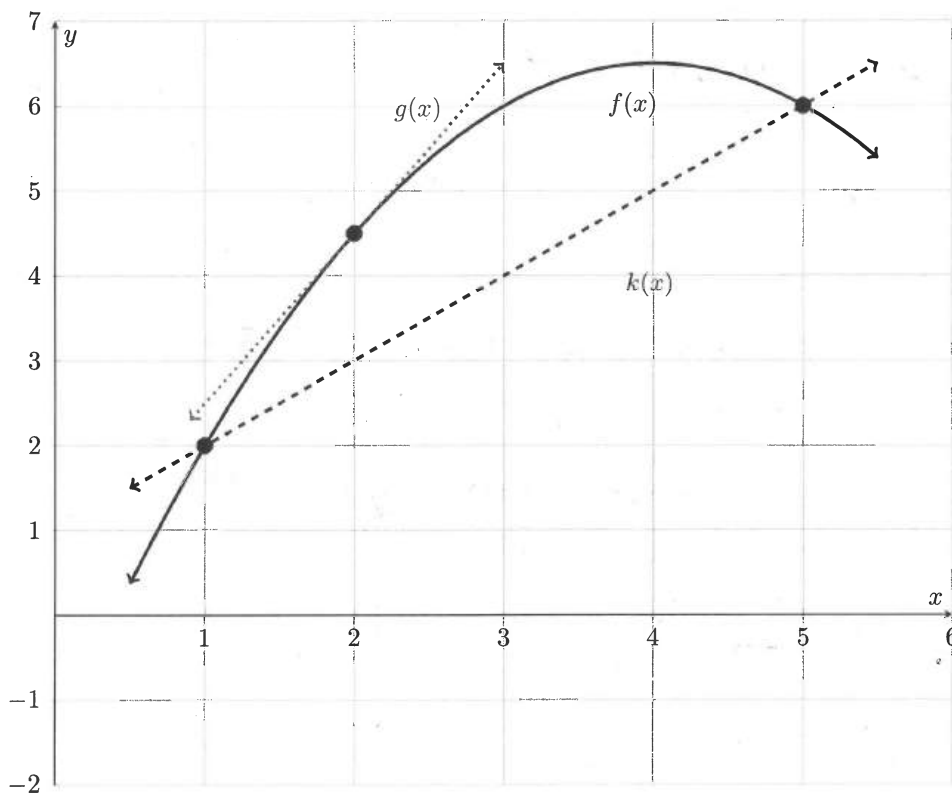


Figure 1

- (a) (5 points) The function $g(x)$ is tangent/~~secant~~ (circle one) to the function $f(x)$ in Figure 1. The slope of $g(x)$ represents an instantaneous/~~average~~ (circle one) rate of change of $f(x)$.
- (b) (10 points) Using limit notation, write the formula for the slope of $g(x)$ at $x = 2$:

$$g'(2) = \lim_{x \rightarrow 2} \frac{g(x) - g(2)}{x - 2}$$

- (c) (5 points) Write the formula used to find the slope of $k(x)$.

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{(6) - (2)}{(5) - (1)} = \frac{4}{4} = 1$$

2. Limits and Rates of Change

- (a) (5 points) Knowing that $4x - 9 \leq f(x) \leq x^2 - 4x + 7$, find $\lim_{x \rightarrow 4} f(x)$. If the limit cannot be determined from the given information, write "not possible" and explain why.

$$\lim_{x \rightarrow 4} 4x - 9 = 4(4) - 9 = 7$$

$$\lim_{x \rightarrow 4} x^2 - 4x + 7 = (4)^2 - 4(4) + 7 = 7$$

$\lim_{x \rightarrow 4} f(x) = 7$, because of the squeeze theorem.

The limits at $x \rightarrow 4$ of $4x - 9$ and $x^2 - 4x + 7$ are equal to 7, meaning that $\lim_{x \rightarrow 4} f(x)$ must fall between that.

- (b) (5 points) Evaluate: $\lim_{x \rightarrow 3} \frac{x-3}{\sqrt{x+1}-2}$

$$\lim_{x \rightarrow 3} \frac{x-3}{\sqrt{x+1}-2} \rightarrow \frac{(3)-3}{\sqrt{(3)+1}-2} \rightarrow \frac{0}{0}$$

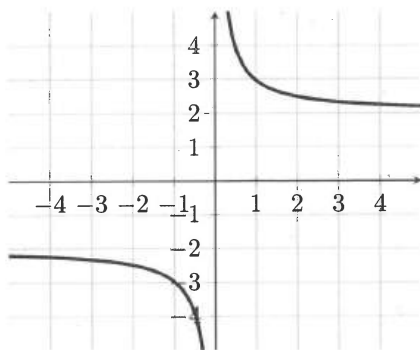
$$\lim_{x \rightarrow 2.9} \frac{x-3}{\sqrt{x+1}-2} = 3.975$$

$$\lim_{x \rightarrow 3.1} \frac{x-3}{\sqrt{x+1}-2} = 4.025$$

this means it probably goes to 4

$$\boxed{\lim_{x \rightarrow 3} \frac{x-3}{\sqrt{x+1}-2} = 4}$$

- (c) (5 points) **Limits from a Graph.** The graph of a function $f(x)$ is shown. Use the graph to determine each limit.



$$1. \lim_{x \rightarrow -\infty} f(x) = \underline{-2}$$

$$2. \lim_{x \rightarrow \infty} f(x) = \underline{2}$$

$$3. \lim_{x \rightarrow 0^-} f(x) = \underline{-\infty}$$

$$4. \lim_{x \rightarrow 0^+} f(x) = \underline{\infty}$$

$$5. \lim_{x \rightarrow 0} f(x) = \underline{DNE}$$

$$(x+h)(x+h) \\ x^2 + 2xh + h^2$$

(d) (10 points) Using the limit definition of the derivative, find the derivative of $f(x) = x^2 + x$.

$$f'(x) = \lim_{h \rightarrow 0} \frac{((x+h)^2 + (x+h)) - (x^2 + x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 + x + h - x^2 - x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2xh + h^2 + h}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(2x + h + 1)}{h}$$

$$= \lim_{h \rightarrow 0} 2x + h + 1 = 2x + (0) + 1 = 2x + 1$$

$$f'(x) = 2x + 1$$

3. Derivative Techniques. Find $\frac{dy}{dx}$ for the following equations. Do not simplify your solution.

(a) (5 points) $y = \ln(3x)$

$$\frac{dy}{dx} = \frac{1}{3x} \cdot 3$$

(b) (5 points) $y = \frac{\cos(x)}{e^x}$

$$\frac{dy}{dx} = \frac{(e^x)(-\sin x) - (\cos x)(1)}{(e^x)^2}$$

(c) (15 points) $y = \frac{(e^x)(x+1)}{x^2-1}$

$$\frac{dy}{dx} = \frac{(x^2-1)(e^x + (e^x)(x+1)) - ((e^x)(x+1))(2x)}{(x^2-1)^2}$$

$$(e^x)(1) + (e^x)(x+1)$$

★ (d) (15 points) $y = \tan(4^{3x})$

$$\frac{dy}{dx} = (\sec^2(4^{3x})) \cdot (4^{3x} \ln 4) \cdot (3)$$

$$\left(\frac{1}{2}\right)(\ln y) + (0)(\ln y)$$

$$\left(\frac{1}{2}\right)\left(\frac{1}{y}y'\right) + (0)(\ln y)$$

- ★ (15 points) **Implicit Differentiation and Tangent Lines.** Find the equation of the line that is tangent to the curve $x - \frac{1}{2}\ln(y) - 1 = 0$ at the point $(1,1)$. **Place your final equation in the provided box.**

$$x - \frac{1}{2}\ln(y) - 1 = 0$$

$$1 - \left(\left(\frac{1}{2}\right)\left(\frac{1}{y}y'\right)\right) = 0$$

$$-\left(\frac{1}{2}\right)\left(\frac{1}{y}y'\right) = -1$$

$$\frac{1}{y}y' = 2$$

$$y' = 2y$$

$$y'(1) = 2(1) = 2$$

$$y - 1 = 2(x - 1)$$

$$y - 1 = 2(x - 1)$$



UNITED STATES MILITARY ACADEMY
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TAB H

WPR #2

Outstanding Student Work

Name (First Last): Katya Withrow
MA104 AY26-2 WPR 2 (100 Points)
Time: 55 min

C Number C19189978
Instructor: CPT Ozga, G
Section: F10

Instructions—Read Carefully

Number of Extra Sheets: _____

1. Print your name on every page and ensure you have all pages. It is highly recommended that you review all questions so that you may formulate a plan of attack.
2. **References.** You are authorized to use your issued calculator and *your* MA104 Reference Card. The reference card must not be used to copy down example problems or gain an unfair advantage and must be in your own handwriting. All reference cards will be turned in with each exam and returned with the graded exam. Any reference card found with example problems or information directly from the exam (ie exam questions) could result in a 0 for the exam.
3. Cell phones, smart watches, and/or music playing devices must be placed in backpacks during the exam.
4. All work written on the WPR is subject to grading unless marked through or clearly marked with words to the effect of “do not grade.”
5. Use a continuation sheet if you need more space for a problem. **Clearly indicate** on the problem and on the continuation sheet which problem is continued, and be sure to put your name on the continuation sheet.
6. Always use proper notation. Assume all numbers are exact. Define any variables that you use which are not explicitly defined in the problem. Unless otherwise stated, assume all angles are given in radians. In problems that involve units of measurement, ensure that your answers include the appropriate units.
7. **SHOW ALL WORK.** Sufficient work that indicates your method of reasoning for the operations performed is required. This work must be done using *calculus concepts* learned in class. **Clearly indicate your final answer.**
8. Early departure is authorized. “Begin work” and “cease work” times will be given by your instructor.
9. **Academic Security.** Cadets are not authorized to discuss the content, structure, or other information about this exam until it has been released from academic security. Discussion includes all forms of written, electronic, and verbal communication.

Question	Points
The Big Idea	15
Limits & Derivatives	20
Min & Max	35
Optimization	30
Total:	100

This WPR will be released from academic security on **04MAR26 at 1610**. Prior to that time, I may not discuss any aspects of this exam with anyone except an MA104 Instructor.

Signature and date to acknowledge: Katya Withrow 27 FEB 2026

1. The Big Idea.

- (a) (5 points) Suppose $f(t)$ is a differentiable function and that its derivative $f'(t)$ is continuous for all real values of t . You are shown two points of $f'(t)$ below.

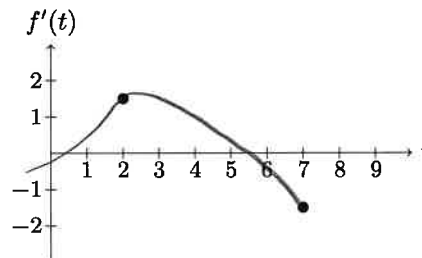
From the graph, you know that

$$f'(2) > 0$$

and

$$f'(7) < 0$$

Aside from the two points shown, you do not know what $f'(t)$ looks like.



Based on the information given, does $f(t)$ have at least one local maximum? Justify your answer.

Yes, it has at least one local max.

$f'(2) > 0$ means the function is increasing and $f'(7) < 0$ means it's decreasing. Because it's continuous and switches from increasing to decreasing ($\wedge$), there IS

- (b) (10 points) Using the following information about the first and second derivatives of a function $h(x)$, sketch a possible graph of $h(x)$ on the coordinate plane provided. Clearly label on the graph any local maximums / minimums and the point(s) of inflection.

at least one local max.

Given information

- $h(x)$ is continuous and differentiable for all real x .
- First derivative:

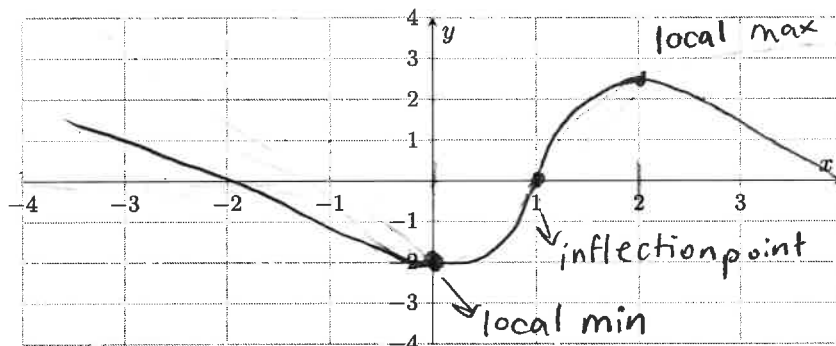
$$h'(x) \begin{cases} < 0 & \text{for } (-\infty, 0) \text{ and } (2, \infty), \text{ decreasing} \\ = 0 & \text{at } x = 0 \text{ and } x = 2, \\ > 0 & \text{for } (0, 2), \text{ increasing} \end{cases}$$

- Second derivative:

$$h''(x) \begin{cases} > 0 & \text{for } (-\infty, 1), \quad \cup \text{ local min} \\ = 0 & \text{at } x = 1, \quad \longrightarrow \text{inflection} \\ < 0 & \text{for } (1, \infty), \quad \cap \text{ local max} \end{cases}$$

- Reference points:

$$h(0) = -2, \quad h(1) = 0.$$



2. Limits & Derivatives.

(a) (10 points) Evaluate $\lim_{x \rightarrow 0} \frac{\cos(x) - 1 + x^2}{x^2}$

$$f'(x) = \lim_{x \rightarrow 0} \frac{-\sin(x) + 2x}{2x} = \frac{0}{0}$$

$$f''(x) = \lim_{x \rightarrow 0} \frac{-\cos(x) + 2}{2} = \frac{-\cos(0) + 2}{2} = \frac{1}{2}$$

$$\lim_{x \rightarrow 0} \frac{\cos x - 1 + x^2}{x^2} = \frac{1}{2}$$

(b) (10 points) **Related Rates (Rectangle Area)**

A rectangle has length l (meters) and width w (meters). Both sides are increasing over time. The area of the rectangle is

$$A = lw.$$

At an instant:

$$l = 8 \text{ m}, \quad w = 5 \text{ m}$$

Rates of change:

$$\frac{dl}{dt} = 0.4 \text{ m/min}$$

$$\frac{dw}{dt} = 0.3 \text{ m/min}$$

Find the rate of change of the area $\left(\frac{dA}{dt}\right)$ at that instant.

$$\frac{dA}{dt} = A'$$

$$A = lw$$

$$A' = \left(1 \frac{dw}{dt}\right)(l) + \left(1 \frac{dl}{dt}\right)(w)$$

$$\frac{dA}{dt} = (0.3)(8) + (0.4)(5)$$

2.4 2

$$\frac{dA}{dt} = 4.4$$

$$\frac{dA}{dt} = 4.4 \text{ m/min}$$

3. Min & Max. (Show all work and justify your answers)

(a) For question (i) and (ii) below $f(x) = x^4 - 4x^3$,

(i) (10 points) List all the critical numbers of $f(x)$.

$$f'(x) = 4x^3 - 12x^2$$

$$f'(x) = 4x^2(x-3) = 0$$

Critical numbers,

$$\downarrow$$

$$3, 0$$

$$x = 3$$

$$x = 0$$

(ii) (10 points) Find all of the inflection points of $f(x)$.

$$f''(x) = 12x^2 - 24x$$

$$f''(x) = 12x(x-2) = 0$$

plug into OG:

$$(2)^4 - 4(2)^3 = -16$$

$$(0)^4 - 4(0)^3 = 0$$

$$2, 0$$

$$12(-1)^2 - 24(-1) = 36 > 0$$

$$12(1)^2 - 24(1) = -12 < 0$$

$$12(3)^2 - 24(3) = 36 > 0$$

$$(2, -16)$$

$$(0, 0)$$

(b) (15 points) **This is a different problem.** What is the value of the absolute minimum of $h(x)$ on the interval $[-1, 2]$? **SHOW ALL WORK.**



$$x = 2$$

$$x = 0$$

changes sign so confirms inflections

$$h(x) = 2x^3 - 15x^2$$

$$h'(x) = 6x^2 - 30x$$

$$h'(x) = 6x(x-5)$$

0, 5 as critical numbers

↓
not with interval

x	f(x)
-1	-17
0	0
2	-44

$(2, -44)$ is abs min.

$$2(-1)^3 - 15(-1)^2 = -17$$

$$2(0)^3 - 15(0)^2 = 0$$

$$2(2)^3 - 15(2)^2 = -44$$

$$(2, -44)$$

4. Optimization.

- (a) You are redesigning the Army's cylindrical fuel tankers to improve material efficiency. Each tanker is modeled as a **closed cylinder** (including the top and bottom). For mission requirements, the tanker must hold a volume of 1000 m^3 . Your goal is to determine the cylinder dimensions that produce the **minimum surface area** while maintaining the required volume.

Volume and Surface Area Formulas are printed on your Note Card.

- (i) (5 points) What is the constraint?

$$V = \pi r^2 h$$

$$\pi r^2 h = 1000 \text{ m}^3$$

- (ii) (10 points) Using your solution from part (i) what is the objective function in terms of one variable?

$$\pi r^2 h = 1000$$

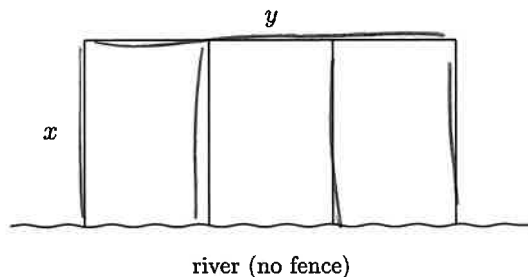
$$h = \frac{1000}{\pi r^2}$$

$$\text{Minimize } SA(r, h) = 2\pi r^2 + 2\pi r(h)$$

$$\text{Minimize } SA(r) = 2\pi r^2 + 2\pi r\left(\frac{1000}{\pi r^2}\right)$$

- (b) (15 points) **This is a new problem:** You are designing a rectangular fenced area with two interior dividers and side lengths x and y in meters. Since these are lengths, $x > 0$ and $y > 0$. One side of the enclosure borders a river, so no fence is required along that side. The area of the fenced in region is given by Equation (1) and is subject to the constraint in Equation (2).

Determine the value of x that maximizes the area. Verify your solution.



Fence with two interior dividers (river on one side)

$$\text{maximize } A(x, y) = xy \quad (1)$$

$$\text{subject to } 4x + y = 1000 \quad (2)$$

$$4x + y = 1000$$

$$y = 1000 - 4x$$

$$y = 1000 - 4(125)$$

$$y = 1000 - 500$$

$$y = 500$$

$$A(x) = x(1000 - 4x)$$

$$A(x) = 1000x - 4x^2$$

$$A'(x) = 1000 - 8x$$

$$1000 - 8x = 0$$

$$\frac{1000}{8} = \frac{8x}{8}$$

$$x = 125$$

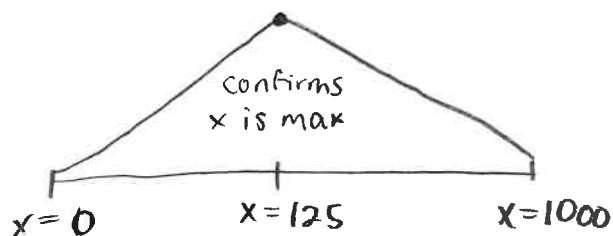
↓
maximizes the area

$$A =$$

$$(125)(500) =$$

$$62500 \text{ m}^2$$

max area



$$1000 - 8(0) = 1000 > 0 \text{ so increasing}$$

$$1000 - 8(1000) = -7000 < 0 \text{ so decreasing}$$

$$x = 125$$

← Verified



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB I

WPR #3

Outstanding Student Work

Start: 9:05

Name (First Last): Grace Houghton
MA104 AY26-2 WPR 3 (100 Points)
Time: 75 min

C Number _____
Instructor: Dr. Libertini
Section: C12

Instructions—Read Carefully

Number of Extra Sheets: _____

1. Print your name on every page and ensure you have all pages. It is highly recommended that you review all questions so that you may formulate a plan of attack.
2. **References.** You are authorized to use your issued calculator and *your* MA104 Reference Card. The reference card must not be used to copy down example problems or gain an unfair advantage and must be in your own handwriting. All reference cards will be turned in with each exam and returned with the graded exam. Any reference card found with example problems or information directly from the exam (ie exam questions) could result in a 0 for the exam.
3. Cell phones, smart watches, and/or music playing devices must be placed in backpacks during the exam.
4. All work written on the WPR is subject to grading unless marked through or clearly marked with words to the effect of “do not grade.”
5. Use a continuation sheet if you need more space for a problem. **Clearly indicate** on the problem and on the continuation sheet which problem is continued, and be sure to put your name on the continuation sheet.
6. Always use proper notation. Assume all numbers are exact. Define any variables that you use which are not explicitly defined in the problem. Unless otherwise stated, assume all angles are given in radians. In problems that involve units of measurement, ensure that your answers include the appropriate units.
7. **SHOW ALL WORK.** Sufficient work that indicates your method of reasoning for the operations performed is required. This work must be done using *calculus concepts* learned in class. **Clearly indicate your final answer.**
8. Early departure is authorized. “Begin work” and “cease work” times will be given by your instructor.
9. **Academic Security.** Cadets are not authorized to discuss the content, structure, or other information about this exam until it has been released from academic security. Discussion includes all forms of written, electronic, and verbal communication.

Question	Points
The Big Idea	20
Derivatives and Integrals	20
Integration Strategies 1	20
Integration Strategies 2	20
Natural Sciences	20
Total:	100

This WPR will be released from academic security on **27MAR26 at 1610**. Prior to that time, I may not discuss any aspects of this exam with anyone except an MA104 Instructor.

Signature and date to acknowledge: _____

[Signature]

27 MAR 26

1. **The Big Idea.** Below is the plot of the velocity $v(t)$ of a model rocket launched straight up from the parade field. Use the plot of velocity in **Figure 1** to answer the questions below. Assume $t = 0$ is when the rocket is launched.

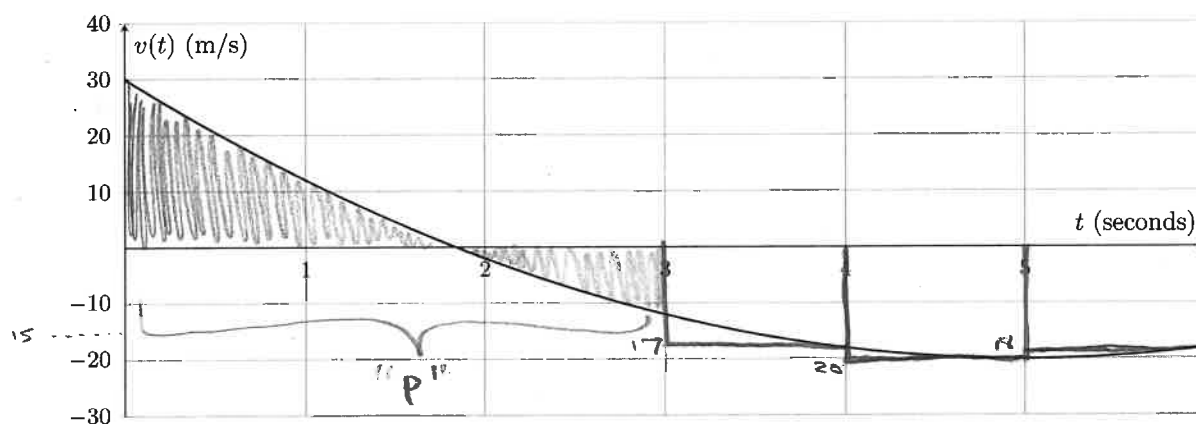


Figure 1: The function for the velocity of the rocket, $v(t)$

- (a) (5 points) Shade the region defined by $\int_0^3 v(t) dt$ on the graph. **Label your sketch as "P".**
- (b) (5 points) Provide your best guess as to whether the region "P" is net positive or net negative? Justify your guess.
- I believe the region has a net positive gain as there is a larger positive shaded region over negative region.
- (c) (5 points) Using **right endpoints** and exactly three rectangles, draw the approximating rectangles on Figure 1 that represent the Riemann sum estimate of $\int_3^6 v(t) dt$.
- (d) (5 points) Using the rectangles from part (c), estimate $\int_3^6 v(t) dt$. Then, use your approximation to estimate the total distance traveled by the rocket on the interval $t = 3$ to $t = 6$. Explain how you determined the total distance.

$$-18(1) - 20(1) - 17(1) = -55$$

By taking the area of each rectangle and adding them all together I was able to approximate the total distance traveled, but distance can only be positive, so it is positive.

$$\int_3^6 v(t) dt \approx 55$$

2. **Derivatives and Integrals.** Evaluate the following. Write your answer in the box provided.

(a) (2 points) $\frac{d}{dx} \left(\frac{4x}{x^4} \right) 4(x^{-3})$

$$\frac{d}{dx} 4x^{-3} = -12x^{-4}$$

$$-12x^{-4}$$

(b) (2 points) $\int \frac{4x}{x^4} dx \quad \int 4(x^{-3}) dx$

$$4 \int x^{-3} dx = 4 \frac{x^{-2}}{-2}$$

$$-2x^{-2} + C$$

(c) (4 points) $\frac{d}{dx} (\cos^3(x^3))$

$$-3\sin^2(x^3)(3x^2)$$

$$-3\sin^2(x^3)(3x^2)$$

(d) (4 points) $\int (5e^x - 3^x) dx$

$$\int 5e^x dx - \int 3^x dx$$

$$5e^x - \frac{1}{\ln 3} 3^x + C$$

$$5e^x - \frac{1}{\ln 3} 3^x + C$$

(e) (4 points) $\int \left(\frac{-5}{\sqrt{1-x^2}} - \sec^2(x) \right) dx$

$$\int \frac{-5}{\sqrt{1-x^2}} dx - \int \sec^2(x) dx$$

$$-5 \int \frac{1}{\sqrt{1-x^2}} dx$$

$$-5 \arcsin(x) - \tan(x) + C$$

$$-5 \arcsin(x) - \tan(x) + C$$

(f) (4 points) $\int (x^3 - \sin(x)) dx$

$$\int x^3 dx - \int \sin(x) dx$$

$$\frac{1}{4} x^4 + \cos(x) + C$$

$$\frac{1}{4} x^4 + \cos(x) + C$$

3. **Integration Strategies 1.** Evaluate the following integrals. You must show all work.

(a) (10 points) $\int x^2 \cos(x^3 + 2) dx$ $u = x^3 + 2$ $dx = \frac{du}{3x^2}$
 $\frac{du}{dx} = 3x^2$

$$\int \cancel{x^2} \cos(u) \frac{du}{3\cancel{x^2}}$$

$$\frac{1}{3} \int \cos(u) du$$

$$\frac{1}{3} \sin(x^3 + 2) + C$$

$$\frac{1}{3} \sin(x^3 + 2) + C$$

(b) (10 points) $\int \ln(x) x^2 dx$

$$uv - \int v du$$

$$u = \ln(x) \quad v = \frac{1}{3}x^3$$

$$du = \frac{1}{x} \quad dv = x^2$$

$$\int \ln(x) x^2 dx = \ln(x) \left(\frac{1}{3}x^3 \right) - \int \frac{1}{3}x^3 \frac{1}{x}$$

$$\ln(x) \left(\frac{1}{3}x^3 \right) - \frac{1}{3} \int x^2$$

$$\ln(x) \left(\frac{1}{3}x^3 \right) - \frac{1}{9}x^3 + C$$

$$\ln x \frac{1}{3}x^3 - \frac{1}{9}x^3 + C$$

4. **Integration Strategies 2.** Evaluate the following integrals. You must show all work to include the **Fundamental Theorem of Calculus, Part 2** to receive credit.

(a) (10 points) $\int_{-1}^1 x e^x dx$

Fundamental theorem of calculus:
 $\int_a^b f(x) dx = F(x)|_a^b = F(b) - F(a)$

$$\begin{aligned} \int_{-1}^1 x e^x dx &= e^x(x-1) \Big|_{-1}^1 = e^1(\cancel{1-1})^0 - (e^{-1}(-1-1)) \\ &= -(e^{-1}(-2)) \end{aligned}$$

$$2e^{-1}$$

(b) (10 points) $\int_1^e \frac{4x}{x^2} dx$

$$\int_1^e 4x^{-1} dx = 4 \int_1^e x^{-1} dx = 4 \int_1^e \frac{1}{x} dx =$$

$$4 \ln|x| \Big|_1^e = 4 \ln \cancel{e}^1 - 4 \ln \cancel{1}^0$$

$$4(1) - 0 = 4$$

$$4$$

5. **Natural Sciences.** A sandbag is thrown straight down from a hovering helicopter with an initial downward velocity of -3 m/s from a height of 50 m above the ground. The sandbag experiences gravity and air resistance according to the acceleration function in Equation (1).

$$a(t) = -8.4t \quad (1)$$

- (a) (15 points) Derive $y(t)$, the vertical position function of the sandbag.

$$\begin{aligned}
 a(t) &= -8.4t & \int a(t) dt &= v(t) \\
 v(t) &= -\frac{8.4t^2}{2} - 3 & \int -8.4t dt &= -\frac{8.4t^2}{2} + C \quad \text{where } C = -3 \\
 y(t) &= -\frac{8.4}{6}t^3 - 3t + 50 & \int v(t) dt &= y(t) \\
 & & \int -\frac{8.4t^2}{2} - 3 dt &= -\frac{8.4}{6}t^3 - 3t + C \\
 & & & \text{where initial height is } 50
 \end{aligned}$$

$$y(t) = -\frac{8.4}{6}t^3 - 3t + 50$$

- (b) (5 points) Write the equation needed to determine how long it takes for the sandbag to hit the ground. **Do not solve.**

I set the position equation to where the height is 0, meaning the sand bag has already hit the ground and is therefore at 0 m. This will allow me to solve for t , time and determine how long it takes for the sand bag to hit the ground.

$$0 = -\frac{8.4}{6}t^3 - 3t + 50$$



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TAB J

WPR #4

Outstanding Student Work

Name (First Last): Franklin Li C Number C96036367
MA104 AY26-2 WPR 4 (100 Points) Instructor: MAJ OZga
Time: 75 min Section: F10

Instructions—Read Carefully

Number of Extra Sheets: _____

1. Print your name on every page and ensure you have all pages. It is highly recommended that you review all questions so that you may formulate a plan of attack.
2. **References.** You are authorized to use your issued calculator and *your* MA104 Reference Card. The reference card must not be used to copy down example problems or gain an unfair advantage and must be in your own handwriting. All reference cards will be turned in with each exam and returned with the graded exam. Any reference card found with example problems or information directly from the exam (ie exam questions) could result in a 0 for the exam.
3. Cell phones, smart watches, and/or music playing devices must be placed in backpacks during the exam.
4. All work written on the WPR is subject to grading unless marked through or clearly marked with words to the effect of “do not grade.”
5. Use a continuation sheet if you need more space for a problem. **Clearly indicate** on the problem and on the continuation sheet which problem is continued, and be sure to put your name on the continuation sheet.
6. Always use proper notation. Assume all numbers are exact. Define any variables that you use which are not explicitly defined in the problem. Unless otherwise stated, assume all angles are given in radians. In problems that involve units of measurement, ensure that your answers include the appropriate units.
7. **SHOW ALL WORK.** Sufficient work that indicates your method of reasoning for the operations performed is required. This work must be done using *calculus concepts* learned in class. **Clearly indicate your final answer.**
8. Early departure is authorized. “Begin work” and “cease work” times will be given by your instructor.
9. **Academic Security.** Cadets are not authorized to discuss the content, structure, or other information about this exam until it has been released from academic security. Discussion includes all forms of written, electronic, and verbal communication.

Question	Points
The Big Idea	20
Derivatives and Integrals	5
Area Between Curves	15
Improper Integrals	15
Integration Application	45
Total:	100

This WPR will be released from academic security on **01MAY26 at 1610**. Prior to that time, I may not discuss any aspects of this exam with anyone except an MA104 Instructor.

Signature and date to acknowledge: Franklin Li 29APR2026

1. **The Big Idea.** Below is the probability density function $f(x)$ representing the distribution of Soldier reaction times (in seconds) in response to an alert signal. **Figure 1** and **Figure 2** are identical figures reproduced on separate pages for your convenience. Please refer to the figure on the same page as each question when answering.

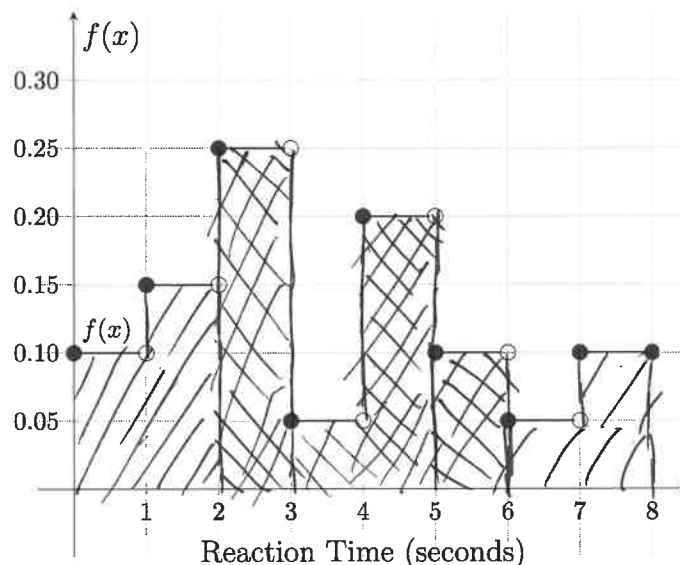


Figure 1: Probability density function $f(x)$ of Soldier reaction times (in seconds).

- (a) (4 points) The region $\int_0^2 f(x) dx$ and the region $\int_6^8 f(x) dx$ represent the probabilities that a Soldier reacts in the first 2 seconds and after 6 seconds, respectively. **Without computing an exact value**, determine which region has a greater probability. Justify your answer.

Both regions have the same probability because they take up the same area under the curve (5 squares). Because the area is equal, the probabilities will be equal.

- (b) (4 points) Use a left Riemann sum with $n = 4$ equal subintervals to approximate $\int_2^6 f(x) dx$. Clearly show all terms of your sum.

$$(3-2)(0.25) + (4-3)(0.05) + (5-4)(0.20) + (6-5)(0.10) = \boxed{0.6}$$

- (c) (4 points) Using your result from part (b), determine whether a randomly selected Soldier is more likely to react between 2 and 6 seconds or more likely to react with some other reaction time. Justify your answer.

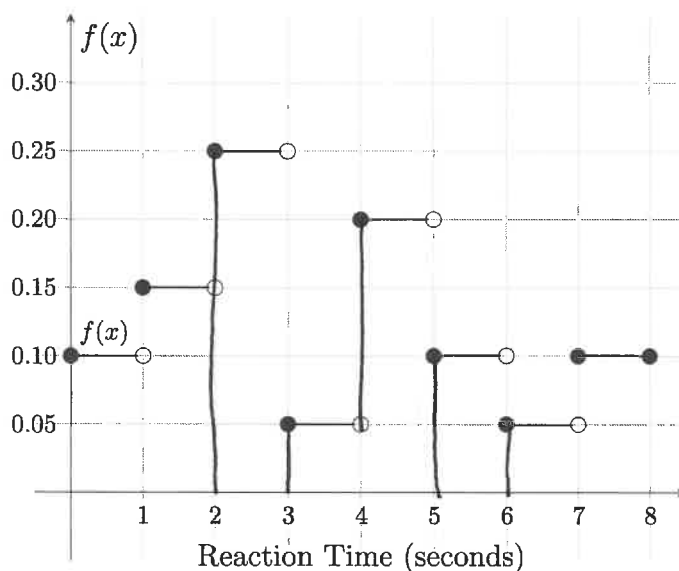
$$(1)(0.10) + 1(0.15) = 0.25$$

$$(1)(0.10) + 1(0.05) + 1(0.10) = 0.25$$

A Soldier ~~has~~ is more likely to react between

2-6 seconds than any other reaction time

because the probability is greater between 2-6 seconds than it is between both 0-2 seconds or 6-8 seconds

Figure 2: Probability density function $f(x)$ of Soldier reaction times (in seconds).

- (d) (4 points) Use a **right Riemann sum** with $n = 4$ equal subintervals to approximate $\int_2^6 f(x) dx$. Clearly show all terms of your sum.

$$(3-2)(0.05) + (4-3)(0.20) + (5-4)(0.10) + (6-5)(0.05) \\ = 0.4$$

- (e) (4 points) What value on the x -axis is the best estimate of the **median** reaction time? Justify your selection.

$$(1-0)(0.10) + (2-1)(0.15) + (3-2)(0.25)$$

$$= 0.25 + 0.25 = 0.5$$

$$\int_0^m f(x) dx = 0.5$$

$x=3$ is the best estimate of the median reaction time because the area from $x=0$ to $x=3$ under the curve is equal to 0.5

2. (5 points) **Derivatives and Integrals.** Evaluate the following.

$$\frac{d}{dx} (\sin(x^2 + x))^3 \\ = 3(\sin(x^2 + x))^2 \cdot (\sin(x^2 + x))' \\ = 3(\sin(x^2 + x))^2 \cdot (2x + 1) \cos(x^2 + x)$$

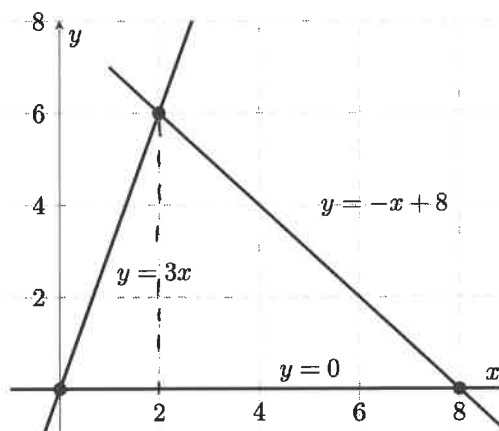
3. (15 points) **Area Between the Curves.**Set up the integral(s) to find the area enclosed by Equations 1, 2, and 3. **Do not solve.**

(1) $y = 0$ (1)

(2) $y = 3x$ (2)

(3) $y = -x + 8$ (3)

$$\int_0^2 3x \, dx + \int_2^8 -x + 8 \, dx$$

4. (15 points) **Improper Integrals.** Determine if the following integral converges or diverges. If it converges, indicate the value.

$$\int_1^{\infty} \frac{1}{x^3} \, dx = \int_1^{\infty} x^{-3} \, dx$$

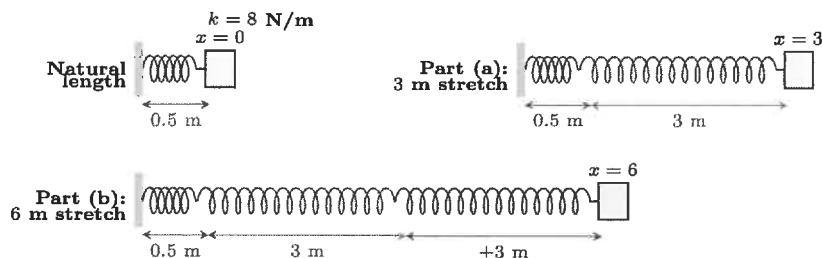
$$= \lim_{t \rightarrow \infty} \int_1^t x^{-3} \, dx = \lim_{t \rightarrow \infty} \left[-\frac{1}{2} x^{-2} \right]_1^t$$

$$= \lim_{t \rightarrow \infty} \left[-\frac{1}{2t^2} - \left(-\frac{1}{2(1)^2} \right) \right] = 0 - \left(-\frac{1}{2} \right)$$

The integral has a value of $\frac{1}{2}$, $= \frac{1}{2}$
 & it converges

5. Integration Applications.

- (a) **Work.** A spring has a natural length of 0.5 meters and a spring constant of $k = 8 \text{ N/m}$. Find the work required to stretch the spring and answer all three parts below.



- i. (6 points) Find the work required to stretch the spring from its natural length to 3 meters beyond its natural length.

$$W = \int_0^3 8x \, dx = \left. \frac{8}{2} x^2 \right|_0^3 = 4(3)^2 - 4(0)^2 = \boxed{36 \text{ J}}$$

$F(x) = kx, \quad k = 8$

- ii. (6 points) Find the work required to stretch the spring from 3 meters beyond its natural length to 6 meters beyond its natural length.

$$W = \int_3^6 8x \, dx = \left. \frac{8}{2} x^2 \right|_3^6 = 4(6)^2 - 4(3)^2 = 108 \text{ J}$$

- iii. (3 points) Do your answers make sense? Each interval has the same length of 3 meters, yet the two values of work are not equal. Explain why this is the case.

My answers make sense because the force ~~is~~ of

the spring is not constant, which means that the area under the curve for force will increase with a non-constant rate, which is why the 2 values of work are not equal.

- (b) (15 points) Average Value. Find the average value of the function on the given interval:

$$\ln(e) = 1$$

$$h(x) = \frac{\ln(x)}{x}, \quad [1, e]$$

$$u = \ln x$$

$$du = \frac{1}{x} dx$$

$$dx = \cancel{\frac{1}{x}} x du$$

$$\ln(1) = 0$$

$$\frac{1}{1-e} \int_1^e \ln x \cdot \frac{1}{x} dx$$

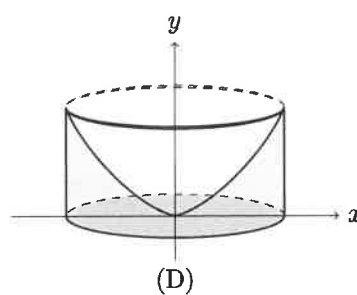
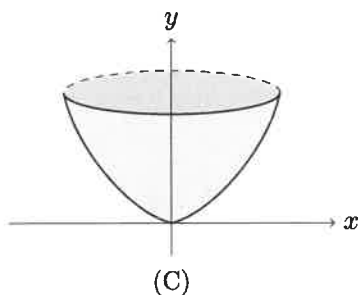
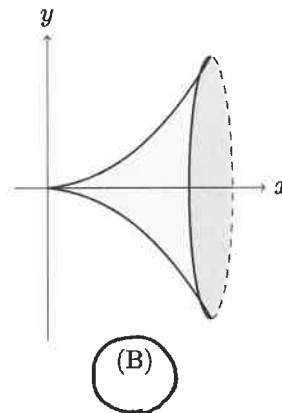
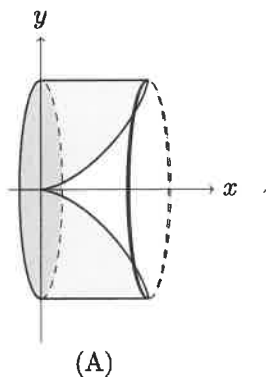
$$= \frac{1}{1-e} \int_0^1 u \cdot \frac{1}{x} \cdot x du$$

$$= \frac{1}{1-e} \int_0^1 u du$$

$$= \frac{1}{1-e} \left[\frac{1}{2} u^2 \right]_0^1 = \frac{1}{1-e} \left[\left(\frac{1}{2} (1)^2 \right) - \left(\frac{1}{2} (0)^2 \right) \right] = \boxed{\frac{1}{1-e} \left(\frac{1}{2} \right)}$$

(c) Volume.

- i. (5 points) Circle the region obtained by rotating about the x -axis the region under the curve $y = x^2$ and above $y = 0$ from $x = 0$ to $x = 1$.



- ii. (10 points) Find the volume of the solid obtained by rotating about the x -axis the region under the curve $y = x^2$ and above $y = 0$ from $x = 0$ to $x = 1$.

$$\begin{aligned}
 & \pi \int_0^1 (x^2)^2 dx \\
 &= \pi \int_0^1 x^4 dx \\
 &= \pi \left[\frac{1}{5} x^5 \right]_0^1 \\
 &= \pi \left[\frac{1}{5} (1)^5 - \frac{1}{5} (0)^5 \right] \\
 &= \pi \cdot \frac{1}{5} = \boxed{\frac{1}{5} \pi}
 \end{aligned}$$



UNITED STATES MILITARY ACADEMY
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TAB K

TERM ENDING EXAM

Outstanding Student Work

C Number: C24837614 Name (First Last): Jacob Shevchuk
MA104 AY26-2 TEE (Points: 200 Points) Instructor: CPT Power
Time: 3.5 Hours Section: D15

Instructions—Read Carefully

Number of Extra Sheets: _____

1. Print your name on every page and ensure you have all pages. It is highly recommended that you review all questions so that you may formulate a plan of attack.
2. **References.** You are authorized your issued calculator and *your* MA104 Reference Card. The reference card will not be used to copy down example problems or gain an unfair advantage and must be in your own handwriting. All reference cards will be turned in with each TEE. Any reference card found with information directly from the exam (ie TEE questions) could result in a 0 for the TEE.
3. Cell phones, smart watches and/or music playing devices must be placed in backpacks during the TEE.
4. All work written on the TEE is subject to grading unless marked through or clearly marked with words to the effect of “do not grade.”
5. Use a continuation sheet if you need more space for a problem. Clearly write “see continuation sheet” at the problem and label the problem number on the continuation sheet. Put your name on the continuation sheet.
6. Always use proper notation. Define any variables that you use which are not explicitly defined in the problem. Unless otherwise stated, assume all angles are given in **RADIANS**. In problems that involve units of measurement, ensure that your answers include the appropriate units.
7. **SHOW ALL WORK.** Sufficient work that indicates your method of reasoning for the operations performed is required. This work must be done using *calculus concepts* learned in class. **Clearly indicate your final answer.** Definite integrals may be verified using a calculator; however, all supporting work must be shown and match your final answer.
8. Early departure is authorized. “Begin work” and “cease work” times will be given by your instructor.
9. If answer boxes are provided, please ensure your answer is written completely in the box using proper notation.

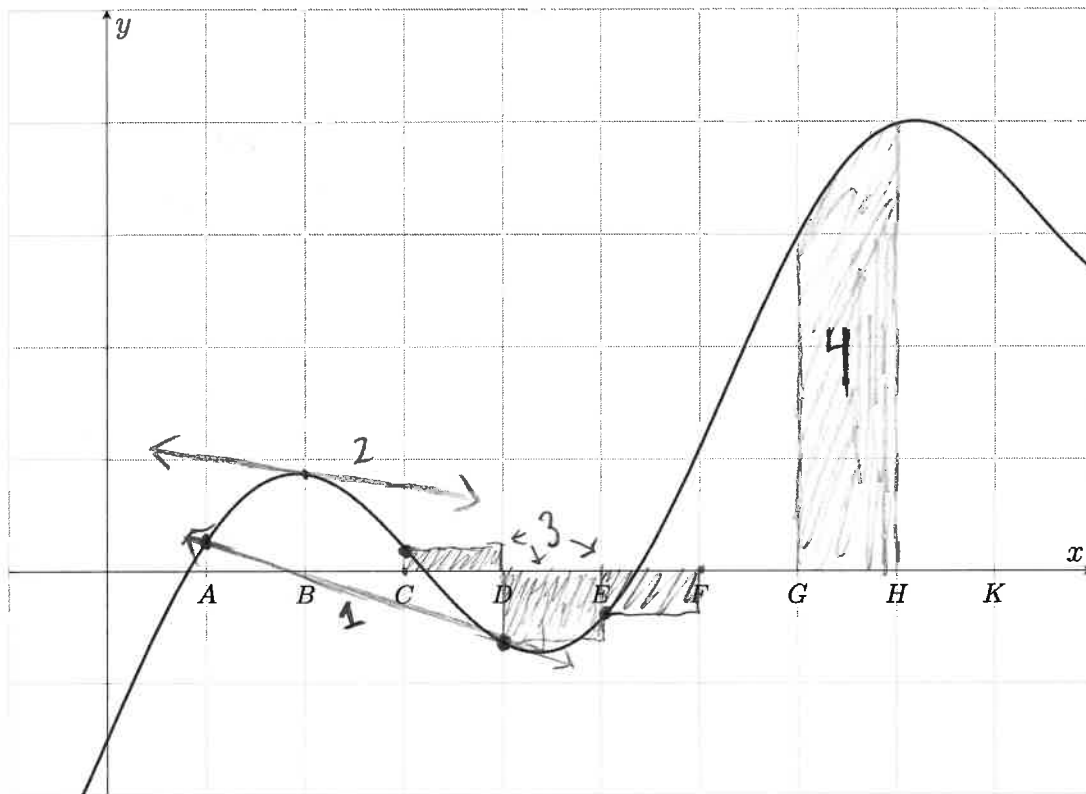
Question	Points
The Big Idea	35
Limits	20
Basic Derivatives	10
Derivative Rules	25
Optimization	25
Basic Integrals	20
Integration Strategies	30
Integration Applications	35
Total:	200

This TEE will not be released from academic security. I may not discuss any aspects of this exam with anyone except an MA104 Instructor.

Signature and date to acknowledge:  14 May 2026

1. (35 Points) **The Big Idea.**

(a) Use figure 1 to answer the following questions on this page.

Figure 1: The function $f(x)$

- i. (5 points) On figure 1, sketch and label as “1” the line
- $y - f(A) = m_1(x - A)$
- where

$$m_1 = \frac{f(D) - f(A)}{D - A}$$

$$\frac{y_2 - y_1}{x_2 - x_1} \left(A, f(A) \right) \left(D, f(D) \right)$$

- ii. (5 points) On figure 1, sketch and label as “2” the line
- $y - f(B) = m_2(x - B)$
- where

$$m_2 = \lim_{x \rightarrow B} \frac{f(x) - f(B)}{x - B}$$

- iii. (5 points) On figure 1, sketch and label as “3” the Riemann Sum (clearly depict the rectangles) indicated by the below equation using the left endpoint method when
- $n = 3$
- .

$$\sum_{i=1}^3 f(x_i) \Delta x \approx \int_C^F f(x) dx$$

- iv. (5 points) On figure 1, sketch and label as “4” the region indicated by
- $\int_G^H f(x) dx$
- .

- (b) (5 points) Find the equation of the tangent line to $f(x) = \frac{5x^3 - 6x}{x}$ at $x = 1$.

$$f(x) = \frac{5x^3 - 6x}{x} \rightarrow 5x^2 - 6$$

$$f'(x) = 10x$$

$$f'(x) = 10(1) = 10 \text{ so } m = 10$$

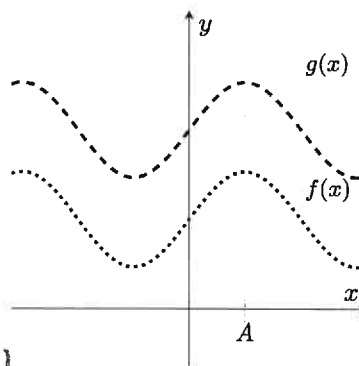
$$-1 = 10(1) + b$$

$$-10 = b$$

$$y = 10x - 11$$

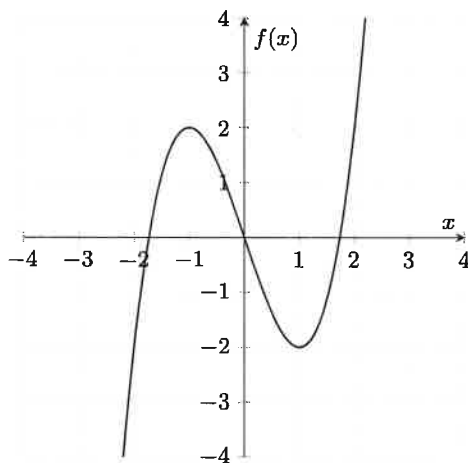
- (c) (5 points) The graph below shows two functions $f(x)$ and $g(x)$. Without calculating, determine whether $f'(A)$ is greater than, less than, or equal to $g'(A)$. Justify your answer.

The $f'(A)$ is equal to that of $g'(A)$ because the derivative is simply the slope at a point, and the functions $f(x)$ and $g(x)$ do not change the actual slope of any point. Since the slopes of these two functions are identical so are $f'(A)$ and $g'(A)$.



- (d) (5 points) The graph of $f(x)$ is shown below. Using the graph, complete the table:

- State all of the inflection point(s).
- State the intervals where $f'(x) > 0$ and $f'(x) < 0$.
- State the intervals where $f''(x) > 0$ and $f''(x) < 0$.



Inflection Point(s):	$x = 0$
$f'(x) > 0$	$(-\infty, -1) \cup (1, \infty)$
$f'(x) < 0$	$(-1, 1)$
$f''(x) > 0$	$(0, \infty)$
$f''(x) < 0$	$(-\infty, 0)$

2. (20 Points) Limits

(a) Evaluate the following limits, you must show all work.

i. (4 points) $\lim_{x \rightarrow 0} \frac{x^2 - 7x + 10}{x^2 - 2x}$

$$\lim_{x \rightarrow 0} \frac{(x-5)(x-2)}{x(x-2)} \quad \lim_{x \rightarrow 0^+} \frac{x-5}{x} = (-) \quad \lim_{x \rightarrow 0^-} \frac{x-5}{x} = (+) \quad \text{limit DNE}$$

$$\lim_{x \rightarrow 0} \frac{x-5}{x}$$

$$\lim_{x \rightarrow 0} \frac{x^2 - 7x + 10}{x^2 - 2x} = \text{DNE}$$

ii. (4 points) $\lim_{x \rightarrow 0} \frac{\cos(4x) - 1}{x}$

$$\lim_{x \rightarrow 0} \frac{-\sin(4x) \cdot 4}{1} \approx -4\sin(4x) \quad \lim_{x \rightarrow 0} \frac{1-1}{0} = 0$$

$$\lim_{x \rightarrow 0} -4\sin(4x) = 0$$

$$\lim_{x \rightarrow 0} \frac{\cos(4x) - 1}{x} = 0$$

iii. (4 points) $\lim_{x \rightarrow 3} \frac{x^2 - 7x + 10}{x^2 - 2x}$

$$\lim_{x \rightarrow 3} \frac{9 - 21 + 10}{9 - 6} = \frac{-2}{3}$$

$$\lim_{x \rightarrow 3} \frac{x^2 - 7x + 10}{x^2 - 2x} = -\frac{2}{3}$$

iv. (4 points) $\lim_{x \rightarrow 2} \frac{x^2 - 7x + 10}{x^2 - 2x}$

$$\lim_{x \rightarrow 2} \frac{2x-7}{2x-2} = \frac{-3}{2}$$

$$\lim_{x \rightarrow 2} \frac{0-4+10}{0-4} = 0 \quad \text{L'H}$$

$$\lim_{x \rightarrow 2} \frac{x^2 - 7x + 10}{x^2 - 2x} = \frac{-3}{2}$$

v. (4 points) $\lim_{x \rightarrow \infty} \frac{2\sqrt{2}(x) + 2x^2}{6x^2 - 2x}$

$$\lim_{x \rightarrow \infty} \frac{\frac{2\sqrt{2}(x)}{x^2} + \frac{2x^2}{x^2}}{\frac{6x^2}{x^2} - \frac{2x}{x^2}} \approx \lim_{x \rightarrow \infty} \frac{\frac{2}{x} + 2}{6 - \frac{2}{x}} \approx \frac{2}{6} = \frac{1}{3}$$

$$\lim_{x \rightarrow \infty} \frac{2\sqrt{2}(x) + 2x^2}{6x^2 - 2x} = \frac{1}{3}$$

3. (10 Points) **Basic Derivatives.** Find the derivative of each of the following functions.

(a) (2 points) $f(x) = 4x^{-3}$

$$= 12x^{-4}$$

(b) (2 points) $h(x) = \sqrt{x}$

$$x^{1/2}$$

$$\frac{1}{2}x^{-1/2}$$

$$f'(x) = -12x^{-4}$$

$$h'(x) = \frac{1}{2}x^{-1/2}$$

(c) (2 points) $p(x) = 9^x$

$$9^x \ln 9$$

(d) (2 points) $y(x) = 10e^\pi$

$$10 \cdot \# = 0$$

$$p'(x) = 9^x \ln 9$$

$$y'(x) = 0$$

(e) (2 points) $q(x) = \arccos(x)$ ~~$\approx (0.5)^{-1}(x)$~~

$$q'(x) = \frac{-1}{\sqrt{1-x^2}}$$

$$q'(x) = \frac{-1}{\sqrt{1-x^2}}$$

4. (25 Points) **Derivative Rules.** Find $\frac{dy}{dx}$ for the following. Simplifying $\frac{dy}{dx}$ is not required.

(a) (5 points) $y = 5x^2 \ln(x)$

Product rule

$$y' = (5x^2)\left(\frac{1}{x}\right) + (\ln(x))(10x)$$

$$\frac{dy}{dx} = (5x^2)\left(\frac{1}{x}\right) + (\ln(x))(10x)$$

(b) (5 points) $y = \frac{4x^2 + 2x}{\tan(x)}$

Quotient rule

$$y' = \frac{(\tan(x))(8x+2) - (4x^2+2x)(\sec^2(x))}{\tan^2(x)}$$

$$(\tan(x))^2 \approx \tan^2(x)$$

$$\frac{dy}{dx} = \frac{(\tan(x))(8x+2) - (4x^2+2x)(\sec^2(x))}{\tan^2(x)}$$

(c) (5 points) $y = e^{3\sin(x)}$

$$y' = e^{3\sin(x)} \cdot 3\cos(x)$$

$$\frac{dy}{dx} = e^{3\sin(x)} \cdot 3\cos(x)$$

Derivative Rules Cont. Find $\frac{dy}{dx}$ for the following. Simplifying $\frac{dy}{dx}$ is not required.

(d) (5 points) $y = (e^x \cos(x))^{19}$

Chain rule

$$y' = 19(e^x \cos(x))^{18} \cdot ((e^x)(-\sin(x)) + (\cos(x))(e^x))$$

$$\frac{dy}{dx} = 19(e^x \cos(x))^{18} \cdot ((e^x)(-\sin(x)) + (\cos(x))(e^x))$$

(e) (5 points) $\frac{1}{3}x^3 - x^2y + y^2 = 9$ (Hint. Use implicit differentiation)

$$x^2 - ((x^2)(y') + (y)(2x)) + 2yy' = 0$$

$$x^2 - x^2y' - y2x + 2yy' = 0$$

$$2yy' - x^2y' = 2xy - x^2$$

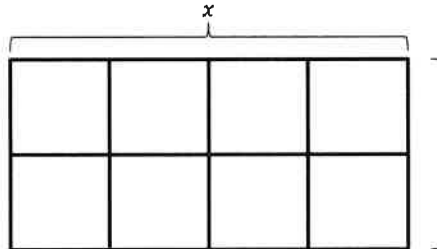
$$y'(2y - x^2) = 2xy - x^2$$

$$y' = \frac{2xy - x^2}{2y - x^2}$$

$$\frac{dy}{dx} = \frac{2xy - x^2}{2y - x^2}$$

5. (25 Points) Optimization.

- (a) A farmer is fencing in a rectangular portion of her land, and she is looking to divide that plot into eight equal sections as shown below. The outer perimeter of the fence costs \$18 per meter to install and the inner dividing sections cost \$10 per meter to install. The total area to be fenced in needs to be $100,000 \text{ m}^2$. What are the dimensions of the fence that minimize the total cost to the farmer?



$outer = 2x + 2y @ \$18$
 $Inner = x + 3y @ \$10$
 or $18(2x + 2y) + 10(x + 3y)$

- i. (5 points) What is the objective function in terms of two variables?

$$Q(x, y) = 18(2x + 2y) + 10(x + 3y)$$

- ii. (5 points) What is the constraint?

$$100,000 \text{ m}^2 = xy$$

- (b) (15 points) This is a different problem. A distribution company must deliver a total of 1000 pallets of equipment to a forward operating base. The total delivery cost (in dollars) is modeled by

$$C(x) = \frac{1000}{x} + 60x \approx 1000x^{-1} + 60x$$

where x is the number of pallets per shipment (partial pallets may be shipped). Assuming x must be positive, determine and verify the value of x that minimizes the total delivery cost.

✓ find crit point

$$C'(x) = -1000x^{-2} + 60 \approx -\frac{1000}{x^2} + 60 = 0$$

$$x^2 = \frac{1000}{60} = 60 \cdot x^2$$

$$\frac{1000}{60} = \frac{60x^2}{60} \quad \sqrt{x^2} = \sqrt{16.667}$$

$$x = 4.08 \rightarrow x = 4.08 \text{ pallets per shipment}$$

$$x = 4.08 \text{ pallets per shipment}$$

Verification:

$$C''(x) = 2000x^{-3}$$

$$C''(x) = \frac{2000}{x^3} \approx \frac{2000}{4.08^3} \approx 71.19 \text{ or } (+) \text{ number so concave up } \curvearrowright$$

not a minimum

6. (20 Points) **Basic Integrals.** Evaluate the following integrals.

(a) (4 points) $\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1}(x)$

$$\sin^{-1}(x) + C$$

(b) (4 points) $\int 4 \cos(x) dx$

$$4 \int \cos(x) = 4 \sin(x) + C$$

$$4 \sin(x) + C$$

(c) (4 points) $\int e^x - 2 dx$

$$= e^x - 2x + C$$

$$e^x - 2x + C$$

Basic Integrals Cont. Evaluate the following integrals.

(d) (4 points) $\int \sqrt[4]{x} + \frac{x^4}{4} - 4^x dx$

$$\int x^{1/4} + \frac{1}{4}x^4 - 4^x dx = \frac{4}{5}x^{5/4} + \frac{1}{20}x^5 - \frac{4^x}{\ln 4} + C$$

$$\frac{4}{5}x^{5/4} + \frac{1}{20}x^5 - \frac{4^x}{\ln 4} + C$$

(e) (4 points) $\int_0^{\pi/2} \frac{1}{2} \sin(x) dx$, where x is in radians. **Show all work.**

$$\frac{1}{2} \int_0^{\pi/2} \sin(x) dx = -\cos(x) \Big|_0^{\pi/2} - (0) - (-1) = 1 \cdot \frac{1}{2} = \frac{1}{2}$$

$$\int_0^{\pi/2} \frac{1}{2} \sin(x) dx = \frac{1}{2}$$

7. (30 Points) **Integration Strategies.**

- (a) (10 points) For each integral,
- circle
- the
- best**
- integration technique (Basic Integration, Substitution, or Integration by Parts).

	Integral	Technique		
(i)	$\int x \sin(x^2) dx$	Basic Integration	<u>Substitution</u>	Integration by Parts
(ii)	$\int x \sin(x) dx$	Basic Integration	Substitution	<u>Integration by Parts</u>
(iii)	$\int \frac{4}{1+x^2} dx$	<u>Basic Integration</u>	Substitution	Integration by Parts
(iv)	$\int \frac{x}{1+x^2} dx$	Basic Integration	<u>Substitution</u>	Integration by Parts
(v)	$\int x^2 e^x dx$	Basic Integration	Substitution	<u>Integration by Parts</u>

- (b) Evaluate the following integrals, you must show all work.

i. (5 points) $\int \frac{1}{x(\ln(x))^3} dx$

$$\begin{aligned}
 & u = \ln(x) \\
 & du = \frac{1}{x} \\
 & \int \frac{1}{x \cdot u^3} \cdot x du \\
 & \int \frac{1}{u^3} du \approx \int u^{-3} du \\
 & \frac{u^{-2}}{-2} \approx -\frac{1}{2u^2} \rightarrow -\frac{1}{2(\ln(x))^2} + C
 \end{aligned}$$

$$\begin{aligned}
 \frac{du}{dx} &= \frac{1}{x} \\
 du &= \frac{1}{x} dx \\
 dx &= x du
 \end{aligned}$$

$$\int \frac{1}{x(\ln(x))^3} dx = -\frac{1}{2(\ln(x))^2} + C$$

Integration Strategies Cont. Evaluate the following integrals, you must show all work.

ii. (5 points) $\int x \sin(x) dx$

IBP
 $u = x \quad dv = 1$

$uv - \int v du$

$du = \sin(x) \quad v = \cos(x)$

$-\int \cos(x) \cdot 1 dx$

$-x \cos(x) - \sin(x)$
 $\downarrow$

$-\sin(x)$

$-x \cos(x) + \sin(x) + C$

$-x \cos(x) + \sin(x) + C$

iii. (10 points) $\int_1^{\infty} \frac{1}{x^4} dx$

Improper Integ

$\frac{x^{-3}}{-3} \approx \frac{-1}{3x^3}$

$\lim_{t \rightarrow \infty} \int_1^t x^{-4} dx$

$\lim_{t \rightarrow \infty} \left. \frac{-1}{3x^3} \right|_1^t = \frac{-1}{3t^3} + \frac{1}{3} \approx \frac{-1}{3(\infty)^3} + \frac{1}{3}$
 so limit $\uparrow$ Converges at $\frac{1}{3}$

Converges @ $\frac{1}{3}$

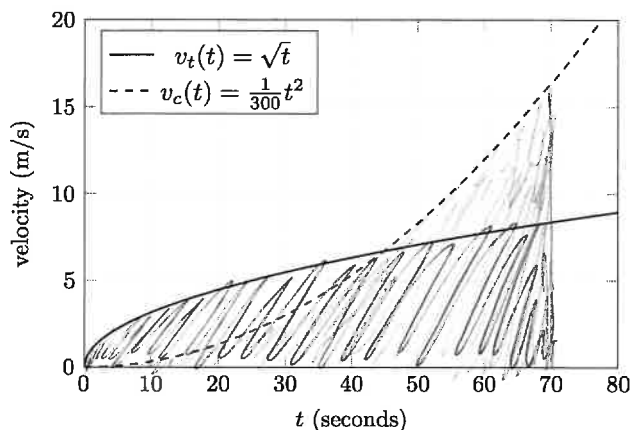
Jacob Shachin

8. (35 points) Integration Applications.

- (a) (10 points) A truck and a car are side-by-side at $t = 0$ and begin accelerating in the same direction at the same time. The truck's velocity (in meters per second) with respect to time is modeled by $v_t(t)$, and the car's velocity (in meters per second) with respect to time is modeled by $v_c(t)$.

$$v_t(t) = \sqrt{t}$$

$$v_c(t) = \frac{1}{300}t^2$$



position $\frac{dy}{dx}$
 velocity $\frac{dy}{dt}$
 acceleration $\frac{d^2y}{dt^2}$

Determine which vehicle is in front after 70 seconds, and by how much (in meters).

$$P_t(t) = \frac{2}{3}\sqrt{t^3} + 0 \quad \text{b/c the car is still at } P_c(t) = 0$$

$$P_c(t) = \frac{t^3}{900} + 0$$

$$P_t(70) = 390.44 \text{ m} \leftarrow \text{Truck is further by } 9.33 \text{ m}$$

$$P_c(70) = 381.11 \text{ m}$$

$$390.44 - 381.11 = 9.33 \text{ m}$$

$$\int t^{1/2} dt = \frac{2}{3}t^{3/2} \approx \frac{2}{3}\sqrt{t^3}$$

$$\frac{1}{900} \int t^2 dt = \frac{1}{900} \cdot \frac{1}{3}t^3 = \frac{t^3}{900}$$

Truck is in front $\rightarrow$ Truck at 390.44m while car only traveled 381.11m.
 Truck is in front by 9.33m.

- (b) The Tactical Officer for your company developed the density function $f(x)$ to describe the number of hours that a random cadet sleeps each night.

$$f(x) = \begin{cases} k(-14 + 15x - x^2) & 2 \leq x \leq 14 \\ 0 & \text{otherwise} \end{cases}$$

Note: For $k \geq 0$, $f(x) \geq 0$

- i. (10 points) What value of k makes $f(x)$ a valid PDF?

$$k \int_2^{14} (-14 + 15x - x^2) dx = 1$$

$$\int_2^{14} (-14 + 15x - x^2) dx = \frac{1}{k}$$

$$\left. -14x + \frac{15}{2}x^2 - \frac{x^3}{3} \right|_2^{14} = 359.333 - 0.6667 = 360 = \frac{1}{k}$$

$$-196 + 1470 - 914.67 = 359\frac{1}{3} + \frac{2}{3} = 360 \Rightarrow k = 0.00278$$

$$-28 + 30 - 2.667$$

$$k = 0.00278$$

- ii. (5 points) Using the k from above, determine the probability that a random Cadet sleeps between 3 and 5 hours in a night.

$$0.00278 \cdot \left. -14x + \frac{15}{2}x^2 - \frac{x^3}{3} \right|_3^5 = 75.8333 - 16.5 = 59.333$$

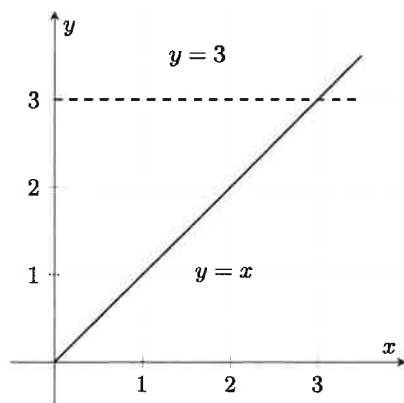
$$-70 + 187.5 - 41.6667 = 0.1649$$

$$-42 + 67.5 - 9$$

$$0.1649 \text{ or } 16.49\%$$

(c) Volume.

- i. (4 points) The region bounded by $y = x$, $y = 3$, and $x = 0$ is shown below. Circle the correct method for find the volume when rotating around each axis.



(a) Rotating about the x -axis:

(A) Disk Method

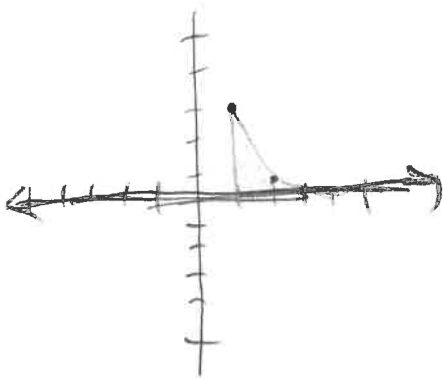
(B) Washer Method

(b) Rotating about the y -axis:

(A) Disk Method

(B) Washer Method

- ii. (6 points) **This is a different problem.** Find the volume of the solid obtained by rotating about the x -axis the region under the curve $y = \frac{3}{x^3}$ and above $y = 0$ from $x = 1$ to $x = \infty$.



$$V = \pi \int_1^{\infty} (3x^{-3})^2 dx$$

$$(3x^{-3})^2 = \int 9x^{-6} dx$$

$$-\frac{9}{5}x^{-5} \text{ or } -\frac{9}{5x^5}$$

$$V = \pi \cdot \left[-\frac{9}{5x^5} \right]_1^{\infty} = \pi \cdot \left(0 - \left(-\frac{9}{5} \right) \right) = \pi \cdot \frac{9}{5}$$

$$V = \frac{9}{5} \pi$$

$$V = \frac{9}{5} \pi$$



UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB K

TERM ENDING EXAM

MA104 AY26–2 Course Report

Prepared by: MAJ Dylan Hyde

18 May 26

Approvals

(a) Calculus PD: _____

(b) Deputy Department Head: _____

Total Students	Number of Students taking TEE	Number not taking TEE
645	645.0	0.0

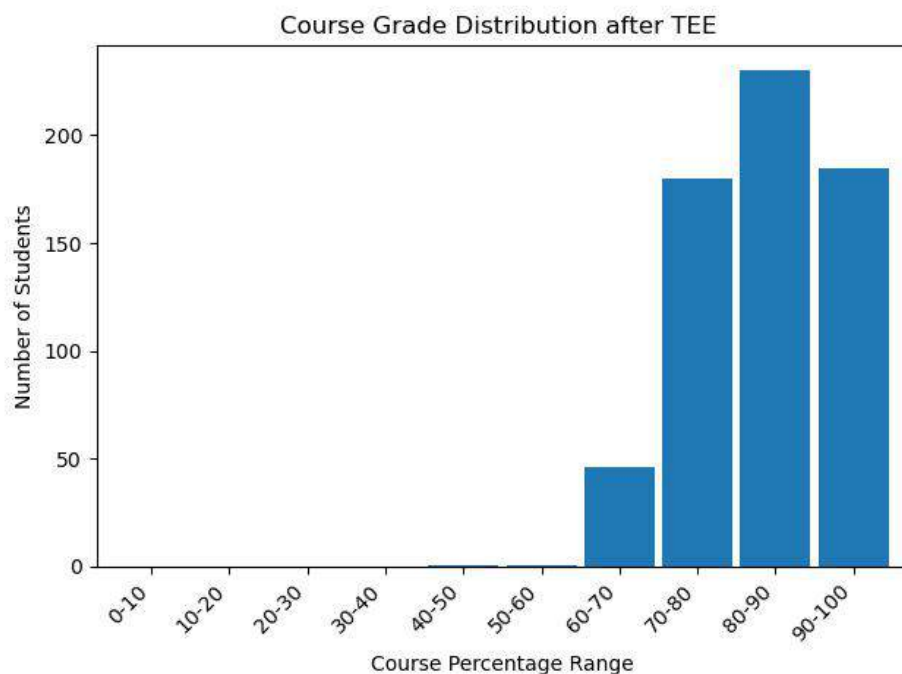


Figure 1: Course Score Distribution After TEE

Mean Score (%)	Min Score (%)	Max Score (%)	Median Score (%)
78.72	32.0	99.5	80.75

Course Grade Distribution Comparison (after TEE)

Term	F	D	C-	C	C+	B-	B	B+	A-	A	A+
<i>Count</i>											
AY25–2	4	29	33	84	65	88	115	84	80	85	15
AY26–2	14*	34	43	79	58	76	95	58	78	83	27

Table 1: Course grade distribution by count and percent after TEE, AY25–2 vs. AY26–2.

**Note: F's include three (3) Students with grades over 65% but scored lower then 50% on the TEE.*

Year-Over-Year Comparison: AY25-2 vs. AY26-2**TEE Summary Statistics**

Term	Mean (%)	Min (%)	Max (%)	Median (%)
AY25-2	85.18	38.5	100.0	86.44
AY26-2	78.72	32.0	99.5	80.75

TEE Enrollment

Term	Total Students	Took TEE	Did Not Take TEE
AY25-2	680	678	2
AY26-2	645	645	0

TEE Grade Distribution Comparison

Term	F	D	C-	C	C+	B-	B	B+	A-	A	A+
<i>Count</i>											
AY25-2	15	31	29	54	53	68	99	84	84	113	48
AY26-2	105	64	50	50	44	47	74	56	52	82	21

Table 2: TEE grade distribution by count and percent, AY25-2 vs. AY26-2.

Course Grade Summary Statistics (after TEE)

Term	Mean (%)	Min (%)	Max (%)	Median (%)
AY25-2	83.84	49.7	100.3	84.4
AY26-2	83.30	43.33	100.37	83.73

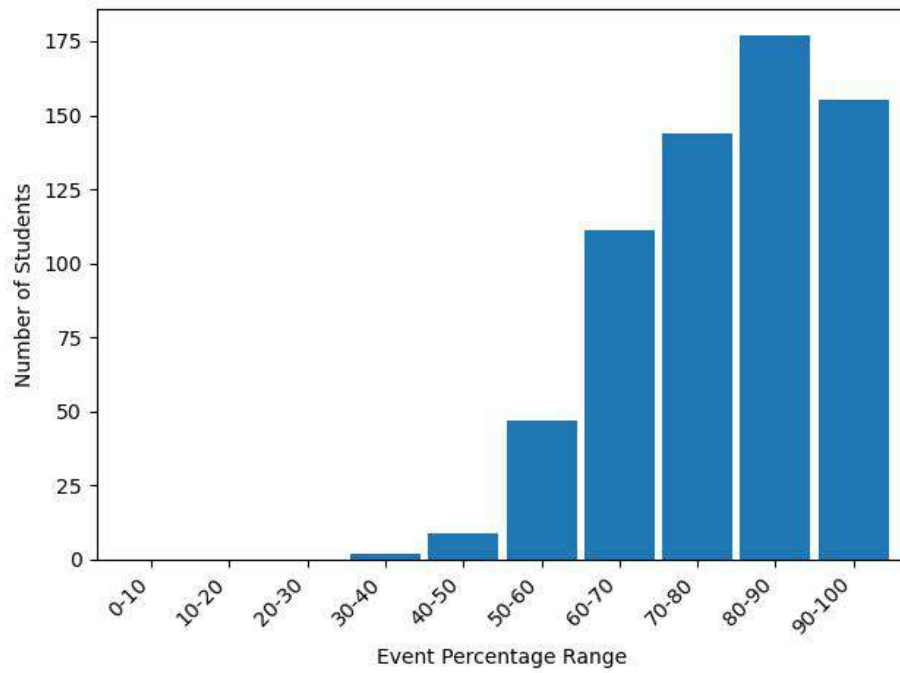
Informal Charts: TEE

Figure 2: Score Distribution

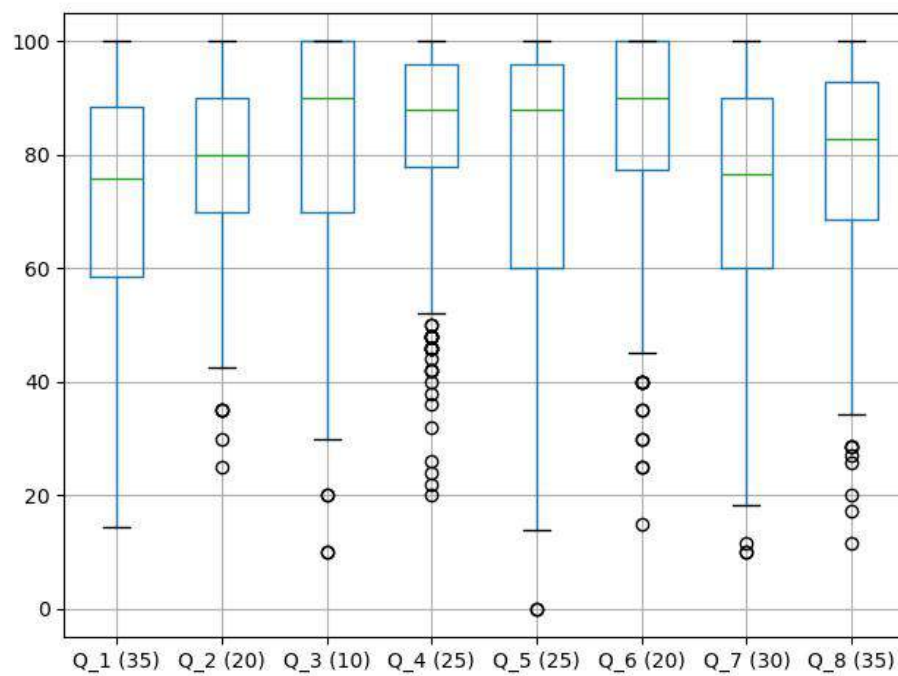


Figure 3: Grade Distribution by Question

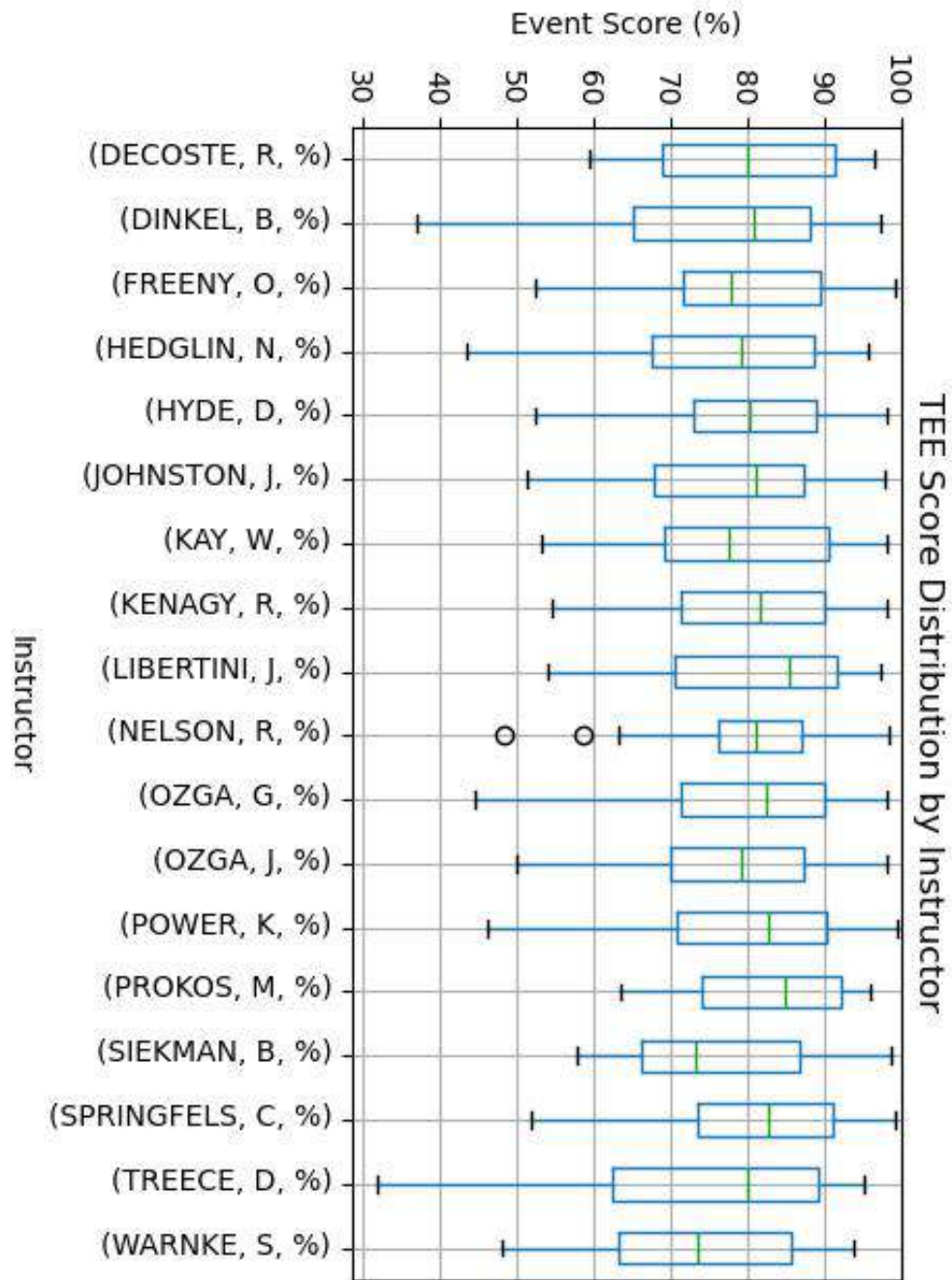


Figure 4: Grade Distribution by Instructor

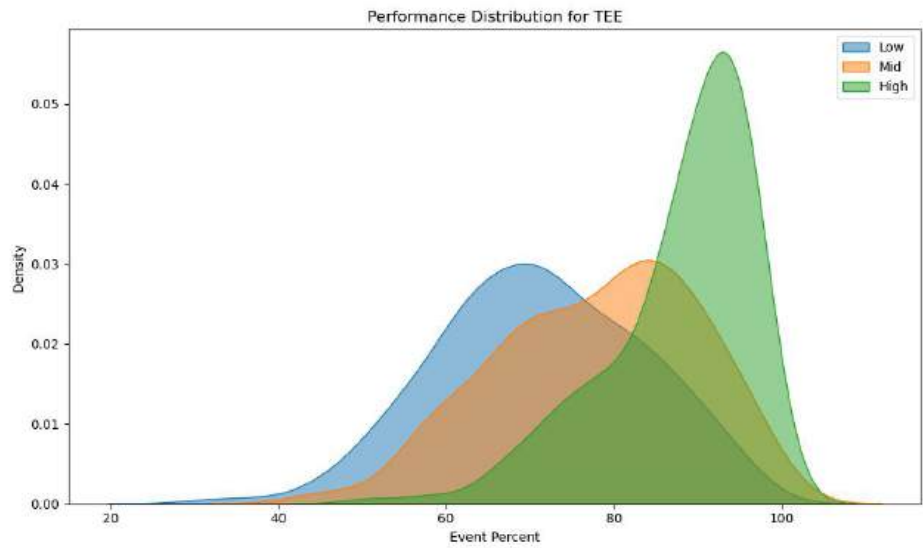


Figure 5: TEE Performance vs Sequence

$$S = 0.85MA103, +0.1FCE + 0.05CALC \tag{1}$$

$$\text{Ability Group} = \begin{cases} \text{High} & 0.815 < S \leq 1 \\ \text{Mid} & 0.75 < S \leq 0.815 \\ \text{Low} & S \leq 0.75 \end{cases} \tag{2}$$

Grades	F	D	C-	C	C+	B-	B	B+	A-	A	A+
Count	105	64	50	50	44	47	74	56	52	82	21
Percent	16.28	9.92	7.75	7.75	6.82	7.29	11.47	8.68	8.06	12.71	3.26

MA104 Course OML after TEE (Top 40 and Bottom 40)

	Student	Instructor	Event Percent	Score	Letter Grade
1	Fiske, Luke	HYDE, D	96.5	100.37	A+
2	Chen, Kuan-Po	LIBERTINI, J	97.5	100.36	A+
3	Lee, Aaron	OZGA, J	98.25	99.78	A+
4	Smith, Silas	NELSON, R	98.5	99.53	A+
5	El-Bashir, Kareem	POWER, K	98.25	99.17	A+
6	Munari, Jameson	TREECE, D	94.75	98.97	A+
7	Sloan, Stephen	SIEKMAN, B	98.75	98.83	A+
8	McCloskey, Lukas	HYDE, D	98.25	98.55	A+
9	Shalom, Prince	DECOSTE, R	94.0	98.25	A+
10	Lundeen, Carter	KAY, W	96.5	98.2	A+
11	Frank, Noah	DECOSTE, R	96.5	98.15	A+
12	Durfee, Kalia	JOHNSTON, J	94.75	97.99	A+
13	Bahar, Shafi	FREENY, O	98.25	97.95	A+
14	Putman, Jaden	POWER, K	95.25	97.82	A+
15	Heath, Roy	DINKEL, B	97.25	97.8	A+
16	Park, Seoyoung	POWER, K	93.0	97.75	A+
17	Rivera Rodriguez, Eduardo	KENAGY, R	96.75	97.7	A+
18	Yang, Jally	FREENY, O	99.25	97.63	A+
19	Meno, Avery	FREENY, O	94.25	97.61	A+
20	Carreras Merced, Juan	KAY, W	95.0	97.59	A+
21	Hokanson, Ivar	KAY, W	97.25	97.47	A+
22	Twomey, Aidan	KAY, W	98.25	97.3	A+
23	Newcomb, Christopher	PROKOS, M	94.25	97.18	A+
24	Li, Franklin	OZGA, G	96.0	97.16	A+
25	Shevchik, Jacob	POWER, K	99.5	97.08	A+
26	Zurowski, Charlie	SIEKMAN, B	91.75	97.08	A+
27	Withrow, Katya	OZGA, G	96.5	97.05	A+
28	Sicat, Sarah	OZGA, J	90.0	96.93	A
29	Hwang, Louis	LIBERTINI, J	96.5	96.66	A
30	Boyer, Jackson	HYDE, D	94.75	96.47	A
31	McCall, Draven	HEDGLIN, N	92.25	96.34	A
32	Edmondson, Eli	JOHNSTON, J	94.5	96.3	A
33	O'Koniewski, Tait	KENAGY, R	96.25	96.29	A
34	Jiang, Daniel	JOHNSTON, J	96.0	96.29	A
35	Leahy, William	DINKEL, B	94.75	96.02	A
36	Tomasits, Cohen	SIEKMAN, B	95.5	95.91	A
37	Ferrell, Maxwell	JOHNSTON, J	98.0	95.84	A
38	Sollie, Hunter	FREENY, O	96.75	95.82	A
39	Phenicie, Max	SPRINGFELS, C	96.5	95.79	A
40	Carter, Stephen	PROKOS, M	93.75	95.6	A
...					
606	Merlano, Alejandro	JOHNSTON, J	53.0	69.12	D
607	Cottrell, Camden	KENAGY, R	58.25	68.88	D
608	Sparks, Noah	SIEKMAN, B	66.75	68.85	D
609	Baka, Hannah	OZGA, J	50.0	68.82	D
610	King, Addison	OZGA, G	44.75	68.79	D
611	Whattcott, Preston	TREECE, D	57.5	68.74	D

Continued on next page

	Student	Instructor	Event Percent	Score	Letter Grade
612	Cheedy, Tekoyia	KAY, W	60.0	68.42	D
613	Ujii, Adam	DINKEL, B	58.25	68.34	D
614	Pena, Miguel	KAY, W	53.25	68.11	D
615	Cicale, Nicholas	LIBERTINI, J	60.0	68.01	D
616	Thom-Rogers, David	SPRINGFELS, C	64.0	68.01	D
617	Messick, Tyler	LIBERTINI, J	63.5	67.96	D
618	Petkus, Samantha	POWER, K	63.0	67.95	D
619	Schaller, Ryan	HYDE, D	52.5	67.88	D
620	Clark, Chase	OZGA, G	58.5	67.79	D
621	Marinucci, Mia	FREENY, O	66.25	67.74	D
622	Swingle, Frederick	TREECE, D	50.0	67.4	D
623	Garris, Kieran	DINKEL, B	53.0	67.25	D
624	Yi, Elliott	DINKEL, B	37.0	67.14	D
625	Stewart, Casper	POWER, K	61.0	67.03	D
626	Daly, Charles	DINKEL, B	60.0	66.85	D
627	Agena, Maya	OZGA, J	83.5	66.84	D
628	Degennaro, Michael	POWER, K	46.25	66.82	D
629	Orr, Harrison	HEDGLIN, N	60.0	66.65	D
630	Ryan, Xavier	DINKEL, B	58.25	66.01	D
631	Mroszczyk, Porter	HYDE, D	64.25	65.76	D
632	Mann, Remo	KENAGY, R	59.75	65.67	D
633	Schiffer, Paige	POWER, K	55.75	65.4	D
634	Green, Jahaan	JOHNSTON, J	53.5	65.17	D
635	Mack, William	SIEKMAN, B	59.5	64.3	F
636	Futch, Hadley	JOHNSTON, J	60.0	64.02	F
637	Barry, Brylee	HEDGLIN, N	43.5	64.0	F
638	Munoz, Emmanuel	OZGA, J	65.75	63.98	F
639	Christie, Caroline	KENAGY, R	61.25	63.74	F
640	Mazzanti, Kadence	TREECE, D	64.25	63.08	F
641	Bishop, Brant	WARNKE, S	48.0	62.39	F
642	Harden, Jadon	TREECE, D	45.25	61.83	F
643	Deapen, Janie	POWER, K	46.75	60.96	F
644	Daniell, Carson	FREENY, O	64.0	58.5	F
645	Kuhn, Jonas	TREECE, D	32.0	43.33	F

MA104 AY26-2 Recommended Failures

A cadet is recommended for failure if they scored **below 50%** on the TEE *or* earned a **final course grade below 65%**. The table below is sorted from highest to lowest final course grade. Blank cells indicate data to be completed by the Course Director.

Name	Final (%)	TEE	WPR 1	WPR 2	WPR 3	WPR 4	WRIT
Recommend Failure							
Delgado, Joshua	69.73	49.75	65.5	88.5	54.0	54.5	70.25
Yi, Elliott	67.14	37.0	65.5	63.0	58.5	52.5	75.0
Degennaro, Michael	66.82	46.25	69.0	70.5	57.5	63.0	69.25
Mack, William	64.30	59.5	71.5	73.0	64.0	58.5	58.25
Futch, Hadley	64.02	60.0	61.5	64.0	38.5	58.5	56.5
Barry, Brylee	64.00	43.5	60.0	56.0	38.5	44.0	74.75
Munoz, Emmanuel	63.98	65.75	73.0	52.5	54.5	65.0	64.0
Christie, Caroline	63.74	61.25	79.0	73.0	34.5	69.5	64.5
Mazzanti, Kadence	63.08	64.25	66.5	73.0	61.0	54.0	79.25
Bishop, Brant	62.39	48.0	67.5	34.5	50.0	60.5	68.5
Harden, Jadon	61.83	45.25	57.0	77.5	54.5	70.0	60.5
Deapen, Janie	60.96	46.75	52.0	59.5	50.0	59.0	59.75
Daniell, Carson	58.50	64.0	73.5	67.0	56.0	61.0	64.75
Kuhn, Jonas	43.33	32.0	43.5	35.0	48.5	31.0	54.5
Recommend Pass (<i>scored below 50% on TEE; overall course performance warrants passing the course</i>)							
Dusza, Tymoteusz	78.28	45.0	76.0	86.5	75.0	80.0	84.75
Hennen, Aidan	74.18	48.5	77.5	80.5	78.5	72.0	84.0
King, Addison	68.79	44.5	68.5	82.5	70.5	62.5	63.25



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SECTION III

ASSIGNMENTS



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TAB M

FINAL PROJECT



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FINAL PROJECT PROMPT

Final Project

Assignment. Your roommate understands differentiation and integration but is struggling to apply these tools to real-world problems.

Your task is to prepare a **15-minute instructional brief** that teaches your roommate how calculus is applied to solve a related applied problem. You and your team will present this brief to your instructor.

The brief may be delivered in any format, but it must clearly **teach and explain** the calculus techniques involved in the project described below.

Mission Context. Your boss has asked your team to design an optimal fuel tank and recommend a steel supplier. Your recommendation must be supported using three applied calculus concepts:

- Optimization (fuel tank design),
- Arc length (route distance and travel time),
- Probability (delivery risk).

Task 1: Teach Optimization (Fuel Tank Design).

Fuel Tank Design Background. The Army is seeking a replacement for the M969 and M967 fuel trailers. The new tank must:

- Be shaped as an **elliptical cylinder**,
- Have a fixed length of **368 inches**,
- Hold **5,000 gallons** of fuel,
- Minimize total material used.

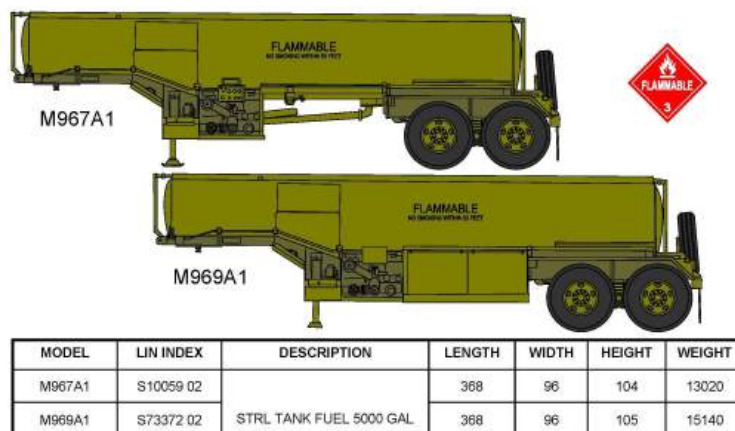
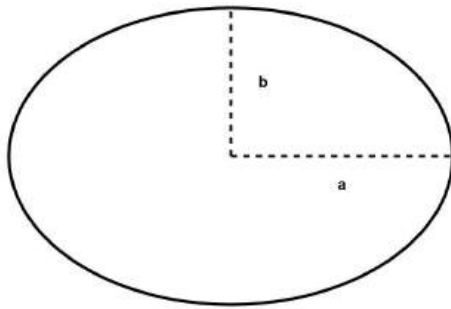


Figure 1: M967 and M969 Current Specifications

In addition to the outer shell, the tank includes **two internal baffles** (half-wall dividers) to reduce fuel slosh. These modeling assumptions are provided to standardize solutions unless explicitly justified otherwise.

Modeling Clarifications.

- The tank cross-section is an ellipse (Figure 2); the length is fixed at $L = 368$ inches.
- “Material used” is modeled as total surface area of steel:
 - Lateral surface area (ellipse perimeter $\times$ length),
 - Two elliptical end caps,
 - Two internal baffles, each modeled as a half-ellipse.
- Use 1 US gallon = 231 in³.
- Ignore material thickness, seams, and manufacturing constraints.

Elliptical Geometry Reference

$$\text{Area} = \pi ab \quad (1)$$

$$\text{Perimeter} \approx \pi(a+b) \left(1 + \frac{3h}{10 + \sqrt{4-h}} \right) \quad (2)$$

$$h = \frac{(a-b)^2}{(a+b)^2} \quad (3)$$

Figure 2: Elliptical cross-section

Instructional Focus (Optimization). Your brief should teach:

- How to define decision variables, an objective function, and constraints,
- How to use a constraint to create a single-variable objective,
- How derivatives identify and justify an optimum.

Simplifying Assumptions (with necessary and reasonable justification).

- **Idealized geometry and negligible thickness.** Model the tank using smooth geometric surfaces and ignore material thickness, weld overlap, seams, and rounded-end effects. This allows use of standard geometric formulas for area and volume. The impact of thickness and fabrication imperfections on total volume and surface area is assumed negligible relative to overall tank dimensions.
- **Continuous dimensions.** Treat a and b as continuous variables. While manufacturers may produce components in discrete sizes, calculus-based optimization requires continuity. This assumption is reasonable because size increments are typically small relative to overall dimensions and can be approximated by continuous values.

This work was completed in the optimization lab and should be **taught and interpreted** rather than re-solved from scratch.

Task 2: Teach Arc Length (Route Distance and Travel Time).

Three suppliers offer identical pricing but follow different delivery routes (Figure 3). Each route is modeled by a function $y = f(x)$ over the domain $5 \leq x \leq 40$.

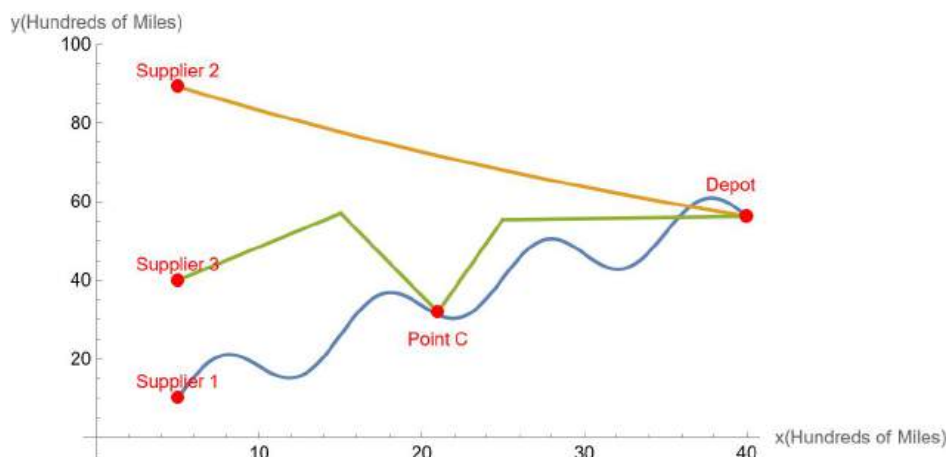


Figure 3: Suppliers' routes to Army Depot

Purpose of this task. In this task, you will use **arc length** to model the total distance traveled along each route and then estimate **travel time** using the given speed ranges.

Arc Length Guidance.

- Treat the arc length of $y = f(x)$ from $x = 5$ to $x = 40$ as the **total route distance** (in miles).
- Convert distance to travel time using the allowed speed ranges (miles/hour). Clearly state and justify your chosen speed.

Route Geometry Models (used for arc length).

Route 1 (Supplier 1). Supplier 1 is located at (5, 10) and delivers to the Army Depot at (40, 56.3). This route follows a winding highway corridor with an allowed speed range of **45–65 mph**.

$$f_1(x) = \frac{52}{\sqrt{1 + \left(\frac{\pi}{70}\right)^2}} \sin\left(\frac{\pi}{100}(x - 5)\right) + \frac{8}{\sqrt{1 + \left(\frac{8\pi}{35}\right)^2}} \sin\left(\frac{7\pi}{35}(x - 5)\right) + 10.0143 \quad (4)$$

Route 2 (Supplier 2). Supplier 2 is located at (5, 89.3) and delivers to the Army Depot at (40, 56.3). This route passes through congested urban areas with an allowed speed range of **25–45 mph**.

$$f_2(x) = 8^{\left(\frac{5-x}{35} + 1\right)} 3^{\left(\frac{5-x}{35} + 2\right)} 13^{\left(\frac{x-5}{35}\right)} + 17.3 \quad (5)$$

Route 3 (Supplier 3). Supplier 3 is located at (5, 40) and delivers to the Army Depot at (40, 56.3). This route uses a mix of rural roads and interstate with an allowed speed range of **35–55 mph**.

$$f_3(x) = \begin{cases} \frac{17}{10}(x - 5) + 40 & 5 \leq x < 15 \\ 57.1 - \frac{21}{5}(x - 15) & 15 \leq x < 21 \\ \frac{59}{10}(x - 20) + 26 & 21 \leq x < 25 \\ 55.4 + \frac{3}{50}(x - 25) & 25 \leq x \leq 40 \end{cases} \quad (6)$$

Instructional Focus (Arc Length & Time). Your brief should teach:

- How arc length models real travel distance,
- How to handle arc length for a piecewise function,
- How distance and speed combine to produce travel time.

Task 3: Teach Probability (Transportation Risk).

After estimating travel time and distance, your boss wants to consider **risk**. For each supplier, risk is modeled using a **probability density function (PDF)**. These PDFs describe the likelihood of delivery failure as a function of either **time** or **distance**.

Purpose of this task. In this task, you will teach how probability can be used to **quantify and compare risk** between routes.

Risk Models (Probability Density Functions).

Route 1 Risk (time-based). Risk depends on time spent on the route (in hours).

$$r_1(t) = \frac{1}{50\sqrt{2\pi}} e^{-\frac{(t-600)^2}{5000}} \quad (7)$$

Route 2 Risk (distance-based). Risk depends on distance traveled from the supplier (in miles).

$$r_2(d) = 0.0001 e^{-\frac{1}{10,000}d} \quad (8)$$

Route 3 Risk (time-based). Risk depends on time spent on the route (in hours).

$$r_3(t) = \frac{1}{250\sqrt{2\pi}} e^{-\frac{(t-550)^2}{2(250^2)}} \quad (9)$$

Risk Interpretation Guidance.

- Clearly state what the variable represents (time or distance).
- Explain how probability is computed as **area under the curve**.
- Use probability to make a quantitative comparison of supplier risk.

Instructional Focus (Probability & Risk). Your brief should teach:

- How to interpret a PDF,
- How probability supports decision-making under uncertainty,
- How risk considerations can change a recommendation.

Summary Tasks (what you must be prepared to teach).

1. **Optimization:** Review and explain the calculus-based optimization used to determine optimal tank dimensions.

2. **Arc Length:** Compute and interpret route distances, then estimate travel time using the provided speed ranges.
3. **Probability:** Interpret the PDFs and use probability to compare route risk in a clear, quantitative way.

Presentation Requirements. This is a group presentation (groups assigned by the instructor). Presentations will occur on 4, 5, and 6 MAY 2026. **Slides due 27 April 2026**

- 15 minutes maximum presentation time,
- 5 minutes for questions.

Your presentation must include:

1. **Project Overview.** A concise summary of the mission context and how optimization, arc length, and probability each contribute to the final recommendation.
2. **Methodology Summary (for each task).** For each of the three tasks, explain the calculus concept used, what it is, why it applies, and how it connects to the problem. This should be a clear, conceptual explanation of the technique (e.g., how optimization uses derivatives to find a minimum, how arc length integrates infinitesimal distances, or how a PDF encodes probability as area).
3. **Fully Worked Problem from This Lab (for each task).** For each of the three tasks, walk through one complete problem drawn directly from this project. This means, for instance, setting up and differentiating the actual surface area objective function from Task 1, computing the arc length integral for one of the three supplier routes in Task 2, and evaluating a probability integral using one of the risk PDFs in Task 3. Your worked example must show all setup, computation, and interpretation in the context of the Army mission.

Outstanding Student Work



UNITED STATES MILITARY ACADEMY
WEST POINT

MA104 Final Project

Shafi Bahar, Aretha CC-Okeke, Daniel Poe





Overview

Our boss has delegated a task to us to design a fuel tank and recommend a steel supplier. Using Calculus and the principles of Optimization, Arc Length, and Probability, we will recommend a steel supplier to provide the materials for the tank.

- Optimization will be used to find the optimal dimensions for the new fuel tank design to minimize production cost.

- Arc Length will be used to determine which route is most optimal from the supplier to our factory.

- Probability will be used quantify and compare risk between each manufacturer.

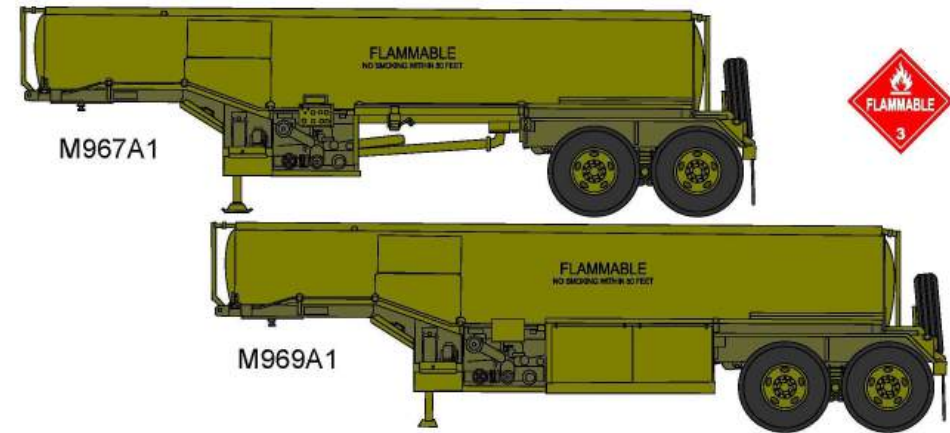




FUEL TANK OPTIMIZATION

Overview:

- The tank must have a volume of 5000 gallons.
- The Length of the tank is fixed, and it is 368 inches.
- Task: Minimize the material used.
Which is basically the Surface Area.



MODEL	LIN INDEX	DESCRIPTION	LENGTH	WIDTH	HEIGHT	WEIGHT
M967A1	S10059 02	STRL TANK FUEL 5000 GAL	368	96	104	13020
M969A1	S73372 02		368	96	105	15140



DEFINING DECISION VARIABLES AND CONSTRAINTS

- a = semi major axis
- b = semi minor axis

Volume of the ellipse,

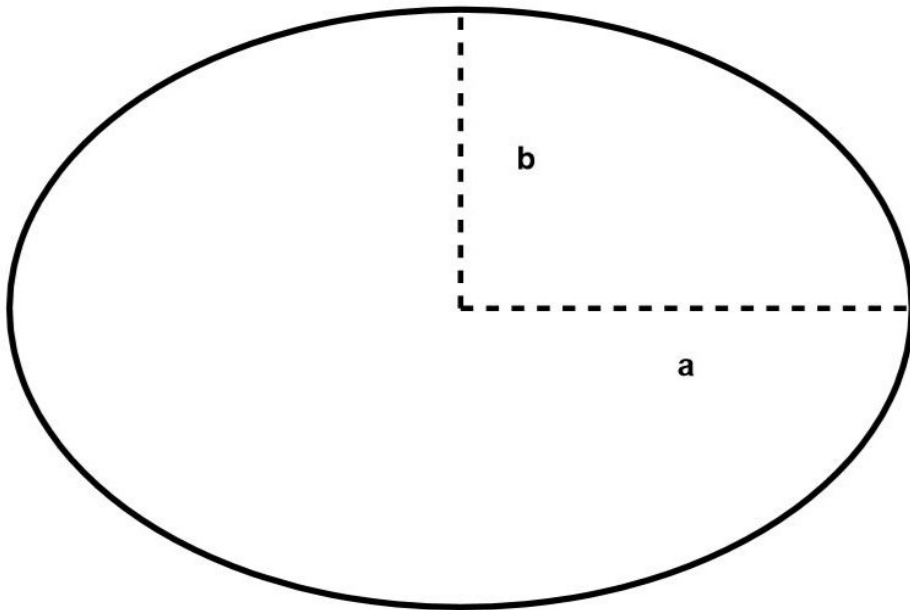
$$V = \pi abL$$

Substituting the values and solving for ab ,

$$ab = \frac{5000}{\pi(368)}$$

Are the units consistent?

NO



$$ab = \frac{5000(231)}{\pi(368)}$$



OUR OBJECTIVE: MINIMIZE THE MATERIAL USED

We can break the surface area into 3 parts:

- Lateral surface area,

$$A_{lateral} = \pi(a + b)L$$

- Two elliptical end caps,

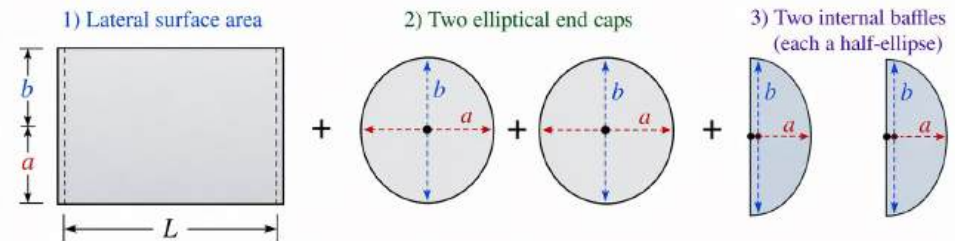
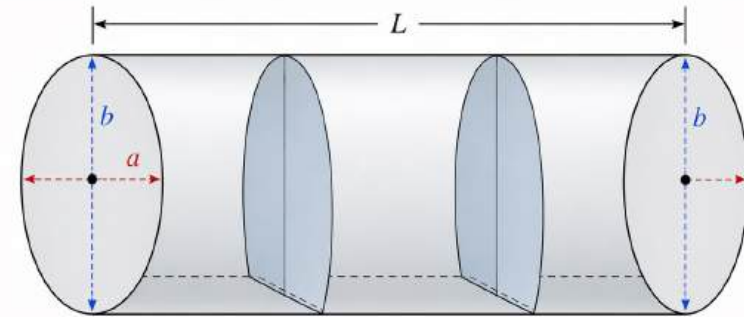
$$A_{ends} = 2\pi ab$$

- Two internal baffles, each modeled as a half-ellipse:

$$A_{baffles} = \pi ab$$

- Minimize total surface area,

$$A_{total} = \pi(a + b)L + 3\pi ab$$





CREATING A SINGLE VARIABLE OBJECTIVE FUNCTION

- From the Constraints,

$$a = \frac{C}{b}, \text{ where } C = \frac{1155000}{\pi(368)}$$

Substituting it in the objective function,

$$S(b) = \pi \left(b + \frac{C}{b} \right) (368) + 3\pi b \frac{C}{b}$$

We reduce it to a single variable so we can use single variable calculus.





LET'S FIND THE MINIMUM!

The first derivative,

$$S'(b) = 368\pi \left(1 - \frac{C}{b^2} \right)$$

For critical points,

$$S'(b) = 0$$

Solving for b ,

$$b = \sqrt{C}$$

How do we know is it a maximum or minimum?





CONFIRMING MAXIMUM OR MINIMUM

- We can do Second Derivative Test to confirm our result,

$$S''(b) = 368\pi \left(\frac{2C}{b^3} \right)$$

$S''(b) > 0$, so the function concaves up and produces a minimum value at the critical point.



OPTIMAL DESIGN FOR FUEL TANK

- Solving for a ,

$$a = \frac{c}{\sqrt{c}}$$

So,

$$a = b$$

The semi axes are **equal**.

Optimal Shape for the fuel tank is a **Circle**.





Overview

Background: Three suppliers offer identical pricing but follow different delivery routes, each route is modeled in the graph below.

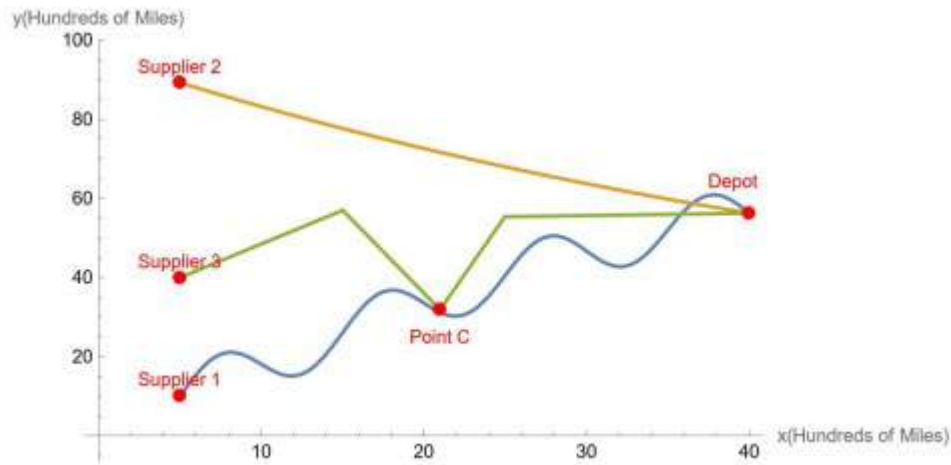


Figure 3: Suppliers' routes to Army Depot

Task: use arc length to model the total distance traveled on each route as well as estimate travel time using the provided speed ranges



Calculus

$$\int_a^b f(x) dx$$

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx$$

$$f_1(x) = \frac{52}{\sqrt{1 + \left(\frac{\pi}{70}\right)^2}} \sin\left(\frac{\pi}{100}(x - 5)\right) + \frac{8}{\sqrt{1 + \left(\frac{8\pi}{35}\right)^2}} \sin\left(\frac{7\pi}{35}(x - 5)\right) + 10.0143$$

$$f_2(x) = 8^{\left(\frac{5-x}{35}+1\right)} 3^{\left(\frac{5-x}{35}+2\right)} 13^{\left(\frac{x-5}{35}\right)} + 17.3$$

$$f_3(x) = \begin{cases} \frac{17}{10}(x - 5) + 40 & 5 \leq x < 15 \\ 57.1 - \frac{21}{5}(x - 15) & 15 \leq x < 21 \\ \frac{59}{10}(x - 20) + 26 & 21 \leq x < 25 \\ 55.4 + \frac{3}{50}(x - 25) & 25 \leq x \leq 40 \end{cases}$$

$$T = \frac{L}{V}$$

$$y = f(x)$$

$$5 \leq x \leq 40$$





Application

How arc length models travel time:

$$dL = \sqrt{dx^2 + dy^2}$$

$$dL = \sqrt{dx^2 \left(1 + \frac{dy^2}{dx^2}\right)}$$

$$dL = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$dL = \sqrt{1 + [f'(x)]^2} dx$$

$$dL = \int_a^b \sqrt{1 + [f'(x)]^2} dx$$

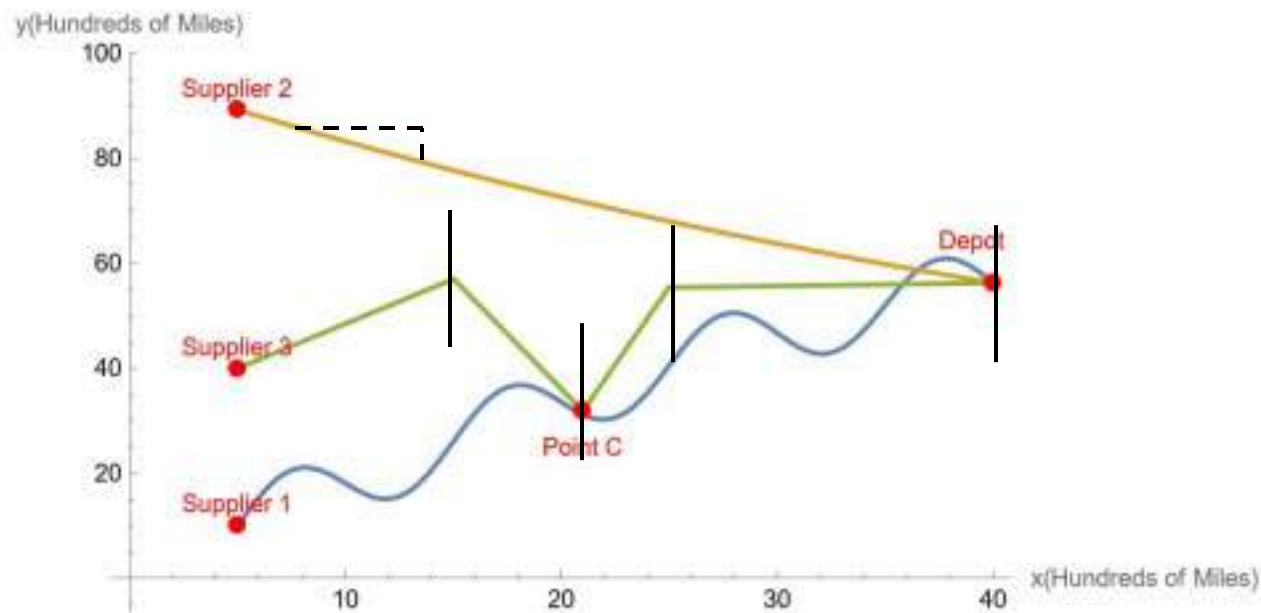


Figure 3: Suppliers' routes to Army Depot

$$f_3(x) = \begin{cases} \frac{17}{10}(x-5) + 40 & 5 \leq x < 15 \\ 57.1 - \frac{21}{5}(x-15) & 15 \leq x < 21 \\ \frac{59}{10}(x-20) + 26 & 21 \leq x < 25 \\ 55.4 + \frac{3}{50}(x-25) & 25 \leq x \leq 40 \end{cases}$$



Application

$$dL = \int_a^b \sqrt{1 + [f'(x)]^2} dx$$

$$f_3(x) = \begin{cases} \frac{17}{10}(x-5) + 40 & 5 \leq x < 15 \\ 57.1 - \frac{21}{5}(x-15) & 15 \leq x < 21 \\ \frac{59}{10}(x-20) + 26 & 21 \leq x < 25 \\ 55.4 + \frac{3}{50}(x-25) & 25 \leq x \leq 40 \end{cases}$$

$$f'(x) = \frac{17}{10}$$

$$\int_5^{15} \sqrt{1 + \left[\frac{17}{10}\right]^2} dx$$

19.72 miles

$$f'(x) = \frac{21}{5}$$

$$\int_{15}^{21} \sqrt{1 + \left[\frac{21}{5}\right]^2} dx$$

25.91 miles

$$f'(x) = \frac{59}{10}$$

$$\int_{21}^{25} \sqrt{1 + \left[\frac{59}{10}\right]^2} dx$$

23.94 miles

$$f'(x) = \frac{3}{50}$$

$$\int_{25}^{40} \sqrt{1 + \left[\frac{3}{50}\right]^2} dx$$

15.03 miles





Travel Time

$$T = \frac{L}{V}$$

Supplier 3 is located at (5, 40) and delivers to the Army Depot at (40, 56.3). This route uses a mix of rural roads and interstate with an allowed speed range of 35–55 mph.

19.72 miles

$5 \leq x \leq 15$

25.91 miles

$15 \leq x \leq 21$

23.94 miles

$21 \leq x \leq 25$

15.03 miles

$25 \leq x \leq 40$

Avg velocity = 45 mph

$$T = \frac{84.6}{45}$$

$$T = 1.88 * 60$$

$$T = 112.8 \text{ minutes}$$

Total = 84.6 miles

$$T = 1 \text{ hr } 53 \text{ minutes}$$



Travel Time

Supplier 1 is located at (5, 10) and delivers to the Army Depot at (40, 56.3). This route uses a highway corridor with some winding sections that allows for a speed of 45-65mph.

Total Distance = 105.27 miles

Avg velocity = 60 mph

T= 1hr 45 minutes

Supplier 2 is located at (5, 90) and delivers to the Army Depot at (40, 56.3). This route goes through a congested urban area with an allowed speed of 25–45 mph.

Total Distance = 48.29 miles

Avg velocity = 35 mph

T= 1hr 23 minutes





Probability & Delivery Risk

Goal: Determine the supplier route that has the lowest risk of delivery failure

- *Probability Density Functions:* Represents the likelihood of a continuous random variable falling within a particular range.

Route 1: $r_1(t)$ = Risk based on time (hours)

$$r_1(t) = \frac{1}{50\sqrt{2\pi}} e^{-\frac{(t-600)^2}{5000}}$$

Route 2: $r_2(d)$ = Risk based on distance (miles)

$$r_2(d) = 0.0001 e^{-\frac{1}{10,000}d}$$

Route 3: $r_3(t)$ = Risk based on time (hours)

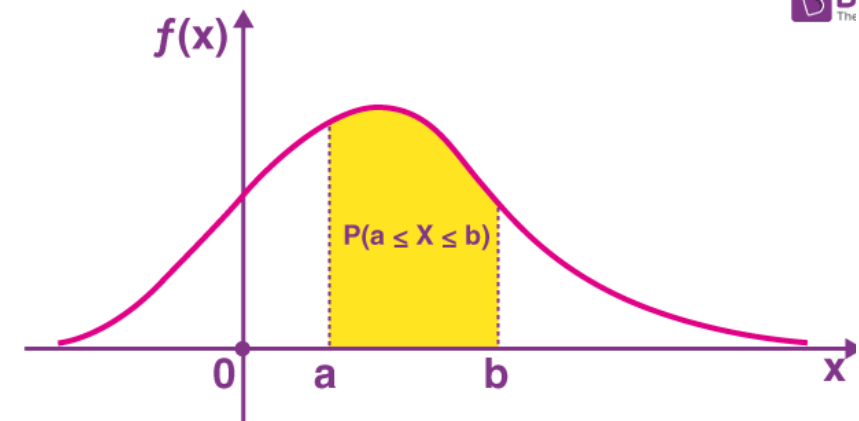
$$r_3(t) = \frac{1}{250\sqrt{2\pi}} e^{-\frac{(t-550)^2}{2(250^2)}}$$



Probability Density Function

Probability = area under the curve

- Probability density = concentration of probability at specific x value
- Describes how risk is distributed
- Higher curve = higher likelihood of failure
- Total area under the curve is 1



$$P(a \leq x \leq b) = \int_a^b r(x) dx$$



Route 3 Probability

$$P(\text{failure during the trip}) = \int_{\text{trip start}}^{\text{trip end}} f(x)dx$$

Route 3: $\int_{\text{trip start}}^{\text{trip end}} r(t)dt$

$$r_3(t) = \frac{1}{250\sqrt{2\pi}} e^{-\frac{(t-550)^2}{2(250^2)}}$$

The distribution is centered at 550 and our trip is only 1.88 hours, the curve is 0 over this integral.

$$P_3 = \int_0^{1.88} r_3(t)dt$$

$$P_3 = \int_0^{1.88} \frac{1}{250\sqrt{2\pi}} e^{-\frac{(t-550)^2}{2(250^2)}} dt$$

$$P_3 = 0.00027$$

The Probability of failure during this trip is nearly zero so it is very low risk.

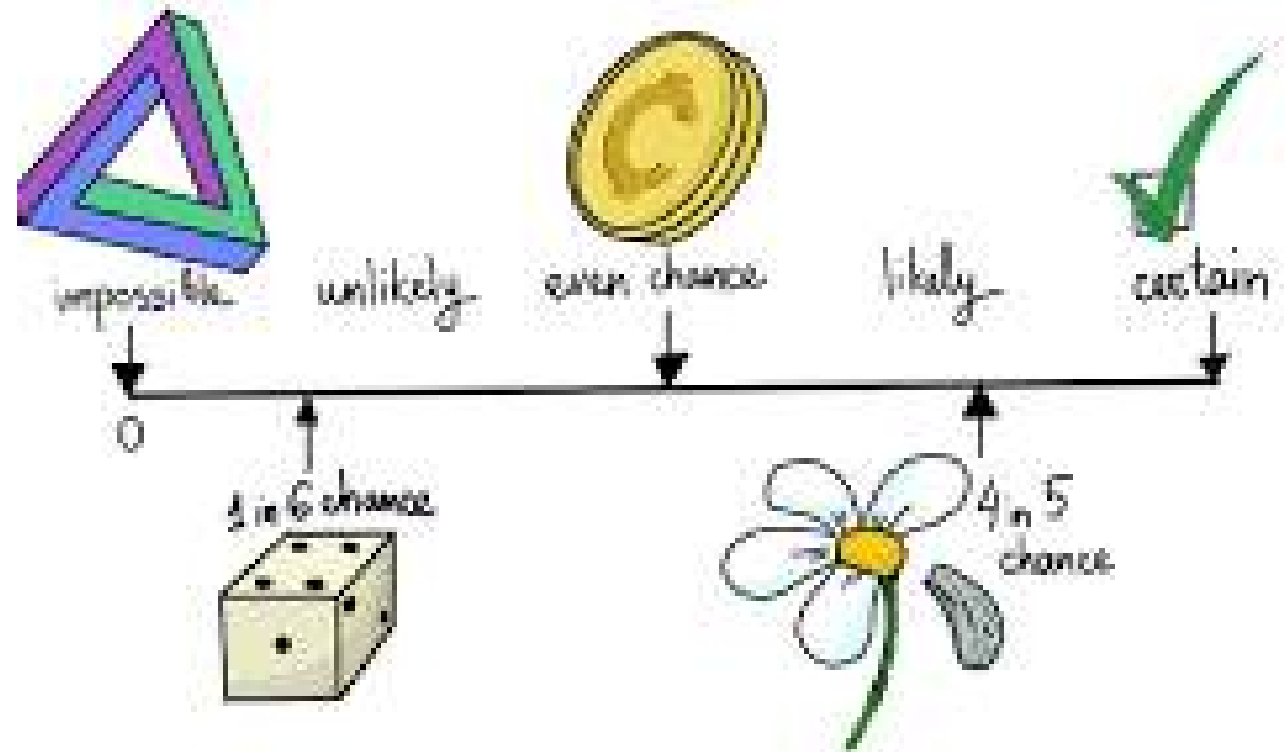




Real Life Application

Probability quantifies uncertainty by analyzing the likelihood of failure which allowed us to compare supplier routes.

Even if one route is faster, a higher probability may make it a worse choice. Risk considerations can change our final recommendations by prioritizing safety over speed.





Probability Conclusion

Route 1: 2.34×10^{-34}

Route 2: 0.00843 (0.843%)

Route 1 is the safest because it has the lowest probability of failure.





Conclusion

- Route one is the best fit for our needs
 - Low probability of risk during transportation
 - 2nd fast time to factory and drives on highways which can be more consistent with traffic patterns than busy urban areas
 - Safer, more reliable choice for our operation

